NORTH
Dakota | Mineral Resources
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29244
NORTH DAKOTA
OIL AND GAS DIVISION
S

May 26, 2020

Mr. Jonathon Travis
Ryan, LLC
2800 Post Oak Boulevard, Suite 4200
Houston, TX 77056

RE: Kline Federal #5300 11-18 3T
Lot 1 Sec. 18, T.153N., R.100W.
McKenzie County, North Dakota
Baker Field
Well File No. 29244
STRIPPER WELL DETERMINATION

Dear Mr. Travis:

Oasis Petroleum North America LLC (Oasis) filed with the North Dakota Industrial Commission – Oil and Gas Division (Commission) on May 12, 2020 an application for a Stripper Well Determination for the above captioned well.

Information contained in the application indicates that the above mentioned well is a stripper well pursuant to statute and rule, and Oasis has elected to designate said well as a stripper well. The well produced from a well depth greater than 10000 feet and was completed after June 30, 2013. During the qualifying period, December 1, 2017 through November 30, 2018, the well produced at a maximum efficient rate or was not capable of exceeding the production threshold. The average daily production from the well was 31.6 barrels of oil per day during this period.

It is therefore determined that the above captioned well qualifies as a "Stripper Well" pursuant to Section 57-51.1-01 of the North Dakota Century Code. This determination is applicable only to the Bakken Pool in and under said well.

The Commission shall have continuing jurisdiction, and shall have the authority to review the matter, and to amend or rescind the determination if such action is supported by additional or newly discovered information. If you have any questions, do not hesitate to contact me.

Sincerely,

David J. McCusker
Petroleum Engineer

Cc: ND Tax Department

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received
SEP 29 2016
ND Oil & Gas Division

Well File No.
29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [ ] Notice of Intent — Approximate Start Date:
- [x] Report of Work Done — Date Work Completed: **September 22, 2016**
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Spill Report
- [ ] Redrilling or Repair
- [ ] Shooting
- [ ] Casing or Liner
- [ ] Acidizing
- [ ] Plug Well
- [ ] Fracture Treatment
- [ ] Supplemental History
- [x] Change Production Method
- [ ] Temporarily Abandon
- [ ] Reclamation
- [ ] Other **Well is now on pump**

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F N L	290 F W L	LOT1	18	153 N	100 W
Field	Pool		County		
Baker	Bakken		McKenzie		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 09/22/2016 the above referenced well was converted to rod pump.

End of Tubing: 2-7/8" L-80 tubing @ 9380'

Pump at 9275.72'

Company		Telephone Number
Oasis Petroleum North America LLC		281-404-9436
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	TX	77002
Signature	Printed Name	
	Jennifer Swenson	
Title	Date	
Regulatory Specialist	September 22, 2016	
Email Address		
jswenson@oasispetroleum.com		

FOR STATE USE ONLY

- [x] Received
- [ ] Approved

Date: 11-2-2016
By: Taylor Roth
Title: TAYLOR ROTH
Engineering Technician

TH

WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

RECEIVED SEP 2015 ND OIL & GAS DIVISION

Well File No. 29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion

☑ Oil Well ☐ EOR Well ☐ Recompletion ☐ Deepened Well ☐ Added Horizontal Leg ☐ Extended Horizontal Leg
☐ Gas Well ☐ SWD Well ☐ Water Supply Well ☐ Other:

Well Name and Number			Spacing Unit Description
Kline Federal 5300 11-18 3T			Sec. 17/18 T153N R100W
Operator: Oasis Petroleum North America		Telephone Number: (281) 404-9591	Field: Baker
Address: 1001 Fannin, Suite 1500			Pool: Bakken
City: Houston	State: TX	Zip Code: 77002	Permit Type: ☐ Wildcat ☑ Development ☐ Extension

LOCATION OF WELL

At Surface		Qtr-Qtr	Section	Township	Range	County
1020 F N L	290 F W L	lot1	18	153 N	100 W	McKenzie

Spud Date	Date TD Reached	Drilling Contractor and Rig Number	KB Elevation (Ft)	Graded Elevation (Ft)
~~March 21, 2015~~ 2-28-15	May 29, 2015	Xtreme 21	2078	2053

Type of Electric and Other Logs Run (See Instructions)
MWD/GR from KOP to TD; CBL from int. TD to surface

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	2130	17 1/2	54.5			1226	0
Vertical Hole	Intermediate	7	0	11101	8 3/4	32			835	2160
Lateral1	Liner	4 1/2	10252	20835	6	13.5			520	10252

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top	Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20840	Perforations	11101	20835	10280		07/17/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft)	Name of Zone (If Different from Pool Name)
Lateral 1- 11101' to 20835'	

Date Well Completed (SEE INSTRUCTIONS)	Producing Method	Pumping-Size & Type of Pump	Well Status (Producing or Shut-In)
August 29, 2015	Flowing		Producing

Date of Test	Hours Tested	Choke Size	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas
08/29/2015	24	48 /64		465	409	6179	°	Sold

Flowing Tubing Pressure (PSI)	Flowing Casing Pressure (PSI)	Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio
	1000		465	409	6179	880

GEOLOGICAL MARKERS

Formation	MD (Ft)	TVD (Ft)
Dakota	5510	5510
Kibbey Lime	8363	8362
First Charles Salt	8513	8512
UB	9129	9128
Base Last Salt	9201	9200
Ratcliffe	9250	9249
Mission Canyon	9432	9431
Lodgepole	9982	9981
False Bakken	10771	10693
Scallion		
Upper Bakken	10789	10703
Middle Bakken	10822	10722
Lower Bakken	10907	10762
Pronghorn	10940	10775
Three Forks	10994	10791

PLUG BACK INFORMATION

Well Bore	Type of Plug	Top (Ft)	Bottom (Ft)	Sacks Cement

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)

Drill Pipe Recovery

Sample Chamber Recovery

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
07/17/2015	Three Forks	11101	20835	36	208220	Barrels

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)
Sand Frac		4183070	9386	76.0

Details
100 Mesh White: 301340
40/70 White: 1556480
30/50 White: 1742280
30/50 Resin Coated: 511100
20/40 Resin Coated: 71870

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units

Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)	Maximum Treatment Rate (BBLS/Min)

Details

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address **jswenson@oasispetroleum.com**	Date **09/29/2015**
Signature	Printed Name **Jennifer Swenson**	Title **Regulatory Specialist**

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

TH

RECEIVED OCT 2015 ND OIL & GAS DIVISION

Well File No. **29244**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [ ] Notice of Intent — Approximate Start Date:
- [x] Report of Work Done — Date Work Completed: **October 7, 2015**
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other **Well is now on pump**
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [x] Change Production Method
- [ ] Reclamation

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F N L	290 F W L	LOT1	18	153 N	100 W
Field	Pool	County			
Baker	Bakken	McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 10/07/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 10215.85'

Pump: ESP @ 9965'

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9436**
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**
Signature	Printed Name	
	Jennifer Swenson	
Title	Date	
Regulatory Specialist	**October 14, 2015**	
Email Address		
jswenson@oasispetroleum.com		

FOR STATE USE ONLY

[x] Received	[ ] Approved
Date 10/26/2015	
By Taylor Roth	
Title **TAYLOR ROTH** Engineering Technician	

AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – Form 8

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

RECEIVED
SEP 2015
ND OIL & GAS DIVISION

TH

Well File No.
29244
NDIC CTB No.
To be assigned

129244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number	Qtr-Qtr	Section	Township	Range	County
KLINE FEDERAL 5300 11-18 3T	LOT1	18	153	100	Mckenzie

Operator	Telephone Number	Field
Oasis Petroleum North America LLC	(281) 404-9573	BAKER

Address	City	State	Zip Code
1001 Fannin, Suite 1500	Houston	TX	77002

Name of First Purchaser	Telephone Number	% Purchased	Date Effective
Oasis Petroleum Marketing LLC	(281) 404-9627	100%	August 22, 2015

Principal Place of Business	City	State	Zip Code
1001 Fannin, Suite 1500	Houston	TX	77002

Field Address	City	State	Zip Code

Transporter	Telephone Number	% Transported	Date Effective
Hiland Crude, LLC	(580) 616-2058	75%	August 22, 2015

Address	City	State	Zip Code
P.O. Box 3886	Enid	OK	73702

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	**% Purchased**	**Date Effective**
Other Transporters Transporting From This Lease	**% Transported**	**Date Effective**
Power Crude Transport	25%	August 22, 2015
Other Transporters Transporting From This Lease	**% Transported**	**Date Effective**
		August 22, 2015

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date August 22, 2015
Signature Dina B	Printed Name Dina Barron	Title Mktg. Contracts Administrator

Above Signature Witnessed By:

Signature	Printed Name	Title
	Jeremy Harris	Marketing Scheduler

FOR STATE USE ONLY

Date Approved
SEP 18 2015
By
Evie Roberson
Title
Oil & Gas Production Analyst

Industrial Commission of North Dakota
Oil and Gas Division
Verbal Approval To Purchase and Transport Oil

Well or Facility No **29244**

Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	**Todd Hanson**	**(701) 577-1632**

WELL INFORMATION

Well Name
KLINE FEDERAL 5300 11-18 3T

Inspector
Richard Dunn

Well Location	QQ	Sec	Twp	Rng
	LOT1	**18**	**153 N**	**100 W**

County
MCKENZIE

Field
BAKER

Footages
1020 Feet From the **N** Line
290 Feet From the **W** Line

Pool
BAKKEN

Date of First Production Through Permanent Wellhead — **This Is Not The First Sales**

PURCHASER / TRANSPORTER

Purchaser	Transporter
Kinder Morgan	**Kinder Morgan**

TANK BATTERY

Single Well Tank Battery Number :

SALES INFORMATION **This Is Not The First Sales**

ESTIMATED BARRELS TO BE SOLD	ACTUAL BARRELS SOLD	DATE
15000 BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	
BBLS	BBLS	

DETAILS

Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date	**8/25/2015**
Date Approved	**8/25/2015**
Approved By	**Richard Dunn**

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED JUL 2015 ND OIL & GAS DIVISION

Well File No. **29244**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **July 14, 2015**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [ ] Other **Waiver from tubing/packer requirement**
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F **N** L	**290** F **W** L	**LOT 1**	**18**	**153 N**	**100 W**
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will ise a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by montioring equipement;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9436**
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**
Signature	Printed Name	
	Jennifer Swenson	
Title	Date	
Regulatory Specialist	**July 14, 2015**	
Email Address		
jswenson@oasispetroleum.com		

FOR STATE USE ONLY

[ ] Received	[x] Approved
Date	July 31, 2015
By	
Title	PETROLEUM ENGINEER

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED
AUG 10 2015
ND OIL & GAS DIVISION

Well File No. 29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: August 9, 2015
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [x] Other: Change well status to CONFIDENTIAL
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F N L	290 F W L	Lot 1	18	153 N	100 W
Field	Pool	County			
Baker	BAKKEN	McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

This well has not been completed

OFF CONFIDENTIAL 2/10/16.

Company	Telephone Number
Oasis Petroleum North America LLC	281-404-9436

Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	TX	77002

Signature	Printed Name
	Jennifer Swenson
Title	Date
Regulatory Specialist	August 10, 2015
Email Address	
jswenson@oasispetroleum.com	

FOR STATE USE ONLY

- [ ] Received
- [x] Approved

Date: 8/12/15

By: Alice D. Webber

Title: Engineering Technician

SUNBURST CONSULTING
A GEOLOGY SERVICE

Oasis Petroleum North America LLC

Kline Federal 5300 11-18 3T

1,020' FNL & 290' FWL
Lot 1 Section 18, T153N, R100W
Baker Field / Three Forks
McKenzie County, North Dakota

BOTTOM HOLE LOCATION:
875.04' south & 9,941.16' east of surface location or approx.
1,895.04' FNL & 314.95' FEL, SE NE Section 17, T153N, R100W

Prepared for:
Curtis Johnson
Oasis Petroleum
North America, LLC.
1001 Fannin, Suite 1500
Houston, TX 77002

Prepared by:
Dillon Johnson, Matt Hegland
PO Box 80507; Billings, MT 59108
(406) 259-4124
geology@sunburstconsulting.com
www.sunburstconsulting.com

WELL EVALUATION

Oasis Petroleum North America, LLC

Kline Federal 5300 11-18 3T

Figure 1. Xtreme 21 drilling the Oasis Petroleum North America, LLC, *Kline Federal 5300 11-18 3T* well during May of 2015 in Williams County, North Dakota.

INTRODUCTION

The Oasis Petroleum, North America LLC, *Kline Federal 5300 11-18 3T* well is located approximately 4 miles south of Williston in McKenzie County, North Dakota [Lot 1 Section 18, T153N, R100W]. The *Kline Federal 5300 11-18 3T* is a horizontal first bench, Three Forks well within the Williston Basin consisting of one 9,739' lateral drilled toward the east. The wells surface, vertical and curve sections were previously drilled, and this evaluation will consist of primarily operations and observations made during the lateral section.

ENGINEERING

The *Kline Federal 5300 11-18 3T* surface, vertical and curve sections were completed earlier in 2015 by a separate drilling rig. Extreme 21 returned to location in May of 2015 to drill the laterals of the *Kline Federal 5300 11-18 3T*, *3T*, *2B* **(Figure 1)**. The *Kline Federal 5300 11-18 3T* was re-entered on May 22, 2015. To start the lateral, a 6″ Smith PDC bit, Ryan Directional Services MWD tool, and a 1.5° adjustable Ryan Directional Services mud motor began drilling operations. The complete bottom-hole assembly summary is tabulated as an appendix (BHA Record) to this report. The well reached total depth at 20,840' MD at 12:45 CDT on May 28, 2015, generating an exposure of 9,739' linear feet of 6″ hole through the Three Forks target interval. The <u>bottom-hole location</u> (BHL) lies 875.04' south & 9,941.16' east of surface location or approximately 1,895.04' FNL & 314.95' FEL, SE NE Section 17, T153N, R100W.

GEOLOGY

Lithology

Sunburst Geology, Inc. was not present for the vertical and curve holes therefore there is no information for formations and members prior to intermediate casing point.

Three Forks Formation

While the curve was being drilled the top of the Three Forks Formation was penetrated at 10,940’ MD, 10,775’ TVD (-8,706’). The *Kline Federal 5300 11-18 3T* only targeted the upper Three Forks, 1st bench. This bench consisted of two main lithology types; most commonly observed was a dolomite mudstone with lesser amounts of shale **(Figure 2)**. The targeted dolomite was most commonly associated with cool gamma markers ranging from 90-50 API counts. The mudstone was light brown to off white in color and was primarily firm to friable but on occasions hard. The mudstone often displayed trace, very fine intercrystalline porosity, trace to rare *light brown spotty oil staining*, and trace amounts of disseminated pyrite and trace to rare amounts of nodular pyrite. Less commonly observed in samples was a light green to light blue-green shale, associated with 110-150 API count gamma signatures. This shale was firm (occasionally soft) with an earthy texture, displayed a sub-blocky texture and contained trace amounts of both disseminated and nodular pyrite, more than that seen throughout the dolomite intervals. Throughout the lateral, samples observed in the upper portion of the target interval displayed more shale than those seen at the base. At the very base of the target zone was a claystone interval. This claystone is what separates the first and second benches of the Three Forks Formation.

Figure 2. Dolomite and trace amounts of shale observed throughout the target interval.

Geosteering

The potential pay zone was identified by evaluating gamma data collected while drilling the *Oasis Petroleum, Kline Federal 5300 41-18 11T2*. The target was determined to be 10' thick, lie 15' below the Pronghorn Member and have the base bounded by a 12' claystone interval. The target zone had a reliable, higher gamma marker at the very top of target, cooler markers throughout the middle of section, and higher gamma counts at the base representing the claystone **(Figure 3)**.

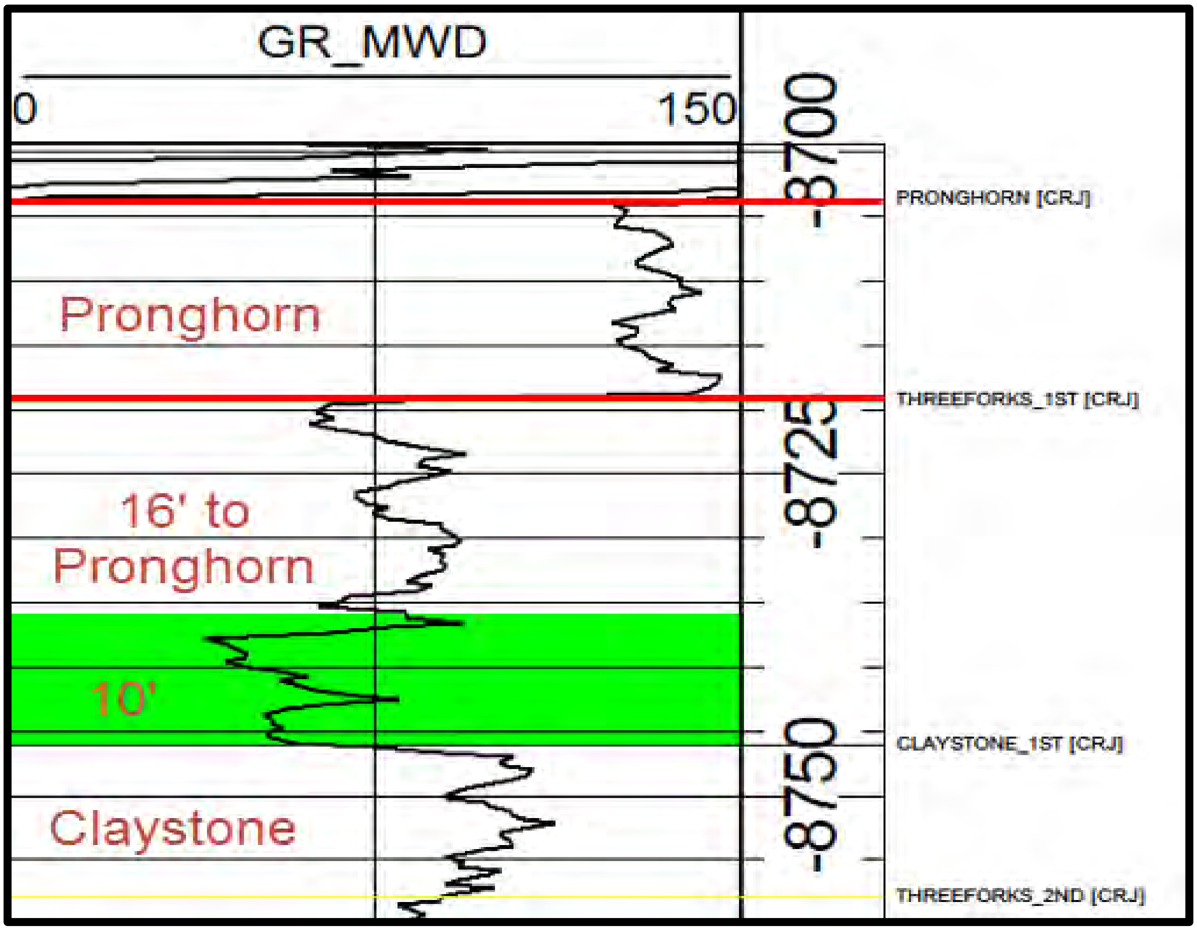


Figure 3. Gamma characteristics observed throughout the Three Forks, first bench while drilling the *Oasis Petroleum, Kline Federal 5300 41-18 11T2*.

Structure maps provided by Oasis Petroleum indicated that for the first several thousand feet dip rates would trend downward before flattening near the halfway point and turning up at the toe of the lateral. Upon drilling out of intermediate casing, the plan was to maintain an inclination so the top of the claystone could be confirmed, and be used throughout the rest of the lateral if stratigraphic location was ever in question. As anticipated, structure dipped downward for the first several thousand feet before leveling and turning up around 15,500' MD. Shortly after the dip reversal, formation began to dip downwards again for approximately 1,000', to a depth of ~17,600' MD. Finally, as structure maps indicated, the formation leveled and began to dip favorably upward through total depth **(Figure 4)**.

Hydrocarbon Shows

Oil and gas shows at the beginning of the lateral were relatively low, but as well-bore exposure increased, from ~13,900' to 20,840' MD, were very promising. Gas values rose gradually, with values exceeding 1,500 units toward the middle and end of the lateral section **(Figure 4)**. The most promising oil staining observed in samples was observed in the upper half of the target. Consistently, throughout the lateral, this interval displayed trace amounts of *spotty to occasionally even, light to medium brown oil staining*. This upper portion also displayed the more favorable fluorescent cuts. Cuts observed in the upper portion of the target zone reacted moderately fast, and streamed pale yellow to light green. The lower half of the target did not display as encouraging hydrocarbon shows as the upper half, but were still favorable. The lower half displayed almost 100% dolomite that displayed *trace light brown spotty oil staining*. The least promising gas and oil shows were observed when making contact with the claystone, with gas shows rarely exceeded 400 units. Samples throughout the claystone also displayed little, and often no, oil staining. Over the course of the lateral there was no oil observed at the shakers and there was never a need vent or circulate fluid through the gas buster.

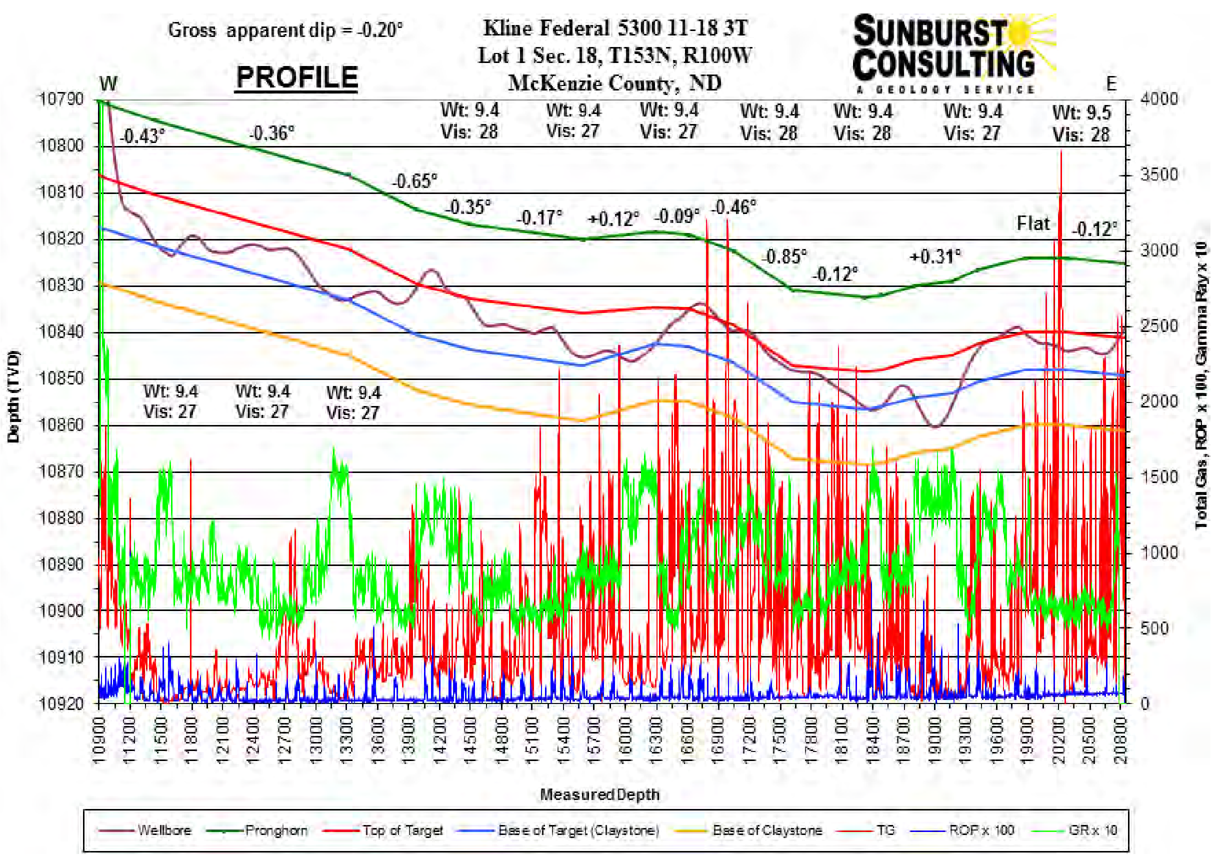


Figure 4. Cross-sectional interpretation of the *Kline Federal 5300 11-18 3T* borehole with estimated dip based on lithology, MWD data, drill rate, and regional structural data.

SUMMARY

The Oasis Petroleum North America, LLC, *Kline Federal 5300 11-18 3T* lateral hole was drilled by Xtreme 21, from re-entry to total depth in 7 days. The well reached a total depth of 20,840' MD on May 28, 2015. Geologic data, hydrocarbon gas measurements, and sample examination indicate an encouraging hydrocarbon bearing system in the Three Forks. Positive background, connection and survey gas shows were recorded throughout the lateral. Samples consisted of cream to light gray dolomite with very fine intercrystalline porosity and trace light brown oil staining, interbedded with blue-green shale. Samples yielded *faint, light green, moderate, diffuse to streaming cut fluorescence.* The well was within the target zone for 84% of the lateral, successful in exposing 9,739' of potentially productive first bench, Three Forks. The well currently awaits completion operations.

Respectfully submitted,
Dillon Johnson
Well-site Geologist
Sunburst Consulting, Inc.
May 28, 2015

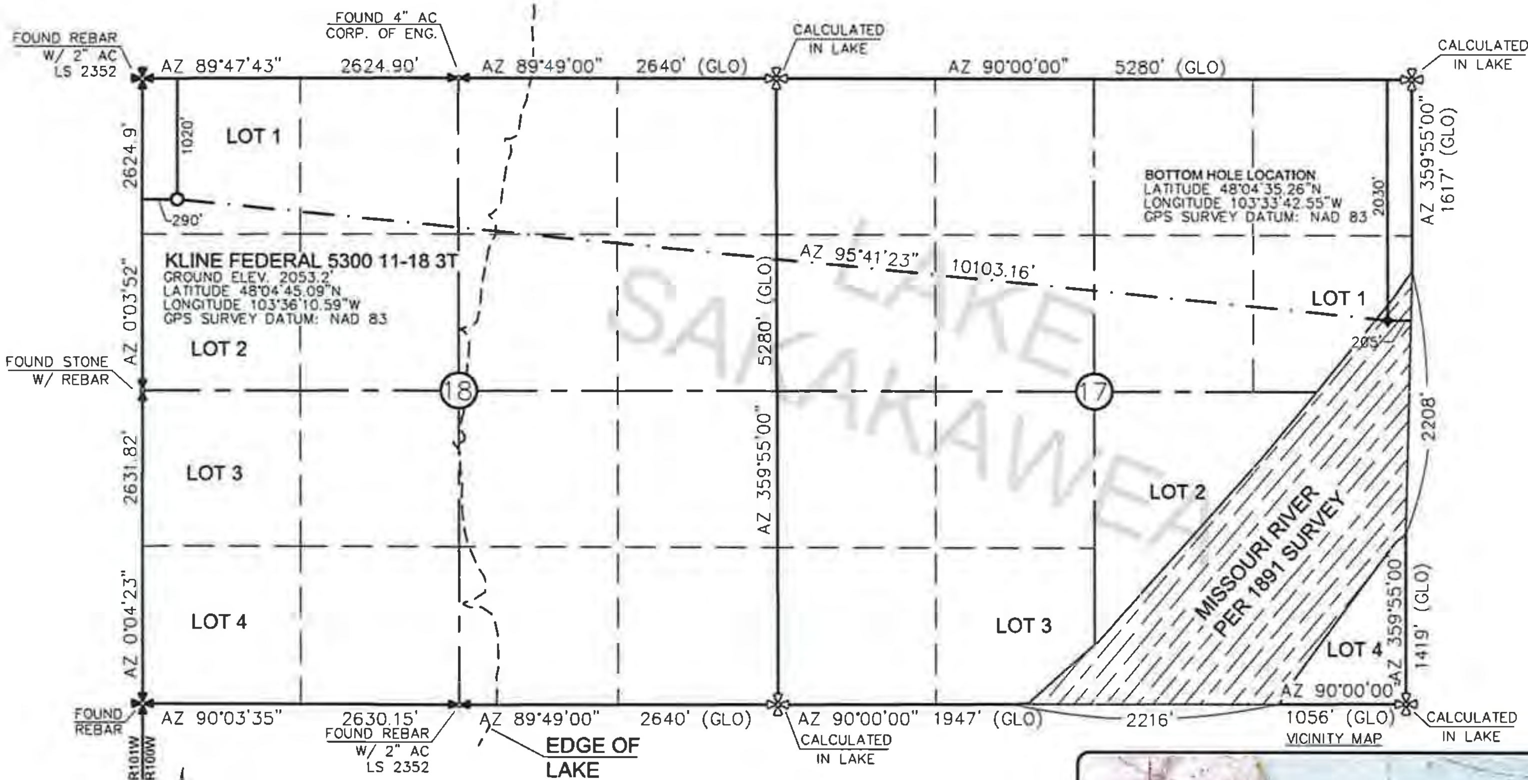
WELL DATA SUMMARY

OPERATOR:	**Oasis Petroleum North America LLC**
ADDRESS:	1001 Fannin, Suite 1500 Houston, TX 77002
WELL NAME:	**Kline Federal 5300 11-18 3T**
API #:	33-053-06225-00-00
WELL FILE #:	29244
SURFACE LOCATION:	1,020' FNL & 290' FWL Lot 1 Section 18, T153N, R100W
FIELD/ OBJECTIVE:	Baker Field / Three Forks
COUNTY, STATE	McKenzie County, North Dakota
BASIN:	Williston
WELL TYPE:	Three Forks Horizontal
ELEVATION:	GL: 2,053' KB: 2,069'
RE-ENTRY DATE:	May 22, 2015
BOTTOM HOLE LOCATION:	875.04' south & 9,941.16' east of surface location or approx. 1,895.04' FNL & 314.95' FEL, SE NE Section 17, T153N, R100W
CLOSURE COORDINATES:	Closure Azimuth: 95.03° Closure Distance: 9,979.59'
TOTAL DEPTH / DATE:	20,840' on May 28, 2015 84% within target interval
TOTAL DRILLING DAYS:	7 days
CONTRACTOR:	Xtreme #21

PUMPS:	Continental-Emsco F1600 (12" stroke length; 5 ½" liners) Output: 0.0838 bbls/stk at 95% efficiency
TOOLPUSHERS:	Josh Barkell, Allen Franklin
FIELD SUPERVISORS:	Miles Gordon, Derek Ramsdell
CHEMICAL COMPANY:	MiSwaco
MUD ENGINEER:	Justin McNicholas
MUD TYPE:	Salt water in the lateral
PROSPECT GEOLOGIST:	Curtis Johnson
WELLSITE GEOLOGISTS:	Dillon Johnson, Matt Hegland
GEOSTEERING SYSTEM:	Sunburst Digital Wellsite Geological System
ROCK SAMPLING:	50' from 11,250' - 20,840'
SAMPLE EXAMINATION:	Binocular microscope & fluoroscope
SAMPLE CUTS:	Trichloroethylene
GAS DETECTION:	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph Serial Number(s): ML-488
DIRECTIONAL DRILLERS:	RPM Consulting, Inc. Miles Gordon, Derek Ramsdell
MWD:	Gyro/data in vertical in curve Ryan Directional Service in lateral Carlo Marrufo, David Foley
CASING:	Surface: 9 5/8" 36# J-55 set to 2,138' Intermediate: 7" 29# & 32# HCP-110 set to 11,101'
SAFETY/ H_2S MONITORING:	Oilind Safety

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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- MONUMENT – RECOVERED
- MONUMENT – NOT RECOVERED

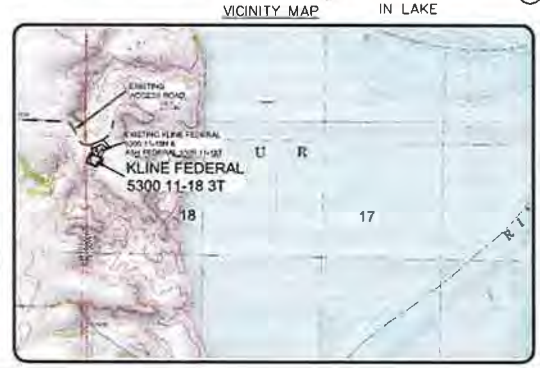
STAKED ON 6/18/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE 6/19/14
NORTH DAKOTA

VICINITY MAP



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425 East Main Street
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OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

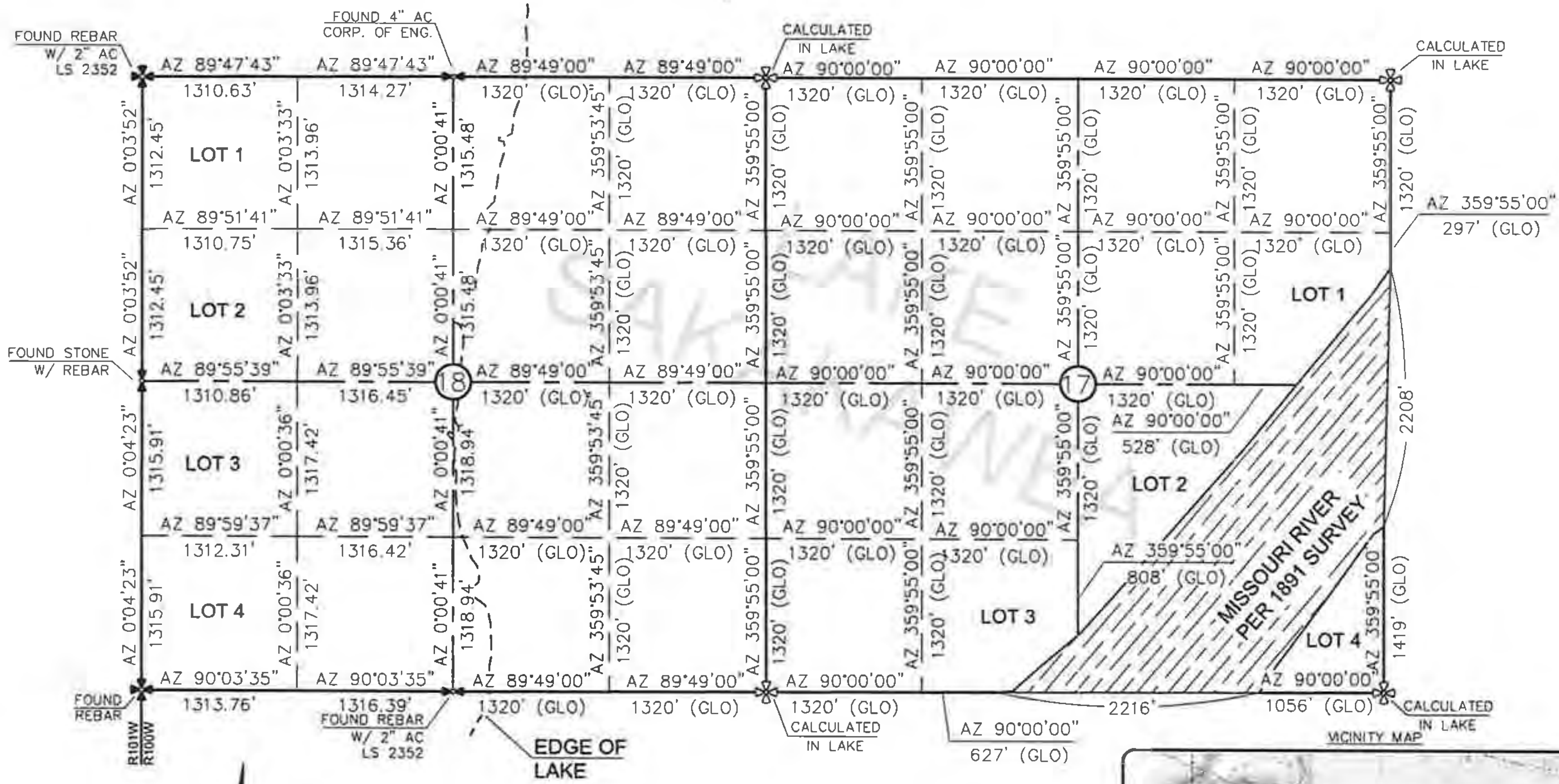
Revision No.	Date	By	Description
REV 1	6/18/14	BHH	MOVED WELLS

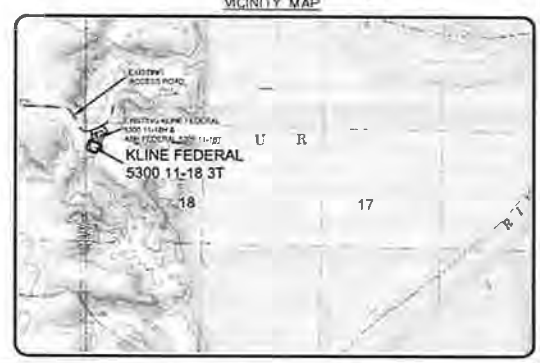
1/8

SHEET NO.

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA





0 1000
1" = 1000'

MONUMENT – RECOVERED
MONUMENT – NOT RECOVERED

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ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE 6/19/14
NORTH DAKOTA

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Sidney, Montana 59270
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Fax (406) 433-5618
www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 17 & 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

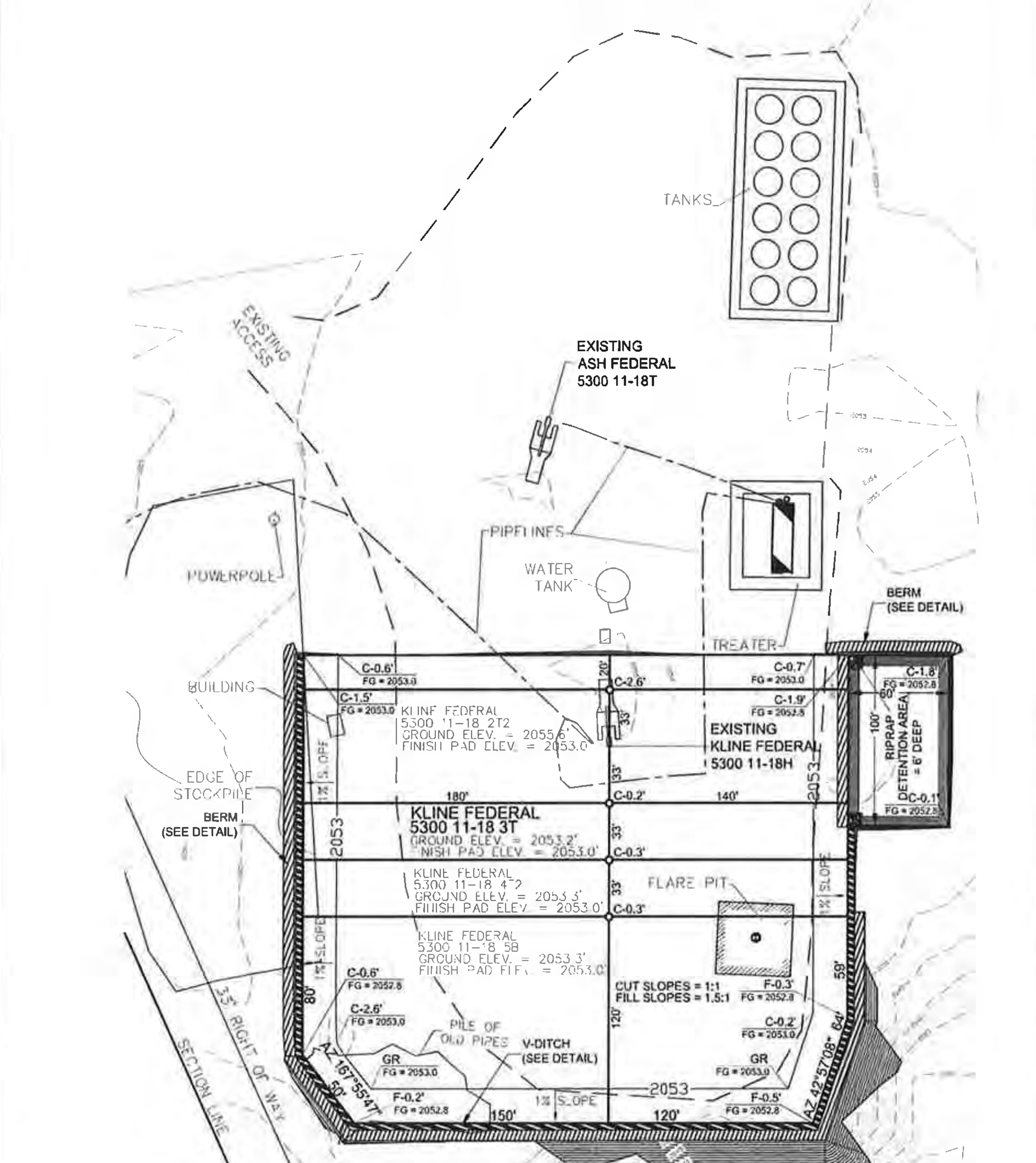
Drawn By: B.H.H. | Project No.: S14-09-127.01
Checked By: D.D.K. | Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/15/14	BHH	MOVED WELLS

INTERSTATE ENGINEERING

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE 1: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE 2: All existing facilities to be removed on construction expansion.

NOTE 3: Cuttings will be hauled to approved disposal site.

V-DITCH DETAIL

2:1 2:1 1.5' min

2.0' min

— BERM

— DITCH

Proposed Contours

Original Contours

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REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE 6/19/14
NORTH DAKOTA

0 60
1" = 60'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

3/8

INTERSTATE ENGINEERING

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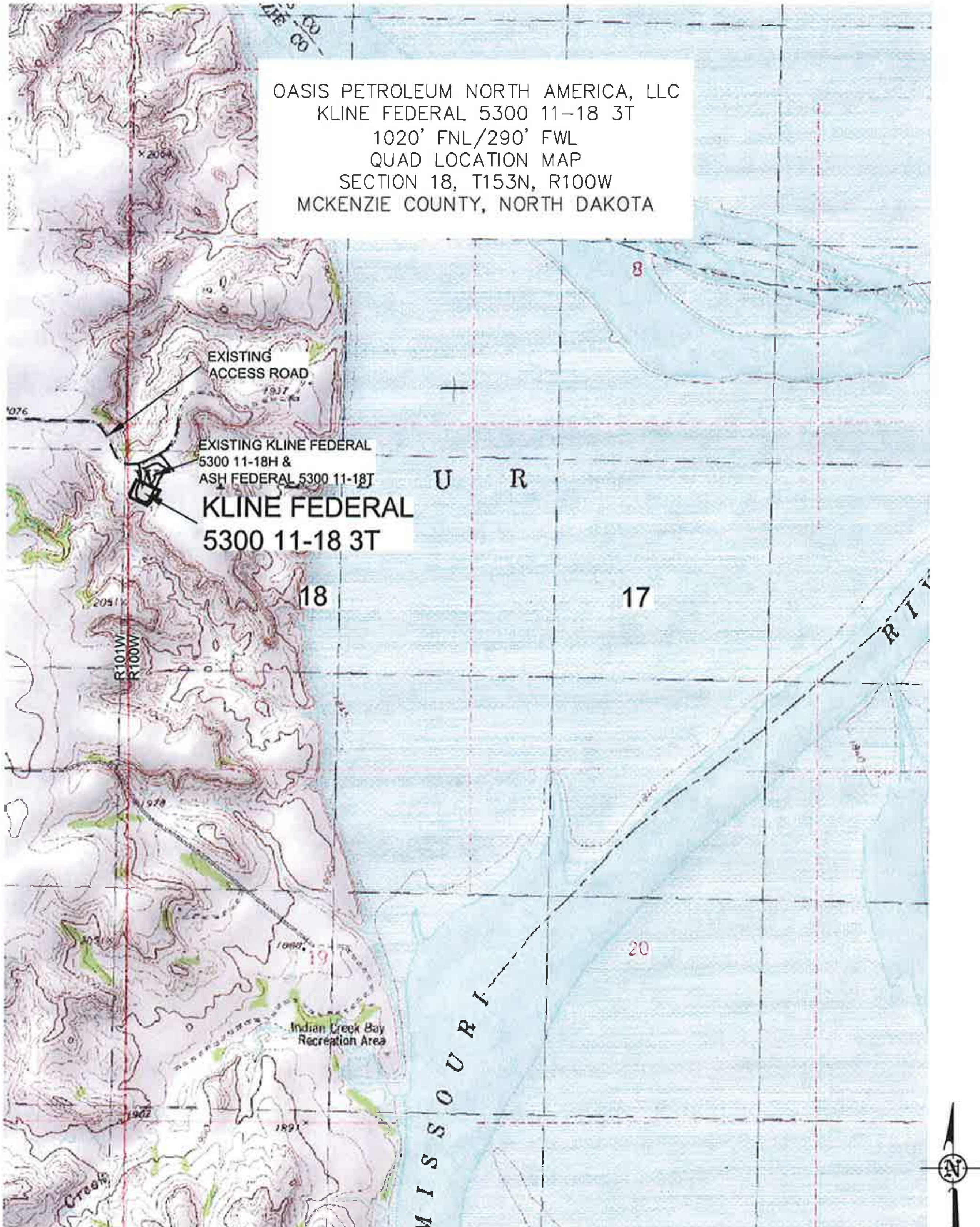
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description



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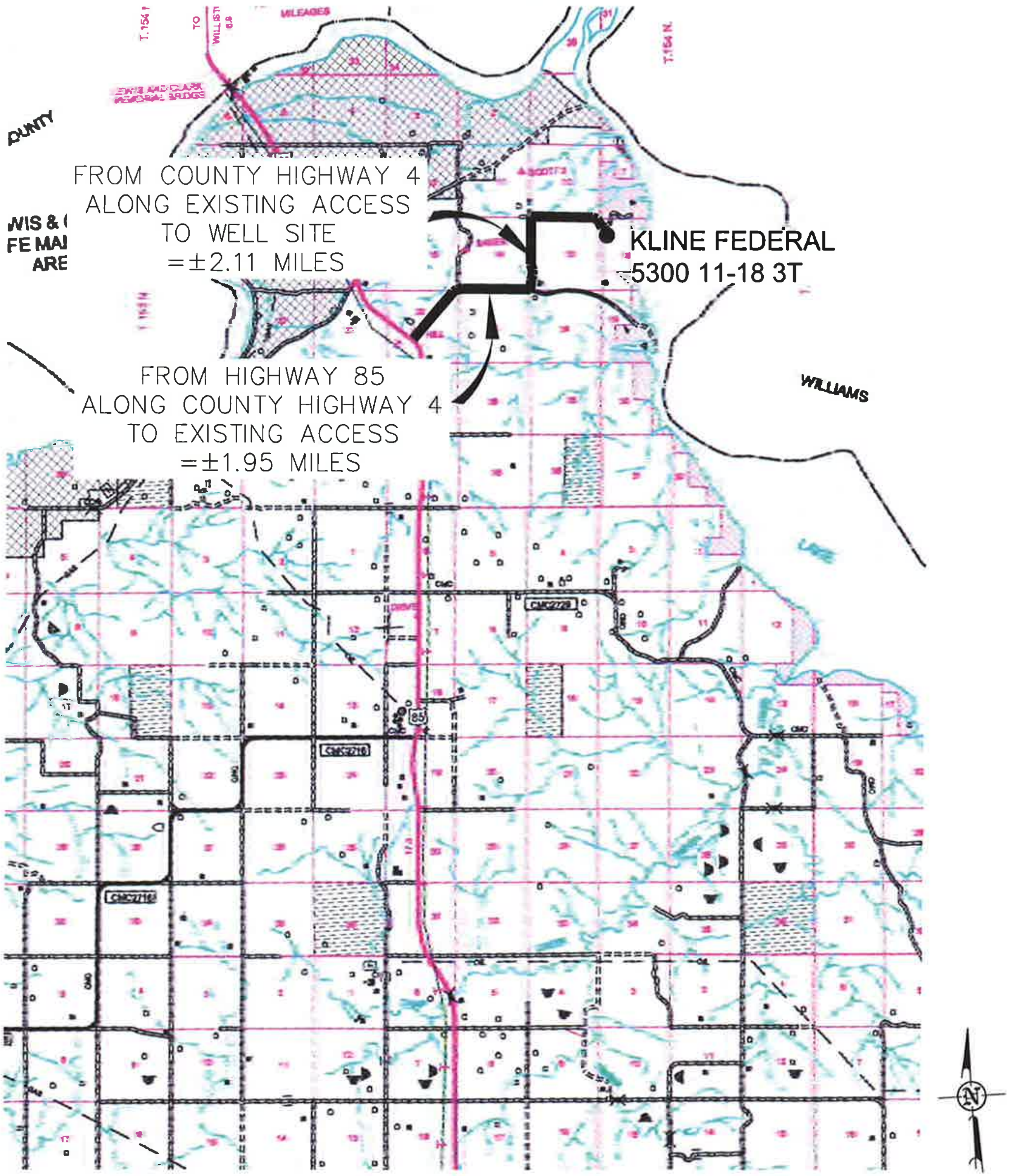
OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description
REV 1	6/18/14	BHH	MOVED WELLS

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

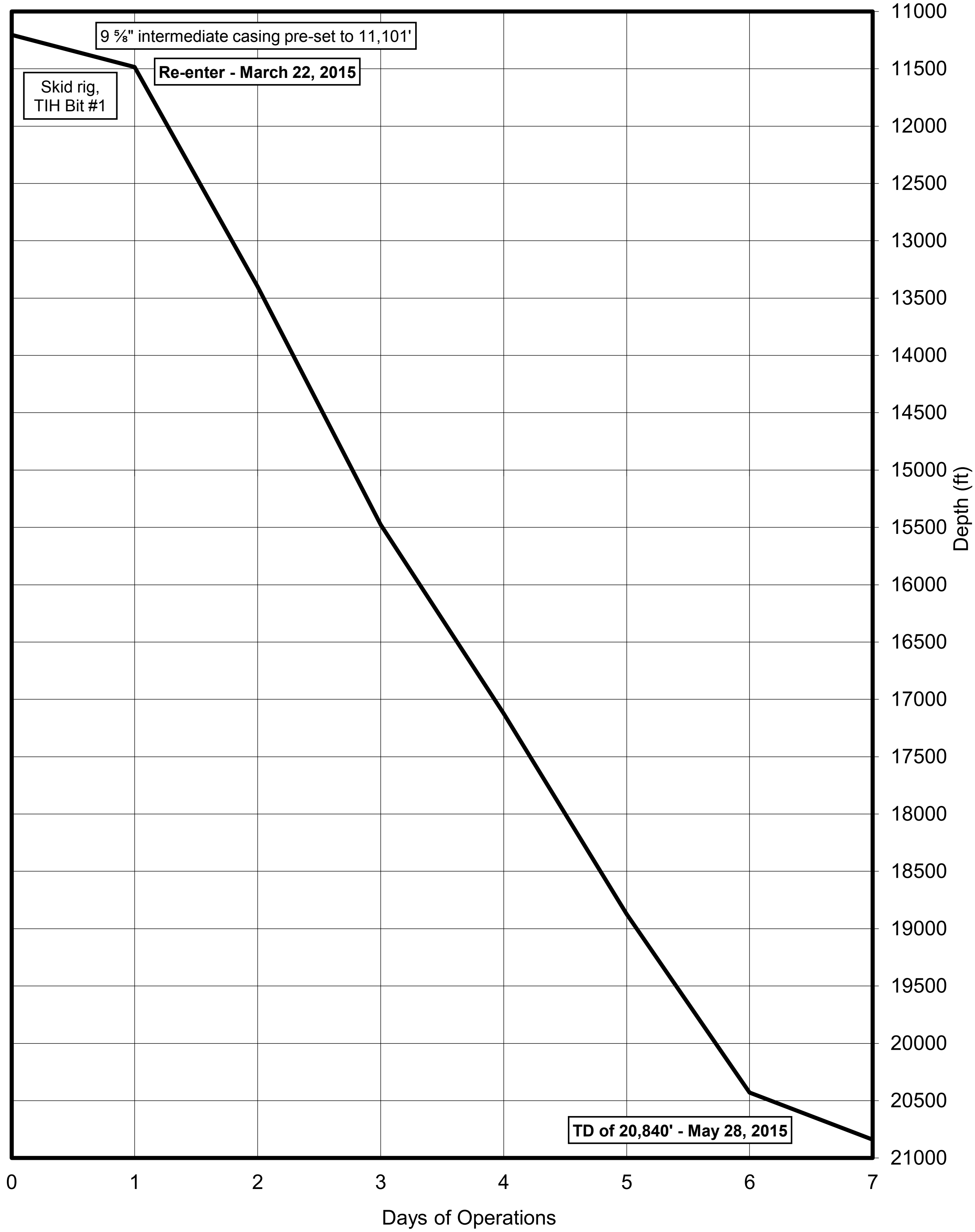
Drawn By: B.H.H. | Project No.: S14-09-127.01
Checked By: D.D.K. | Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	8/10/14	BHH	MOVED WELLS

TIME VS. DEPTH

Oasis Petroleum North America LLC

Kline Federal 5300 11-18 3T



MORNING REPORT SUMMARY

Day	Date 2015	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity Summary	Formation
0	5/22	11,204'	-	-	-	-	-	-	-	-	-	-	Skid rig from Kline Federal 11-18 5B.	-
1	5/23	11,485'	281'	1	20	40	20	260	3500	90	0	325	Skid rig. Test BOP. Pick up BHA. TIH. Drill out cement. FIT test. Slip and cut. Drill F/11,204'-11,485'.	Three Forks
2	5/24	13,400'	1,915'	1	19	40	40	259	4050	0	90	324	Drill F/11,485'-12,170'. Lubricate rig. Drill F/12,170'-13,400'.	Three Forks
3	5/25	15,476'	2,076'	1	25	40	50	253	3900	0	88	316	Drill F/13,400'-14,355'. Lubricate rig. Drill 14,355'-15,476'.	Three Forks
4	5/26	17,124'	1,648'	1	27	40	57	253	4300	0	88	316	Drill F/15,476'-16,231'. Lubricate rig. Drill F/16,231'-17,189'.	Three Forks
5	5/27	18,875'	1,751'	1	28	39	70	244	4000	85	0	305	Drill F/17,189'-18,148'. Lubricate rig. Drill 18,148'-18,875'.	Three Forks
6	5/28	20,430'	1,555'	1	28	40	75	238	4000	0	83	298	Drill F/18,875'-19,490'. Lubricate rig. Drill F/19,490'-20,430'.	Three Forks
7	5/29	20,840'	410'	1	28	40	75	238	4000	0	83	298	Drill F/ 20,430'-20,840'. Circulate and condition mud. TOOH for short trip. TIH. Circulate and conditon mud. TOOH for liner run	Three Forks

DAILY MUD SUMMARY

Date 2015	Mud Depth	Mud WT (ppg)	VIS (sec/ qt)	PV (cP)	YP (lbs/ 100 ft^2)	Gels (lbs/ 100 ft^2)	600/ 300	NAP/ H_20 (ratio)	NAP/ H_20 (% by vol)	Cor. Solids (%)	Oil/ H_20 (%)	pH	Cl^- (mg/L)	LGS/ HGS (%)	Gain/ Loss (bbls)
05/23	11,605'	9.5	27	1	1	1/1/1	3/2	-/100	-/94	0.8	-/94	7	118k	0.8/7.6	-
05/24	13,200'	9.4	27	-	-	-		-	-	-	-	-	-	-	-
05/25	15,828'	9.5	27	1	1	1/1/1	3/2	-/100	-/94	1	-/94	7.5	115k	1/9.5	-
05/26	17,600'	9.4	28	-	-	-		-	-	-	-	-	-	-	-
05/27	19,107'	9.45	27	1	1	1/1/1	3/2	-/100	-/94	1.1	-/94	8	108k	1.1/10.2	-
05/28	20,200'	9.4	27	-	-	-		-	-	-	-	-	-	-	-

BOTTOM HOLE ASSEMBLY RECORD

Bit Data											Motor Data				Reason For Removal
Bit #	Size (in.)	Type	Make	Model	Depth In	Depth Out	Footage	Hours	Σ hrs	Vert. Dev.	Make	Model	Bend	Rev/Gal	
1	6	PDC	Smith	Z613	11,204'	20,840'	9,636'	111	111	Lateral	Ryan	-	1.5	0.8	TD lateral

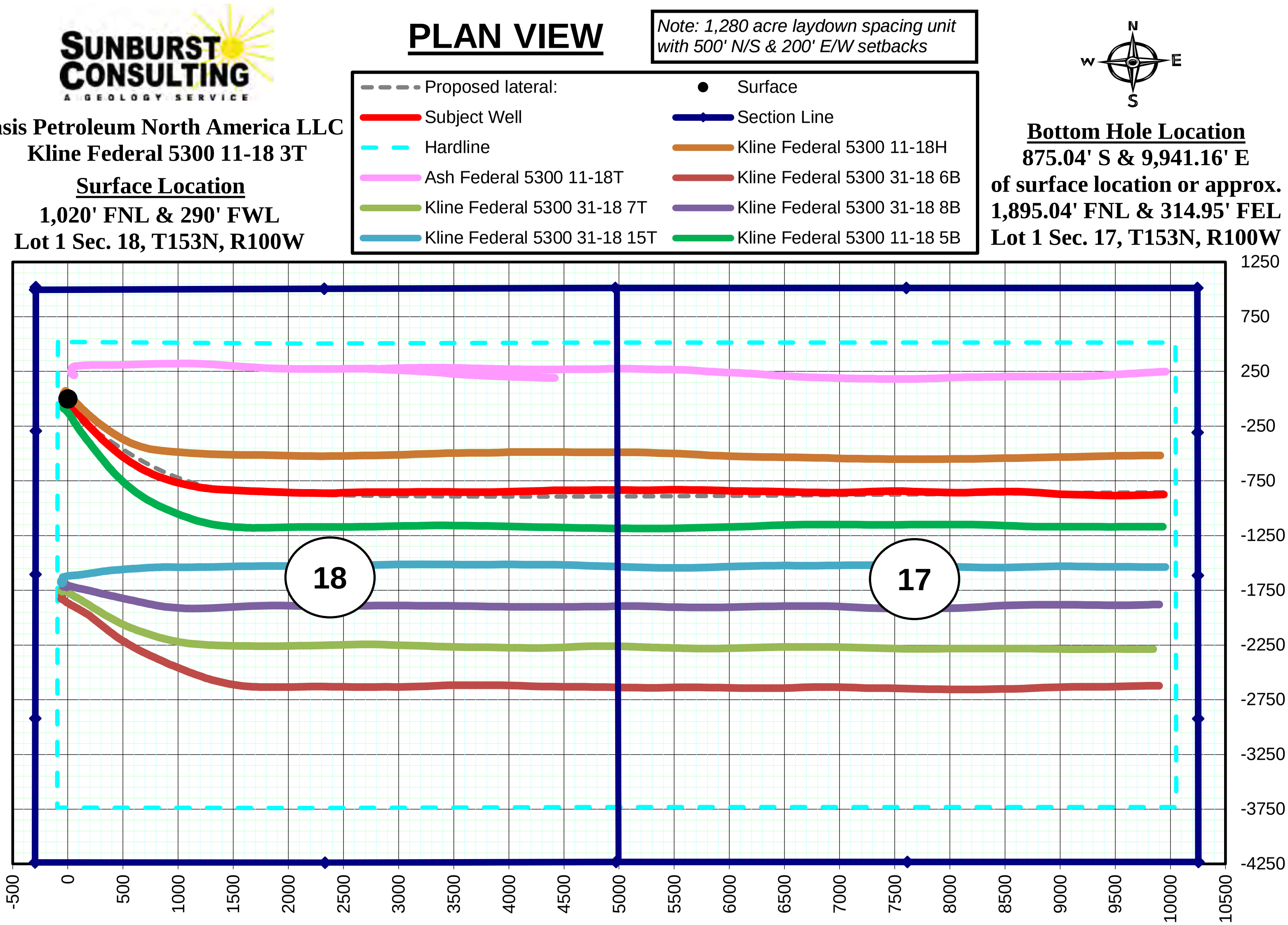
Oasis Petroleum North America LLC
Kline Federal 5300 11-18 3T

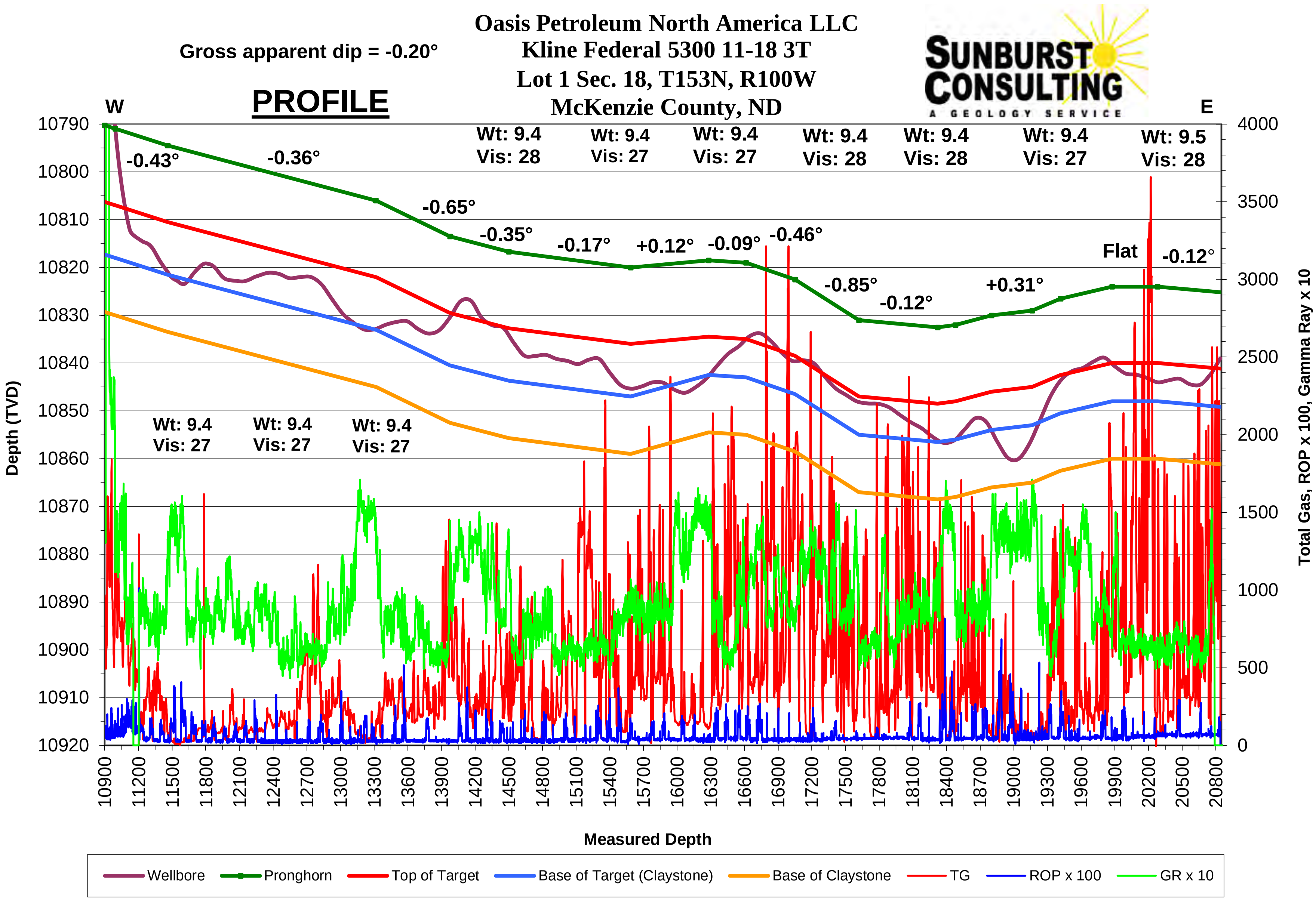
Surface Location
1,020' FNL & 290' FWL
Lot 1 Sec. 18, T153N, R100W

PLAN VIEW

Note: 1,280 acre laydown spacing unit with 500' N/S & 200' E/W setbacks

Bottom Hole Location
875.04' S & 9,941.16' E
of surface location or approx.
1,895.04' FNL & 314.95' FEL
Lot 1 Sec. 17, T153N, R100W

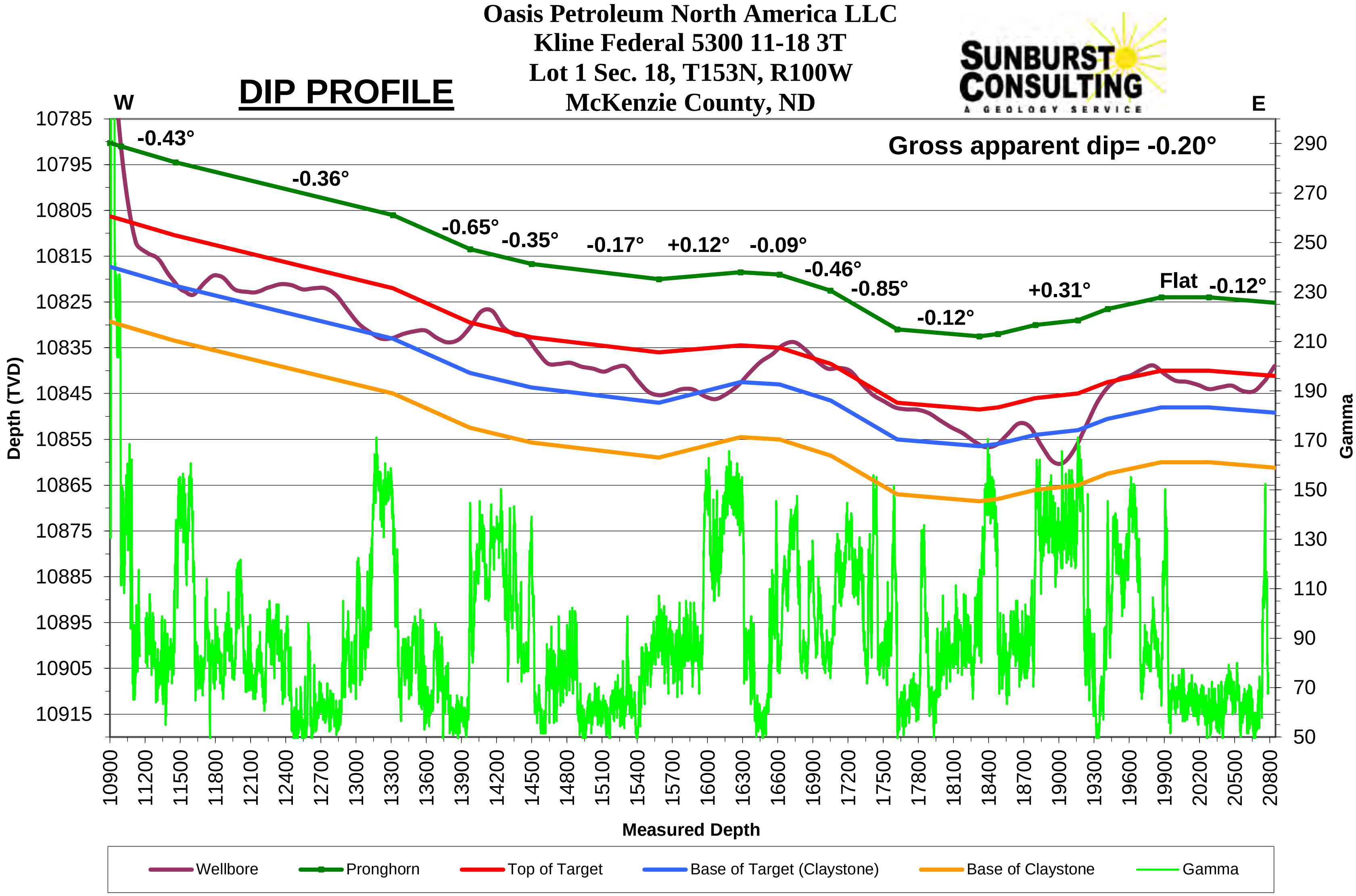



Oasis Petroleum North America LLC
Kline Federal 5300 11-18 3T
Lot 1 Sec. 18, T153N, R100W
McKenzie County, ND
Gross apparent dip = -0.20°
PROFILE
SUNBURST CONSULTING
A GEOLOGY SERVICE
W
E
-0.43°
-0.36°
-0.65°
-0.35°
-0.17°
+0.12°
-0.09°
-0.46°
-0.85°
-0.12°
+0.31°
Flat
-0.12°
Wt: 9.4 Vis: 28
Wt: 9.4 Vis: 27
Wt: 9.4 Vis: 27
Wt: 9.4 Vis: 28
Wt: 9.4 Vis: 28
Wt: 9.4 Vis: 27
Wt: 9.5 Vis: 28
Wt: 9.4 Vis: 27
Wt: 9.4 Vis: 27
Wt: 9.4 Vis: 27
Depth (TVD)
Total Gas, ROP x 100, Gamma Ray x 10
Measured Depth
Wellbore
Pronghorn
Top of Target
Base of Target (Claystone)
Base of Claystone
TG
ROP x 100
GR x 10

FORMATION MARKERS & DIP ESTIMATES

Oasis Petroleum North America LLC - Kline Federal 5300 11-18 3T

Dip Change Points Projected to Pronghorn	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker							
Target entry	11,075'	10,791.60					Gamma
Top of claystone	11,460'	10,794.50	2.90	385.00	***-0.43***	Down	Gamma
Top of claystone	13,315'	10,806.00	11.50	1855.00	***-0.36***	Down	Gamma
Top of claystone	13,975'	10,813.50	7.50	660.00	***-0.65***	Down	Gamma
Top of claystone	14,500'	10,816.70	3.20	525.00	***-0.35***	Down	Gamma
Warm just above claystone	15,585'	10,820.00	3.30	1085.00	***-0.17***	Down	Gamma
Top of claystone	16,281'	10,818.50	-1.50	696.00	***0.12***	Up	Gamma
Warm at top of target	16,614'	10,819.00	0.50	333.00	***-0.09***	Down	Gamma
Warm at top of target	17,050'	10,822.50	3.50	436.00	***-0.46***	Down	Gamma
Warm at top of target	17,620'	10,831.00	8.50	570.00	***-0.85***	Down	Gamma
Top of claystone	18,320'	10,832.50	1.50	700.00	***-0.12***	Down	Gamma
Top of claystone	18,480'	10,832.00	-0.50	160.00	***0.18***	Up	Gamma
Top of claystone	18,800'	10,830.00	-2.00	320.00	***0.36***	Up	Gamma
Top of claystone	19,162'	10,829.00	-1.00	362.00	***0.16***	Up	Gamma
Warm at top of target	19,417'	10,826.50	-2.50	255.00	***0.56***	Up	Gamma
Warm at top of target	19,875'	10,824.00	-2.50	458.00	***0.31***	Up	Gamma
Cool near the center of target	20,280'	10,824.00	0.00	405.00	***0.00***	Flat	Gamma
Warm at top of target	20,840'	10,825.20	1.20	560.00	***-0.12***	Down	Gamma
Gross Dip							
Initial Target Contact	11,075'	10,791.60					
Projected Final Target Contact	20,840'	10,825.20	33.60	9765.00	***-0.20***	Down	Projection



DIP PROFILE
Oasis Petroleum North America LLC
Kline Federal 5300 11-18 3T
Lot 1 Sec. 18, T153N, R100W
McKenzie County, ND
SUNBURST CONSULTING
A GEOLOGY SERVICE
W
E
Gross apparent dip= -0.20°
-0.43°
-0.36°
-0.65°
-0.35°
-0.17°
+0.12°
-0.09°
-0.46°
-0.85°
-0.12°
+0.31°
Flat
-0.12°
Depth (TVD)
Gamma
Measured Depth
10785
10795
10805
10815
10825
10835
10845
10855
10865
10875
10885
10895
10905
10915
290
270
250
230
210
190
170
150
130
110
90
70
50
10900
11200
11500
11800
12100
12400
12700
13000
13300
13600
13900
14200
14500
14800
15100
15400
15700
16000
16300
16600
16900
17200
17500
17800
18100
18400
18700
19000
19300
19600
19900
20200
20500
20800
Wellbore
Pronghorn
Top of Target
Base of Target (Claystone)
Base of Claystone
Gamma

< **SUNBURST CONSULTING, INC.** >

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
Tie	2084.00	0.50	95.30	2083.79	3.41	-4.87	-5.14	0.00
1	2176.00	1.10	95.00	2175.78	3.30	-3.59	-3.86	0.65
2	2270.00	0.50	245.30	2269.78	3.05	-3.06	-3.31	1.65
3	2365.00	0.80	289.20	2364.77	3.09	-4.07	-4.32	0.59
4	2459.00	1.20	301.20	2458.76	3.82	-5.53	-5.83	0.48
5	2553.00	1.60	317.30	2552.73	5.29	-7.26	-7.69	0.59
6	2647.00	1.10	275.70	2646.70	6.34	-9.05	-9.56	1.13
7	2742.00	1.90	286.60	2741.67	6.89	-11.47	-12.01	0.89
8	2836.00	1.10	300.50	2835.64	7.79	-13.74	-14.35	0.93
9	2930.00	1.00	227.50	2929.63	7.69	-15.12	-15.72	1.33
10	3025.00	1.50	247.70	3024.60	6.66	-16.88	-17.39	0.69
11	3119.00	1.60	247.60	3118.57	5.69	-19.23	-19.65	0.11
12	3213.00	1.50	255.00	3212.53	4.88	-21.63	-21.97	0.24
13	3307.00	1.30	255.00	3306.51	4.28	-23.85	-24.13	0.21
14	3401.00	1.20	249.80	3400.48	3.66	-25.81	-26.02	0.16
15	3496.00	1.10	261.80	3495.47	3.19	-27.64	-27.81	0.27
16	3590.00	1.10	244.70	3589.45	2.68	-29.35	-29.47	0.35
17	3684.00	0.90	247.30	3683.43	2.01	-30.85	-30.91	0.22
18	3778.00	0.80	248.50	3777.42	1.48	-32.14	-32.15	0.11
19	3872.00	0.80	253.40	3871.41	1.05	-33.38	-33.35	0.07
20	3966.00	0.70	247.90	3965.41	0.65	-34.54	-34.47	0.13
21	4060.00	0.50	263.20	4059.40	0.38	-35.48	-35.38	0.27
22	4154.00	0.40	283.10	4153.40	0.41	-36.21	-36.11	0.20
23	4249.00	0.50	270.30	4248.40	0.49	-36.94	-36.85	0.15
24	4343.00	0.50	291.40	4342.39	0.64	-37.73	-37.65	0.19
25	4437.00	0.50	283.70	4436.39	0.89	-38.51	-38.45	0.07
26	4531.00	0.30	306.20	4530.39	1.13	-39.11	-39.07	0.27
27	4626.00	0.50	299.70	4625.38	1.48	-39.67	-39.65	0.22
28	4720.00	0.40	291.00	4719.38	1.80	-40.34	-40.34	0.13
29	4813.00	0.40	349.80	4812.38	2.24	-40.70	-40.74	0.42
30	4907.00	0.20	327.40	4906.38	2.70	-40.84	-40.92	0.24
31	5001.00	0.50	307.70	5000.38	3.09	-41.26	-41.37	0.34
32	5096.00	0.60	273.70	5095.37	3.37	-42.08	-42.21	0.35
33	5190.00	0.20	210.80	5189.37	3.26	-42.65	-42.78	0.57
34	5284.00	0.70	243.80	5283.37	2.87	-43.25	-43.34	0.58
35	5378.00	0.60	214.10	5377.36	2.21	-44.05	-44.07	0.37
36	5473.00	0.80	203.20	5472.35	1.19	-44.59	-44.52	0.25
37	5567.00	1.10	195.00	5566.34	-0.29	-45.08	-44.89	0.35

<   **SUNBURST CONSULTING, INC.**   >

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
38	5661.00	1.10	197.70	5660.32	-2.02	-45.59	-45.25	0.06
39	5756.00	0.60	151.60	5755.31	-3.32	-45.63	-45.18	0.85
40	5850.00	0.40	212.30	5849.31	-4.03	-45.57	-45.06	0.57
41	5944.00	0.40	178.80	5943.31	-4.64	-45.74	-45.17	0.25
42	6038.00	0.10	283.40	6037.31	-4.95	-45.81	-45.22	0.46
43	6132.00	0.40	352.00	6131.31	-4.61	-45.93	-45.37	0.40
44	6226.00	0.70	17.60	6225.30	-3.73	-45.81	-45.32	0.41
45	6320.00	0.90	3.60	6319.29	-2.45	-45.59	-45.21	0.30
46	6414.00	1.30	0.40	6413.28	-0.65	-45.53	-45.31	0.43
47	6509.00	1.00	357.40	6508.26	1.26	-45.56	-45.50	0.32
48	6603.00	1.00	2.40	6602.24	2.90	-45.57	-45.65	0.09
49	6697.00	0.80	25.70	6696.23	4.31	-45.25	-45.45	0.44
50	6791.00	0.40	161.10	6790.23	4.59	-44.86	-45.08	1.19
51	6885.00	0.20	310.80	6884.23	4.39	-44.87	-45.08	0.62
52	6979.00	0.70	56.50	6978.22	4.81	-44.52	-44.77	0.83
53	7074.00	0.70	55.20	7073.22	5.46	-43.56	-43.87	0.02
54	7168.00	0.80	54.70	7167.21	6.17	-42.55	-42.92	0.11
55	7262.00	0.80	43.00	7261.20	7.03	-41.57	-42.02	0.17
56	7356.00	0.60	107.30	7355.19	7.36	-40.65	-41.13	0.81
57	7450.00	0.20	49.70	7449.19	7.32	-40.06	-40.53	0.55
58	7544.00	0.40	25.70	7543.19	7.72	-39.79	-40.30	0.25
59	7639.00	0.60	165.40	7638.19	7.54	-39.52	-40.02	0.99
60	7733.00	0.30	185.80	7732.19	6.82	-39.42	-39.86	0.36
61	7827.00	0.10	24.70	7826.19	6.65	-39.41	-39.83	0.42
62	7921.00	0.00	8.80	7920.19	6.72	-39.38	-39.81	0.11
63	8016.00	0.20	346.50	8015.19	6.89	-39.41	-39.86	0.21
64	8109.00	0.30	36.30	8108.19	7.24	-39.31	-39.78	0.25
65	8204.00	0.30	50.00	8203.18	7.60	-38.97	-39.48	0.08
66	8298.00	0.20	133.80	8297.18	7.65	-38.66	-39.17	0.36
67	8392.00	0.40	89.90	8391.18	7.53	-38.22	-38.72	0.31
68	8486.00	0.50	86.20	8485.18	7.56	-37.48	-37.99	0.11
69	8580.00	0.70	67.40	8579.17	7.81	-36.54	-37.07	0.30
70	8674.00	0.60	58.50	8673.17	8.29	-35.59	-36.17	0.15
71	8769.00	0.80	52.70	8768.16	8.95	-34.64	-35.28	0.22
72	8863.00	0.70	61.00	8862.15	9.62	-33.61	-34.31	0.16
73	8957.00	1.10	225.40	8956.15	9.27	-33.76	-34.42	1.90
74	9051.00	1.20	232.70	9050.13	8.04	-35.18	-35.74	0.19
75	9146.00	0.80	240.80	9145.11	7.11	-36.55	-37.02	0.45

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
76	9240.00	0.70	259.80	9239.11	6.69	-37.69	-38.12	0.28
77	9334.00	0.60	245.30	9333.10	6.38	-38.70	-39.10	0.20
78	9428.00	0.40	284.40	9427.10	6.26	-39.47	-39.86	0.41
79	9523.00	0.20	279.40	9522.10	6.37	-39.95	-40.35	0.21
80	9617.00	0.20	291.70	9616.10	6.46	-40.27	-40.67	0.05
81	9711.00	0.20	314.00	9710.10	6.63	-40.54	-40.95	0.08
82	9805.00	0.20	345.20	9804.09	6.90	-40.70	-41.14	0.11
83	9899.00	0.10	252.70	9898.09	7.04	-40.82	-41.27	0.24
84	9993.00	0.20	46.50	9992.09	7.13	-40.78	-41.23	0.31
85	10087.00	0.50	10.80	10086.09	7.64	-40.58	-41.08	0.38
86	10182.00	0.70	20.10	10181.09	8.59	-40.30	-40.89	0.23
87	10226.00	1.10	22.00	10225.08	9.24	-40.05	-40.69	0.91
88	10263.00	1.00	26.00	10262.08	9.86	-39.78	-40.47	0.33
89	10294.00	2.50	101.40	10293.06	9.97	-39.00	-39.70	7.89
90	10326.00	6.50	115.60	10324.96	9.05	-36.68	-37.31	12.88
91	10357.00	10.80	121.30	10355.60	6.78	-32.61	-33.07	14.13
92	10388.00	14.80	124.80	10385.82	3.01	-26.88	-27.03	13.14
93	10420.00	19.20	128.30	10416.42	-2.59	-19.39	-19.09	14.11
94	10451.00	23.80	131.50	10445.25	-9.90	-10.70	-9.81	15.31
95	10483.00	28.00	134.90	10474.04	-19.48	-0.53	1.13	13.92
96	10514.00	31.30	135.90	10500.97	-30.41	10.23	12.78	10.76
97	10545.00	33.00	138.40	10527.22	-42.50	21.44	24.98	6.96
98	10577.00	35.20	140.80	10553.72	-56.17	33.05	37.72	8.06
99	10608.00	37.70	142.20	10578.65	-70.58	44.51	50.37	8.50
100	10639.00	40.20	142.90	10602.76	-86.06	56.36	63.49	8.19
101	10671.00	43.40	142.40	10626.61	-103.01	69.30	77.83	10.05
102	10702.00	46.70	140.90	10648.51	-120.20	82.92	92.86	11.18
103	10734.00	50.20	140.80	10669.73	-138.77	98.03	109.51	10.94
104	10765.00	53.10	141.20	10688.96	-157.67	113.33	126.36	9.41
105	10796.00	55.10	140.50	10707.14	-177.14	129.18	143.81	6.71
106	10828.00	57.80	139.80	10724.82	-197.61	146.27	162.59	8.63
107	10859.00	60.50	138.30	10740.72	-217.70	163.72	181.68	9.65
108	10890.00	64.40	136.50	10755.06	-237.92	182.32	201.94	13.59
109	10922.00	67.10	135.70	10768.20	-258.94	202.55	223.89	8.74
110	10953.00	70.90	135.60	10779.30	-279.63	222.78	245.81	12.26
111	10984.00	74.60	136.60	10788.50	-300.96	243.30	268.07	12.33
112	11016.00	77.50	135.90	10796.21	-323.39	264.78	291.38	9.31
113	11047.00	80.20	134.90	10802.20	-345.05	286.13	314.50	9.27

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
114	11079.00	82.10	134.20	10807.13	-367.23	308.66	338.84	6.32
115	11110.00	84.00	134.10	10810.88	-388.66	330.74	362.67	6.14
116	11140.00	88.40	133.50	10812.87	-409.37	352.34	385.96	14.80
117	11236.00	89.60	129.10	10814.54	-472.71	424.43	463.18	4.75
118	11267.00	89.30	128.10	10814.84	-492.05	448.66	488.97	3.37
119	11298.00	89.10	127.50	10815.27	-511.04	473.15	514.99	2.04
120	11329.00	87.80	125.50	10816.11	-529.48	498.06	541.38	7.69
121	11360.00	87.60	125.20	10817.36	-547.40	523.33	568.08	1.16
122	11391.00	87.70	124.80	10818.63	-565.16	548.70	594.88	1.33
123	11421.00	88.30	122.40	10819.67	-581.75	573.67	621.17	8.24
124	11452.00	88.10	122.00	10820.65	-598.26	599.89	648.70	1.44
125	11483.00	87.90	121.90	10821.73	-614.66	626.17	676.29	0.72
126	11514.00	89.60	119.40	10822.41	-630.46	652.83	704.20	9.75
127	11545.00	89.00	117.50	10822.78	-645.22	680.09	732.61	6.43
128	11576.00	89.00	117.10	10823.33	-659.44	707.63	761.27	1.29
129	11607.00	90.50	114.90	10823.46	-673.03	735.49	790.18	8.59
130	11638.00	91.60	114.60	10822.89	-686.00	763.64	819.33	3.68
131	11669.00	92.00	113.50	10821.92	-698.63	791.93	848.60	3.77
132	11699.00	91.50	110.50	10821.00	-709.86	819.73	877.26	10.13
133	11730.00	91.30	109.80	10820.24	-720.54	848.82	907.16	2.35
134	11761.00	91.30	108.80	10819.54	-730.78	878.07	937.17	3.22
135	11792.00	90.10	107.20	10819.16	-740.36	907.55	967.36	6.45
136	11822.00	89.60	106.90	10819.24	-749.16	936.24	996.69	1.94
137	11853.00	89.50	106.10	10819.48	-757.96	965.96	1027.05	2.60
138	11883.00	88.30	104.00	10820.06	-765.75	994.92	1056.57	8.06
139	11915.00	88.30	103.90	10821.01	-773.46	1025.96	1088.16	0.31
140	11946.00	88.40	103.70	10821.90	-780.85	1056.06	1118.77	0.72
141	11978.00	89.70	101.60	10822.43	-787.86	1087.27	1150.47	7.72
142	12010.00	89.60	101.50	10822.63	-794.26	1118.62	1182.26	0.44
143	12042.00	90.00	101.60	10822.74	-800.67	1149.98	1214.04	1.29
144	12074.00	89.80	99.10	10822.80	-806.42	1181.45	1245.89	7.84
145	12105.00	89.80	98.60	10822.90	-811.19	1212.08	1276.82	1.61
146	12137.00	90.20	97.90	10822.90	-815.78	1243.75	1308.76	2.52
147	12168.00	90.50	96.20	10822.72	-819.58	1274.52	1339.74	5.57
148	12200.00	90.60	96.00	10822.41	-822.98	1306.33	1371.73	0.70
149	12232.00	90.80	94.80	10822.02	-826.00	1338.19	1403.73	3.80
150	12264.00	90.10	93.30	10821.77	-828.26	1370.11	1435.72	5.17
151	12359.00	90.70	92.80	10821.10	-833.31	1464.97	1530.67	0.82

< **SUNBURST CONSULTING, INC.** >

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
152	12453.00	89.00	92.70	10821.35	-837.82	1558.86	1624.60	1.81
153	12547.00	89.90	92.50	10822.25	-842.08	1652.75	1718.52	0.98
154	12642.00	90.40	92.00	10822.00	-845.81	1747.68	1813.42	0.74
155	12738.00	89.60	92.60	10822.00	-849.67	1843.60	1909.32	1.04
156	12832.00	88.50	92.70	10823.56	-854.01	1937.49	2003.23	1.18
157	12928.00	87.70	91.70	10826.74	-857.69	2033.36	2099.07	1.33
158	13024.00	88.70	90.90	10829.76	-859.87	2129.29	2194.83	1.33
159	13120.00	89.10	90.50	10831.60	-861.04	2225.26	2290.56	0.59
160	13213.00	89.20	90.60	10832.98	-861.94	2318.25	2383.28	0.15
161	13308.00	90.90	88.70	10832.90	-861.36	2413.24	2477.88	2.68
162	13403.00	90.20	87.90	10831.99	-858.54	2508.19	2572.25	1.12
163	13498.00	90.50	88.20	10831.41	-855.31	2603.13	2666.57	0.45
164	13593.00	89.70	90.20	10831.24	-853.98	2698.12	2761.09	2.27
165	13689.00	88.40	89.20	10832.83	-853.48	2794.10	2856.68	1.71
166	13783.00	90.40	90.00	10833.82	-852.82	2888.09	2950.27	2.29
167	13879.00	90.40	90.10	10833.15	-852.90	2984.08	3045.93	0.10
168	13975.00	92.80	89.40	10830.47	-852.49	3080.04	3141.50	2.60
169	14070.00	91.30	89.20	10827.07	-851.33	3174.97	3235.98	1.59
170	14164.00	88.80	90.10	10826.99	-850.75	3268.96	3329.58	2.83
171	14258.00	86.90	90.10	10830.51	-850.92	3362.89	3423.18	2.02
172	14353.00	91.20	90.70	10832.09	-851.58	3457.85	3517.85	4.57
173	14449.00	88.20	90.70	10832.59	-852.75	3553.83	3613.59	3.13
174	14543.00	88.00	90.20	10835.71	-853.49	3647.78	3707.25	0.57
175	14638.00	88.70	89.80	10838.44	-853.49	3742.73	3801.87	0.85
176	14733.00	91.20	89.10	10838.52	-852.58	3837.72	3896.43	2.73
177	14828.00	89.10	88.50	10838.28	-850.59	3932.70	3990.89	2.30
178	14925.00	89.90	88.50	10839.12	-848.05	4029.66	4087.28	0.82
179	15022.00	89.60	88.50	10839.55	-845.51	4126.62	4183.68	0.31
180	15117.00	89.60	88.80	10840.21	-843.27	4221.59	4278.11	0.32
181	15212.00	91.50	87.60	10839.30	-840.29	4316.54	4372.45	2.37
182	15306.00	88.70	88.10	10839.13	-836.76	4410.46	4465.74	3.03
183	15401.00	87.80	90.20	10842.03	-835.35	4505.40	4560.21	2.40
184	15496.00	89.10	90.70	10844.60	-836.10	4600.36	4654.89	1.47
185	15592.00	90.00	89.50	10845.36	-836.27	4696.35	4750.55	1.56
186	15687.00	90.60	88.40	10844.86	-834.53	4791.34	4845.03	1.32
187	15782.00	90.40	89.60	10844.03	-832.87	4886.32	4939.53	1.28
188	15877.00	89.50	90.60	10844.11	-833.03	4981.31	5034.19	1.42
189	15973.00	88.80	90.90	10845.54	-834.29	5077.29	5129.93	0.79

< **SUNBURST CONSULTING, INC.** >

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
190	16069.00	90.40	90.40	10846.21	-835.38	5173.28	5225.66	1.75
191	16166.00	91.00	89.30	10845.02	-835.13	5270.27	5322.28	1.29
192	16262.00	91.20	88.70	10843.18	-833.45	5366.24	5417.75	0.66
193	16358.00	92.00	89.70	10840.50	-832.11	5462.19	5513.24	1.33
194	16454.00	90.90	90.40	10838.07	-832.19	5558.16	5608.87	1.36
195	16551.00	91.00	90.20	10836.46	-832.70	5655.14	5705.54	0.23
196	16647.00	91.50	90.90	10834.37	-833.62	5751.11	5801.24	0.90
197	16742.00	89.20	91.70	10833.79	-835.78	5846.08	5896.05	2.56
198	16838.00	88.80	92.10	10835.46	-838.96	5942.01	5991.90	0.59
199	16935.00	88.30	91.80	10837.92	-842.26	6038.93	6088.74	0.60
200	17030.00	89.70	90.70	10839.58	-844.33	6133.89	6183.53	1.87
201	17124.00	90.50	90.50	10839.41	-845.32	6227.88	6277.27	0.88
202	17220.00	88.70	91.00	10840.08	-846.57	6323.86	6373.01	1.95
203	17317.00	88.00	91.40	10842.88	-848.60	6420.80	6469.77	0.83
204	17412.00	89.10	91.10	10845.28	-850.68	6515.75	6564.55	1.20
205	17508.00	89.20	91.10	10846.70	-852.52	6611.72	6660.32	0.10
206	17603.00	89.20	91.00	10848.03	-854.26	6706.69	6755.10	0.11
207	17698.00	90.30	90.70	10848.44	-855.67	6801.68	6849.86	1.20
208	17794.00	89.60	90.80	10848.53	-856.93	6897.67	6945.61	0.74
209	17891.00	89.50	88.90	10849.29	-856.67	6994.66	7042.23	1.96
210	17987.00	88.60	88.70	10850.88	-854.66	7090.63	7137.67	0.96
211	18083.00	89.60	87.80	10852.39	-851.73	7186.57	7233.02	1.40
212	18180.00	88.90	88.80	10853.66	-848.85	7283.52	7329.37	1.26
213	18276.00	89.00	88.20	10855.42	-846.34	7379.47	7424.75	0.63
214	18372.00	89.50	88.70	10856.67	-843.74	7475.42	7520.14	0.74
215	18468.00	91.10	91.60	10856.17	-843.99	7571.41	7615.80	3.45
216	18562.00	91.70	91.90	10853.88	-846.86	7665.34	7709.63	0.71
217	18658.00	91.10	91.60	10851.53	-849.79	7761.26	7805.45	0.70
218	18753.00	88.00	92.20	10852.28	-852.94	7856.20	7900.31	3.32
219	18851.00	87.20	91.80	10856.38	-856.36	7954.05	7998.10	0.91
220	18947.00	88.70	90.40	10859.81	-858.20	8049.96	8093.82	2.14
221	19041.00	91.00	88.00	10860.06	-856.89	8143.94	8187.34	3.54
222	19138.00	92.70	87.70	10856.93	-853.25	8240.82	8283.56	1.78
223	19234.00	93.30	88.60	10851.90	-850.16	8336.64	8378.76	1.13
224	19329.00	92.80	89.20	10846.85	-848.34	8431.48	8473.11	0.82
225	19425.00	91.30	89.30	10843.42	-847.08	8527.41	8568.58	1.57
226	19520.00	90.80	91.60	10841.67	-847.83	8622.39	8663.27	2.48
227	19615.00	90.00	93.60	10841.01	-852.14	8717.28	8758.19	2.27

< **SUNBURST CONSULTING, INC.** >

Operator:	Oasis Petroleum North America LLC	Kick-off:	5/22/2015
Well:	Kline Federal 5300 11-18 3T	Finish:	5/28/2015
Surface Coordinates:	1,020' FNL & 290' FWL		
Surface Location:	Lot 1 Sec. 18, T153N, R100W	Directional Supervision:	
County State:	McKenzie County, ND	RPM Directional Inc.	

Minimum Curvature Method (SPE-3362) Proposed dir 94.89

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
228	19710.00	91.60	94.50	10839.68	-858.85	8812.03	8853.17	1.93
229	19805.00	89.40	93.50	10838.86	-865.47	8906.79	8948.14	2.54
230	19900.00	88.40	93.20	10840.68	-871.02	9001.61	9043.09	1.10
231	19996.00	89.80	91.50	10842.19	-874.96	9097.51	9138.98	2.29
232	20091.00	89.90	91.90	10842.44	-877.78	9192.47	9233.83	0.43
233	20187.00	89.30	91.10	10843.11	-880.29	9288.43	9329.66	1.04
234	20283.00	89.60	90.80	10844.03	-881.88	9384.41	9425.43	0.44
235	20379.00	90.90	91.20	10843.61	-883.55	9480.39	9521.20	1.42
236	20475.00	89.50	89.50	10843.27	-884.14	9576.39	9616.90	2.29
237	20571.00	89.10	88.70	10844.45	-882.63	9672.37	9712.40	0.93
238	20668.00	90.90	88.80	10844.45	-880.52	9769.34	9808.84	1.86
239	20764.00	92.00	88.10	10842.02	-877.92	9865.27	9904.20	1.36
240	20775.00	92.30	87.80	10841.60	-877.53	9876.26	9915.11	3.86
241	20840.00	92.30	87.80	10839.00	-875.04	9941.16	9979.56	0.00

DEVIATION SURVEYS

Depth	Inclination	Azimuth
0	0.00	0.00
132	0.90	144.50
224	1.20	153.80
314	1.70	133.30
399	0.60	159.80
389	0.80	265.10
579	1.00	273.20
663	0.90	266.60
751	0.40	297.80
840	0.80	258.90
927	0.40	291.60
1015	0.50	281.30
1101	1.00	276.90
1188	0.40	308.00
1277	0.50	287.90
1365	0.60	270.00
1450	0.60	284.90
1539	0.80	329.60
1628	1.20	18.40
1714	0.70	31.60
1803	0.70	24.10
1889	1.10	23.80
1987	1.00	23.10

FORMATION TOPS & STRUCTURAL RELATIONSHIPS

	Subject Well:							
Operator:	**Oasis Petroleum North America LLC**							
Well Name:	**Kline Federal 5300 11-18 3T**							
Location:	1,020' FNL & 290' FWL							
	Lot 1 Section 18, T153N, R100W							
Elevation:	GL: 2,053'	Sub: 16'	KB: 2,069'					
Formation/ Marker	**Prog. Top**	**Prog. Datum (MSL)**	**Driller's Depth Top (MD)**	**Driller's Depth Top (TVD)**	**Datum (MSL)**	**Interval Thickness**	**Thickness to Target**	**Dip To Prog.**
Kibbey Lime	8,366'	-6,297'	8,363'	8,362'	-6,293'	150'	2,451'	4'
First Charles Salts	8,516'	-6,447'	8,513'	8,512'	-6,443'	616'	2,301'	4'
Upper Berentson	9,167'	-7,098'	9,129'	9,128'	-7,059'	72'	1,685'	39'
Base Last Salt	9,215'	-7,146'	9,201'	9,200'	-7,131'	49'	1,613'	15'
Ratcliffe	9,263'	-7,194'	9,250'	9,249'	-7,180'	182'	1,564'	14'
Mission Canyon	9,439'	-7,370'	9,432'	9,431'	-7,362'	550'	1,382'	8'
Lodgepole	10,001'	-7,932'	9,982'	9,981'	-7,912'	712'	832'	20'
False Bakken	10,697'	-8,628'	10,771'	10,693'	-8,624'	10'	120'	4'
Upper Bakken Shale	10,707'	-8,638'	10,789'	10,703'	-8,634'	19'	110'	4'
Middle Bakken	10,721'	-8,652'	10,822'	10,722'	-8,653'	40'	91'	-1'
Lower Bakken Shale	10,766'	-8,697'	10,907'	10,762'	-8,693'	13'	51'	4'
Pronghorn	10,780'	-8,711'	10,940'	10,775'	-8,706'	16'	38'	5'
Three Forks	10,792'	-8,723'	10,994'	10,791'	-8,722'	-	22'	1'

LITHOLOGY

Rig crews caught lagged samples in 50' intervals under the supervision of a Sunburst geologist. A detailed list of sampling intervals is included in the well data summary page. Samples were examined wet and dry conditions under a binocular microscope and checked for hydrocarbon cut fluorescence with Entron. Sample descriptions began in the Three Forks. The drilling fluid was salt water brine solution in the lateral.

(Re-Enter Lateral)

Three Forks Formation 10,994' MD; 10,791' TVD (-8,722')

11,204-11,250 DOLOMITE: mudstone, light tan, light orange, off white, light gray-brown, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: light blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite, trace nodular pyrite; slow white streaming becoming diffuse cut fluorescence

11,250-11,300 DOLOMITE: mudstone, off white, light tan, light orange, medium gray-brown, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow white streaming becoming diffuse cut fluorescence

11,300-11,350 DOLOMITE: mudstone, cream, light brown-gray, off white, medium gray-brown, microcrystalline, firm, rare friable, dense, earthy, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow white diffuse cut fluorescence

11,350-11,400 DOLOMITE: mudstone, light gray, off white, cream, trace pink, microcrystalline, firm, dense, earthy, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; slow white diffuse cut fluorescence

11,400-11,450 DOLOMITE: mudstone, tan, cream, off white, microcrystalline, firm, dense, earthy, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; slow light yellow diffuse cut fluorescence

11,450-11,500 DOLOMITE: mudstone, off white-tan, microcrystalline, firm, dense, occasional earthy, trace intercrystalline porosity, trace light brown spotty oil stain; CLAYSTONE: light blue gray, cream, rare light brown-gray, firm-friable, platy, smooth-silty texture, common disseminated pyrite, trace nodular pyrite, no visible oil stain; slightly pale yellow diffuse cut fluorescence

11,500-11,550 CLAYSTONE: light gray, light blue, firm, platy, smooth-silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; slow poor pale yellow diffuse cut fluorescence

11,550-11,600 CLAYSTONE: cream, light gray, light blue, firm, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; slow poor pale yellow diffuse cut fluorescence

11,600-11,650 CLAYSTONE: light blue, cream, light gray, firm, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, intercrystalline porosity, no visible oil stain; DOLOMITE: mudstone, pink, light brown-gray, cream, microcrystalline, firm, dense, occasional earthy, trace intercrystalline porosity, trace light brown spotty oil stain; slow poor pale yellow diffuse cut fluorescence

11,650-11,700 DOLOMITE: mudstone, cream, off white, light gray, microcrystalline, firm, rare friable, dense, earthy, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; slow white occasional pal yellow diffuse cut fluorescence

11,700-11,750 DOLOMITE: mudstone, off white, light pink, light tan, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light to medium brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow white diffuse cut fluorescence

11,750-11,800 DOLOMITE: mudstone, light gray, tan, cream, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: pale green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

11,800-11,850 DOLOMITE: mudstone, cream, off white, light to medium gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, pale green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow off white diffuse cut fluorescence

11,850-11,900 DOLOMITE: mudstone, tan, light brown, cream, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; slow white diffuse cut fluorescence

11,900-11,950 DOLOMITE: mudstone, light to medium gray, tan, light brown, cream, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light to medium brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; slow white diffuse cut fluorescence

11,950-12,000 DOLOMITE: mudstone, off white, light pink, light tan, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light to medium brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow white diffuse cut fluorescence

12,000-12,050 DOLOMITE: mudstone, light to medium gray, tan, cream, occasional pink, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: pale green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,050-12,100 DOLOMITE: mudstone, light tan, light orange, off white, gray-brown, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; very slow pale yellow streaming becoming diffuse cut fluorescence

12,100-12,150 DOLOMITE: mudstone, light brown, tan, light orange, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; very slow pale yellow streaming becoming diffuse cut fluorescence

12,150-12,200 DOLOMITE: mudstone, tan, light orange-light brown, off white, gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; very slow pale yellow streaming becoming diffuse cut fluorescence

12,200-12,250 DOLOMITE: mudstone, light brown-gray, tan, light orange, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,250-12,300 DOLOMITE: mudstone, tan, light orange, medium brown-gray, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,300-12,350 DOLOMITE: mudstone, orange, tan, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,350-12,400 DOLOMITE: mudstone, orange, tan, medium brown-gray, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,400-12,450 DOLOMITE: mudstone, orange, tan, light brown-gray, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,450-12,500 DOLOMITE: mudstone, orange, tan, off white light brown-gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,500-12,550 DOLOMITE: mudstone, orange, tan, off white, light brown-gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; very slow pale yellow diffuse cut fluorescence

12,550-12,600 DOLOMITE: mudstone, orange, tan, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,600-12,650 DOLOMITE: mudstone, orange, tan-light brown, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,650-12,700 DOLOMITE: mudstone, orange, tan-light brown, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,700-12,750 DOLOMITE: mudstone, light to medium brown, orange, tan, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,750-12,800 DOLOMITE: mudstone, light to medium brown-gray, off white, tan, light orange, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,800-12,850 DOLOMITE: mudstone, orange, tan to medium brown, off white, gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,850-12,900 DOLOMITE: mudstone, orange, tan to medium brown, off white, gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,900-12,950 DOLOMITE: mudstone, light to medium brown-gray, cream, orange, tan to medium brown, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

12,950-13,000 DOLOMITE: mudstone, light to medium brown-gray, cream, orange, tan, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,000-13,050 DOLOMITE: mudstone, light to medium brown-gray, off white, orange, tan to medium brown, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,050-13,100 DOLOMITE: mudstone, light orange, tan to medium brown, brown-gray, off white, orange, tan to medium brown, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,100-13,150 DOLOMITE: mudstone, medium brown-gray, orange, tan to medium brown, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,150-13,200 CLAYSTONE: cream, light gray, light blue, firm, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; very slow poor pale yellow diffuse cut fluorescence

13,200-13,250 CLAYSTONE: cream, light blue, firm, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; very slow poor pale yellow diffuse cut fluorescence

13,250-13,300 CLAYSTONE: cream, light blue, firm, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; very slow poor pale yellow diffuse cut fluorescence

13,300-13,350 DOLOMITE: mudstone, cream, light gray, medium brown, off white, microcrystalline, firm, dense, earthy texture, trace intercrystalline porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,350-13,400 DOLOMITE: mudstone, tan, off white, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,400-13,450 DOLOMITE: mudstone, off white, cream, light to medium gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace medium brown spotty oil stain; occasional SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,450-13,500 DOLOMITE: mudstone, off white, cream, light to medium gray, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: pale green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,500-13,550 DOLOMITE: mudstone, light gray, off white, tan, light brown-gray, medium-dark gray, microcrystalline, firm, dense, earthy texture, trace intercrystalline porosity, trace medium brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow-moderate pale yellow streaming cut fluorescence

13,550-13,600 DOLOMITE: mudstone, off white, tan, light brown-gray, medium gray, microcrystalline, firm, dense, earthy texture, trace intercrystalline porosity, trace medium brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow-moderate pale yellow diffuse cut fluorescence

13,600-13,650 DOLOMITE: mudstone, off white, tan, light brown-gray, medium gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace medium brown spotty oil stain; trace SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

13,700-13,750 DOLOMITE: mudstone, off white, cream, microcrystalline, firm, dense, earthy texture, trace disseminated pry, trace nodular pyrite, trace intercrystalline porosity, trace medium brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow-moderate pale yellow streaming cut fluorescence

13,700-13,750 DOLOMITE: mudstone, light gray, off white, cream, microcrystalline, firm, dense, earthy texture, trace disseminated pry, rare nodular pyrite, trace intercrystalline porosity, trace medium brown spotty oil stain; slow-moderate pale yellow streaming cut fluorescence

13,750-13,800 DOLOMITE: mudstone, cream, light brown, light to medium gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pry, trace nodular pyrite, trace intercrystalline porosity, trace medium brown spotty oil stain; slow-moderate pale yellow streaming cut fluorescence

13,800-13,850 DOLOMITE: mudstone, light brown, off white, cream, trace light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pry, trace intercrystalline porosity, trace medium brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,850-13,900 DOLOMITE: mudstone, light brown, off white, cream, trace light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pry, trace intercrystalline porosity, trace medium brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

13,900-13,950 DOLOMITE: mudstone, light brown, light brown-gray, off white, tan, trace pink, microcrystalline, firm, dense, earthy, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow streaming cut fluorescence

13,950-14,000 DOLOMITE: mudstone, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; slow pale yellow diffuse cut fluorescence

14,000-14,050 DOLOMITE: mudstone, light brown, light brown-gray, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; slow pale yellow diffuse cut fluorescence

14,050-14,100 DOLOMITE: mudstone, light brown, off white, cream, light gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; moderate pale yellow streaming becoming diffuse cut fluorescence

14,100-14,150 DOLOMITE: mudstone, off white, cream, light gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow streaming becoming diffuse cut fluorescence

14,150-14,200 DOLOMITE: mudstone, light brown, cream, tan, trace light gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; moderate pale yellow streaming becoming diffuse cut fluorescence

14,200-14,250 DOLOMITE: mudstone, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,250-14,300 DOLOMITE: mudstone, tan, light orange, off white, gray-brown, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,300-14,350 DOLOMITE: mudstone, tan, light orange-medium brown, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,350-14,400 DOLOMITE: mudstone, light to medium brown, tan, light orange, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,400-14,450 DOLOMITE: mudstone, off white, light gray-brown, tan, light orange, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,450-14,500 DOLOMITE: mudstone, off white, light gray-brown, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; abundant SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,500-14,550 DOLOMITE: mudstone, light to medium gray, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,550-14,600 DOLOMITE: mudstone, light to medium gray, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,600-14,650 DOLOMITE: mudstone, light to medium gray, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,650-14,700 DOLOMITE: mudstone, light to medium gray-brown, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,700-14,750 DOLOMITE: mudstone, off white, light to medium gray-brown, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,750-14,800 DOLOMITE: mudstone, orange, tan, off white, light to medium gray-brown, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: pale green, light blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,800-14,850 DOLOMITE: mudstone, orange, tan, medium brown, off white, gray-brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,850-14,900 DOLOMITE: mudstone, orange, tan, light to medium brown, off white, gray-brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,900-14,950 DOLOMITE: mudstone, orange, tan, light to medium brown, off white, gray-brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence

14,950-15,000 DOLOMITE: mudstone, light to medium gray-brown, off white, orange, tan, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence, sample moderately contaminated with lube

15,000-15,050 DOLOMITE: mudstone, light to medium gray-brown, off white, orange, tan, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence, sample moderately contaminated with lube

15,050-15,100 DOLOMITE: mudstone, off white, light to medium gray-brown, orange, tan, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence, sample moderately contaminated with lube

15,100-15,150 DOLOMITE: mudstone, light to medium gray-brown, orange, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence, sample moderately contaminated with lube

15,150-15,200 DOLOMITE: mudstone, medium brown, light gray, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence, sample moderately contaminated with lube

15,200-15,250 DOLOMITE: mudstone, tan, light orange, off white, gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; moderate pale yellow diffuse cut fluorescence, sample moderately contaminated with lube

15,250-15,300 DOLOMITE: mudstone, light to medium gray-brown, cream, tan, light orange, gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

15,300-15,350 DOLOMITE: mudstone, light brown, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

15,350-15,400 DOLOMITE: mudstone, light brown, light to medium gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

15,400-15,450 DOLOMITE: mudstone, light gray, off white, cream, light brown, trace dark gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

15,450-15,500 DOLOMITE: mudstone, off white, cream, light brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

15,500-15,550 DOLOMITE: mudstone, tan, light brown, cream, off white, trace light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

15,550-15,600 DOLOMITE: mudstone, light brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

15,600-15,650 DOLOMITE: mudstone, off white, cream, tan, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

15,650-15,700 DOLOMITE: mudstone, light brown, light to medium gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

15,700-15,750 DOLOMITE: mudstone, light brown, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

15,750-15,800 DOLOMITE: mudstone, light to medium gray, light brown, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, trace disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

15,800-15,850 DOLOMITE: mudstone, off white, cream, tan, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

15,850-15,900 DOLOMITE: mudstone, light brown, light to medium gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

15,900-15,950 DOLOMITE: mudstone, light brown, light brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

15,950-16,000 CLAYSTONE: light blue, light brown-gray, cream, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,000-16,050 CLAYSTONE: light blue, light brown-gray, cream, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,050-16,100 CLAYSTONE: light to medium blue gray, light brown, off white, firm, platy, silty texture, rare disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,100-16,150 CLAYSTONE: light to medium blue gray, off white, firm, platy, silty texture, rare disseminated pyrite, trace nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,150-16,200 CLAYSTONE: gray-brown, light blue gray, off white, firm, platy, silty texture, rare disseminated pyrite, trace nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,200-16,250 CLAYSTONE: light to medium gray-brown, off white, blue gray, firm, platy, silty texture, rare disseminated pyrite, trace nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,250-16,300 CLAYSTONE: light blue gray, medium gray-brown, off white, firm, platy, silty texture, occasional disseminated pyrite, trace nodular pyrite, no visible oil stain; sample moderately contaminated with lube

16,300-16,350 DOLOMITE: mudstone, light brown, light brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

16,350-16,400 DOLOMITE: mudstone, medium brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

16,400-16,450 DOLOMITE: mudstone, orange, tan, light brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

16,450-16,500 DOLOMITE: mudstone, light orange, tan, off white, medium brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,500-16,550 DOLOMITE: mudstone, medium brown-gray, tan, off white, medium brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,550-16,600 DOLOMITE: mudstone, medium brown-gray, orange, tan, off white, medium brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,600-16,650 DOLOMITE: mudstone, medium brown-gray, orange, tan, off white, medium brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,650-16,700 DOLOMITE: mudstone, light to medium brown, tan, gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,700-16,750 DOLOMITE: mudstone, off white, light brown-gray, orange, tan, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,750-16,800 DOLOMITE: mudstone, medium gray-brown, orange, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,800-16,850 DOLOMITE: mudstone, medium gray-brown, orange, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,850-16,900 DOLOMITE: mudstone, orange, tan, off white, gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

16,900-16,950 DOLOMITE: mudstone, light brown, gray, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

16,950-17,000 DOLOMITE: mudstone, light orange, tan, off white, medium brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

17,000-17,050 DOLOMITE: mudstone, light brown, cream, off white, light brown-gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

17,050-17,100 DOLOMITE: mudstone, light brown, cream, off white, light brown-gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

17,100-17,150 DOLOMITE: mudstone, light orange, tan, off white, medium brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, occasional disseminated pyrite; sample moderately contaminated with lube

17,150-17,200 DOLOMITE: mudstone, tan, cream, light brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

17,200-17,250 DOLOMITE: mudstone, light brown, light to medium gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

17,250-17,300 DOLOMITE: mudstone, light brown, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; sample moderately contaminated with lube

17,300-17,350 DOLOMITE: mudstone, light brown, light brown-gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: green, blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; sample moderately contaminated with lube

17,350-17,400 DOLOMITE: mudstone, off white, medium brown, trace light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

17,400-17,450 DOLOMITE: mudstone, light brown, off white, cream, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub-blocky, trace disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

17,450-17,500 DOLOMITE: mudstone, light brown, light to medium gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

17,500-17,550 DOLOMITE: mudstone, light to medium gray, off white, light brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

17,550-17,600 DOLOMITE: mudstone, tan, cream, light brown, light to medium gray, trace dark gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: pale green, firm, smooth, sub-blocky, trace disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

17,600-17,650 DOLOMITE: mudstone, light gray, light brown, cream, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

17,650-17,700 DOLOMITE: mudstone, light gray, light brown, cream, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

17,700-17,750 DOLOMITE: mudstone, off white, medium brown, trace light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

17,750-17,800 DOLOMITE: mudstone, medium brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

17,800-17,850 DOLOMITE: mudstone, tan, cream, light brown, light to medium gray, dark gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: pale green, firm, smooth, sub-blocky, trace disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

17,850-17,900 DOLOMITE: mudstone, tan, cream, light brown, light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: pale green, firm, smooth, sub-blocky, trace disseminated pyrite, rare nodular pyrite; sample moderately contaminated with lube

17,900-17,950 DOLOMITE: mudstone, off white, cream, light brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, pale green, firm, smooth, sub-blocky, rare disseminated pyrite trace nodular pyrite; sample moderately contaminated with lube

17,950-18,000 DOLOMITE: mudstone, light brown, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, trace disseminated pyrite; sample moderately contaminated with lube

18,000-18,050 DOLOMITE: mudstone, light brown, gray, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,050-18,100 DOLOMITE: mudstone, light brown-gray, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,100-18,150 DOLOMITE: mudstone, light brown-gray, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,150-18,200 DOLOMITE: mudstone, tan, light orange, light brown-gray, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,200-18,250 DOLOMITE: mudstone, off white, tan, light orange, light brown-gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,250-18,300 DOLOMITE: mudstone, medium gray, light to medium brown, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,300-18,350 DOLOMITE: mudstone, medium gray, light to medium brown, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,350-18,400 CLAYSTONE: light blue, light brown-gray, cream, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, intercrystalline porosity, no visible oil stain; sample moderately contaminated with lube

18,400-18,450 CLAYSTONE: light blue gray, cream, medium brown-gray, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, intercrystalline porosity, no visible oil stain; sample moderately contaminated with lube

18,450-18,500 DOLOMITE: mudstone, medium gray, light to medium brown, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,500-18,550 DOLOMITE: mudstone, tan, light to medium brown-gray, off white, microcrystalline, firm, dense, earthy texture, occasional disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,550-18,600 DOLOMITE: mudstone, light to medium brown-gray, off white, orange, tan, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,600-18,650 DOLOMITE: mudstone, tan, tan, light orange, off white, medium gray, microcrystalline, firm, dense, earthy texture, occasional disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,650-18,700 DOLOMITE: mudstone, gray-brown, off white, tan, microcrystalline, firm, dense, earthy texture, occasional disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,700-18,750 DOLOMITE: mudstone, light to medium gray, tan, off white, microcrystalline, firm, dense, earthy texture, occasional disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

18,750-18,800 DOLOMITE: mudstone, light to medium gray, tan, off white, microcrystalline, firm, dense, earthy texture, occasional disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional CLAYSTONE: light blue gray, cream, medium brown-gray, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, no visible oil stain; sample moderately contaminated with lube

18,800-18,850 CLAYSTONE: light blue gray, cream, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

18,850-18,900 CLAYSTONE: light blue gray, cream, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

18,900-18,950 CLAYSTONE: light blue gray, cream, firm, occasional hard, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, no visible oil stain; sample moderately contaminated with lube

18,950-19,000 CLAYSTONE: light blue gray, cream, firm, occasional hard, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; sample moderately contaminated with lube

19,000-19,050 CLAYSTONE: light to medium gray, off white, light blue-green, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

19,050-19,100 CLAYSTONE: light to medium gray, off white, light blue-green, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

19,100-19,150 CLAYSTONE: light to medium gray, off white, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

19,150-19,200 CLAYSTONE: light to medium gray, off white, light blue-green, firm, friable, trace hard, platy, silty texture, trace disseminated pyrite, trace nodular pyrite, no visible oil stain; sample moderately contaminated with lube

19,200-19,250 DOLOMITE: mudstone, off white, light brown, light gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace CLAYSTONE: light blue gray, cream, medium brown-gray, firm, platy, silty texture, trace disseminated pyrite, rare nodular pyrite, no visible oil stain; sample moderately contaminated with lube

19,250-19,300 DOLOMITE: mudstone, medium brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

19,300-19,350 DOLOMITE: mudstone, medium brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

19,350-19,400 DOLOMITE: mudstone, light brown, medium gray, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

19,400-19,450 DOLOMITE: mudstone, light brown, medium gray, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample moderately contaminated with lube

19,450-19,500 DOLOMITE: mudstone, light brown, medium gray, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,500-19,550 DOLOMITE: mudstone, tan, light brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,550-19,600 DOLOMITE: mudstone, gray, tan, light brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,600-19,650 DOLOMITE: mudstone, medium brown, off white, tan, light brown-gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,650-19,700 DOLOMITE: mudstone, off white, tan, light brown-gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,700-19,750 DOLOMITE: mudstone, off white, tan, light brown-gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,750-19,800 DOLOMITE: mudstone, tan-medium brown, gray, off white, tan, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,800-19,850 DOLOMITE: mudstone, light gray-brown, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,850-19,900 DOLOMITE: mudstone, light gray-brown, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,900-19,950 DOLOMITE: mudstone, medium brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

19,950-20,000 DOLOMITE: mudstone, tan, light orange, light to medium gray-brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub-blocky, rare disseminated pyrite; sample heavily contaminated with lube

20,000-20,050 DOLOMITE: mudstone, light to medium gray-brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample heavily contaminated with lube

20,050-20,100 DOLOMITE: mudstone, tan, medium gray, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample heavily contaminated with lube

20,100-20,150 DOLOMITE: mudstone, light brown, medium gray, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample heavily contaminated with lube

20,150-20,200 DOLOMITE: mudstone, tan, off white, gray-brown, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, occasional disseminated pyrite; sample heavily contaminated with lube

20,200-20,250 DOLOMITE: mudstone, light to medium brown-gray, tan, off white, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, occasional disseminated pyrite; sample heavily contaminated with lube

20,250-20,300 DOLOMITE: mudstone, light brown, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample moderately contaminated with lube

20,300-20,350 DOLOMITE: mudstone, light brown, tan, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample moderately contaminated with lube

20,350-20,400 DOLOMITE: mudstone, tan, off white, gray-brown, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, trace disseminated pyrite; sample heavily contaminated with lube

20,400-20,450 DOLOMITE: mudstone, light brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub blocky, trace disseminated pyrite; sample heavily contaminated with lube

20,450-20,500 DOLOMITE: mudstone, off white, light gray, rare light brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: pale green, firm, smooth, sub blocky, trace disseminated pyrite; sample heavily contaminated with lube

20,500-20,550 DOLOMITE: mudstone, off white, light gray, rare light brown, off white, microcrystalline, firm, occasional hard, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: light blue-green, firm, smooth, sub blocky, trace disseminated pyrite, rare nodular pyrite; sample heavily contaminated with lube

20,550-20,600 DOLOMITE: mudstone, medium brown-gray, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, rare nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; trace SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample moderately contaminated with lube

20,600-20,650 DOLOMITE: mudstone, light to medium gray-brown, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample heavily contaminated with lube

20,650-20,700 DOLOMITE: mudstone, tan, off white, medium gray, microcrystalline, firm, dense, earthy texture, occasional disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; rare SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample moderately contaminated with lube

20,700-20,750 DOLOMITE: mudstone, light brown, light gray, cream, off white, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: blue-green, firm, smooth, sub blocky, trace disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

20,750-20,800 DOLOMITE: mudstone, off white, cream, light brown, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; occasional SHALE: pale green, light blue-green, firm, smooth, sub blocky, rare disseminated pyrite, trace nodular pyrite; sample moderately contaminated with lube

20,800-20,840 DOLOMITE: mudstone, medium gray, light to medium brown, off white, tan, light orange, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, trace nodular pyrite, trace intercrystalline porosity, trace light brown spotty oil stain; common SHALE: blue-green, firm, smooth, sub blocky, rare disseminated pyrite; sample moderately contaminated with lube

ENSECO
Directional Drilling

Directional Survey Certification

Operator: Oasis Petroleum LLC **Well Name:** Kline Federal 5300 11-18 3T **API:** 33-053-06225

Enseco Job#: S15091-02 **Job Type:** MWD D&I **County, State:** McKenzie County, N. Dakota

Well Surface Hole Location (SHL): Lot 1, Sec. 18, T153N, R100W (1020' FNL & 290 FWL)

Latitude: 48° 04' 45.090 N **Longitude:** 103° 36' 10.590 W **Datum:** Nad 83

Final MWD Report Date: March 3, 2015 **MWD Survey Run Date:** March 1, 2015 to March 2, 2015

Tied In to Surveys Provided By: Enseco Directional Drilling D&I MWD **MD:** Surface

MWD Surveyed from 00 **ft to** 2,084.0 **ft MD** **Survey Type:** Positive Pulse D&I MWD **Sensor to Bit:** 63 **ft**

Rig Contractor: Noble **Rig Number:** 2 **RKB Height:** 2,056.0 **ft** **GL Elevation:** 2,056.0 **ft**

MWD Surveyor Name: David Hopper

"The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Enseco Energy Services USA Corp. I am authorized and qualified to review the data, calculations and this report and that the report represents a true and correct Directional Survey of this well based on the original data corrected to True North and obtained at the well site. Wellbore coordinates are calculated using the minimum curvature method."

Jonathan Hovland, Well Planner	Jonathan Hovland	March 6th 2015
Enseco Representative Name, Title	**Signature**	**Date Signed**

On this the __day of ___, 20__, before me personally appeared First & Last Name, to me known as the person described in and who executed the foregoing instrument and acknowledged the (s)he executed the same as his/her free act and deed.

Seal: ____________________ ____________________

Notary Public **Commission Expiry**

Enseco Survey Report

06 March, 2015

Oasis Petroleum LLC

McKenzie County, North Dakota
Lot 1 Sec.18 Twp.153N Rge.100W
Kline Federal 5300 11-18 3T
Job # S15091-02
API#: 33-053-06225

Survey: Final Surveys Vertical Section

ENSECO
Directional Drilling

Company:	Oasis Petroleum LLC	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	McKenzie County, North Dakota	**Ground Level Elevation:**	2,056.00usft
Site:	Lot 1 Sec.18 Twp.153N Rge.100W	**Wellhead Elevation:**	WELL @ 2056.00usft (Original Well Elev)
Well:	Kline Federal 5300 11-18 3T	**North Reference:**	True
Wellbore:	Job # S15091-02	**Survey Calculation Method:**	Minimum Curvature
Design:	Final Surveys Vertical Section		
		Database:	EDM5000

Project	McKenzie County, North Dakota		
Map System:	US State Plane 1983	**System Datum:**	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		Using geodetic scale factor

Site	Lot 1 Sec.18 Twp.153N Rge.100W				
Site Position:		**Northing:**	408,992.30 usft	**Latitude:**	48° 4' 45.680 N
From:	Lat/Long	**Easting:**	1,210,243.30 usft	**Longitude:**	103° 36' 10.190 W
Position Uncertainty:	0.00 usft	**Slot Radius:**	13-3/16 "	**Grid Convergence:**	-2.309 °

Well	Kline Federal 5300 11-18 3T | API#: 33-053-06225					
Well Position	**+N/-S**	-59.79 usft	**Northing:**	408,933.65 usft	**Latitude:**	48° 4' 45.090 N
	+E/-W	-27.16 usft	**Easting:**	1,210,213.75 usft	**Longitude:**	103° 36' 10.590 W
Position Uncertainty		0.00 usft	**Wellhead Elevation:**	usft	**Ground Level:**	2,056.00usft

Wellbore	Job # S15091-02				
Magnetics	**Model Name**	**Sample Date**	**Declination (°)**	**Dip Angle (°)**	**Field Strength (nT)**
	IGRF2015	3/6/2015	8.302	72.945	56,270

Design:	Final Surveys Vertical Section				
Audit Notes:			**Survey Error Model:**	Standard ISCWSA MWD Tool	
Version:	1.0	**Phase:**	ACTUAL	**Tie On Depth:**	0.00
Vertical Section:		**Depth From (TVD) (usft)**	**+N/-S (usft)**	**+E/-W (usft)**	**Direction (°)**
		0.00	0.00	0.00	305.01

Company:	Oasis Petroleum LLC	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	McKenzie County, North Dakota	**Ground Level Elevation:**	2,056.00usft
Site:	Lot 1 Sec.18 Twp.153N Rge.100W	**Wellhead Elevation:**	WELL @ 2056.00usft (Original Well Elev)
Well:	Kline Federal 5300 11-18 3T	**North Reference:**	True
Wellbore:	Job # S15091-02	**Survey Calculation Method:**	Minimum Curvature
Design:	Final Surveys Vertical Section	**Database:**	EDM5000

Survey

MD (usft)	Inc (°)	Azi (°)	TVD (usft)	SS (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
Tie-in from Surface										
0.00	**0.00**	**0.00**	**0.00**	**2,056.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
132.00	0.90	144.50	131.99	1,924.01	-0.84	0.60	-0.98	0.68	0.68	0.00
224.00	1.20	153.80	223.98	1,832.02	-2.30	1.45	-2.50	0.37	0.33	10.11
314.00	1.70	133.30	313.95	1,742.05	-4.06	2.83	-4.65	0.79	0.56	-22.78
399.00	0.60	159.80	398.93	1,657.07	-5.34	3.91	-6.26	1.40	-1.29	31.18
489.00	0.80	265.10	488.93	1,567.07	-5.84	3.44	-6.17	1.24	0.22	117.00
579.00	1.00	273.20	578.92	1,477.08	-5.85	2.03	-5.02	0.26	0.22	9.00
663.00	0.90	266.60	662.91	1,393.09	-5.84	0.64	-3.88	0.18	-0.12	-7.86
751.00	0.40	297.80	750.90	1,305.10	-5.74	-0.32	-3.03	0.68	-0.57	35.45
840.00	0.80	258.90	839.90	1,216.10	-5.72	-1.20	-2.29	0.62	0.45	-43.71
927.00	0.40	291.60	926.89	1,129.11	-5.72	-2.08	-1.58	0.59	-0.46	37.59
1,015.00	0.50	281.30	1,014.89	1,041.11	-5.53	-2.74	-0.93	0.15	0.11	-11.70
1,101.00	1.00	276.90	1,100.88	955.12	-5.37	-3.86	0.08	0.58	0.58	-5.12
1,188.00	0.40	308.00	1,187.87	868.13	-5.09	-4.85	1.05	0.79	-0.69	35.75
1,277.00	0.50	287.90	1,276.87	779.13	-4.78	-5.46	1.73	0.21	0.11	-22.58
1,365.00	0.60	270.00	1,364.87	691.13	-4.66	-6.29	2.48	0.22	0.11	-20.34
1,450.00	0.60	284.90	1,449.86	606.14	-4.55	-7.17	3.26	0.18	0.00	17.53
1,539.00	0.80	329.60	1,538.86	517.14	-3.89	-7.93	4.26	0.63	0.22	50.22
1,628.00	1.20	18.40	1,627.84	428.16	-2.47	-7.95	5.09	1.01	0.45	54.83
1,714.00	0.70	31.60	1,713.83	342.17	-1.17	-7.39	5.38	0.63	-0.58	15.35
1,803.00	0.70	24.10	1,802.83	253.17	-0.21	-6.88	5.52	0.10	0.00	-8.43
1,889.00	1.10	23.80	1,888.81	167.19	1.02	-6.34	5.78	0.47	0.47	-0.35
1,987.00	1.00	23.10	1,986.80	69.20	2.67	-5.62	6.14	0.10	-0.10	-0.71
Last MWD Survey										
2,084.00	**0.50**	**95.30**	**2,083.79**	**-27.79**	**3.41**	**-4.87**	**5.94**	**1.00**	**-0.52**	**74.43**

Survey Annotations

MD (usft)	TVD (usft)	Local Coordinates +N/-S (usft)	Local Coordinates +E/-W (usft)	Comment
0.00	0.00	0.00	0.00	Tie-in from Surface
2,084.00	2,083.79	3.41	-4.87	Last MWD Survey

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Thursday, June 11, 2015

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Kline Federal 5300 11-18 3T
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

Surveyor Name	*Surveyor Title*	*Borehole Number*	*Start Depth*	*End Depth*	*Start Date*	*End Date*	*Type of*	*TD Straight Line Projection*
David Foley	MWD Operator	O.H.	2084'	20775'	03/27/15	05/28/15	MWD	20840'

If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson

Douglas Hudson
Well Planner

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Thursday, May 28, 2015

State of North Dakota
County of McKenzie

Subject: **Survey Certification Letter**

Survey Company:	**Ryan Directional Services, Inc.**		
Job Number:	**8982**	Surface:	**48° 4' 45.090 N / 103° 36' 10.590 W**
Survey Job Type:	**Ryan MWD**	A.P.I. No:	**33-053-06225**
Customer:	**Oasis Petroleum North America LLC**	Location:	**McKenzie, ND**
Well Name:	**Kline Federal 5300 11-18 3T**	RKB Height:	**2069'**
Rig Name:	**Xtreme 21**	Distance to Bit:	**65'**

Surveyor Name	*Surveyor Title*	*Borehole Number*	*Start Depth*	*End Depth*	*Start Date*	*End Date*	*Type of*	*TD Straight Line Projection*
David Foley Jr	MWD Supervisor	OH	2084'	20775'	03/27/15	05/28/15	MWD	20840'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

David Foley Jr
MWD Supervisor
Ryan Directional Services, Inc.

Report #: 1

Date: 28-May-15

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # 8982

SURVEY REPORT

Customer:	Oasis Petroleum, Inc	MWD Operator:	David Foley / David Unger
Well Name:	Kline Federal 5300 11-18 3T	Directional Drillers:	RPM
Rig #:	Xtreme 21	Survey Corrected To:	True North
API #:	33-053-06225	Vertical Section Direction:	94.89
Calculation Method:	Minimum Curvature Calculation	Total Correction:	8.30
		Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	**2084**	**0.50**	**95.30**		**2083.79**	**-5.14**	**3.41**	**-4.87**	**1.00**
1	2176	1.10	95.00	64.00	2175.78	-3.86	3.30	-3.59	0.65
2	2270	0.50	245.30	66.00	2269.78	-3.31	3.05	-3.06	1.65
3	2365	0.80	289.20	68.00	2364.77	-4.32	3.09	-4.07	0.59
4	2459	1.20	301.20	69.00	2458.76	-5.83	3.82	-5.53	0.48
5	**2553**	**1.60**	**317.30**	**69.00**	**2552.73**	**-7.69**	**5.29**	**-7.26**	**0.59**
6	2647	1.10	275.70	71.00	2646.70	-9.56	6.34	-9.05	1.13
7	2742	1.90	286.60	73.00	2741.67	-12.01	6.89	-11.47	0.89
8	2836	1.10	300.50	73.00	2835.64	-14.35	7.79	-13.74	0.93
9	2930	1.00	227.50	75.00	2929.63	-15.72	7.69	-15.12	1.33
10	**3025**	**1.50**	**247.70**	**77.00**	**3024.60**	**-17.39**	**6.66**	**-16.88**	**0.69**
11	3119	1.60	247.60	78.00	3118.57	-19.65	5.69	-19.23	0.11
12	3213	1.50	255.00	80.00	3212.53	-21.97	4.88	-21.63	0.24
13	3307	1.30	255.00	82.00	3306.51	-24.13	4.28	-23.85	0.21
14	3401	1.20	249.80	84.00	3400.48	-26.02	3.66	-25.81	0.16
15	**3496**	**1.10**	**261.80**	**86.00**	**3495.47**	**-27.81**	**3.19**	**-27.64**	**0.27**
16	3590	1.10	244.70	87.00	3589.45	-29.47	2.68	-29.35	0.35
17	3684	0.90	247.30	87.00	3683.43	-30.91	2.01	-30.85	0.22
18	3778	0.80	248.50	89.00	3777.42	-32.15	1.48	-32.14	0.11
19	3872	0.80	253.40	91.00	3871.41	-33.35	1.05	-33.38	0.07
20	**3966**	**0.70**	**247.90**	**93.00**	**3965.41**	**-34.47**	**0.65**	**-34.54**	**0.13**
21	4060	0.50	263.20	93.00	4059.40	-35.38	0.38	-35.48	0.27
22	4154	0.40	283.10	95.00	4153.40	-36.11	0.41	-36.21	0.20
23	4249	0.50	270.30	96.00	4248.40	-36.85	0.49	-36.94	0.15
24	4343	0.50	291.40	98.00	4342.39	-37.65	0.64	-37.73	0.19
25	**4437**	**0.50**	**283.70**	**100.00**	**4436.39**	**-38.45**	**0.89**	**-38.51**	**0.07**
26	4531	0.30	306.20	102.00	4530.39	-39.07	1.13	-39.11	0.27
27	4626	0.50	299.70	104.00	4625.38	-39.65	1.48	-39.67	0.22
28	4720	0.40	291.00	105.00	4719.38	-40.34	1.80	-40.34	0.13
29	4813	0.40	349.80	105.00	4812.38	-40.74	2.24	-40.70	0.42
30	**4907**	**0.20**	**327.40**	**107.00**	**4906.38**	**-40.92**	**2.70**	**-40.84**	**0.24**
31	5001	0.50	307.70	109.00	5000.38	-41.37	3.09	-41.26	0.34
32	5096	0.60	273.70	111.00	5095.37	-42.21	3.37	-42.08	0.35
33	5190	0.20	210.80	111.00	5189.37	-42.78	3.26	-42.65	0.57
34	5284	0.70	243.80	113.00	5283.37	-43.34	2.87	-43.25	0.58
35	**5378**	**0.60**	**214.10**	**113.00**	**5377.36**	**-44.07**	**2.21**	**-44.05**	**0.37**
36	5473	0.80	203.20	113.00	5472.35	-44.52	1.19	-44.59	0.25
37	5567	1.10	195.00	114.00	5566.34	-44.89	-0.29	-45.08	0.35
38	5661	1.10	197.70	116.00	5660.32	-45.25	-2.02	-45.59	0.06
39	5756	0.60	151.60	116.00	5755.31	-45.18	-3.32	-45.63	0.85
40	**5850**	**0.40**	**212.30**	**118.00**	**5849.31**	**-45.06**	**-4.03**	**-45.57**	**0.57**
41	5944	0.40	178.80	120.00	5943.31	-45.17	-4.64	-45.74	0.25
42	6038	0.10	283.40	122.00	6037.31	-45.22	-4.95	-45.81	0.46
43	6132	0.40	352.00	123.00	6131.31	-45.37	-4.61	-45.93	0.40
44	6226	0.70	17.60	125.00	6225.30	-45.32	-3.73	-45.81	0.41
45	**6320**	**0.90**	**3.60**	**127.00**	**6319.29**	**-45.21**	**-2.45**	**-45.59**	**0.30**
46	6414	1.30	0.40	129.00	6413.28	-45.31	-0.65	-45.53	0.43
47	6509	1.00	357.40	131.00	6508.26	-45.50	1.26	-45.56	0.32
48	6603	1.00	2.40	125.00	6602.24	-45.65	2.90	-45.57	0.09
49	6697	0.80	25.70	127.00	6696.23	-45.45	4.31	-45.25	0.44
50	**6791**	**0.40**	**161.10**	**132.00**	**6790.23**	**-45.08**	**4.59**	**-44.86**	**1.19**
51	6885	0.20	310.80	134.00	6884.23	-45.08	4.39	-44.87	0.62
52	6979	0.70	56.50	136.00	6978.22	-44.77	4.81	-44.52	0.83
53	7074	0.70	55.20	138.00	7073.22	-43.87	5.46	-43.56	0.02
54	7168	0.80	54.70	140.00	7167.21	-42.92	6.17	-42.55	0.11
55	**7262**	**0.80**	**43.00**	**141.00**	**7261.20**	**-42.02**	**7.03**	**-41.57**	**0.17**
56	7356	0.60	107.30	141.00	7355.19	-41.13	7.36	-40.65	0.81
57	7450	0.20	49.70	145.00	7449.19	-40.53	7.32	-40.06	0.55
58	7544	0.40	25.70	147.00	7543.19	-40.30	7.72	-39.79	0.25
59	7639	0.60	165.40	149.00	7638.19	-40.02	7.54	-39.52	0.99
60	**7733**	**0.30**	**185.80**	**150.00**	**7732.19**	**-39.86**	**6.82**	**-39.42**	**0.36**

Report #: 1

Date: 28-May-15

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # 8982

SURVEY REPORT

Customer:	Oasis Petroleum, Inc	MWD Operator:	David Foley / David Unger
Well Name:	Kline Federal 5300 11-18 3T	Directional Drillers:	RPM
Rig #:	Xtreme 21	Survey Corrected To:	True North
API #:	33-053-06225	Vertical Section Direction:	94.89
Calculation Method:	Minimum Curvature Calculation	Total Correction:	8.30
		Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
61	7827	0.10	24.70	154.00	7826.19	-39.83	6.65	-39.41	0.42
62	7921	0.00	8.80	154.00	7920.19	-39.81	6.72	-39.38	0.11
63	8016	0.20	346.50	154.00	8015.19	-39.86	6.89	-39.41	0.21
64	8109	0.30	36.30	154.00	8108.19	-39.78	7.24	-39.31	0.25
65	**8204**	**0.30**	**50.00**	**158.00**	**8203.18**	**-39.48**	**7.60**	**-38.97**	**0.08**
66	8298	0.20	133.80	159.00	8297.18	-39.17	7.65	-38.66	0.36
67	8392	0.40	89.90	161.00	8391.18	-38.72	7.53	-38.22	0.31
68	8486	0.50	86.20	161.00	8485.18	-37.99	7.56	-37.48	0.11
69	8580	0.70	67.40	143.00	8579.17	-37.07	7.81	-36.54	0.30
70	**8674**	**0.60**	**58.50**	**149.00**	**8673.17**	**-36.17**	**8.29**	**-35.59**	**0.15**
71	8769	0.80	52.70	154.00	8768.16	-35.28	8.95	-34.64	0.22
72	8863	0.70	61.00	158.00	8862.15	-34.31	9.62	-33.61	0.16
73	8957	1.10	225.40	161.00	8956.15	-34.42	9.27	-33.76	1.90
74	9051	1.20	232.70	167.00	9050.13	-35.74	8.04	-35.18	0.19
75	**9146**	**0.80**	**240.80**	**167.00**	**9145.11**	**-37.02**	**7.11**	**-36.55**	**0.45**
76	9240	0.70	259.80	170.00	9239.11	-38.12	6.69	-37.69	0.28
77	9334	0.60	245.30	168.00	9333.10	-39.10	6.38	-38.70	0.20
78	9428	0.40	284.40	174.00	9427.10	-39.86	6.26	-39.47	0.41
79	9523	0.20	279.40	174.00	9522.10	-40.35	6.37	-39.95	0.21
80	**9617**	**0.20**	**291.70**	**177.00**	**9616.10**	**-40.67**	**6.46**	**-40.27**	**0.05**
81	9711	0.20	314.00	176.00	9710.10	-40.95	6.63	-40.54	0.08
82	9805	0.20	345.20	177.00	9804.09	-41.14	6.90	-40.70	0.11
83	9899	0.10	252.70	179.00	9898.09	-41.27	7.04	-40.82	0.24
84	9993	0.20	46.50	183.00	9992.09	-41.23	7.13	-40.78	0.31
85	**10087**	**0.50**	**10.80**	**183.00**	**10086.09**	**-41.08**	**7.64**	**-40.58**	**0.38**
86	10182	0.70	20.10	185.00	10181.09	-40.89	8.59	-40.30	0.23
87	10226	1.10	22.00	185.00	10225.08	-40.69	9.24	-40.05	0.91
88	10263	1.00	26.00	172.00	10262.08	-40.47	9.86	-39.78	0.33
89	10294	2.50	101.40	172.00	10293.06	-39.70	9.97	-39.00	7.89
90	**10326**	**6.50**	**115.60**	**176.00**	**10324.96**	**-37.31**	**9.05**	**-36.68**	**12.88**
91	10357	10.80	121.30	176.00	10355.60	-33.07	6.78	-32.61	14.13
92	10388	14.80	124.80	176.00	10385.82	-27.03	3.01	-26.88	13.14
93	10420	19.20	128.30	181.00	10416.42	-19.09	-2.59	-19.39	14.11
94	10451	23.80	131.50	183.00	10445.25	-9.81	-9.90	-10.70	15.31
95	**10483**	**28.00**	**134.90**	**185.00**	**10474.04**	**1.13**	**-19.48**	**-0.53**	**13.92**
96	10514	31.30	135.90	185.00	10500.97	12.78	-30.41	10.23	10.76
97	10545	33.00	138.40	186.00	10527.22	24.98	-42.50	21.44	6.96
98	10577	35.20	140.80	186.00	10553.72	37.72	-56.17	33.05	8.06
99	10608	37.70	142.20	186.00	10578.65	50.37	-70.58	44.51	8.50
100	**10639**	**40.20**	**142.90**	**188.00**	**10602.76**	**63.49**	**-86.06**	**56.36**	**8.19**
101	10671	43.40	142.40	188.00	10626.61	77.83	-103.01	69.30	10.05
102	10702	46.70	140.90	190.00	10648.51	92.86	-120.20	82.92	11.18
103	10734	50.20	140.80	190.00	10669.73	109.51	-138.77	98.03	10.94
104	10765	53.10	141.20	190.00	10688.96	126.36	-157.67	113.33	9.41
105	**10796**	**55.10**	**140.50**	**192.00**	**10707.14**	**143.81**	**-177.14**	**129.18**	**6.71**
106	10828	57.80	139.80	192.00	10724.82	162.59	-197.61	146.27	8.63
107	10859	60.50	138.30	192.00	10740.72	181.68	-217.70	163.72	9.65
108	10890	64.40	136.50	192.00	10755.06	201.94	-237.92	182.32	13.59
109	10922	67.10	135.70	192.00	10768.20	223.89	-258.94	202.55	8.74
110	**10953**	**70.90**	**135.60**	**192.00**	**10779.30**	**245.81**	**-279.63**	**222.78**	**12.26**
111	10984	74.60	136.60	194.00	10788.50	268.07	-300.96	243.30	12.33
112	11016	77.50	135.90	194.00	10796.21	291.38	-323.39	264.78	9.31
113	11047	80.20	134.90	194.00	10802.20	314.50	-345.05	286.13	9.27
114	11079	82.10	134.20	195.00	10807.13	338.84	-367.23	308.66	6.32
115	**11110**	**84.00**	**134.10**	**195.00**	**10810.88**	**362.67**	**-388.66**	**330.74**	**6.14**
116	11140	88.40	133.50	197.00	10812.87	385.96	-409.37	352.34	14.80
117	11236	89.60	129.10	235.00	10814.54	463.18	-472.71	424.43	4.75
118	11267	89.30	128.10	235.00	10814.84	488.97	-492.05	448.66	3.37
119	11298	89.10	127.50	235.00	10815.27	514.99	-511.04	473.15	2.04
120	**11329**	**87.80**	**125.50**	**235.00**	**10816.11**	**541.38**	**-529.48**	**498.06**	**7.69**

Report #: 1

Date: 28-May-15

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # 8982

SURVEY REPORT

Customer:	Oasis Petroleum, Inc	MWD Operator:	David Foley / David Unger
Well Name:	Kline Federal 5300 11-18 3T	Directional Drillers:	RPM
Rig #:	Xtreme 21	Survey Corrected To:	True North
API #:	33-053-06225	Vertical Section Direction:	94.89
Calculation Method:	Minimum Curvature Calculation	Total Correction:	8.30
		Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
121	11360	87.60	125.20	237.00	10817.36	568.08	-547.40	523.33	1.16
122	11391	87.70	124.80	237.00	10818.63	594.88	-565.16	548.70	1.33
123	11421	88.30	122.40	237.00	10819.67	621.17	-581.75	573.67	8.24
124	11452	88.10	122.00	233.00	10820.65	648.70	-598.26	599.89	1.44
125	**11483**	**87.90**	**121.90**	**235.00**	**10821.73**	**676.29**	**-614.66**	**626.17**	**0.72**
126	11514	89.60	119.40	237.00	10822.41	704.20	-630.46	652.83	9.75
127	11545	89.00	117.50	233.00	10822.78	732.61	-645.22	680.09	6.43
128	11576	89.00	117.10	235.00	10823.33	761.27	-659.44	707.63	1.29
129	11607	90.50	114.90	237.00	10823.46	790.18	-673.03	735.49	8.59
130	**11638**	**91.60**	**114.60**	**239.00**	**10822.89**	**819.33**	**-686.00**	**763.64**	**3.68**
131	11669	92.00	113.50	239.00	10821.92	848.60	-698.63	791.93	3.77
132	11699	91.50	110.50	239.00	10821.00	877.26	-709.86	819.73	10.13
133	11730	91.30	109.80	237.00	10820.24	907.16	-720.54	848.82	2.35
134	11761	91.30	108.80	239.00	10819.54	937.17	-730.78	878.07	3.22
135	**11792**	**90.10**	**107.20**	**240.00**	**10819.16**	**967.36**	**-740.36**	**907.55**	**6.45**
136	11822	89.60	106.90	240.00	10819.24	996.69	-749.16	936.24	1.94
137	11853	89.50	106.10	240.00	10819.48	1027.05	-757.96	965.96	2.60
138	11883	88.30	104.00	240.00	10820.06	1056.57	-765.75	994.92	8.06
139	11915	88.30	103.90	240.00	10821.01	1088.16	-773.46	1025.96	0.31
140	**11946**	**88.40**	**103.70**	**240.00**	**10821.90**	**1118.77**	**-780.85**	**1056.06**	**0.72**
141	11978	89.70	101.60	242.00	10822.43	1150.47	-787.86	1087.27	7.72
142	12010	89.60	101.50	244.00	10822.63	1182.26	-794.26	1118.62	0.44
143	12042	90.00	101.60	244.00	10822.74	1214.04	-800.67	1149.98	1.29
144	12074	89.80	99.10	244.00	10822.80	1245.89	-806.42	1181.45	7.84
145	**12105**	**89.80**	**98.60**	**244.00**	**10822.90**	**1276.82**	**-811.19**	**1212.08**	**1.61**
146	12137	90.20	97.90	244.00	10822.90	1308.76	-815.78	1243.75	2.52
147	12168	90.50	96.20	244.00	10822.72	1339.74	-819.58	1274.52	5.57
148	12200	90.60	96.00	246.00	10822.41	1371.73	-822.98	1306.33	0.70
149	12232	90.80	94.80	244.00	10822.02	1403.73	-826.00	1338.19	3.80
150	**12264**	**90.10**	**93.30**	**244.00**	**10821.77**	**1435.72**	**-828.26**	**1370.11**	**5.17**
151	12359	90.70	92.80	248.00	10821.10	1530.67	-833.31	1464.97	0.82
152	12453	89.00	92.70	249.00	10821.35	1624.60	-837.82	1558.86	1.81
153	12547	89.90	92.50	251.00	10822.25	1718.52	-842.08	1652.75	0.98
154	12642	90.40	92.00	253.00	10822.00	1813.42	-845.81	1747.68	0.74
155	**12738**	**89.60**	**92.60**	**251.00**	**10822.00**	**1909.32**	**-849.67**	**1843.60**	**1.04**
156	12832	88.50	92.70	251.00	10823.56	2003.23	-854.01	1937.49	1.18
157	12928	87.70	91.70	255.00	10826.74	2099.07	-857.69	2033.36	1.33
158	13024	88.70	90.90	255.00	10829.76	2194.83	-859.87	2129.29	1.33
159	13119	89.10	90.50	257.00	10831.58	2289.56	-861.03	2224.26	0.60
160	**13213**	**89.20**	**90.60**	**255.00**	**10832.98**	**2383.28**	**-861.93**	**2318.25**	**0.15**
161	13308	90.90	88.70	257.00	10832.89	2477.88	-861.35	2413.24	2.68
162	13403	90.20	87.90	258.00	10831.98	2572.25	-858.54	2508.19	1.12
163	13498	90.50	88.20	257.00	10831.40	2666.57	-855.30	2603.13	0.45
164	13593	89.70	90.20	258.00	10831.24	2761.09	-853.98	2698.12	2.27
165	**13689**	**88.40**	**89.20**	**260.00**	**10832.83**	**2856.68**	**-853.47**	**2794.10**	**1.71**
166	13783	90.40	90.00	258.00	10833.81	2950.27	-852.82	2888.09	2.29
167	13879	90.40	90.10	260.00	10833.14	3045.93	-852.90	2984.09	0.10
168	13975	92.80	89.40	262.00	10830.46	3141.50	-852.48	3080.04	2.60
169	14070	91.30	89.20	257.00	10827.06	3235.98	-851.32	3174.97	1.59
170	**14164**	**88.80**	**90.10**	**258.00**	**10826.98**	**3329.58**	**-850.75**	**3268.96**	**2.83**
171	14258	86.90	90.10	260.00	10830.51	3423.18	-850.91	3362.89	2.02
172	14353	91.20	90.70	260.00	10832.08	3517.85	-851.58	3457.85	4.57
173	14449	88.20	90.70	260.00	10832.59	3613.59	-852.75	3553.83	3.13
174	14543	88.00	90.20	262.00	10835.70	3707.25	-853.49	3647.78	0.57
175	**14638**	**88.70**	**89.80**	**262.00**	**10838.44**	**3801.87**	**-853.49**	**3742.73**	**0.85**
176	14733	91.20	89.10	264.00	10838.52	3896.43	-852.58	3837.72	2.73
177	14828	89.10	88.50	264.00	10838.27	3990.89	-850.59	3932.70	2.30
178	14925	89.90	88.50	266.00	10839.12	4087.28	-848.05	4029.66	0.82
179	15022	89.60	88.50	264.00	10839.54	4183.68	-845.51	4126.62	0.31
180	**15117**	**89.60**	**88.80**	**266.00**	**10840.20**	**4278.11**	**-843.27**	**4221.59**	**0.32**

Report #: 1
Date: 28-May-15

RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # 8982

SURVEY REPORT

Customer:	Oasis Petroleum, Inc	MWD Operator:	David Foley / David Unger
Well Name:	Kline Federal 5300 11-18 3T	Directional Drillers:	RPM
Rig #:	Xtreme 21	Survey Corrected To:	True North
API #:	33-053-06225	Vertical Section Direction:	94.89
Calculation Method:	Minimum Curvature Calculation	Total Correction:	8.30
		Temperature Forecasting Model (Chart Only):	Logarithmic

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
181	15212	91.50	87.60	266.00	10839.29	4372.45	-840.29	4316.54	2.37
182	15306	88.70	88.10	264.00	10839.13	4465.74	-836.76	4410.46	3.03
183	15401	87.80	90.20	264.00	10842.03	4560.21	-835.35	4505.40	2.40
184	15496	89.10	90.70	262.00	10844.60	4654.89	-836.10	4600.36	1.47
185	**15592**	**90.00**	**89.50**	**264.00**	**10845.35**	**4750.55**	**-836.26**	**4696.35**	**1.56**
186	15687	90.60	88.40	266.00	10844.86	4845.03	-834.52	4791.34	1.32
187	15782	90.40	89.60	266.00	10844.03	4939.53	-832.87	4886.32	1.28
188	15877	89.50	90.60	266.00	10844.11	5034.19	-833.03	4981.31	1.42
189	15973	88.80	90.90	267.00	10845.53	5129.93	-834.29	5077.29	0.79
190	**16069**	**90.40**	**90.40**	**269.00**	**10846.20**	**5225.66**	**-835.38**	**5173.28**	**1.75**
191	16166	91.00	89.30	269.00	10845.02	5322.28	-835.12	5270.27	1.29
192	16262	91.20	88.70	271.00	10843.18	5417.75	-833.45	5366.24	0.66
193	16358	92.00	89.70	269.00	10840.50	5513.24	-832.11	5462.19	1.33
194	16454	90.90	90.40	269.00	10838.07	5608.87	-832.19	5558.16	1.36
195	**16551**	**91.00**	**90.20**	**269.00**	**10836.46**	**5705.54**	**-832.70**	**5655.14**	**0.23**
196	16647	91.50	90.90	271.00	10834.36	5801.24	-833.62	5751.11	0.90
197	16742	89.20	91.70	271.00	10833.78	5896.05	-835.78	5846.08	2.56
198	16838	88.80	92.10	273.00	10835.46	5991.90	-838.96	5942.01	0.59
199	16935	88.30	91.80	271.00	10837.91	6088.74	-842.26	6038.93	0.60
200	**17030**	**89.70**	**90.70**	**273.00**	**10839.57**	**6183.53**	**-844.33**	**6133.89**	**1.87**
201	17124	90.50	90.50	273.00	10839.41	6277.27	-845.31	6227.88	0.88
202	17220	88.70	91.00	273.00	10840.08	6373.01	-846.57	6323.86	1.95
203	17317	88.00	91.40	273.00	10842.87	6469.77	-848.60	6420.80	0.83
204	17412	89.10	91.10	273.00	10845.27	6564.55	-850.67	6515.75	1.20
205	**17508**	**89.20**	**91.10**	**275.00**	**10846.70**	**6660.32**	**-852.52**	**6611.72**	**0.10**
206	17603	89.20	91.00	275.00	10848.03	6755.10	-854.26	6706.69	0.11
207	17698	90.30	90.70	275.00	10848.44	6849.86	-855.67	6801.68	1.20
208	17794	89.60	90.80	275.00	10848.52	6945.61	-856.92	6897.67	0.74
209	17891	89.50	88.90	273.00	10849.29	7042.23	-856.67	6994.66	1.96
210	**17987**	**88.60**	**88.70**	**273.00**	**10850.88**	**7137.67**	**-854.66**	**7090.63**	**0.96**
211	18083	89.60	87.80	267.00	10852.38	7233.02	-851.73	7186.57	1.40
212	18180	88.90	88.80	273.00	10853.65	7329.37	-848.85	7283.52	1.26
213	18276	89.00	88.20	273.00	10855.41	7424.75	-846.34	7379.47	0.63
214	18372	89.50	88.70	273.00	10856.67	7520.14	-843.74	7475.42	0.74
215	**18468**	**91.10**	**91.60**	**271.00**	**10856.17**	**7615.79**	**-843.99**	**7571.41**	**3.45**
216	18562	91.70	91.90	273.00	10853.87	7709.63	-846.86	7665.34	0.71
217	18658	91.10	91.60	273.00	10851.53	7805.45	-849.79	7761.26	0.70
218	18753	88.00	92.20	273.00	10852.27	7900.31	-852.94	7856.20	3.32
219	18851	87.20	91.80	271.00	10856.38	7998.10	-856.36	7954.05	0.91
220	**18947**	**88.70**	**90.40**	**271.00**	**10859.81**	**8093.82**	**-858.20**	**8049.96**	**2.14**
221	19041	91.00	88.00	271.00	10860.06	8187.34	-856.89	8143.94	3.54
222	19138	92.70	87.70	275.00	10856.92	8283.56	-853.25	8240.82	1.78
223	19234	93.30	88.60	275.00	10851.90	8378.76	-850.16	8336.64	1.13
224	19329	92.80	89.20	275.00	10846.85	8473.11	-848.33	8431.48	0.82
225	**19425**	**91.30**	**89.30**	**273.00**	**10843.41**	**8568.58**	**-847.08**	**8527.41**	**1.57**
226	19520	90.80	91.60	275.00	10841.67	8663.27	-847.82	8622.39	2.48
227	19615	90.00	93.60	276.00	10841.01	8758.19	-852.13	8717.28	2.27
228	19710	91.60	94.50	276.00	10839.68	8853.17	-858.84	8812.03	1.93
229	19805	89.40	93.50	275.00	10838.85	8948.14	-865.47	8906.79	2.54
230	**19900**	**88.40**	**93.20**	**276.00**	**10840.68**	**9043.09**	**-871.02**	**9001.61**	**1.10**
231	19996	89.80	91.50	275.00	10842.18	9138.98	-874.95	9097.51	2.29
232	20091	89.90	91.90	276.00	10842.43	9233.83	-877.77	9192.47	0.43
233	20187	89.30	91.10	276.00	10843.10	9329.66	-880.29	9288.43	1.04
234	20283	89.60	90.80	276.00	10844.02	9425.43	-881.88	9384.41	0.44
235	**20379**	**90.90**	**91.20**	**276.00**	**10843.60**	**9521.20**	**-883.55**	**9480.39**	**1.42**
236	20475	89.50	89.50	275.00	10843.27	9616.90	-884.14	9576.39	2.29
237	20571	89.10	88.70	276.00	10844.44	9712.40	-882.63	9672.37	0.93
238	20668	90.90	88.80	276.00	10844.44	9808.84	-880.51	9769.34	1.86
239	20764	92.00	88.10	276.00	10842.01	9904.20	-877.92	9865.27	1.36
240	**20775**	**92.30**	**87.80**	**276.00**	**10841.60**	**9915.11**	**-877.53**	**9876.26**	**3.86**
Projection	20840	92.30	87.80		10838.99	9979.56	-875.03	9941.16	0.00

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED APR 2015 ND OIL & GAS DIVISION

Well File No. **29244**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **April 9, 2015**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [x] Other **Suspension of Drilling**
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F N L	290 F W L	11	18	153 N	100 W
Field: Baker	Pool: Bakken	County: McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum requests permission for suspension of drilling for approximately 55 days for the referenced well under NDAC 43-02-03-055. Oasis would like to suspend drilling on this well in order to drill the approved Carson SWD 5301 12-24 (well file #90329). The current rig will move to the Carson SWD pad once the vertical well bores have been drilled for all 3 wells on the Kline 11-18 pad. Oasis will return to the Kline 11-18 pad with a second rig on approximately June 6, 2015, to drill the lateral portion of the referenced well to TD.

Company		Telephone Number
Oasis Petroleum North America LLC		281-404-9575
Address: 1001 Fannin St, Suite 1500		
City: Houston	State: TX	Zip Code: 77002
Signature: Michael Kukuk	Printed Name: Michael Kukuk	
Title: Regulatory Supervisor	Date: March 26, 2015	
Email Address: mkukuk@oasispetroleum.com		

FOR STATE USE ONLY

[ ] Received	[x] Approved
Date: 4/8/15	
By: Todd L. Holweger	
Title: Mineral Resources Permit Manager	

Holweger, Todd L.

From: Michael Kukuk <mkukuk@oasispetroleum.com>
Sent: Thursday, March 26, 2015 6:40 PM
To: Holweger, Todd L.
Cc: Regulatory; APD; Karyme Martin; Jason Swaren
Subject: SOD sundries for the Kline Federal 5300 11-18 pad
Attachments: Kline Federal 11-18 SOD sundries.pdf; ATT00001.txt

Importance: High

Good Evening Todd,

Per our conversation I have attached the SOD sundries for the 3 wells on the Kline pad. A few key points:

1) **We will move the rig to the Carson SWD pad once we have finished drilling the vertical portions of all 3 wells on this pad.**
2) **We will finish drilling the vertical portion of the third well, the Kline Federal 5300 11-18 2B, on April 9th.**
3) **We will utilize a 2nd rig to drill the lateral portions of the 3 wells on this pad.**
4) **The 2nd rig is currently drilling wells on a different pad and is scheduled to reach TD on the final well in late May/early June, leaving a gap of approximately 55 days.**
5) **We will be able to return to the Kline Federal 5300 11-18 pad on or before June 6th.**

Given the time sensitive nature of this request, expedited review of these sundries would be greatly appreciated.

Thank you for your consideration,

Michael P. Kukuk
Regulatory Supervisor
1001 Fannin, Suite 1500
Houston, Texas 77002
281-404-9575
281-382-5877 (cell)

mkukuk@oasispetroleum.com

OASIS PETROLEUM

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: August 22, 2014
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Redrilling or Repair
- [ ] Casing or Liner
- [ ] Plug Well
- [ ] Supplemental History
- [ ] Temporarily Abandon
- [x] Other Address
- [ ] Spill Report
- [ ] Shooting
- [ ] Acidizing
- [ ] Fracture Treatment
- [ ] Change Production Method
- [ ] Reclamation

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F N L	290 F W L	LOT1	18	153 N	100 W
Field: Baker	Pool: Bakken	County: McKenzie			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully submits the below physical addres for the above referenced well:

13798 46th Street NW, Pad 2
Alexander, ND 58831

Company: Oasis Petroleum North America LLC	Telephone Number: 281-404-9563
Address: 1001 Fannin, Suite 1500	
City: Houston	State: TX — Zip Code: 77002
Signature: Heather McCowan	Printed Name: Heather McCowan
Title: Regulatory Assistant	Date: August 22, 2014
Email Address: hmccowan@oasispetroleum.com	

FOR STATE USE ONLY

[x] Received	[ ] Approved
Date: 3/17/2015	
By: Matthew Messana	
Title: ENGINEERING TECHNICIAN	

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

FEB 2015
RECEIVED
ND OIL & GAS DIVISION

Well File No.
29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **April 14, 2015**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Spill Report
- [ ] Redrilling or Repair
- [ ] Shooting
- [ ] Casing or Liner
- [ ] Acidizing
- [ ] Plug Well
- [ ] Fracture Treatment
- [ ] Supplemental History
- [ ] Change Production Method
- [ ] Temporarily Abandon
- [ ] Reclamation
- [x] Other **Change to Original APD**

Well Name and Number				
Kline Federal 5300 11-18 3T				
Footages		Qtr-Qtr	Section	Township	Range
1020 F N L	290 F W L	NWNW	18	153 N	100 W

Field	Pool	County
Baker	Bakken	McKenzie

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests approval to make the following changes to the original APD as follows:

255' NDIC Calc

BHL change: 1875' FNL & 250' FEL Sec 17 T153N R100W
(previously: 2030' FNL & 206' FEL)

Surface casing design:
Surface Casing of 13 3/8" set at 2068' (previously 9 5/8")
Contingency Casing of 9 5/8" set at 6447'
Intermediate Casing of 7" with weight of 32 set at 11,080' (previously set at 11,085)
Production liner of 4 1/2" set from 10,283' to 20924' (previously set from 10,287' to 21,039')

See attached supporting documents.

Company	Telephone Number
Oasis Petroleum North America LLC	(281) 404-9652

Address		
1001 Fannin Suite 1500		
City	State	Zip Code
Houston	TX	77002

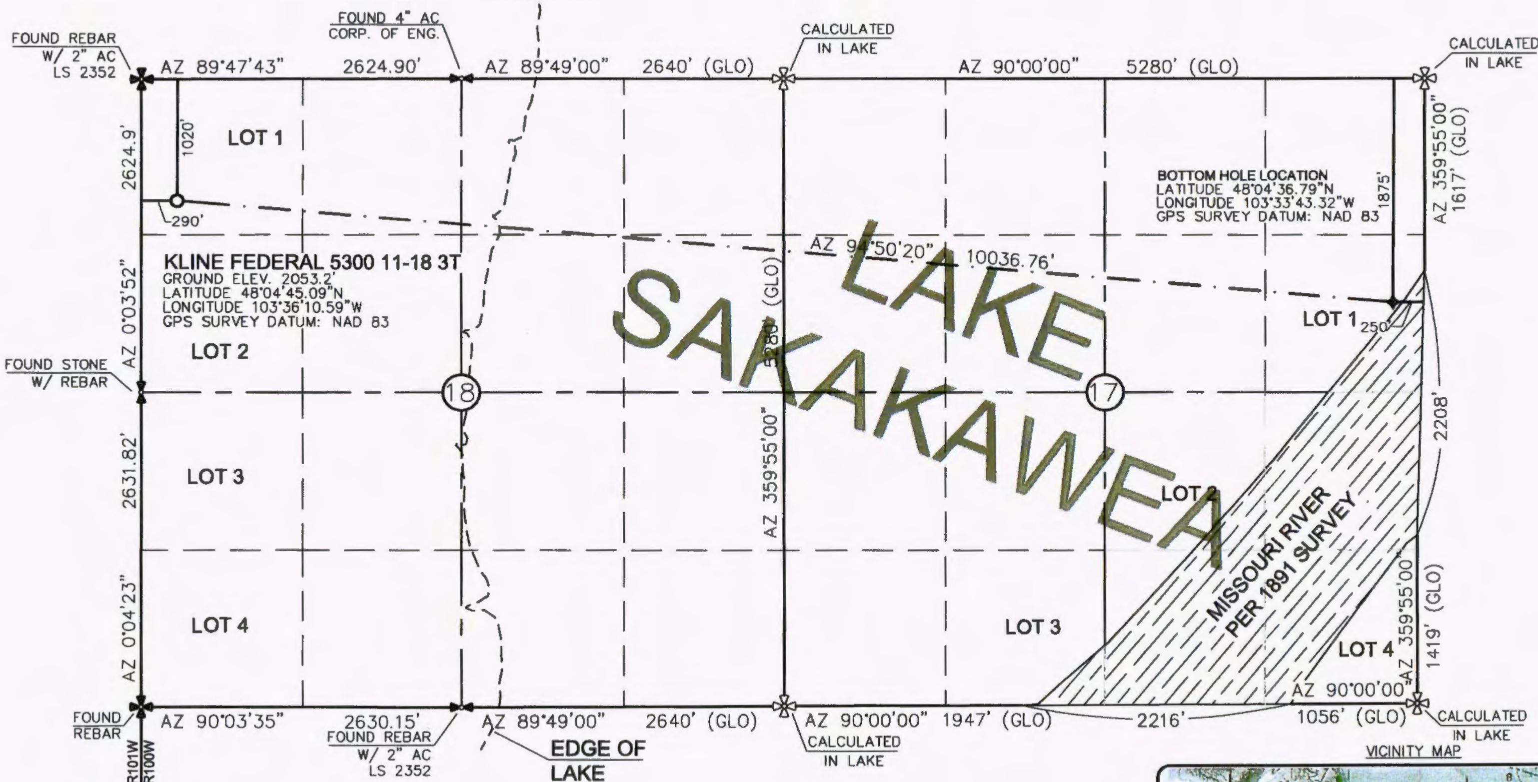
Signature	Printed Name
[signature]	Victoria Siemieniewski
Title	Date
Regulatory Specialist	February 5, 2015
Email Address	
vsiemieniewski@oasispetroleum.com	

FOR STATE USE ONLY

Received	Approved
[ ]	[x]
Date	02-17-2015
By	[signature]
Title	David Burns Engineering Tech.

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 1/29/15 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

- MONUMENT – RECOVERED
- MONUMENT – NOT RECOVERED

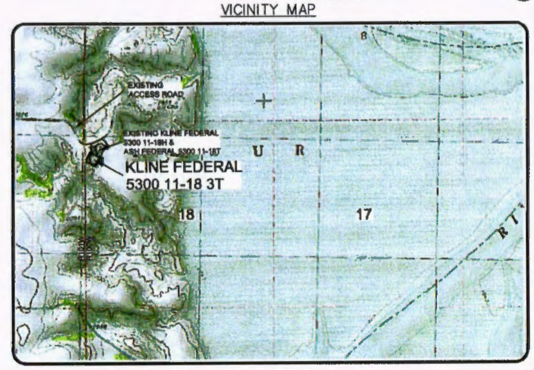
STAKED ON 6/18/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 1/29/15
NORTH DAKOTA

VICINITY MAP



OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.H.H. Project No.: S14-09-127.01
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS
REV 2	12/30/14	JJS	CHANGED WELL NAME & BH
REV 3	1/27/15	BHH	CHANGED WELL NAMES & BHL

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

INTERSTATE ENGINEERING
Professionals you need, people you trust

SHEET NO. 1/8

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Section 18 T153N R100W
McKenzie County, ND

Contingency INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' - 6447'	36	HCL-80	LTC	8.835"	8.75"	5450	7270	9090

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 6447'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.16	3520 / 1.28	453 / 1.53

API Rating & Safety Factor

a) Collapse based on full casing evacuation with 10.4 ppg fluid on backside (6447' setting depth).

b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 15.2#/ft fracture gradient. Backup of 9 ppg fluid..

c) Tension based on string weight in 10.4 ppg fluid at 6447' TVD plus 100k# overpull. (Buoyed weight equals 195k lbs.)

Cement volumes are based on 9-5/8" casing set in 12-1/4 " hole with 10% excess to circulate cement back to surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **568 sks** (293 bbls), 2.90 ft3/sk, 11.5 lb/gal Conventional system with 94 lb/sk cement, 4% D079 extender, 2% D053 expanding agent, 2% CaCl2 and 0.250 lb/sk D130 lost circulation control agent.

Tail Slurry: **610 sks** (126 bbls), 1.16 ft3/sk 15.8 lb/gal Conventional system with 94 lb/sk cement, 0.25% CaCl2, and 0.250 lb/sk lost circulation control agent

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Section 18 T153N R100W
McKenzie County, ND

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift****	**Minimum**	**Optimum**	**Max**
7"	0' - 11080'	32	HCP-110	LTC	6.094"	6.000"**	6730	8970	9870

****Special Drift 7"32# to 6.0"**

Interval	**Length**	**Description**	**Collapse**	**Burst**	**Tension**
			(psi) a	(psi) b	(1000 lbs) / c
0' - 11080'	11080'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.10*	12460 / 1.28	897 / 2.24
6785' - 9216'	2431'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.06**	12460 / 1.30	

API Rating & Safety Factor

a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.

b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10810' TVD.

c) Based on string weight in 10 ppg fluid, (300k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.
Mix and pump the following slurry

Pre-flush (Spacer): **100 bbls** Saltwater
20 bbls Tuned Spacer III

Lead Slurry: **176 sks** (81 bbls), 11.8 ppg, 2.55 cu. ft./sk Econocem Cement with .3% Fe-2 and .25 lb/sk Lost Circulation Additive

Tail Slurry: **586 sks** (171 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendcem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Section 18 T153N R100W
McKenzie County, ND

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10283' - 20924'	13.5	P-110	BTC	3.920"	3.795"	2270	3020	3780

Interval	Length	Desctiption	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10283' - 20924'	10641	4-1/2", 13.5 lb, P-110, BTC	10670 / 1.98	12410 / 1.28	443 / 1.98

API Rating & Safety Factor

a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10862' TVD.

b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10862' TVD.

c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 123k lbs.) plus 100k lbs overpull.

DRILLING PLAN

OPERATOR	Oasis Petroleum	**COUNTY/STATE**	McKenzie Co., ND
WELL NAME	Kline Federal 5300 11-18 3T	**RIG**	0
WELL TYPE	Horizontal Upper Three Forks		
LOCATION	NWNW 18-153N-100W Surface Location (survey plat): 1020' fnl	290' fwl	
EST. T.D.	20,924'	**GROUND ELEV:**	2053 Finished Pad Elev **Sub Height:** 25
	TOTAL LATERA 9,844'	**KB ELEV:**	2078

PROGNOSIS: Based on 2,078' KB(est)

MARKER		**DEPTH (Surf Loc)**	**DATUM (Surf Loc)**
Pierre	NDIC MAP	1,968	110
Greenhorn		4,615	(2,537)
Mowry		5,021	(2,943)
Dakota		5,448	(3,370)
Rierdon		6,447	(4,369)
Dunham Salt		6,785	(4,707)
Dunham Salt Base		6,896	(4,818)
Spearfish		6,993	(4,915)
Pine Salt		7,248	(5,170)
Pine Salt Base		7,296	(5,218)
Opeche Salt		7,341	(5,263)
Opeche Salt Base		7,371	(5,293)
Broom Creek (Top of Minnelusa Gp.)		7,573	(5,495)
Amsden		7,653	(5,575)
Tyler		7,821	(5,743)
Otter (Base of Minnelusa Gp.)		8,012	(5,934)
Kibbey Lime		8,367	(6,289)
Charles Salt		8,517	(6,439)
UB		9,141	(7,063)
Base Last Salt		9,216	(7,138)
Ratcliffe		9,264	(7,186)
Mission Canyon		9,440	(7,362)
Lodgepole		10,002	(7,924)
Lodgepole Fracture Zone		10,208	(8,130)
False Bakken		10,698	(8,620)
Upper Bakken		10,708	(8,630)
Middle Bakken		10,722	(8,644)
Lower Bakken		10,767	(8,689)
Pronghorn		10,781	(8,703)
Three Forks		10,793	(8,715)
TF Target Top		10,805	(8,727)
TF Target Base		10,815	(8,737)
Claystone		10,816	(8,738)

Dip Rate: -0.3

Max. Anticipated BHP: 4682

LOGS:

	Type	**Interval**
OH Logs:	Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota	
CBL/GR:	Above top of cement/GR to base of casing	
MWD GR:	KOP to lateral TD	

DEVIATION:

Surf: 3 deg. max., 1 deg / 100'; srvy every 500'
Prod: 5 deg. max., 1 deg / 100'; srvy every 100'

DST'S:

None planned

CORES:

None planned

MUDLOGGING:

Two-Man: 8,317'
~200' above the Charles (Kibbey) to Casing point; Casing point to TD
30' samples at direction of wellsite geologist; 10' through target @ curve land

BOP:

11" 5000 psi blind, pipe & annular

Surface Formation: *Glacial till*

MUD:	**Interval**		**Type**	**WT**	**Vis**	**WL**	**Remarks**
Surface:	0' -	2,068'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	2,068' -	11,080'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	11,080' -	20,924'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks

CASING:	**Size**	**Wt ppf**	**Hole**	**Depth**		**Cement**	**WOC**	**Remarks**
Surface:	13-3/8"	54.5#	17-1/2"	2,068'		To Surface	12	100' into Pierre
Inermediate (Dakota):	9-5/8"	36#	12-1/4"	6,447'		To Surface	24	Set Casing across Dakota
Intermediate:	7"	32#	8-3/4"	11,080'		4948	24	500' above Dakota
Production Liner:	4-1/2"	13.5#	6"	20,924'	TOL @ 10,283'			50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:	**MD**	**TVD**	**FNL/FSL**	**FEL/FWL**	**S-T-R**	**AZI**	
Surface:	2,068	2,068	1020' FNL	290' FWL	SEC 18-T153N-R100W		Survey Company:
KOP:	10,333'	10,333'	1020'FNL	240' FWL	SEC 18-T153N-R100W		Build Rate: 12 deg /100'
EOC:	11,080'	10,810'	1348' FNL	584' FWL	SEC 18-T153N-R100W	133.60	
Casing Point:	11,080'	10,810'	1348' FNL	584' FWL	SEC 18-T153N-R100W	133.60	
Threeforks Lateral TD:	20,924'	10,862'	1875' FNL	250' FEL	SEC 17-T153N-R100W	90.00	

Comments:

Request a Sundry for an Open Hole Log Waiver

Exception well: Oasis Petroleum's Kline 5300 11-18H.

Completion Notes: 35 packers, 35 sleeves, no frac string

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)

68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)

OASIS PETROLEUM

Geology: M. Steed (4/3/2014) **Engineering:** hlbader rpm 5/30/14

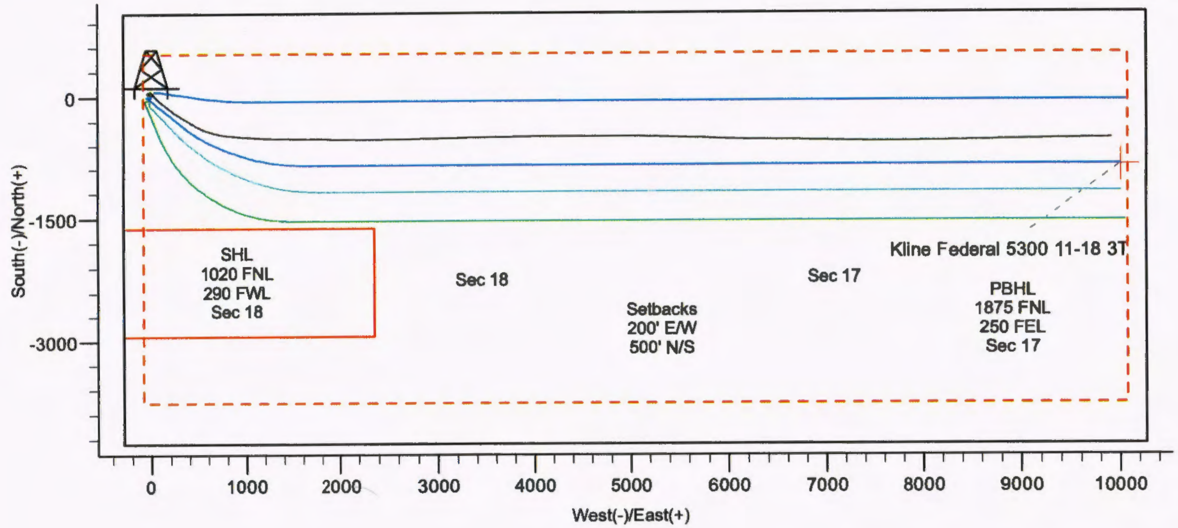
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 11-18 3T
Wellbore: Kline Federal 5300 11-18 3T
Design: Design #5

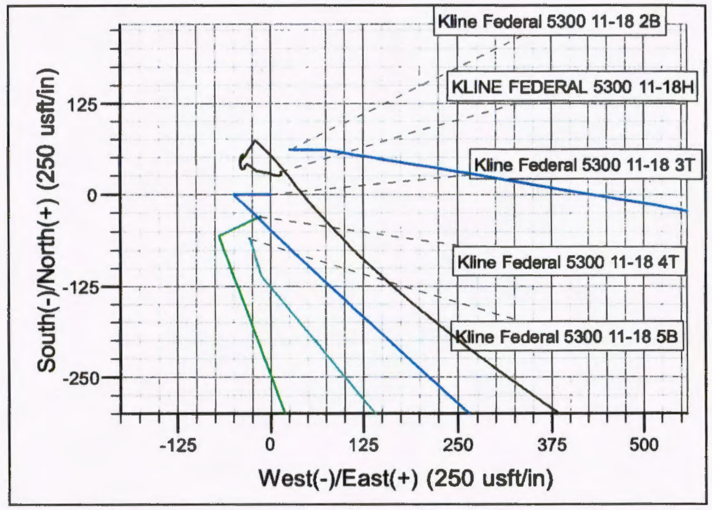
OASIS PETROLEUM

WELL DETAILS: Kline Federal 5300 11-18 3T

Ground Level: 2053.0

Northing	Easting	Latittude	Longitude
408933.65	1210213.75	48° 4' 45.090 N	103° 36' 10.590 W





SECTION DETAILS

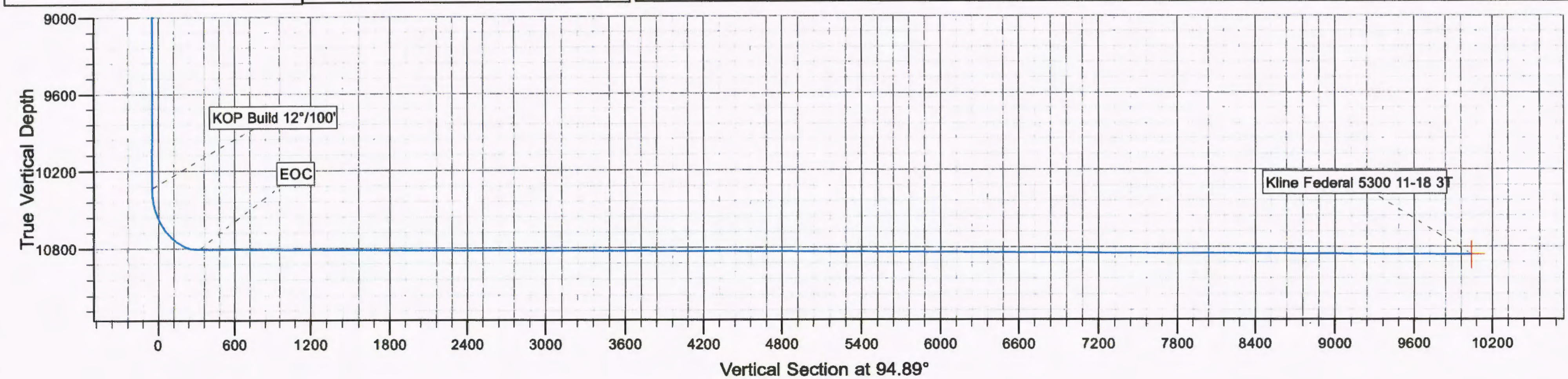
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2500.0	0.00	0.00	2500.0	0.0	0.0	0.00	
2516.7	0.50	270.00	2516.7	0.0	-0.1	3.00	
8226.7	0.50	270.00	8226.4	0.0	-49.9	0.00	
8243.3	0.00	0.00	8243.1	0.0	-50.0	3.00	
10332.7	0.00	0.00	10332.5	0.0	-50.0	0.00	
11080.2	89.70	133.60	10810.0	-327.5	294.0	12.00	
12533.4	89.70	90.00	10818.0	-854.3	1610.9	3.00	
20923.6	89.70	90.00	10862.0	-855.0	10001.0	0.00	Kline Federal 5300 11-18 3T

Azimuths to True North
Magnetic North: 8.28°

Magnetic Field
Strength: 56450.1snT
Dip Angle: 73.00°
Date: 4/22/2014
Model: IGRF2010

CASING DETAILS

TVD	MD	Name	Size
2068.0	2068.0	13 3/8"	13.375
6446.9	6447.0	9 5/8"	9.625
10810.0	11080.2	7"	7.000



Oasis

Indian Hills
153N-100W-17/18
Kline Federal 5300 11-18 3T

Plan: Design #5

Standard Planning Report

27 January, 2015

Planning Report

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Project	Indian Hills		
Map System:	US State Plane 1983	**System Datum:**	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18						
Site Position:			**Northing:**	408,962.44 usft	**Latitude:**		48° 4' 45.380 N
From:	Lat/Long		**Easting:**	1,210,229.18 usft	**Longitude:**		103° 36' 10.380 W
Position Uncertainty:		0.0 usft	**Slot Radius:**	13.200 in	**Grid Convergence:**		-2.31 °

Well	Kline Federal 5300 11-18 3T					
Well Position	**+N/-S**	-29.4 usft	**Northing:**	408,933.65 usft	**Latitude:**	48° 4' 45.090 N
	+E/-W	-14.3 usft	**Easting:**	1,210,213.75 usft	**Longitude:**	103° 36' 10.590 W
Position Uncertainty		2.0 usft	**Wellhead Elevation:**		**Ground Level:**	2,053.0 usft

Wellbore	Kline Federal 5300 11-18 3T				
Magnetics	**Model Name**	**Sample Date**	**Declination (°)**	**Dip Angle (°)**	**Field Strength (nT)**
	IGRF2010	4/22/2014	8.28	73.00	56,450

Design	Design #5				
Audit Notes:					
Version:		**Phase:**	PROTOTYPE	**Tie On Depth:**	0.0
Vertical Section:	**Depth From (TVD) (usft)**	**+N/-S (usft)**	**+E/-W (usft)**	**Direction (°)**	
	0.0	0.0	0.0	94.89	

Plan Sections

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,516.7	0.50	270.00	2,516.7	0.0	-0.1	3.00	3.00	0.00	270.00	
8,226.7	0.50	270.00	8,226.4	0.0	-49.9	0.00	0.00	0.00	0.00	
8,243.3	0.00	0.00	8,243.1	0.0	-50.0	3.00	-3.00	0.00	180.00	
10,332.7	0.00	0.00	10,332.5	0.0	-50.0	0.00	0.00	0.00	0.00	
11,080.2	89.70	133.60	10,810.0	-327.5	294.0	12.00	12.00	0.00	133.60	
12,533.4	89.70	90.00	10,818.0	-854.3	1,610.9	3.00	0.00	-3.00	-90.12	
20,923.6	89.70	90.00	10,862.0	-855.0	10,001.0	0.00	0.00	0.00	0.00	Kline Federal 5300 11

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.0	0.00	0.00	0.00
2,516.7	0.50	270.00	2,516.7	0.0	-0.1	-0.1	3.00	3.00	0.00
2,600.0	0.50	270.00	2,600.0	0.0	-0.8	-0.8	0.00	0.00	0.00
2,700.0	0.50	270.00	2,700.0	0.0	-1.7	-1.7	0.00	0.00	0.00
2,800.0	0.50	270.00	2,800.0	0.0	-2.5	-2.5	0.00	0.00	0.00
2,900.0	0.50	270.00	2,900.0	0.0	-3.4	-3.4	0.00	0.00	0.00
3,000.0	0.50	270.00	3,000.0	0.0	-4.3	-4.3	0.00	0.00	0.00
3,100.0	0.50	270.00	3,100.0	0.0	-5.2	-5.1	0.00	0.00	0.00
3,200.0	0.50	270.00	3,200.0	0.0	-6.0	-6.0	0.00	0.00	0.00
3,300.0	0.50	270.00	3,300.0	0.0	-6.9	-6.9	0.00	0.00	0.00
3,400.0	0.50	270.00	3,400.0	0.0	-7.8	-7.8	0.00	0.00	0.00
3,500.0	0.50	270.00	3,500.0	0.0	-8.7	-8.6	0.00	0.00	0.00
3,600.0	0.50	270.00	3,600.0	0.0	-9.5	-9.5	0.00	0.00	0.00
3,700.0	0.50	270.00	3,700.0	0.0	-10.4	-10.4	0.00	0.00	0.00
3,800.0	0.50	270.00	3,800.0	0.0	-11.3	-11.2	0.00	0.00	0.00
3,900.0	0.50	270.00	3,899.9	0.0	-12.1	-12.1	0.00	0.00	0.00
4,000.0	0.50	270.00	3,999.9	0.0	-13.0	-13.0	0.00	0.00	0.00
4,100.0	0.50	270.00	4,099.9	0.0	-13.9	-13.8	0.00	0.00	0.00
4,200.0	0.50	270.00	4,199.9	0.0	-14.8	-14.7	0.00	0.00	0.00
4,300.0	0.50	270.00	4,299.9	0.0	-15.6	-15.6	0.00	0.00	0.00
4,400.0	0.50	270.00	4,399.9	0.0	-16.5	-16.4	0.00	0.00	0.00
4,500.0	0.50	270.00	4,499.9	0.0	-17.4	-17.3	0.00	0.00	0.00
4,600.0	0.50	270.00	4,599.9	0.0	-18.3	-18.2	0.00	0.00	0.00
4,700.0	0.50	270.00	4,699.9	0.0	-19.1	-19.1	0.00	0.00	0.00
4,800.0	0.50	270.00	4,799.9	0.0	-20.0	-19.9	0.00	0.00	0.00
4,900.0	0.50	270.00	4,899.9	0.0	-20.9	-20.8	0.00	0.00	0.00
5,000.0	0.50	270.00	4,999.9	0.0	-21.7	-21.7	0.00	0.00	0.00
5,100.0	0.50	270.00	5,099.9	0.0	-22.6	-22.5	0.00	0.00	0.00
5,200.0	0.50	270.00	5,199.9	0.0	-23.5	-23.4	0.00	0.00	0.00
5,300.0	0.50	270.00	5,299.9	0.0	-24.4	-24.3	0.00	0.00	0.00
5,400.0	0.50	270.00	5,399.9	0.0	-25.2	-25.1	0.00	0.00	0.00
5,500.0	0.50	270.00	5,499.9	0.0	-26.1	-26.0	0.00	0.00	0.00
5,600.0	0.50	270.00	5,599.9	0.0	-27.0	-26.9	0.00	0.00	0.00
5,700.0	0.50	270.00	5,699.9	0.0	-27.9	-27.8	0.00	0.00	0.00
5,800.0	0.50	270.00	5,799.9	0.0	-28.7	-28.6	0.00	0.00	0.00
5,900.0	0.50	270.00	5,899.9	0.0	-29.6	-29.5	0.00	0.00	0.00
6,000.0	0.50	270.00	5,999.9	0.0	-30.5	-30.4	0.00	0.00	0.00
6,100.0	0.50	270.00	6,099.9	0.0	-31.3	-31.2	0.00	0.00	0.00
6,200.0	0.50	270.00	6,199.9	0.0	-32.2	-32.1	0.00	0.00	0.00
6,300.0	0.50	270.00	6,299.9	0.0	-33.1	-33.0	0.00	0.00	0.00
6,400.0	0.50	270.00	6,399.9	0.0	-34.0	-33.8	0.00	0.00	0.00
6,447.0	0.50	270.00	6,446.9	0.0	-34.4	-34.2	0.00	0.00	0.00
9 5/8"									
6,500.0	0.50	270.00	6,499.8	0.0	-34.8	-34.7	0.00	0.00	0.00
6,600.0	0.50	270.00	6,599.8	0.0	-35.7	-35.6	0.00	0.00	0.00
6,700.0	0.50	270.00	6,699.8	0.0	-36.6	-36.4	0.00	0.00	0.00
6,800.0	0.50	270.00	6,799.8	0.0	-37.5	-37.3	0.00	0.00	0.00
6,900.0	0.50	270.00	6,899.8	0.0	-38.3	-38.2	0.00	0.00	0.00
7,000.0	0.50	270.00	6,999.8	0.0	-39.2	-39.1	0.00	0.00	0.00
7,100.0	0.50	270.00	7,099.8	0.0	-40.1	-39.9	0.00	0.00	0.00
7,200.0	0.50	270.00	7,199.8	0.0	-40.9	-40.8	0.00	0.00	0.00
7,300.0	0.50	270.00	7,299.8	0.0	-41.8	-41.7	0.00	0.00	0.00
7,400.0	0.50	270.00	7,399.8	0.0	-42.7	-42.5	0.00	0.00	0.00
7,500.0	0.50	270.00	7,499.8	0.0	-43.6	-43.4	0.00	0.00	0.00
7,600.0	0.50	270.00	7,599.8	0.0	-44.4	-44.3	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,700.0	0.50	270.00	7,899.8	0.0	-45.3	-45.1	0.00	0.00	0.00
7,800.0	0.50	270.00	7,799.8	0.0	-46.2	-46.0	0.00	0.00	0.00
7,900.0	0.50	270.00	7,899.8	0.0	-47.1	-46.9	0.00	0.00	0.00
8,000.0	0.50	270.00	7,999.8	0.0	-47.9	-47.7	0.00	0.00	0.00
8,100.0	0.50	270.00	8,099.8	0.0	-48.8	-48.6	0.00	0.00	0.00
8,200.0	0.50	270.00	8,199.8	0.0	-49.7	-49.5	0.00	0.00	0.00
8,226.7	0.50	270.00	8,226.4	0.0	-49.9	-49.7	0.00	0.00	0.00
8,243.3	0.00	0.00	8,243.1	0.0	-50.0	-49.8	3.00	-3.00	0.00
8,300.0	0.00	0.00	8,299.8	0.0	-50.0	-49.8	0.00	0.00	0.00
8,400.0	0.00	0.00	8,399.8	0.0	-50.0	-49.8	0.00	0.00	0.00
8,500.0	0.00	0.00	8,499.8	0.0	-50.0	-49.8	0.00	0.00	0.00
8,600.0	0.00	0.00	8,599.8	0.0	-50.0	-49.8	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	0.0	-50.0	-49.8	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	0.0	-50.0	-49.8	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,000.0	0.00	0.00	8,999.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,200.0	0.00	0.00	9,199.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,300.0	0.00	0.00	9,299.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,400.0	0.00	0.00	9,399.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,500.0	0.00	0.00	9,499.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	0.0	-50.0	-49.8	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	0.0	-50.0	-49.8	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	0.0	-50.0	-49.8	0.00	0.00	0.00
10,100.0	0.00	0.00	10,099.8	0.0	-50.0	-49.8	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	0.0	-50.0	-49.8	0.00	0.00	0.00
10,300.0	0.00	0.00	10,299.8	0.0	-50.0	-49.8	0.00	0.00	0.00
10,332.7	0.00	0.00	10,332.5	0.0	-50.0	-49.8	0.00	0.00	0.00
KOP Build 12°/100'									
10,350.0	2.07	133.60	10,349.8	-0.2	-49.7	-49.5	11.98	11.98	0.00
10,375.0	5.07	133.60	10,374.7	-1.3	-48.8	-48.3	12.00	12.00	0.00
10,400.0	8.07	133.60	10,399.6	-3.3	-46.5	-46.1	12.00	12.00	0.00
10,425.0	11.07	133.60	10,424.2	-6.1	-43.5	-42.9	12.00	12.00	0.00
10,450.0	14.07	133.60	10,448.6	-9.9	-39.6	-38.6	12.00	12.00	0.00
10,475.0	17.07	133.60	10,472.7	-14.5	-34.7	-33.4	12.00	12.00	0.00
10,500.0	20.07	133.60	10,496.4	-20.0	-29.0	-27.2	12.00	12.00	0.00
10,525.0	23.07	133.60	10,519.6	-26.3	-22.3	-20.0	12.00	12.00	0.00
10,550.0	26.07	133.60	10,542.4	-33.5	-14.8	-11.9	12.00	12.00	0.00
10,575.0	29.07	133.60	10,564.5	-41.5	-6.4	-2.9	12.00	12.00	0.00
10,600.0	32.07	133.60	10,586.0	-50.3	2.8	7.1	12.00	12.00	0.00
10,825.0	35.07	133.60	10,606.9	-59.8	12.8	17.9	12.00	12.00	0.00
10,650.0	38.07	133.60	10,626.9	-70.1	23.6	29.5	12.00	12.00	0.00
10,675.0	41.07	133.60	10,646.2	-81.0	35.1	41.9	12.00	12.00	0.00
10,700.0	44.07	133.60	10,664.6	-92.7	47.4	55.1	12.00	12.00	0.00
10,725.0	47.07	133.60	10,682.1	-105.0	60.3	69.0	12.00	12.00	0.00
10,750.0	50.07	133.60	10,698.7	-117.9	73.9	83.7	12.00	12.00	0.00
10,775.0	53.07	133.60	10,714.2	-131.4	88.1	98.9	12.00	12.00	0.00
10,800.0	56.07	133.60	10,728.7	-145.5	102.8	114.8	12.00	12.00	0.00
10,825.0	59.07	133.60	10,742.1	-160.0	118.1	131.3	12.00	12.00	0.00
10,850.0	62.07	133.60	10,754.4	-175.1	133.9	148.3	12.00	12.00	0.00
10,875.0	65.07	133.60	10,765.5	-190.5	150.1	165.7	12.00	12.00	0.00
10,900.0	68.07	133.60	10,775.4	-206.3	166.7	183.6	12.00	12.00	0.00
10,925.0	71.07	133.60	10,784.2	-222.5	183.6	201.9	12.00	12.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,950.0	74.07	133.60	10,791.6	-238.9	200.9	220.5	12.00	12.00	0.00
10,975.0	77.07	133.60	10,797.9	-255.6	218.4	239.4	12.00	12.00	0.00
11,000.0	80.07	133.60	10,802.8	-272.5	236.2	258.5	12.00	12.00	0.00
11,025.0	83.07	133.60	10,806.5	-289.6	254.1	277.8	12.00	12.00	0.00
11,050.0	86.07	133.60	10,808.9	-306.7	272.1	297.2	12.00	12.00	0.00
11,075.0	89.07	133.60	10,809.9	-323.9	290.2	316.7	12.00	12.00	0.00
11,080.2	89.70	133.60	10,810.0	-327.5	294.0	320.8	12.00	12.00	0.00
EOC - 7"									
11,100.0	89.70	133.01	10,810.1	-341.1	308.4	336.3	3.00	0.01	-3.00
11,200.0	89.69	130.01	10,810.6	-407.4	383.2	416.6	3.00	-0.01	-3.00
11,300.0	89.69	127.01	10,811.1	-469.6	461.5	499.8	3.00	0.00	-3.00
11,400.0	89.68	124.01	10,811.7	-527.7	542.9	585.9	3.00	0.00	-3.00
11,500.0	89.68	121.01	10,812.2	-581.4	627.2	674.5	3.00	0.00	-3.00
11,600.0	89.68	118.01	10,812.8	-630.7	714.2	765.4	3.00	0.00	-3.00
11,700.0	89.68	115.01	10,813.4	-675.3	803.7	858.3	3.00	0.00	-3.00
11,800.0	89.68	112.01	10,813.9	-715.2	895.4	953.1	3.00	0.00	-3.00
11,900.0	89.68	109.01	10,814.5	-750.2	989.0	1,049.4	3.00	0.00	-3.00
12,000.0	89.68	106.01	10,815.1	-780.3	1,084.4	1,146.9	3.00	0.00	-3.00
12,100.0	89.68	103.01	10,815.6	-805.3	1,181.2	1,245.5	3.00	0.00	-3.00
12,200.0	89.68	100.01	10,816.2	-825.3	1,279.2	1,344.8	3.00	0.00	-3.00
12,300.0	89.69	97.01	10,816.7	-840.1	1,378.1	1,444.6	3.00	0.00	-3.00
12,400.0	89.69	94.01	10,817.3	-849.7	1,477.6	1,544.6	3.00	0.00	-3.00
12,500.0	89.70	91.01	10,817.8	-854.0	1,577.5	1,644.5	3.00	0.01	-3.00
12,533.4	89.70	90.00	10,818.0	-854.3	1,610.9	1,677.8	3.00	0.01	-3.00
12,600.0	89.70	90.00	10,818.3	-854.3	1,677.5	1,744.2	0.00	0.00	0.00
12,700.0	89.70	90.00	10,818.8	-854.3	1,777.5	1,843.8	0.00	0.00	0.00
12,800.0	89.70	90.00	10,819.4	-854.3	1,877.5	1,943.4	0.00	0.00	0.00
12,900.0	89.70	90.00	10,819.9	-854.4	1,977.5	2,043.1	0.00	0.00	0.00
13,000.0	89.70	90.00	10,820.4	-854.4	2,077.5	2,142.7	0.00	0.00	0.00
13,100.0	89.70	90.00	10,820.9	-854.4	2,177.5	2,242.3	0.00	0.00	0.00
13,200.0	89.70	90.00	10,821.5	-854.4	2,277.5	2,342.0	0.00	0.00	0.00
13,300.0	89.70	90.00	10,822.0	-854.4	2,377.5	2,441.6	0.00	0.00	0.00
13,400.0	89.70	90.00	10,822.5	-854.4	2,477.5	2,541.2	0.00	0.00	0.00
13,500.0	89.70	90.00	10,823.0	-854.4	2,577.5	2,640.9	0.00	0.00	0.00
13,600.0	89.70	90.00	10,823.6	-854.4	2,677.5	2,740.5	0.00	0.00	0.00
13,700.0	89.70	90.00	10,824.1	-854.4	2,777.5	2,840.1	0.00	0.00	0.00
13,800.0	89.70	90.00	10,824.6	-854.4	2,877.5	2,939.8	0.00	0.00	0.00
13,900.0	89.70	90.00	10,825.1	-854.4	2,977.5	3,039.4	0.00	0.00	0.00
14,000.0	89.70	90.00	10,825.7	-854.4	3,077.5	3,139.1	0.00	0.00	0.00
14,100.0	89.70	90.00	10,826.2	-854.4	3,177.5	3,238.7	0.00	0.00	0.00
14,200.0	89.70	90.00	10,826.7	-854.5	3,277.5	3,338.3	0.00	0.00	0.00
14,300.0	89.70	90.00	10,827.2	-854.5	3,377.5	3,438.0	0.00	0.00	0.00
14,400.0	89.70	90.00	10,827.8	-854.5	3,477.5	3,537.6	0.00	0.00	0.00
14,500.0	89.70	90.00	10,828.3	-854.5	3,577.5	3,637.2	0.00	0.00	0.00
14,600.0	89.70	90.00	10,828.8	-854.5	3,677.5	3,736.9	0.00	0.00	0.00
14,700.0	69.70	90.00	10,829.3	-854.5	3,777.5	3,836.5	0.00	0.00	0.00
14,800.0	89.70	90.00	10,829.9	-854.5	3,877.4	3,936.1	0.00	0.00	0.00
14,900.0	89.70	90.00	10,830.4	-854.5	3,977.4	4,035.8	0.00	0.00	0.00
15,000.0	89.70	90.00	10,830.9	-854.5	4,077.4	4,135.4	0.00	0.00	0.00
15,100.0	89.70	90.00	10,831.4	-854.5	4,177.4	4,235.1	0.00	0.00	0.00
15,200.0	89.70	90.00	10,832.0	-854.5	4,277.4	4,334.7	0.00	0.00	0.00
15,300.0	89.70	90.00	10,832.5	-854.5	4,377.4	4,434.3	0.00	0.00	0.00
15,400.0	89.70	90.00	10,833.0	-854.6	4,477.4	4,534.0	0.00	0.00	0.00
15,500.0	89.70	90.00	10,833.5	-854.6	4,577.4	4,633.6	0.00	0.00	0.00
15,600.0	89.70	90.00	10,834.1	-854.6	4,677.4	4,733.2	0.00	0.00	0.00

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
15,700.0	89.70	90.00	10,834.6	-854.6	4,777.4	4,832.9	0.00	0.00	0.00
15,800.0	89.70	90.00	10,835.1	-854.6	4,877.4	4,932.5	0.00	0.00	0.00
15,900.0	89.70	90.00	10,835.6	-854.6	4,977.4	5,032.1	0.00	0.00	0.00
16,000.0	89.70	90.00	10,836.2	-854.6	5,077.4	5,131.8	0.00	0.00	0.00
16,100.0	89.70	90.00	10,836.7	-854.6	5,177.4	5,231.4	0.00	0.00	0.00
16,200.0	89.70	90.00	10,837.2	-854.6	5,277.4	5,331.0	0.00	0.00	0.00
16,300.0	89.70	90.00	10,837.7	-854.6	5,377.4	5,430.7	0.00	0.00	0.00
16,400.0	89.70	90.00	10,838.3	-854.6	5,477.4	5,530.3	0.00	0.00	0.00
16,500.0	89.70	90.00	10,838.8	-854.6	5,577.4	5,630.0	0.00	0.00	0.00
16,600.0	89.70	90.00	10,839.3	-854.6	5,677.4	5,729.6	0.00	0.00	0.00
16,700.0	89.70	90.00	10,839.8	-854.7	5,777.4	5,829.2	0.00	0.00	0.00
16,800.0	89.70	90.00	10,840.4	-854.7	5,877.4	5,928.9	0.00	0.00	0.00
16,900.0	89.70	90.00	10,840.9	-854.7	5,977.4	6,028.5	0.00	0.00	0.00
17,000.0	89.70	90.00	10,841.4	-854.7	6,077.4	6,128.1	0.00	0.00	0.00
17,100.0	89.70	90.00	10,841.9	-854.7	6,177.4	6,227.8	0.00	0.00	0.00
17,200.0	89.70	90.00	10,842.5	-854.7	6,277.4	6,327.4	0.00	0.00	0.00
17,300.0	89.70	90.00	10,843.0	-854.7	6,377.4	6,427.0	0.00	0.00	0.00
17,400.0	89.70	90.00	10,843.5	-854.7	6,477.4	6,526.7	0.00	0.00	0.00
17,500.0	89.70	90.00	10,844.0	-854.7	6,577.4	6,626.3	0.00	0.00	0.00
17,600.0	89.70	90.00	10,844.6	-854.7	6,677.4	6,725.9	0.00	0.00	0.00
17,700.0	89.70	90.00	10,845.1	-854.7	6,777.4	6,825.6	0.00	0.00	0.00
17,800.0	89.70	90.00	10,845.6	-854.7	6,877.4	6,925.2	0.00	0.00	0.00
17,900.0	89.70	90.00	10,846.1	-854.6	6,977.4	7,024.9	0.00	0.00	0.00
18,000.0	89.70	90.00	10,846.7	-854.8	7,077.4	7,124.5	0.00	0.00	0.00
18,100.0	89.70	90.00	10,847.2	-854.8	7,177.4	7,224.1	0.00	0.00	0.00
18,200.0	89.70	90.00	10,847.7	-854.8	7,277.4	7,323.8	0.00	0.00	0.00
18,300.0	89.70	90.00	10,848.2	-854.8	7,377.4	7,423.4	0.00	0.00	0.00
18,400.0	89.70	90.00	10,848.8	-854.8	7,477.4	7,523.0	0.00	0.00	0.00
18,500.0	89.70	90.00	10,849.3	-854.8	7,577.4	7,622.7	0.00	0.00	0.00
18,600.0	89.70	90.00	10,849.8	-854.8	7,677.4	7,722.3	0.00	0.00	0.00
18,700.0	89.70	90.00	10,850.3	-854.8	7,777.4	7,821.9	0.00	0.00	0.00
18,800.0	89.70	90.00	10,850.9	-854.8	7,877.4	7,921.6	0.00	0.00	0.00
18,900.0	89.70	90.00	10,851.4	-854.8	7,977.4	8,021.2	0.00	0.00	0.00
19,000.0	89.70	90.00	10,651.9	-854.8	8,077.4	8,120.8	0.00	0.00	0.00
19,100.0	89.70	90.00	10,852.4	-854.8	8,177.4	8,220.5	0.00	0.00	0.00
19,200.0	89.70	90.00	10,853.0	-854.9	8,277.4	8,320.1	0.00	0.00	0.00
19,300.0	89.70	90.00	10,853.5	-854.9	8,377.4	8,419.8	0.00	0.00	0.00
19,400.0	89.70	90.00	10,654.0	-854.9	8,477.4	8,519.4	0.00	0.00	0.00
19,500.0	89.70	90.00	10,854.5	-854.9	8,577.4	8,619.0	0.00	0.00	0.00
19,600.0	89.70	90.00	10,855.1	-854.9	8,677.4	8,718.7	0.00	0.00	0.00
19,700.0	89.70	90.00	10,855.6	-854.9	8,777.4	8,818.3	0.00	0.00	0.00
19,800.0	89.70	90.00	10,856.1	-854.9	8,877.4	8,917.9	0.00	0.00	0.00
19,900.0	89.70	90.00	10,856.6	-854.9	8,977.4	9,017.6	0.00	0.00	0.00
20,000.0	89.70	90.00	10,857.2	-854.9	9,077.4	9,117.2	0.00	0.00	0.00
20,100.0	89.70	90.00	10,857.7	-854.9	9,177.4	9,216.8	0.00	0.00	0.00
20,200.0	89.70	90.00	10,858.2	-854.9	9,277.4	9,316.5	0.00	0.00	0.00
20,300.0	89.70	90.00	10,858.7	-854.9	9,377.4	9,416.1	0.00	0.00	0.00
20,400.0	89.70	90.00	10,859.3	-855.0	9,477.4	9,515.8	0.00	0.00	0.00
20,500.0	89.70	90.00	10,859.8	-855.0	9,577.4	9,615.4	0.00	0.00	0.00
20,600.0	89.70	90.00	10,860.3	-855.0	9,677.4	9,715.0	0.00	0.00	0.00
20,700.0	89.70	90.00	10,860.8	-855.0	9,777.4	9,814.7	0.00	0.00	0.00
20,800.0	89.70	90.00	10,861.4	-855.0	9,877.4	9,914.3	0.00	0.00	0.00
20,900.0	89.70	90.00	10,861.9	-855.0	9,977.4	10,013.9	0.00	0.00	0.00
20,923.6	89.70	90.00	10,862.0	-855.0	10,001.0	10,037.5	0.00	0.00	0.00

Kline Federal 5300 11-18 3T

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
Kline Federal 5300 11-1! - plan hits target center - Point	0.00	0.00	10,862.0	-855.0	10,001.0	407,676.42	1,220,172.19	48° 4' 36.626 N	103° 33' 43.308 W

Casing Points

Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,068.0	2,068.0	13 3/8"	13.375	17.500
6,447.0	6,446.9	9 5/8"	9.625	12.250
11,080.2	10,810.0	7"	7.000	8.750

Database:	OpenWellsCompass - EDM Prod	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis	**TVD Reference:**	RKB @ 2078.0usft
Project:	Indian Hills	**MD Reference:**	RKB @ 2078.0usft
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Kline Federal 5300 11-18 3T		
Design:	Design #5		

Formations

Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,968.0	1,968.0	Pierre			
4,615.1	4,615.0	Greenhorn			
5,021.1	5,021.0	Mowry			
5,448.1	5,448.0	Dakota			
6,447.1	6,447.0	Rierdon			
6,785.2	6,785.0	Dunham Salt			
6,896.2	6,896.0	Dunham Salt Base			
6,993.2	6,993.0	Spearfish			
7,248.2	7,248.0	Pine Salt			
7,296.2	7,296.0	Pine Salt Base			
7,341.2	7,341.0	Opeche Salt			
7,371.2	7,371.0	Opeche Salt Base			
7,573.2	7,573.0	Broom Creek (Top of Minnelusa Gp.)			
7,653.2	7,653.0	Amsden			
7,821.2	7,821.0	Tyler			
8,012.2	8,012.0	Otter (Base of Minnelusa Gp.)			
8,367.2	8,367.0	Kibbey Lime			
8,517.2	8,517.0	Charles Salt			
9,141.2	9,141.0	UB			
9,216.2	9,216.0	Base Last Salt			
9,264.2	9,264.0	Ratcliffe			
9,440.2	9,440.0	Mission Canyon			
10,002.2	10,002.0	Lodgepole			
10,208.2	10,208.0	Lodgepole Fracture Zone			
10,749.0	10,698.0	False Bakken			
10,764.8	10,708.0	Upper Bakken			
10,788.2	10,722.0	Middle Bakken			
10,878.6	10,767.0	Lower Bakken			
10,915.5	10,781.0	Pronghorn			
10,955.0	10,793.0	Three Forks			
11,013.7	10,805.0	TF Target Top			
11,989.7	10,815.0	TF Target Base			
12,168.6	10,816.0	Claystone			

Plan Annotations

Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates +N/-S (usft)	+E/-W (usft)	Comment
10,332.7	10,332.5	0.0	-50.0	KOP Build 12°/100'
11,080.2	10,810.0	-327.5	294.0	EOC

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Section 18 T153N R100W
McKenzie County, ND

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
13-3/8"	0' - 2068'	54.5	J-55	STC	12.615"	12.459"	4100	5470	6840

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 2068'	13-3/8", 54.5#, J-55, LTC, 8rd	1130 / 1.16	2730 / 1.95	514 / 2.60

API Rating & Safety Factor

a) Collapse based on full casing evacuation with 9 ppg fluid on backside (2068' setting depth).
b) Burst pressure based on 13 ppg fluid with no fluid on backside (2068' setting depth).
c) Based on string weight in 9 ppg fluid at 2068' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2 " hole with 60% excess to circulate cement back to surface. Mix and pump the following slurry.

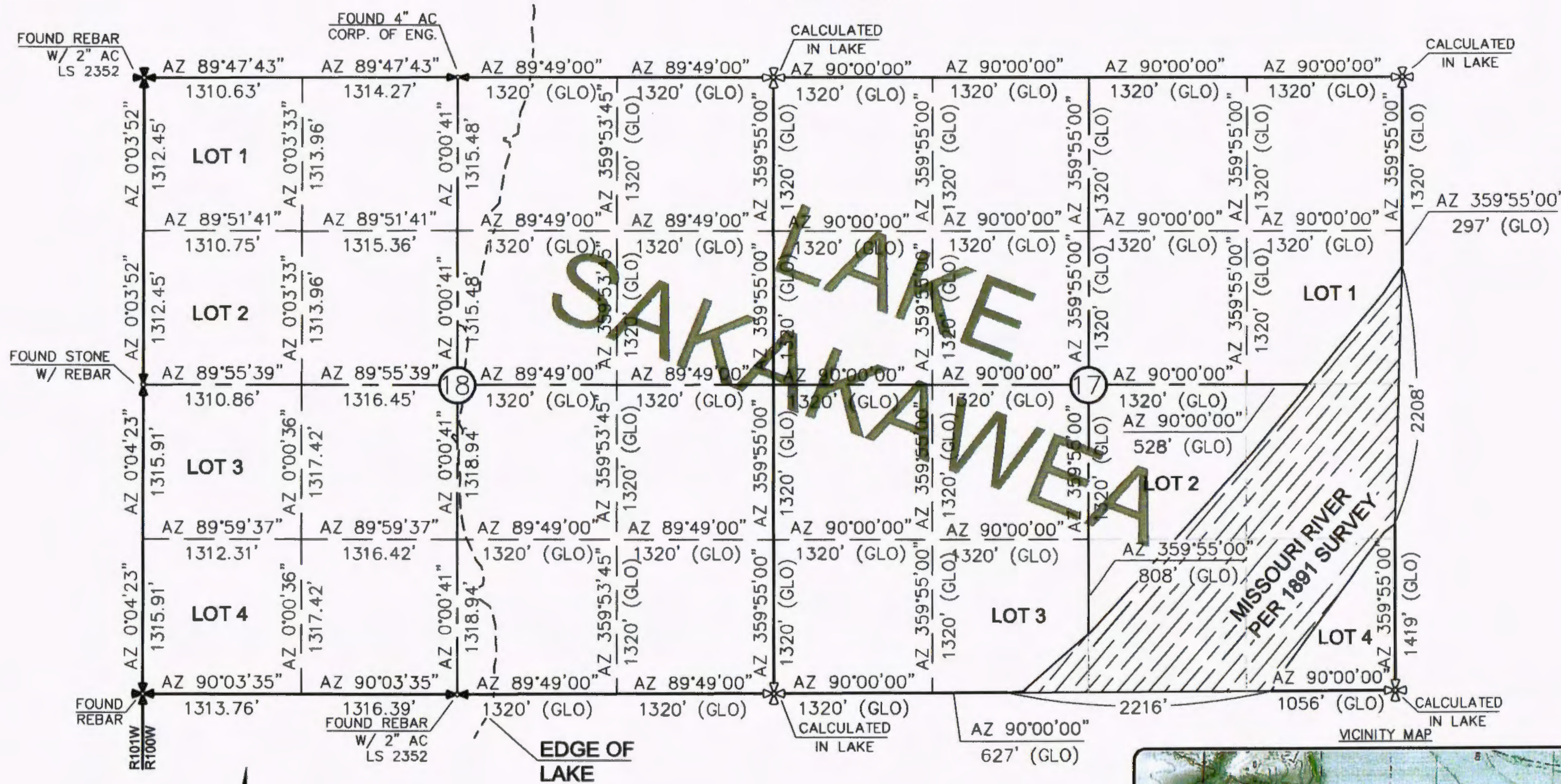
Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **694 sks** (358 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: **300 sks** (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



0 1000
1" = 1000'

- MONUMENT – RECOVERED
- MONUMENT – NOT RECOVERED

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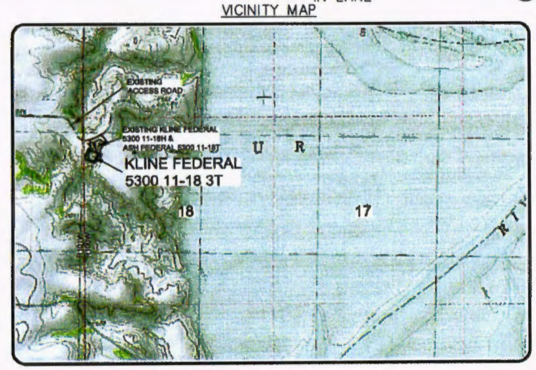
ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 1/29/15
NORTH DAKOTA

VICINITY MAP



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OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 17 & 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	6/18/14	BHH	MOVED WELLS
REV 2	12/30/14	JJS	CHANGED WELL NAME & BH
REV 3	1/27/15	BHH	CHANGED WELL NAMES & BHL

SHEET NO. 2/8

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
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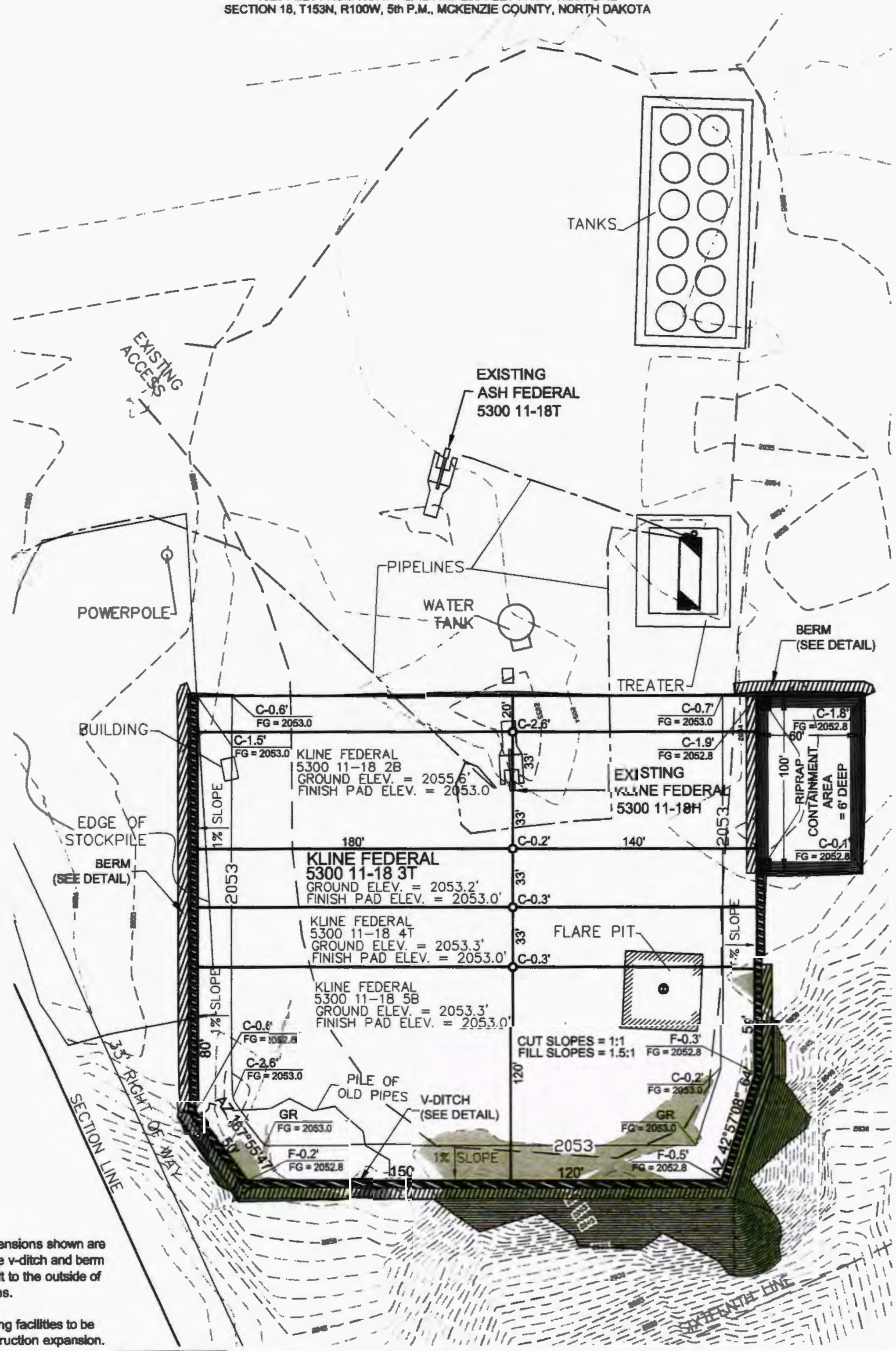
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

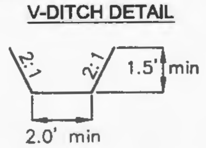


NOTE 1: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE 2: All existing facilities to be removed on construction expansion.

NOTE 3: Cuttings will be hauled to approved disposal site.

V-DITCH DETAIL



- BERM
- DITCH

Proposed Contours
Original Contours

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DARYL D. KASEMAN
LS-3880
DATE: 1/29/15
NORTH DAKOTA

0 60
1" = 60'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

3/8
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OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
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ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
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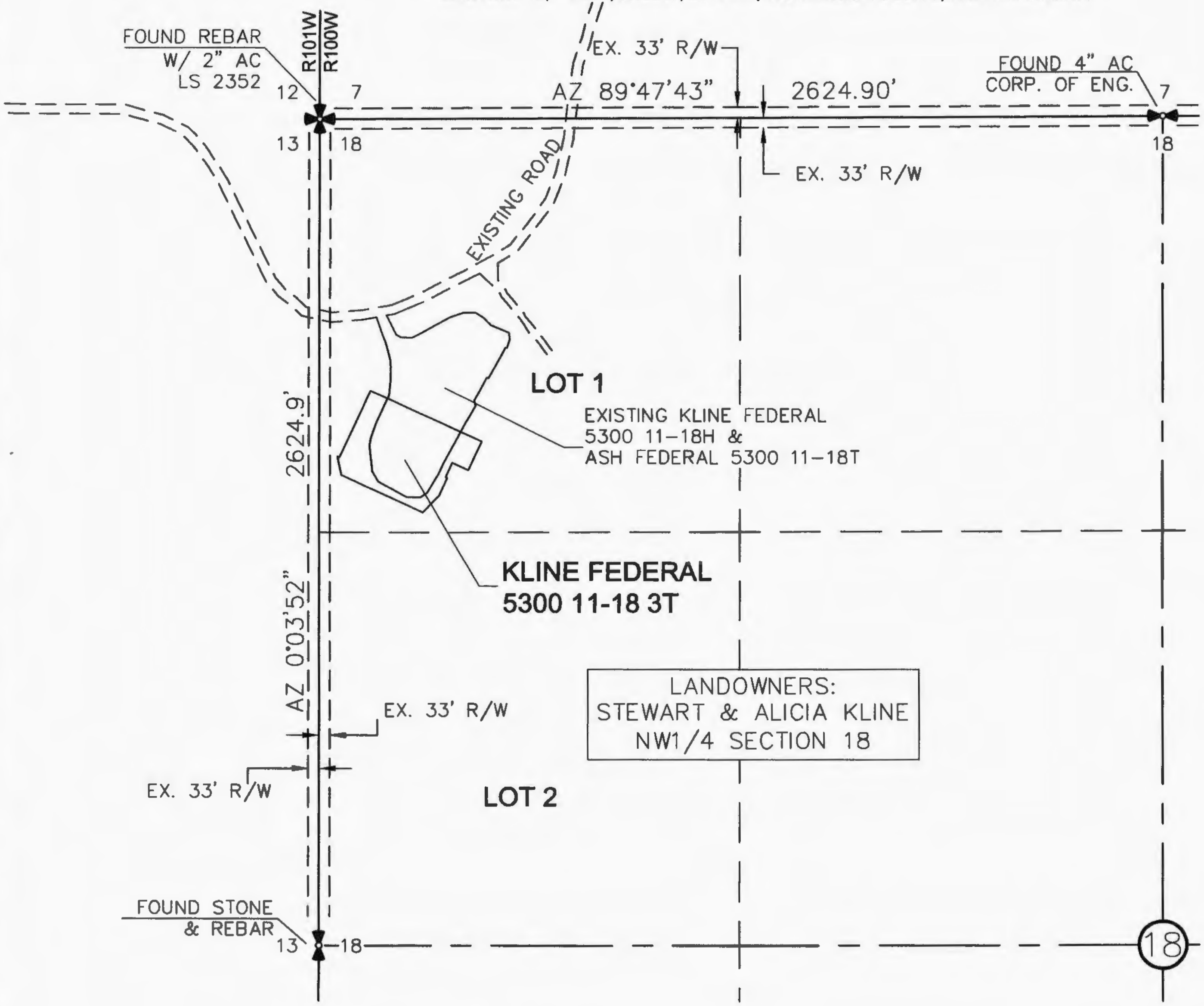

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DARYL D. KASEMAN
LS-3880
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NORTH DAKOTA

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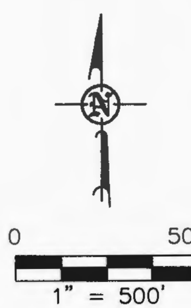

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ACCESS APPROACH
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

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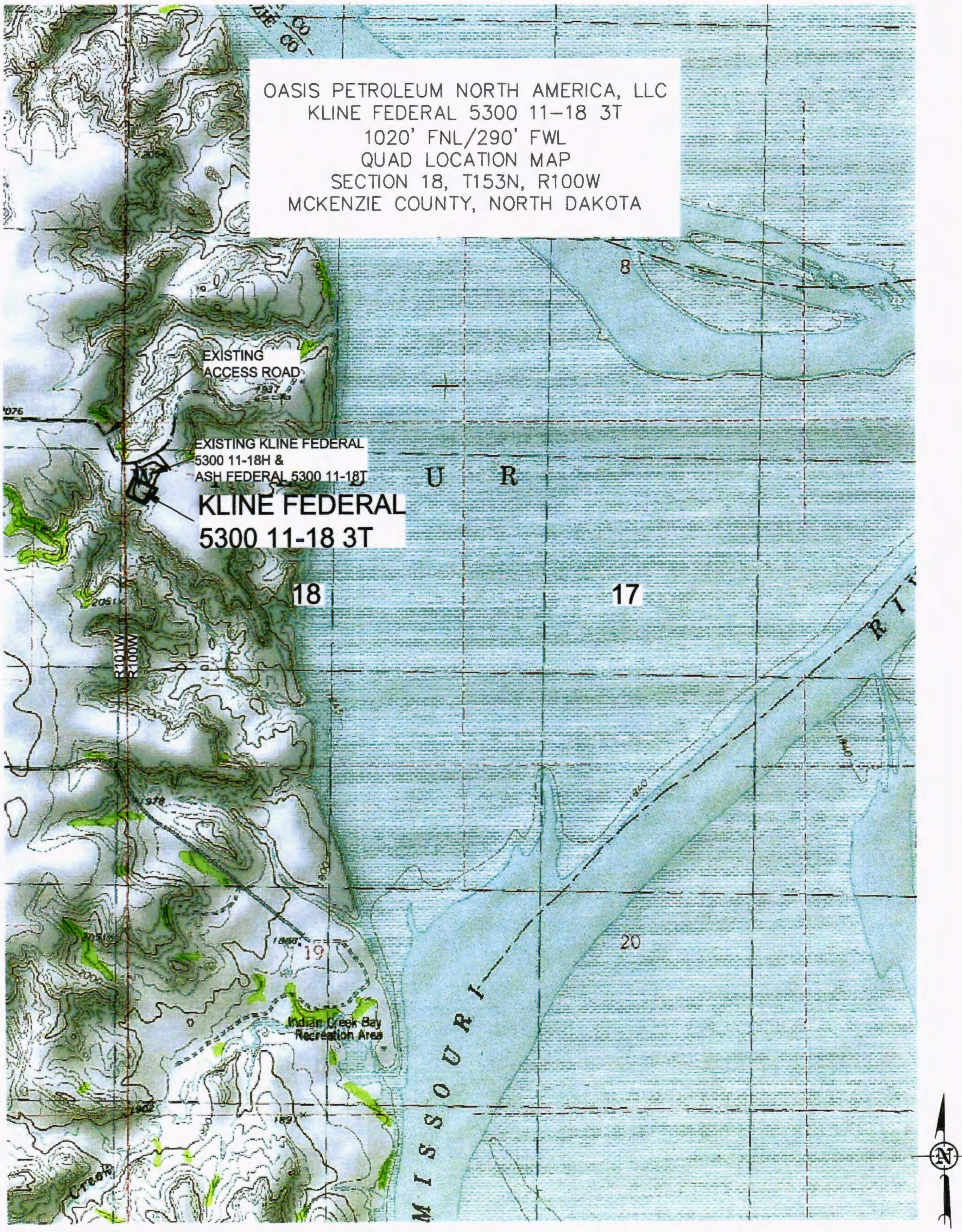
SHEET NO. 4/8



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OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

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Checked By: D.D.K. | Date: APRIL 2014

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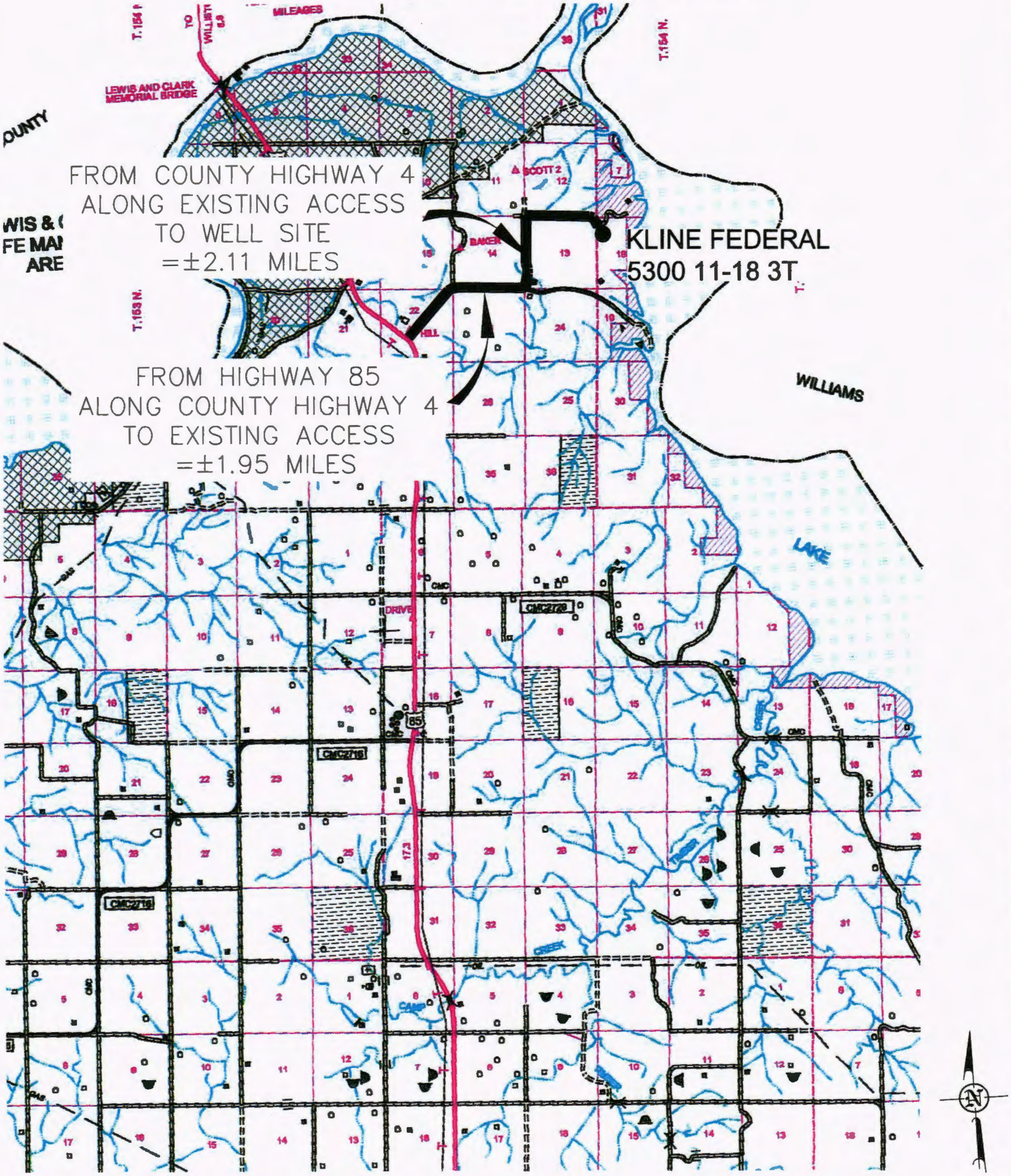
COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

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CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
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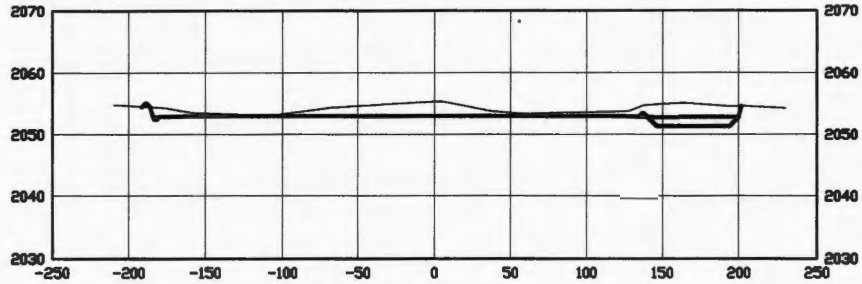

STA. 2+82

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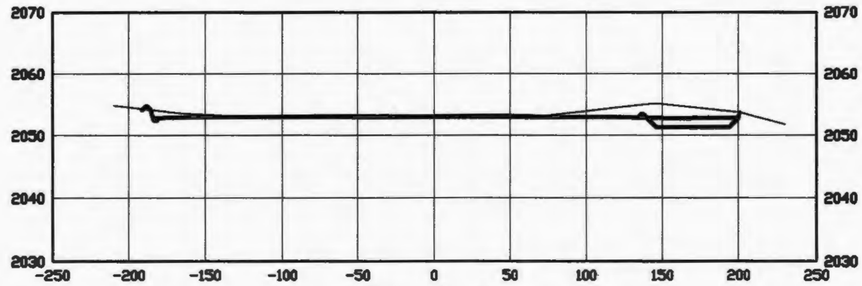
DARYL D. KASEMAN
LS-3880
DATE: 1/29/15
NORTH DAKOTA



STA. 2+00

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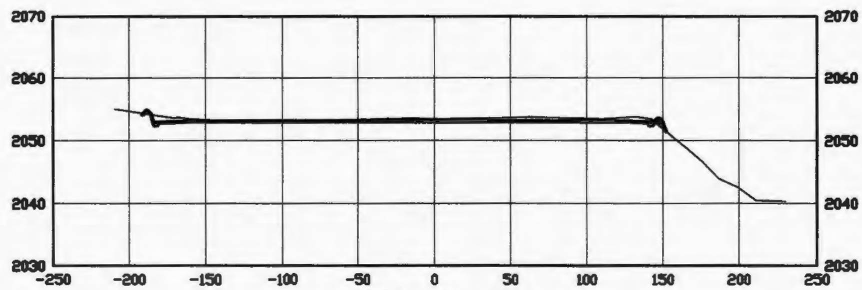

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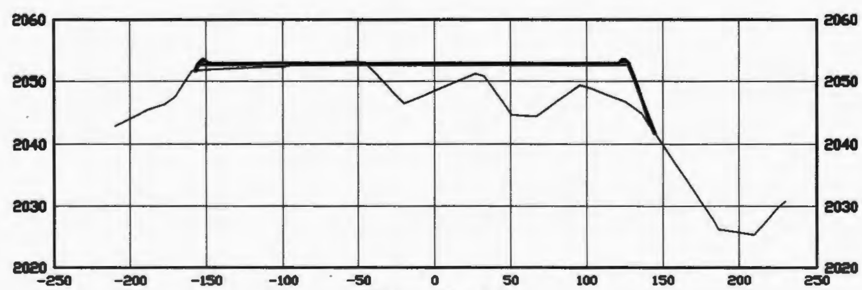

STA. 0+00

SCALE
HORIZ 1"=120'
VERT 1"=30'

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OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
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WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

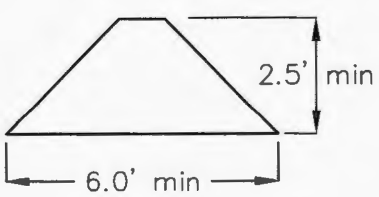
WELL SITE ELEVATION	2053.2
WELL PAD ELEVATION	2053.0

EXCAVATION	1,906
PLUS PIT	0
	1,906
EMBANKMENT	869
PLUS SHRINKAGE (30%)	261
	1,130
STOCKPILE PIT	0
STOCKPILE TOP SOIL (6")	1,934
BERMS	883 LF = 286 CY
DITCHES	727 LF = 111 CY
CONTAINMENT AREA	1,112 CY
ADDITIONAL MATERIAL NEEDED	221
DISTURBED AREA FROM PAD	2.40 ACRES

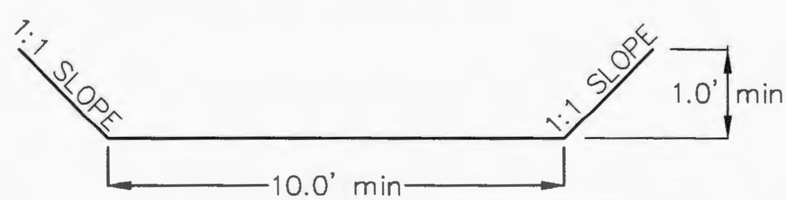
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
1020' FNL
290' FWL

BERM DETAIL



DIVERSION DITCH DETAIL



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OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
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NORTH DAKOTA
OIL AND GAS DIVISION

Oil and Gas Division 29244

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: **9/2/2014**

RE: <u>CORES AND SAMPLES</u>

Well Name:	**KLINE FEDERAL 5300 11-18 3T** Well File No.: **29244**
Location:	**LOT1 18-153-100** County: MCKENZIE
Permit Type:	**Development - HORIZONTAL**
Field:	**BAKER** Target Horizon: THREE FORKS B1

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.

2) <u>Samples</u>: The Operator is to begin collecting sample drill cuttings no lower than the:
<u>Base of the Last Charles Salt</u>
   - Sample cuttings shall be collected at:
     - 30' maximum intervals through all vertical and build sections.
     - 100' maximum intervals through any horizontal sections.
   - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
   - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.

3) <u>Cores:</u> Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed $12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Stephen Fried
Geologist

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: July 2, 2014
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [ ] Drilling Prognosis
- [ ] Spill Report
- [ ] Redrilling or Repair
- [ ] Shooting
- [ ] Casing or Liner
- [ ] Acidizing
- [ ] Plug Well
- [ ] Fracture Treatment
- [ ] Supplemental History
- [ ] Change Production Method
- [ ] Temporarily Abandon
- [ ] Reclamation
- [x] Other: Suspension of Drilling

Well Name and Number					
KLINE FEDERAL 5300 11-18 3T					
Footages		**Qtr-Qtr**	**Section**	**Township**	**Range**
1020 F N L	290 F W L	LOT 1	18	153 N	100 W
Field	**Pool**		**County**		
BAKER	BAKKEN		McKenzie		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Advanced Energy Services			
Address	**City**	**State**	**Zip Code**

DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

NOTIFY NDIC INSPECTOR RICHARD DUNN AT (701) 770-3554 WITH SPUD & TD INFO

Company			Telephone Number
Oasis Petroleum North America LLC			(281) 404-9563
Address			
1001 Fannin, Suite 1500			
City		**State**	**Zip Code**
Houston		TX	77002
Signature	**Printed Name**		
Heather McCowan	Heather McCowan		
Title	**Date**		
Regulatory Assistant	July 2, 2014		
Email Address			
hmccowan@oasispetroleum.com			

FOR STATE USE ONLY

[ ] Received	[x] Approved
Date	8/26/2014
By	Matthew Messana
Title	ENGINEERING TECHNICIAN

SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 29244

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

- [x] Notice of Intent — Approximate Start Date: **July 2, 2014**
- [ ] Report of Work Done — Date Work Completed:
- [ ] Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. — Approximate Start Date:

- [x] Drilling Prognosis
- [ ] Spill Report
- [ ] Redrilling or Repair
- [ ] Shooting
- [ ] Casing or Liner
- [ ] Acidizing
- [ ] Plug Well
- [ ] Fracture Treatment
- [ ] Supplemental History
- [ ] Change Production Method
- [ ] Temporarily Abandon
- [ ] Reclamation
- [x] Other: **Waiver to rule Rule 43-02-03-31**

Well Name and Number					
Kline Federal 5300 11-18 3T					
Footages		Qtr-Qtr	Section	Township	Range
1020 F **N** L	**290** F **W** L	**NWNW**	**18**	**153 N**	**100 W**
Field	Pool	County			
Baker	**Bakken**	**McKenzie**			

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum/ Kline Federal 5300 11-18H located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Approved Per #20275

Company		Telephone Number
Oasis Petroleum North America LLC		**281-404-9563**
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	**TX**	**77002**
Signature: Heather McCowan	Printed Name: **Heather McCowan**	
Title: **Regulatory Assistant**	Date: **July 2, 2014**	
Email Address: **hmccowan@oasispetroleum.com**		

FOR STATE USE ONLY

[ ] Received	[X] Approved
Date	8/26/2014
By	Matthew Messana
Title	Engineering Tech

Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

August 26, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

RE: HORIZONTAL WELL
KLINE FEDERAL 5300 11-18 3T
LOT1 Section 18-153N-100W
McKenzie County
Well File # 29244

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 18 & 17, T153N R100W.

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. Also, OASIS PETROLEUM NORTH AMERICA LLC must take into consideration NDAC 43-02-03-28 (Safety Regulation) when contemplating simultaneous operations on the above captioned location. Pursuant to NDAC 43-02-03-28 (Safety Regulation) "No boiler, portable electric lighting generator, or treater shall be placed nearer than 150 feet to any producing well or oil tank." Lastly, OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit
NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines
Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 10028E.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of $100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.
Survey points shall be of such frequency to accurately determine the entire location of the well bore.
Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov
Thank you for your cooperation.

Sincerely,

Matt Messana
Engineering Technician

APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work	Type of Well	Approximate Date Work Will Start	Confidential Status
New Location	Oil & Gas	10 / 1 / 2013	No

Operator	Telephone Number
OASIS PETROLEUM NORTH AMERICA LLC	281-404-9563

Address	City	State	Zip Code
1001 Fannin Suite 1500	Houston	TX	77002

☑ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☑ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name	Well Number
KLINE FEDERAL	5300 11-18 3T

	Qtr-Qtr	Section	Township	Range	County
Surface Footages: 1020 F N L, 290 F W L	LOT1	18	153 N	100 W	McKenzie
Longstring Casing Point Footages: 1367 F N L, 562 F W L	LOT1	18	153 N	100 W	McKenzie
Bottom Hole Footages From Nearest Section Line: 2030 F N L, 206 F E L	SENE	17	153 N	100 W	McKenzie

Longstring Casing Point Coordinates From Well Head	Azimuth	Longstring Total Depth
347 S From WH, 272 E From WH	137 °	11084 Feet MD, 10813 Feet TVD

Bottom Hole Coordinates From Well Head	KOP Lateral 1	Azimuth Lateral 1	Estimated Total Depth Lateral 1
1010 S From WH, 10051 E From WH	10337 Feet MD	90 °	21039 Feet MD, 10865 Feet TVD

Latitude of Well Head	Longitude of Well Head	NAD Reference	Description of Spacing Unit: (Subject to NDIC Approval)
48 ° 04 ' 45.09 "	-103 ° 36 ' 10.59 "	NAD83	Sections 18 & 17 T153N R100W

Ground Elevation	Acres in Spacing/Drilling Unit	Spacing/Drilling Unit Setback Requirement	Industrial Commission Order
2053 Feet Above S.L.	1280	500 Feet N/S, 200 Feet E/W	23752

North Line of Spacing/Drilling Unit	South Line of Spacing/Drilling Unit	East Line of Spacing/Drilling Unit	West Line of Spacing/Drilling Unit
10544 Feet	10489 Feet	5244 Feet	5256 Feet

Objective Horizons	Pierre Shale Top
Three Forks B1	1967

Proposed Surface Casing	Size	Weight	Depth	Cement Volume	
	9 - 5/8 "	36 Lb./Ft.	2068 Feet	984 Sacks	**NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.**

Proposed Longstring Casing	Size	Weight(s)	Longstring Total Depth	Cement Volume	Cement Top	Top Dakota Sand
	7 - "	29/32 Lb./Ft.	11084 Feet MD, 10813 Feet TVD	832 Sacks	3947 Feet	5447 Feet

Base Last Charles Salt (If Applicable)	
9215 Feet	**NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.**

Proposed Logs
Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota

Drilling Mud Type (Vertical Hole - Below Surface Casing)	Drilling Mud Type (Lateral)
Invert	Salt Water Gel

Survey Type in Vertical Portion of Well	Survey Frequency: Build Section	Survey Frequency: Lateral	Survey Contractor
MWD Every 100 Feet	30 Feet	90 Feet	Ryan

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, $100 fee.

See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2	Azimuth Lateral 2	Estimated Total Depth Lateral 2		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	
Bottom Hole Footages From Nearest Section Line	**Qtr-Qtr**	**Section**	**Township**	**Range**	**County**
F L F L			N	W	

Lateral 3

KOP Lateral 3	Azimuth Lateral 3	Estimated Total Depth Lateral 3		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	
Bottom Hole Footages From Nearest Section Line	**Qtr-Qtr**	**Section**	**Township**	**Range**	**County**
F L F L			N	W	

Lateral 4

KOP Lateral 4	Azimuth Lateral 4	Estimated Total Depth Lateral 4		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	
Bottom Hole Footages From Nearest Section Line	**Qtr-Qtr**	**Section**	**Township**	**Range**	**County**
F L F L			N	W	

Lateral 5

KOP Lateral 5	Azimuth Lateral 5	Estimated Total Depth Lateral 5		KOP Coordinates From Well Head	
Feet MD	°	Feet MD	Feet TVD	From WH	From WH

Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head	
From WH	From WH	From WH	From WH

KOP Footages From Nearest Section Line	Qtr-Qtr	Section	Township	Range	County
F L F L			N	W	
Bottom Hole Footages From Nearest Section Line	**Qtr-Qtr**	**Section**	**Township**	**Range**	**County**
F L F L			N	W	

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date **7 / 02 / 2014**
ePermit	Printed Name **Heather McCowan**	Title **Regulatory Assistant**

FOR STATE USE ONLY

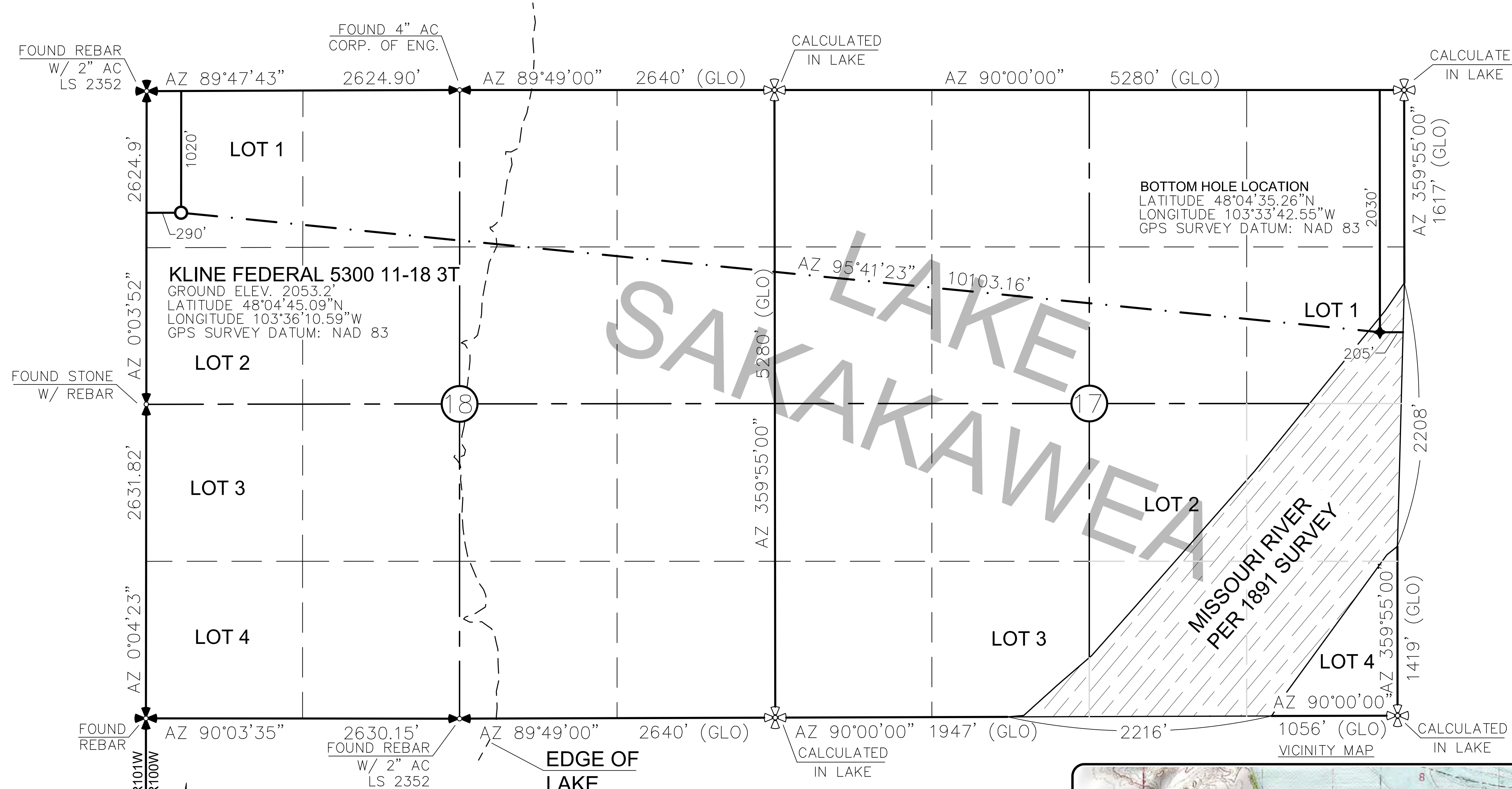
Permit and File Number	API Number
29244	**33 - 053 - 06225**
Field **BAKER**	
Pool **BAKKEN**	Permit Type **DEVELOPMENT**

FOR STATE USE ONLY

Date Approved
8 / 26 / 2014
By **Matt Messana**
Title **Engineering Technician**

WELL LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



N

0 1000

1" = 1000'

MONUMENT – RECOVERED

MONUMENT – NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 6/19/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

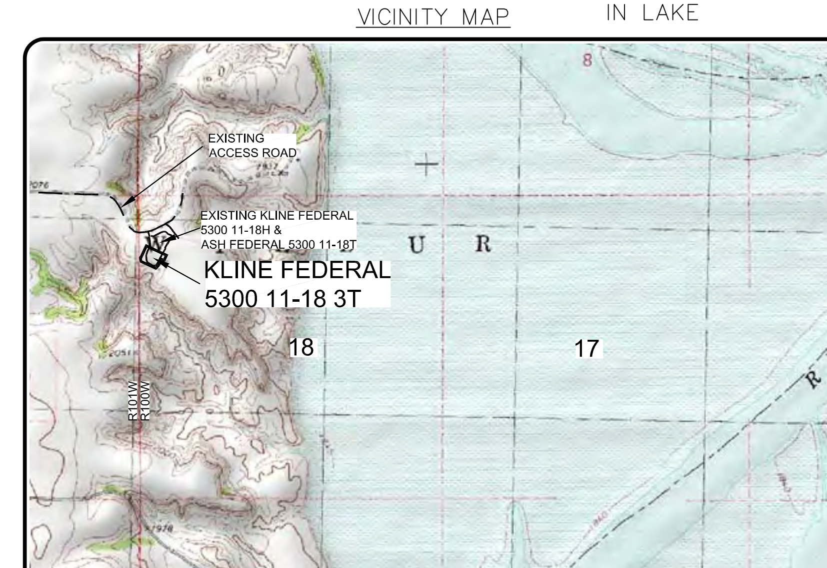
STAKED ON 6/18/14
VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 6/19/14
NORTH DAKOTA

VICINITY MAP



Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS

Q:\2014\S14-09-127.01 Oasis Petroleum – 2nd Well of 4 on Quad Pad for Kline Infills\CAD\Kline Federal 5300 11-18 3T.dwg – 6/19/2014 8:44 AM josh schmierer

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
Checked By: D.D.K. Date: APRIL 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

INTERSTATE ENGINEERING
Professionals you need, people you trust

SHEET NO. 1/8

Oil and Gas Division

Lynn D. Helms - Director   Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

RE: **Filter Socks and Other Filter Media**
Leakproof Container Required
Oil and Gas Wells

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

Sincerely,

Bruce E. Hicks
Assistant Director

600 E Boulevard Ave - Dept 405, Bismarck, North Dakota 58505-0840 Phone (701)328-8020 Fax (701)328-8022

DRILLING PLAN

OPERATOR	Oasis Petroleum	**COUNTY/STATE**	McKenzie Co., ND
WELL NAME	Kline Federal 5300 11-18 3T	**RIG**	0
WELL TYPE	Horizontal Upper Three Forks		
LOCATION	NWNW 18-153N-100W Surface Location (survey plat): 1020' fnl	290' fwl	
EST. T.D.	21,039'	**GROUND ELEV:**	2052 Finished Pad Elev. **Sub Height:** 25
	TOTAL LATERA 9,954'	**KB ELEV:**	2077

PROGNOSIS: Based on 2,077' KB(est)

MARKER		**DEPTH (Surf Loc)**	**DATUM (Surf Loc)**
Pierre	NDIC MAP	1,967	110
Greenhorn		4,614	(2,537)
Mowry		5,020	(2,943)
Dakota		5,447	(3,370)
Rierdon		6,446	(4,369)
Dunham Salt		6,784	(4,707)
Dunham Salt Base		6,895	(4,818)
Spearfish		6,992	(4,915)
Pine Salt		7,247	(5,170)
Pine Salt Base		7,295	(5,218)
Opeche Salt		7,340	(5,263)
Opeche Salt Base		7,370	(5,293)
Broom Creek (Top of Minnelusa Gp.)		7,572	(5,495)
Amsden		7,652	(5,575)
Tyler		7,820	(5,743)
Otter (Base of Minnelusa Gp.)		8,011	(5,934)
Kibbey Lime		8,366	(6,289)
Charles Salt		8,516	(6,439)
UB		9,140	(7,063)
Base Last Salt		9,215	(7,138)
Ratcliffe		9,263	(7,186)
Mission Canyon		9,439	(7,362)
Lodgepole		10,001	(7,924)
Lodgepole Fracture Zone		10,207	(8,130)
False Bakken		10,697	(8,620)
Upper Bakken		10,707	(8,630)
Middle Bakken		10,721	(8,644)
Lower Bakken		10,766	(8,689)
Pronghorn		10,780	(8,703)
Three Forks		10,792	(8,715)
TF Target Top		10,804	(8,727)
TF Target Base		10,814	(8,737)
Claystone		10,815	(8,738)

Dip Rate: -0.3

Max. Anticipated BHP: 4682

LOGS:

	Type	**Interval**
OH Logs:	Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota	
CBL/GR:	Above top of cement/GR to base of casing	
MWD GR:	KOP to lateral TD	

DEVIATION:

Surf: 3 deg. max., 1 deg / 100'; srvy every 500'
Prod: 5 deg. max., 1 deg / 100'; srvy every 100'

DST'S:

None planned

CORES:

None planned

MUDLOGGING:

Two-Man: 8,316'
~200' above the Charles (Kibbey) to Casing point; Casing point to TD
30' samples at direction of wellsite geologist; 10' through target @ curve land

BOP:

11" 5000 psi blind, pipe & annular

Surface Formation: *Glacial till*

MUD:	**Interval**		**Type**	**WT**	**Vis**	**WL**	**Remarks**
Surface:	0' -	2,068'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	2,068' -	11,085'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	11,085' -	21,039'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks

CASING:	**Size**	**Wt ppf**	**Hole**	**Depth**		**Cement**	**WOC**	**Remarks**
Surface:	13-3/8"	54.5#	17-1/2"	2,068'		To Surface	12	100' into Pierre
Inermediate (Dakota):	9-5/8"	40#	12-1/4"	6,100'		To Surface	24	Set Casing across Dakota
Intermediate:	7"	29/32#	8-3/4"	11,085'		3947	24	1500' above Dakota
Production Liner:	4-1/2"	13.5#	6"	21,039'	TOL @ 10,287'			50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:	**MD**	**TVD**	**FNL/FSL**	**FEL/FWL**	**S-T-R**	**AZI**
Surface:	2,068	2,068	1020' FNL	290' FWL	SEC 18-T153N-R100W	
KOP:	10,337'	10,335'	1020'FNL	238' FWL	SEC 18-T153N-R100W	
EOC:	11,085'	10,813'	1367' FNL	562' FWL	SEC 18-T153N-R100W	137.00
Casing Point:	11,085'	10,813'	1367' FNL	562' FWL	SEC 18-T153N-R100W	137.00
Threeforks Lateral TD:	21,039'	10,865'	2030' FNL	205' FEL	SEC 17-T153N-R100W	90.00

Survey Company:
Build Rate: 12 deg /100'

Comments:

Request a Sundry for an Open Hole Log Waiver

Exception well: Oasis Petroleum's Kline 5300 11-18H

Completion Notes: 35 packers, 35 sleeves, no frac string

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)

OASIS PETROLEUM

Geology: M. Steed (4/3/2014) **Engineering:** hlbader rpm 5/30/14

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Sec. 18 T153N R100W
McKenzie County, North Dakota

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
13-3/8”	0’ to 2,068’	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0’ to 2,068’	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.17	2730 / 2.82	514 / 2.61

API Rating & Safety Factor

a) Collapse pressure based on full casing evacuation with 9 ppg fluid on backside (2068' setting depth).
b) Burst pressure based on 9 ppg fluid with no fluid on backside (2068' setting depth).
c) Based on string weight in 9 ppg fluid at 2068' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **635 sks** (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCL2, 4% D079 Extender, and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25 lb/sk Lost Circulation Control Agent, and .25% CaCL2.

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Sec. 18 T153N R100W
McKenzie County, North Dakota

CONTINGENCY INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
9-5/8"	0' - 6101'	40	HCL-80	LTC	8.835"	8.75"**	5,450	7,270	9,090

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6101'	9-5/8", 40#, HCL-80, LTC, 8rd	4230 / 5.33	5750 / 1.23	837 / 2.73

API Rating & Safety Factor

a. Collapse pressure based on 11.5ppg fluid on backside and 9ppg fluid inside of casing.
b. Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
c. Yield based on string weight in 10 ppg fluid, (207k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 9-5/8" casing set in a 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **598 sks (309 bbls)** Conventional system with 75 lb/sk cement, 0.5lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl2, 0.2% anti foam, and 0.4% fluid loss

Tail Slurry: **349 sks (72 bbls)** Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid lo agent, 0.3 lb/sk lost circulation control agent, 0.2% anti foam, and 0.1% retarder

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Sec. 18 T153N R100W
McKenzie County, North Dakota

INTERMEDIATE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	**Interval**	**Weight**	**Grade**	**Coupling**	**I.D.**	**Drift**	**Minimum**	**Optimum**	**Max**
7"	0' - 6634'	29	P-110	LTC	6.184"	6.059"	5980	7970	8770
7"	6634' - 10337'	32	HCP-110	LTC	6.094"	6.000"**	6730	8970	9870
7"	10337' - 11085'	29	P-110	LTC	6.184"	6.059"	5980	7970	8770

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6634'	7", 29#, P-110, LTC, 8rd	8530 / 2.47*	11220 / 1.19	797 / 2.08
6634' - 10337'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.19*	12460 / 1.29	
6634' - 10337'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.28**	12460 / 1.29	
10337' - 11085'	7", 29#, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.15	

API Rating & Safety Factor

a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
b. Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,813' TVD.
c. Based on string weight in 10 ppg fluid, (301k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **207 sks (81 bbls)** 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 Extender, 3.0% KCl, 3.0% D154 Extender, 0.3% D208 Viscosifier, 0.07% Retarder, 0.2% Anti Foam, 0.5lb/sk D130 LCM

Tail Slurry: **625 sks (171 bbls)** 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% Retarder, 0.2% Fluid Loss, 0.2% Anti Foam, 0.5 lb/sk LCM

Oasis Petroleum
Well Summary
Kline Federal 5300 11-18 3T
Sec. 18 T153N R100W
McKenzie County, North Dakota

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Torque
4-1/2”	10,287' – 21,039'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse	Burst	Tension
		(psi) **a**	(psi) **b**	(1000 lbs) **c**
10,287' – 21,039'	4-1/2”, 13.5 lb, P-110, BTC, 8rd	10680 / 1.99	12410 / 1.28	443 / 1.98

API Rating & Safety Factor

a) Collapse pressure based on full casing evacuation with 9.5 ppg fluid on backside @ 10865' TVD.
b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10865' TVD.
c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 124k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

Company: Oasis Petroleum
Field: Indian Hills
Location: 153N-100W-17/18
Well: Kline Federal 5300 11-18 3T
Wellbore #1
Plan: Design #6 (Kline Federal 5300 11-18 3T/Wellbore #1)

SECTION DETAILS

Sec	MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	TFace	VSect	Target
1	0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.0	
2	2500.0	0.00	0.00	2500.0	0.0	0.0	0.00	0.00	0.0	
3	2650.0	3.00	270.00	2649.9	0.0	-3.9	2.00	270.00	-3.9	
4	3501.2	3.00	270.00	3500.0	0.0	-48.5	0.00	0.00	-48.2	
5	3651.2	0.00	0.00	3649.9	0.0	-52.4	2.00	180.00	-52.1	
6	10336.6	0.00	0.00	10335.3	0.0	-52.4	0.00	0.00	-52.1	
7	11084.1	89.70	137.00	10812.8	-347.4	271.5	12.00	137.00	304.9	
8	11099.1	89.70	137.00	10812.8	-358.3	281.8	0.00	0.00	316.2	
9	12655.6	89.70	90.30	10821.5	-965.6	1668.4	3.00	-90.13	1756.6	PBHL Kline Federal 5300 11-18 3T
10	21038.5	89.70	90.30	10865.4	-1010.0	10051.0	0.00	0.00	10101.6	PBHL Kline Federal 5300 11-18 3T

WELL DETAILS: Kline Federal 5300 11-18 3T

Ground Level: 2053.0

+N/-S	+E/-W	Northing	Easting	Latittude	Longitude	Slot
0.0	0.0	408933.65	1210213.75	48° 4' 45.090 N	103° 36' 10.590 W	

WELLBORE TARGET DETAILS

Name	TVD	+N/-S	+E/-W	Latitude	Longitude	Shape
PBHL Kline Federal 5300 11-18 3T	10865.0	-1010.0	10051.0	48° 4' 35.096 N	103° 33' 42.573 W	Point

ANNOTATIONS

TVD	MD	Annotation
2500.0	2500.0	Start Build 2.00
2649.9	2650.0	Start 851.2 hold at 2650.0 MD
3500.0	3501.2	Start DLS 2.00 TFO 180.00
3649.9	3651.2	Start 6685.4 hold at 3651.2 MD
10335.3	10336.6	Start Build 12.00
10812.8	11084.1	Start 15.0 hold at 11084.1 MD
10812.8	11099.1	Start DLS 3.00 TFO -90.13
10821.5	12655.6	Start 8382.8 hold at 12655.6 MD
10865.4	21038.5	TD at 21038.5

CASING DETAILS

TVD	MD	Name	Size
2068.0	2068.0	13 3/8" Casing	13-3/8
10812.8	11085.0	7" Casing	7
6100.0	6101.3	9 5/8" Casing	9-5/8

Plan: Design #6 (Kline Federal 5300 11-18 3T/Wellbore #1)

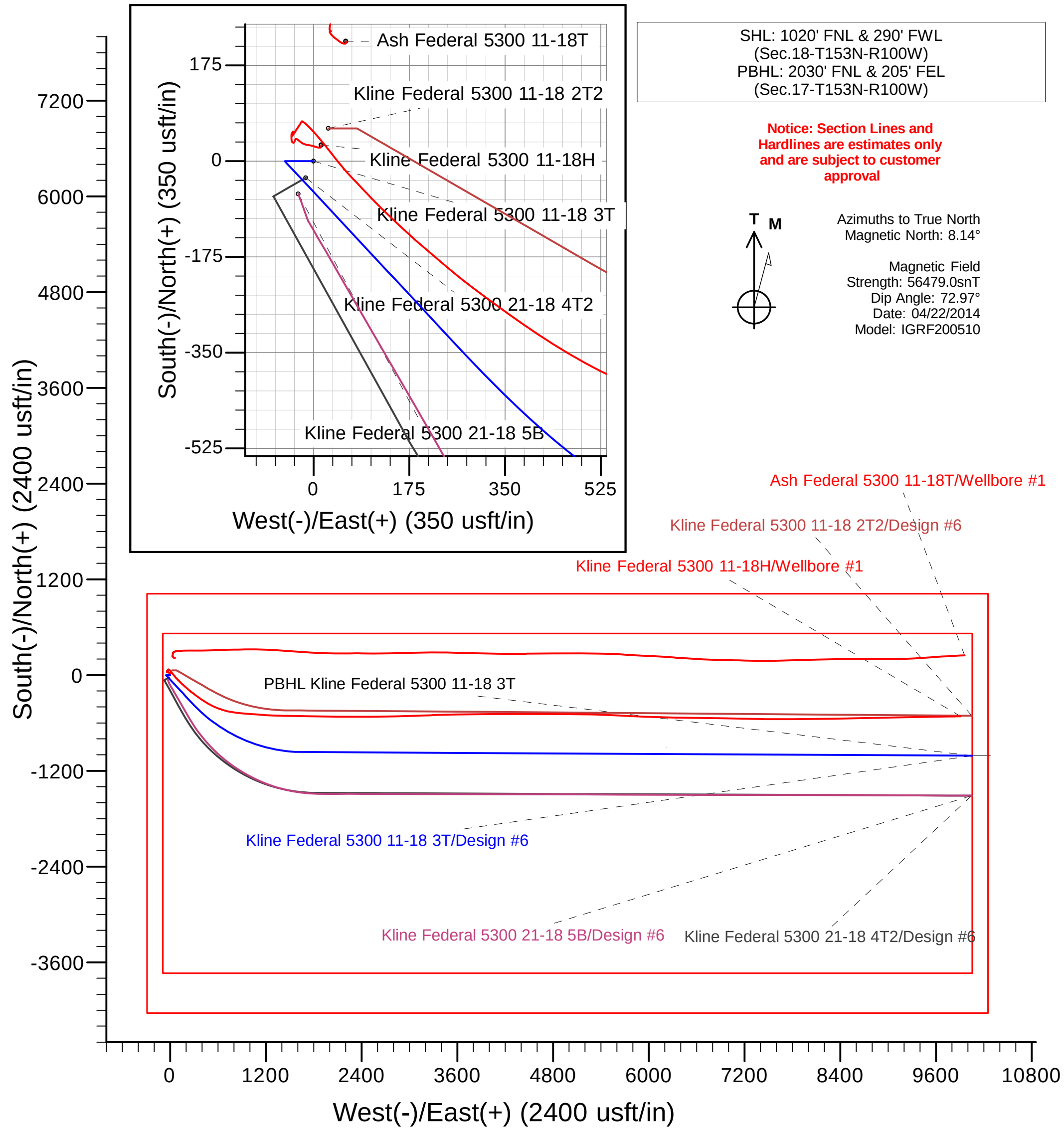
Created By: M. Loucks Date: 15:28, June 30 2014

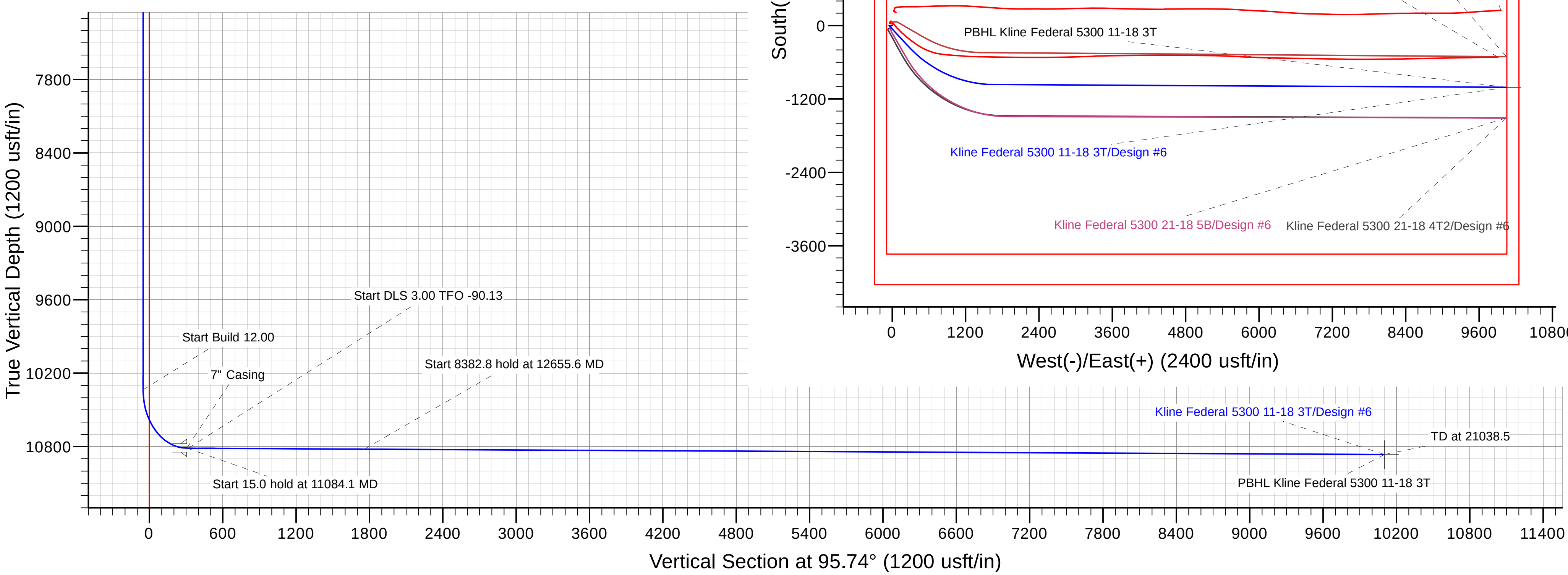
SHL: 1020' FNL & 290' FWL
(Sec.18-T153N-R100W)
PBHL: 2030' FNL & 205' FEL
(Sec.17-T153N-R100W)

Notice: Section Lines and Hardlines are estimates only and are subject to customer approval

Azimuths to True North
Magnetic North: 8.14°

Magnetic Field
Strength: 56479.0snT
Dip Angle: 72.97°
Date: 04/22/2014
Model: IGRF200510





Oasis Petroleum

Indian Hills
153N-100W-17/18
Kline Federal 5300 11-18 3T

Wellbore #1

Plan: Design #6

Standard Planning Report

30 June, 2014

Precision Wellbore Placement

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Project	Indian Hills		
Map System:	US State Plane 1983	**System Datum:**	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18				
Site Position:		**Northing:**	408,992.30 usft	**Latitude:**	48° 4' 45.680 N
From:	Lat/Long	**Easting:**	1,210,243.30 usft	**Longitude:**	103° 36' 10.190 W
Position Uncertainty:	0.0 usft	**Slot Radius:**	13-3/16 "	**Grid Convergence:**	-2.31 °

Well	Kline Federal 5300 11-18 3T					
Well Position	**+N/-S**	-59.8 usft	**Northing:**	408,933.65 usft	**Latitude:**	48° 4' 45.090 N
	+E/-W	-27.2 usft	**Easting:**	1,210,213.75 usft	**Longitude:**	103° 36' 10.590 W
Position Uncertainty		0.0 usft	**Wellhead Elevation:**		**Ground Level:**	2,053.0 usft

Wellbore Wellbore #1

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	04/22/14	8.14	72.97	56,479

Design	Design #6				
Audit Notes:					
Version:		**Phase:**	PLAN	**Tie On Depth:**	0.0

Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	95.74

Plan Sections

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,650.0	3.00	270.00	2,649.9	0.0	-3.9	2.00	2.00	0.00	270.00	
3,501.2	3.00	270.00	3,500.0	0.0	-48.5	0.00	0.00	0.00	0.00	
3,651.2	0.00	0.00	3,649.9	0.0	-52.4	2.00	-2.00	60.00	180.00	
10,336.6	0.00	0.00	10,335.3	0.0	-52.4	0.00	0.00	0.00	0.00	
11,084.1	89.70	137.00	10,812.8	-347.4	271.5	12.00	12.00	0.00	137.00	
11,099.1	89.70	137.00	10,812.8	-358.3	281.8	0.00	0.00	0.00	0.00	
12,655.6	89.70	90.30	10,821.5	-965.6	1,668.4	3.00	0.00	-3.00	-90.13	PBHL Kline Federal 5
21,038.5	89.70	90.30	10,865.4	-1,010.0	10,051.0	0.00	0.00	0.00	0.00	PBHL Kline Federal 5

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,968.0	0.00	0.00	1,968.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,068.0	0.00	0.00	2,068.0	0.0	0.0	0.0	0.00	0.00	0.00
13 3/8" Casing									
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,300.0	0.00	0.00	2,300.0	0.0	0.0	0.0	0.00	0.00	0.00
2,400.0	0.00	0.00	2,400.0	0.0	0.0	0.0	0.00	0.00	0.00
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 2.00									
2,600.0	2.00	270.00	2,600.0	0.0	-1.7	-1.7	2.00	2.00	0.00
2,650.0	3.00	270.00	2,649.9	0.0	-3.9	-3.9	2.00	2.00	0.00
Start 851.2 hold at 2650.0 MD									
2,700.0	3.00	270.00	2,699.9	0.0	-6.5	-6.5	0.00	0.00	0.00
2,800.0	3.00	270.00	2,799.7	0.0	-11.8	-11.7	0.00	0.00	0.00
2,900.0	3.00	270.00	2,899.6	0.0	-17.0	-16.9	0.00	0.00	0.00
3,000.0	3.00	270.00	2,999.5	0.0	-22.2	-22.1	0.00	0.00	0.00
3,100.0	3.00	270.00	3,099.3	0.0	-27.5	-27.3	0.00	0.00	0.00
3,200.0	3.00	270.00	3,199.2	0.0	-32.7	-32.5	0.00	0.00	0.00
3,300.0	3.00	270.00	3,299.0	0.0	-37.9	-37.8	0.00	0.00	0.00
3,400.0	3.00	270.00	3,398.9	0.0	-43.2	-43.0	0.00	0.00	0.00
3,501.2	3.00	270.00	3,500.0	0.0	-48.5	-48.2	0.00	0.00	0.00
Start DLS 2.00 TFO 180.00									
3,600.0	1.02	270.00	3,598.7	0.0	-51.9	-51.7	2.00	-2.00	0.00
3,651.2	0.00	0.00	3,649.9	0.0	-52.4	-52.1	2.00	-2.00	175.66
Start 6685.4 hold at 3651.2 MD									
3,700.0	0.00	0.00	3,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
3,800.0	0.00	0.00	3,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
3,900.0	0.00	0.00	3,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,000.0	0.00	0.00	3,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,100.0	0.00	0.00	4,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,200.0	0.00	0.00	4,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
4,300.0	0.00	0.00	4,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,400.0	0.00	0.00	4,398.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,500.0	0.00	0.00	4,498.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,600.0	0.00	0.00	4,598.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,616.3	0.00	0.00	4,615.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Greenhorn									
4,700.0	0.00	0.00	4,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,800.0	0.00	0.00	4,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
4,900.0	0.00	0.00	4,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,000.0	0.00	0.00	4,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,022.3	0.00	0.00	5,021.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Mowry									
5,100.0	0.00	0.00	5,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,200.0	0.00	0.00	5,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,300.0	0.00	0.00	5,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,400.0	0.00	0.00	5,398.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,449.3	0.00	0.00	5,448.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Dakota									
5,500.0	0.00	0.00	5,498.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,600.0	0.00	0.00	5,598.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,700.0	0.00	0.00	5,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,800.0	0.00	0.00	5,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
5,900.0	0.00	0.00	5,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,000.0	0.00	0.00	5,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,100.0	0.00	0.00	6,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,101.3	0.00	0.00	6,100.0	0.0	-52.4	-52.1	0.00	0.00	0.00
9 5/8" Casing									
6,200.0	0.00	0.00	6,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,300.0	0.00	0.00	6,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,400.0	0.00	0.00	6,398.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,448.3	0.00	0.00	6,447.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Rierdon									
6,500.0	0.00	0.00	6,498.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,600.0	0.00	0.00	6,598.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,700.0	0.00	0.00	6,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,786.3	0.00	0.00	6,785.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Dunham Salt									
6,800.0	0.00	0.00	6,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,897.3	0.00	0.00	6,896.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Dunham Salt Base									
6,900.0	0.00	0.00	6,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
6,994.3	0.00	0.00	6,993.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Spearfish									
7,000.0	0.00	0.00	6,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,100.0	0.00	0.00	7,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,200.0	0.00	0.00	7,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,249.3	0.00	0.00	7,248.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Pine Salt									
7,297.3	0.00	0.00	7,296.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Pine Salt Base									
7,300.0	0.00	0.00	7,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,342.3	0.00	0.00	7,341.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Opeche Salt									

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
7,372.3	0.00	0.00	7,371.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Opeche Salt Base									
7,400.0	0.00	0.00	7,398.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,500.0	0.00	0.00	7,498.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,574.3	0.00	0.00	7,573.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,600.0	0.00	0.00	7,598.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,654.3	0.00	0.00	7,653.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Amsden									
7,700.0	0.00	0.00	7,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,800.0	0.00	0.00	7,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
7,822.3	0.00	0.00	7,821.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Tyler									
7,900.0	0.00	0.00	7,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,000.0	0.00	0.00	7,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,013.3	0.00	0.00	8,012.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,100.0	0.00	0.00	8,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,200.0	0.00	0.00	8,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,300.0	0.00	0.00	8,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,368.3	0.00	0.00	8,367.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Kibbey									
8,400.0	0.00	0.00	8,398.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,500.0	0.00	0.00	8,498.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,518.3	0.00	0.00	8,517.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,598.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,700.0	0.00	0.00	8,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,800.0	0.00	0.00	8,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
8,900.0	0.00	0.00	8,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,000.0	0.00	0.00	8,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,100.0	0.00	0.00	9,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,142.3	0.00	0.00	9,141.0	0.0	-52.4	-52.1	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,217.3	0.00	0.00	9,216.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Base Last Salt									
9,265.3	0.00	0.00	9,264.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Ratcliffe									
9,300.0	0.00	0.00	9,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,400.0	0.00	0.00	9,398.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,441.3	0.00	0.00	9,440.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,498.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,600.0	0.00	0.00	9,598.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,700.0	0.00	0.00	9,698.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,800.0	0.00	0.00	9,798.7	0.0	-52.4	-52.1	0.00	0.00	0.00
9,900.0	0.00	0.00	9,898.7	0.0	-52.4	-52.1	0.00	0.00	0.00
10,000.0	0.00	0.00	9,998.7	0.0	-52.4	-52.1	0.00	0.00	0.00
10,003.3	0.00	0.00	10,002.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,098.7	0.0	-52.4	-52.1	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,200.0	0.00	0.00	10,198.7	0.0	-52.4	-52.1	0.00	0.00	0.00
10,209.3	0.00	0.00	10,208.0	0.0	-52.4	-52.1	0.00	0.00	0.00
Lodgepole Fracture Zone (LP_B)									
10,300.0	0.00	0.00	10,298.7	0.0	-52.4	-52.1	0.00	0.00	0.00
10,336.6	0.00	0.00	10,335.3	0.0	-52.4	-52.1	0.00	0.00	0.00
Start Build 12.00									
10,350.0	1.61	137.00	10,348.7	-0.1	-52.3	-52.0	12.00	12.00	1,022.67
10,375.0	4.61	137.00	10,373.7	-1.1	-51.4	-51.0	12.00	12.00	0.00
10,400.0	7.61	137.00	10,398.5	-3.1	-49.5	-49.0	12.00	12.00	0.00
10,425.0	10.61	137.00	10,423.2	-6.0	-46.8	-46.0	12.00	12.00	0.00
10,450.0	13.61	137.00	10,447.6	-9.8	-43.3	-42.1	12.00	12.00	0.00
10,475.0	16.61	137.00	10,471.8	-14.6	-38.8	-37.2	12.00	12.00	0.00
10,500.0	19.61	137.00	10,495.5	-20.2	-33.5	-31.3	12.00	12.00	0.00
10,525.0	22.61	137.00	10,518.8	-26.8	-27.4	-24.6	12.00	12.00	0.00
10,550.0	25.61	137.00	10,541.7	-34.3	-20.4	-16.9	12.00	12.00	0.00
10,575.0	28.61	137.00	10,563.9	-42.6	-12.6	-8.3	12.00	12.00	0.00
10,600.0	31.61	137.00	10,585.5	-51.8	-4.1	1.1	12.00	12.00	0.00
10,625.0	34.61	137.00	10,606.5	-61.8	5.2	11.4	12.00	12.00	0.00
10,650.0	37.61	137.00	10,626.7	-72.6	15.3	22.4	12.00	12.00	0.00
10,675.0	40.61	137.00	10,646.1	-84.1	26.0	34.3	12.00	12.00	0.00
10,700.0	43.61	137.00	10,664.6	-96.3	37.4	46.9	12.00	12.00	0.00
10,725.0	46.61	137.00	10,682.3	-109.3	49.5	60.2	12.00	12.00	0.00
10,748.5	49.43	137.00	10,698.0	-122.1	61.5	73.4	12.00	12.00	0.00
False Bakken									
10,750.0	49.61	137.00	10,698.9	-122.9	62.2	74.2	12.00	12.00	0.00
10,764.2	51.31	137.00	10,708.0	-130.9	69.7	82.4	12.00	12.00	0.00
Upper Bakken									
10,775.0	52.61	137.00	10,714.6	-137.1	75.5	88.8	12.00	12.00	0.00
10,787.3	54.09	137.00	10,722.0	-144.4	82.2	96.2	12.00	12.00	0.00
Middle Bakken									
10,800.0	55.61	137.00	10,729.3	-151.9	89.3	104.0	12.00	12.00	0.00
10,825.0	58.61	137.00	10,742.9	-167.3	103.6	119.8	12.00	12.00	0.00
10,850.0	61.61	137.00	10,755.3	-183.2	118.4	136.1	12.00	12.00	0.00
10,875.0	64.61	137.00	10,766.6	-199.5	133.6	152.9	12.00	12.00	0.00
10,875.8	64.71	137.00	10,767.0	-200.0	134.1	153.4	12.00	12.00	0.00
Lower Bakken									
10,900.0	67.61	137.00	10,776.8	-216.2	149.2	170.0	12.00	12.00	0.00
10,911.5	68.98	137.00	10,781.0	-224.0	156.4	178.0	12.00	12.00	0.00
Pronghorn									
10,925.0	70.61	137.00	10,785.7	-233.2	165.1	187.6	12.00	12.00	0.00
10,948.7	73.46	137.00	10,793.0	-249.8	180.5	204.6	12.00	12.00	0.00
Three Forks									
10,950.0	73.61	137.00	10,793.4	-250.6	181.3	205.5	12.00	12.00	0.00
10,975.0	76.61	137.00	10,799.8	-268.3	197.8	223.6	12.00	12.00	0.00
11,000.0	79.61	137.00	10,804.9	-286.2	214.5	242.0	12.00	12.00	0.00
11,000.4	79.65	137.00	10,805.0	-286.5	214.7	242.3	12.00	12.00	0.00
Three Forks Target Top									
11,025.0	82.61	137.00	10,808.8	-304.3	231.3	260.6	12.00	12.00	0.00
11,050.0	85.61	137.00	10,811.4	-322.5	248.3	279.3	12.00	12.00	0.00
11,075.0	88.61	137.00	10,812.6	-340.7	265.3	298.1	12.00	12.00	0.00
11,084.1	89.70	137.00	10,812.8	-347.4	271.5	304.9	12.00	12.00	0.00
Start 15.0 hold at 11084.1 MD									

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
11,085.0	89.70	137.00	10,812.8	-348.0	272.1	305.6	0.00	0.00	0.00
7" Casing									
11,099.1	89.70	137.00	10,812.8	-358.3	281.8	316.2	0.00	0.00	0.00
Start DLS 3.00 TFO -90.13									
11,200.0	89.69	133.97	10,813.4	-430.3	352.5	393.7	3.00	-0.01	-3.00
11,300.0	89.69	130.97	10,813.9	-497.8	426.2	473.9	3.00	-0.01	-3.00
11,400.0	89.68	127.97	10,814.5	-561.4	503.4	557.0	3.00	0.00	-3.00
11,497.5	89.68	125.05	10,815.0	-619.3	581.7	640.8	3.00	0.00	-3.00
Three Forks Target Base									
11,500.0	89.68	124.97	10,815.0	-620.8	583.8	643.0	3.00	0.00	-3.00
11,600.0	89.68	121.97	10,815.6	-675.9	667.2	731.5	3.00	0.00	-3.00
11,675.0	89.68	119.72	10,816.0	-714.4	731.6	799.4	3.00	0.00	-3.00
Base Target/Claystone									
11,700.0	89.67	118.97	10,816.1	-726.6	753.4	822.3	3.00	0.00	-3.00
11,800.0	89.67	115.97	10,816.7	-772.8	842.1	915.2	3.00	0.00	-3.00
11,900.0	89.67	112.97	10,817.3	-814.2	933.1	1,009.8	3.00	0.00	-3.00
12,000.0	89.67	109.97	10,817.9	-850.8	1,026.2	1,106.1	3.00	0.00	-3.00
12,100.0	89.68	106.97	10,818.4	-882.5	1,121.0	1,203.6	3.00	0.00	-3.00
12,200.0	89.68	103.97	10,819.0	-909.1	1,217.4	1,302.2	3.00	0.00	-3.00
12,300.0	89.68	100.97	10,819.5	-930.7	1,315.0	1,401.5	3.00	0.00	-3.00
12,400.0	89.69	97.97	10,820.1	-947.2	1,413.6	1,501.2	3.00	0.00	-3.00
12,500.0	89.69	94.97	10,820.6	-958.5	1,513.0	1,601.2	3.00	0.01	-3.00
12,600.0	89.70	91.97	10,821.2	-964.5	1,612.8	1,701.1	3.00	0.01	-3.00
12,655.6	89.70	90.30	10,821.5	-965.6	1,668.4	1,756.6	3.00	0.01	-3.00
Start 8382.8 hold at 12655.6 MD									
12,700.0	89.70	90.30	10,821.7	-965.9	1,712.7	1,800.7	0.00	0.00	0.00
12,800.0	89.70	90.30	10,822.2	-966.4	1,812.7	1,900.3	0.00	0.00	0.00
12,900.0	89.70	90.30	10,822.7	-966.9	1,912.7	1,999.8	0.00	0.00	0.00
13,000.0	89.70	90.30	10,823.3	-967.5	2,012.7	2,099.4	0.00	0.00	0.00
13,100.0	89.70	90.30	10,823.8	-968.0	2,112.7	2,198.9	0.00	0.00	0.00
13,200.0	89.70	90.30	10,824.3	-968.5	2,212.7	2,298.5	0.00	0.00	0.00
13,300.0	89.70	90.30	10,824.8	-969.0	2,312.7	2,398.0	0.00	0.00	0.00
13,400.0	89.70	90.30	10,825.4	-969.6	2,412.7	2,497.6	0.00	0.00	0.00
13,500.0	89.70	90.30	10,825.9	-970.1	2,512.7	2,597.1	0.00	0.00	0.00
13,600.0	89.70	90.30	10,826.4	-970.6	2,612.7	2,696.7	0.00	0.00	0.00
13,700.0	89.70	90.30	10,826.9	-971.2	2,712.7	2,796.2	0.00	0.00	0.00
13,800.0	89.70	90.30	10,827.5	-971.7	2,812.7	2,895.8	0.00	0.00	0.00
13,900.0	89.70	90.30	10,828.0	-972.2	2,912.7	2,995.3	0.00	0.00	0.00
14,000.0	89.70	90.30	10,828.5	-972.7	3,012.7	3,094.9	0.00	0.00	0.00
14,100.0	89.70	90.30	10,829.0	-973.3	3,112.7	3,194.4	0.00	0.00	0.00
14,200.0	89.70	90.30	10,829.6	-973.8	3,212.7	3,294.0	0.00	0.00	0.00
14,300.0	89.70	90.30	10,830.1	-974.3	3,312.7	3,393.5	0.00	0.00	0.00
14,400.0	89.70	90.30	10,830.6	-974.9	3,412.7	3,493.1	0.00	0.00	0.00
14,500.0	89.70	90.30	10,831.1	-975.4	3,512.7	3,592.6	0.00	0.00	0.00
14,600.0	89.70	90.30	10,831.6	-975.9	3,612.7	3,692.2	0.00	0.00	0.00
14,700.0	89.70	90.30	10,832.2	-976.5	3,712.7	3,791.7	0.00	0.00	0.00
14,800.0	89.70	90.30	10,832.7	-977.0	3,812.7	3,891.3	0.00	0.00	0.00
14,900.0	89.70	90.30	10,833.2	-977.5	3,912.7	3,990.8	0.00	0.00	0.00
15,000.0	89.70	90.30	10,833.7	-978.0	4,012.7	4,090.4	0.00	0.00	0.00
15,100.0	89.70	90.30	10,834.3	-978.6	4,112.7	4,189.9	0.00	0.00	0.00
15,200.0	89.70	90.30	10,834.8	-979.1	4,212.7	4,289.5	0.00	0.00	0.00
15,300.0	89.70	90.30	10,835.3	-979.6	4,312.7	4,389.0	0.00	0.00	0.00
15,400.0	89.70	90.30	10,835.8	-980.2	4,412.7	4,488.6	0.00	0.00	0.00
15,500.0	89.70	90.30	10,836.4	-980.7	4,512.7	4,588.1	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
15,600.0	89.70	90.30	10,836.9	-981.2	4,612.7	4,687.7	0.00	0.00	0.00
15,700.0	89.70	90.30	10,837.4	-981.7	4,712.7	4,787.2	0.00	0.00	0.00
15,800.0	89.70	90.30	10,837.9	-982.3	4,812.7	4,886.8	0.00	0.00	0.00
15,900.0	89.70	90.30	10,838.5	-982.8	4,912.7	4,986.3	0.00	0.00	0.00
16,000.0	89.70	90.30	10,839.0	-983.3	5,012.7	5,085.9	0.00	0.00	0.00
16,100.0	89.70	90.30	10,839.5	-983.9	5,112.7	5,185.4	0.00	0.00	0.00
16,200.0	89.70	90.30	10,840.0	-984.4	5,212.6	5,285.0	0.00	0.00	0.00
16,300.0	89.70	90.30	10,840.5	-984.9	5,312.6	5,384.5	0.00	0.00	0.00
16,400.0	89.70	90.30	10,841.1	-985.4	5,412.6	5,484.0	0.00	0.00	0.00
16,500.0	89.70	90.30	10,841.6	-986.0	5,512.6	5,583.6	0.00	0.00	0.00
16,600.0	89.70	90.30	10,842.1	-986.5	5,612.6	5,683.1	0.00	0.00	0.00
16,700.0	89.70	90.30	10,842.6	-987.0	5,712.6	5,782.7	0.00	0.00	0.00
16,800.0	89.70	90.30	10,843.2	-987.6	5,812.6	5,882.2	0.00	0.00	0.00
16,900.0	89.70	90.30	10,843.7	-988.1	5,912.6	5,981.8	0.00	0.00	0.00
17,000.0	89.70	90.30	10,844.2	-988.6	6,012.6	6,081.3	0.00	0.00	0.00
17,100.0	89.70	90.30	10,844.7	-989.2	6,112.6	6,180.9	0.00	0.00	0.00
17,200.0	89.70	90.30	10,845.3	-989.7	6,212.6	6,280.4	0.00	0.00	0.00
17,300.0	89.70	90.30	10,845.8	-990.2	6,312.6	6,380.0	0.00	0.00	0.00
17,400.0	89.70	90.30	10,846.3	-990.7	6,412.6	6,479.5	0.00	0.00	0.00
17,500.0	89.70	90.30	10,846.8	-991.3	6,512.6	6,579.1	0.00	0.00	0.00
17,600.0	89.70	90.30	10,847.4	-991.8	6,612.6	6,678.6	0.00	0.00	0.00
17,700.0	89.70	90.30	10,847.9	-992.3	6,712.6	6,778.2	0.00	0.00	0.00
17,800.0	89.70	90.30	10,848.4	-992.9	6,812.6	6,877.7	0.00	0.00	0.00
17,900.0	89.70	90.30	10,848.9	-993.4	6,912.6	6,977.3	0.00	0.00	0.00
18,000.0	89.70	90.30	10,849.5	-993.9	7,012.6	7,076.8	0.00	0.00	0.00
18,100.0	89.70	90.30	10,850.0	-994.4	7,112.6	7,176.4	0.00	0.00	0.00
18,200.0	89.70	90.30	10,850.5	-995.0	7,212.6	7,275.9	0.00	0.00	0.00
18,300.0	89.70	90.30	10,851.0	-995.5	7,312.6	7,375.5	0.00	0.00	0.00
18,400.0	89.70	90.30	10,851.5	-996.0	7,412.6	7,475.0	0.00	0.00	0.00
18,500.0	89.70	90.30	10,852.1	-996.6	7,512.6	7,574.6	0.00	0.00	0.00
18,600.0	89.70	90.30	10,852.6	-997.1	7,612.6	7,674.1	0.00	0.00	0.00
18,700.0	89.70	90.30	10,853.1	-997.6	7,712.6	7,773.7	0.00	0.00	0.00
18,800.0	89.70	90.30	10,853.6	-998.2	7,812.6	7,873.2	0.00	0.00	0.00
18,900.0	89.70	90.30	10,854.2	-998.7	7,912.6	7,972.8	0.00	0.00	0.00
19,000.0	89.70	90.30	10,854.7	-999.2	8,012.6	8,072.3	0.00	0.00	0.00
19,100.0	89.70	90.30	10,855.2	-999.7	8,112.6	8,171.9	0.00	0.00	0.00
19,200.0	89.70	90.30	10,855.7	-1,000.3	8,212.6	8,271.4	0.00	0.00	0.00
19,300.0	89.70	90.30	10,856.3	-1,000.8	8,312.6	8,371.0	0.00	0.00	0.00
19,400.0	89.70	90.30	10,856.8	-1,001.3	8,412.6	8,470.5	0.00	0.00	0.00
19,500.0	89.70	90.30	10,857.3	-1,001.9	8,512.6	8,570.1	0.00	0.00	0.00
19,600.0	89.70	90.30	10,857.8	-1,002.4	8,612.6	8,669.6	0.00	0.00	0.00
19,700.0	89.70	90.30	10,858.4	-1,002.9	8,712.6	8,769.2	0.00	0.00	0.00
19,800.0	89.70	90.30	10,858.9	-1,003.4	8,812.5	8,868.7	0.00	0.00	0.00
19,900.0	89.70	90.30	10,859.4	-1,004.0	8,912.5	8,968.3	0.00	0.00	0.00
20,000.0	89.70	90.30	10,859.9	-1,004.5	9,012.5	9,067.8	0.00	0.00	0.00
20,100.0	89.70	90.30	10,860.4	-1,005.0	9,112.5	9,167.4	0.00	0.00	0.00
20,200.0	89.70	90.30	10,861.0	-1,005.6	9,212.5	9,266.9	0.00	0.00	0.00
20,300.0	89.70	90.30	10,861.5	-1,006.1	9,312.5	9,366.5	0.00	0.00	0.00
20,400.0	89.70	90.30	10,862.0	-1,006.6	9,412.5	9,466.0	0.00	0.00	0.00
20,500.0	89.70	90.30	10,862.5	-1,007.1	9,512.5	9,565.6	0.00	0.00	0.00
20,600.0	89.70	90.30	10,863.1	-1,007.7	9,612.5	9,665.1	0.00	0.00	0.00
20,700.0	89.70	90.30	10,863.6	-1,008.2	9,712.5	9,764.7	0.00	0.00	0.00
20,800.0	89.70	90.30	10,864.1	-1,008.7	9,812.5	9,864.2	0.00	0.00	0.00
20,900.0	89.70	90.30	10,864.6	-1,009.3	9,912.5	9,963.8	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
21,000.0	89.70	90.30	10,865.2	-1,009.8	10,012.5	10,063.3	0.00	0.00	0.00
21,038.5	89.70	90.30	10,865.4	-1,010.0	10,051.0	10,101.6	0.00	0.00	0.00
TD at 21038.5									

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
PBHL Kline Federal 530	0.00	0.00	10,865.0	-1,010.0	10,051.0	407,519.52	1,220,215.90	48° 4' 35.096 N	103° 33' 42.573 W

- plan misses target center by 0.4usft at 21038.5usft MD (10865.4 TVD, -1010.0 N, 10051.0 E)
- Point

Casing Points

Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")
2,068.0	2,068.0	13 3/8" Casing	13-3/8	17-1/2
6,101.3	6,100.0	9 5/8" Casing	9-5/8	12-1/4
11,085.0	10,812.8	7" Casing	7	8-3/4

Database:	EDM 5000.1 Single User Db	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Company:	Oasis Petroleum	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Project:	Indian Hills	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site:	153N-100W-17/18	**North Reference:**	True
Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #6		

Formations

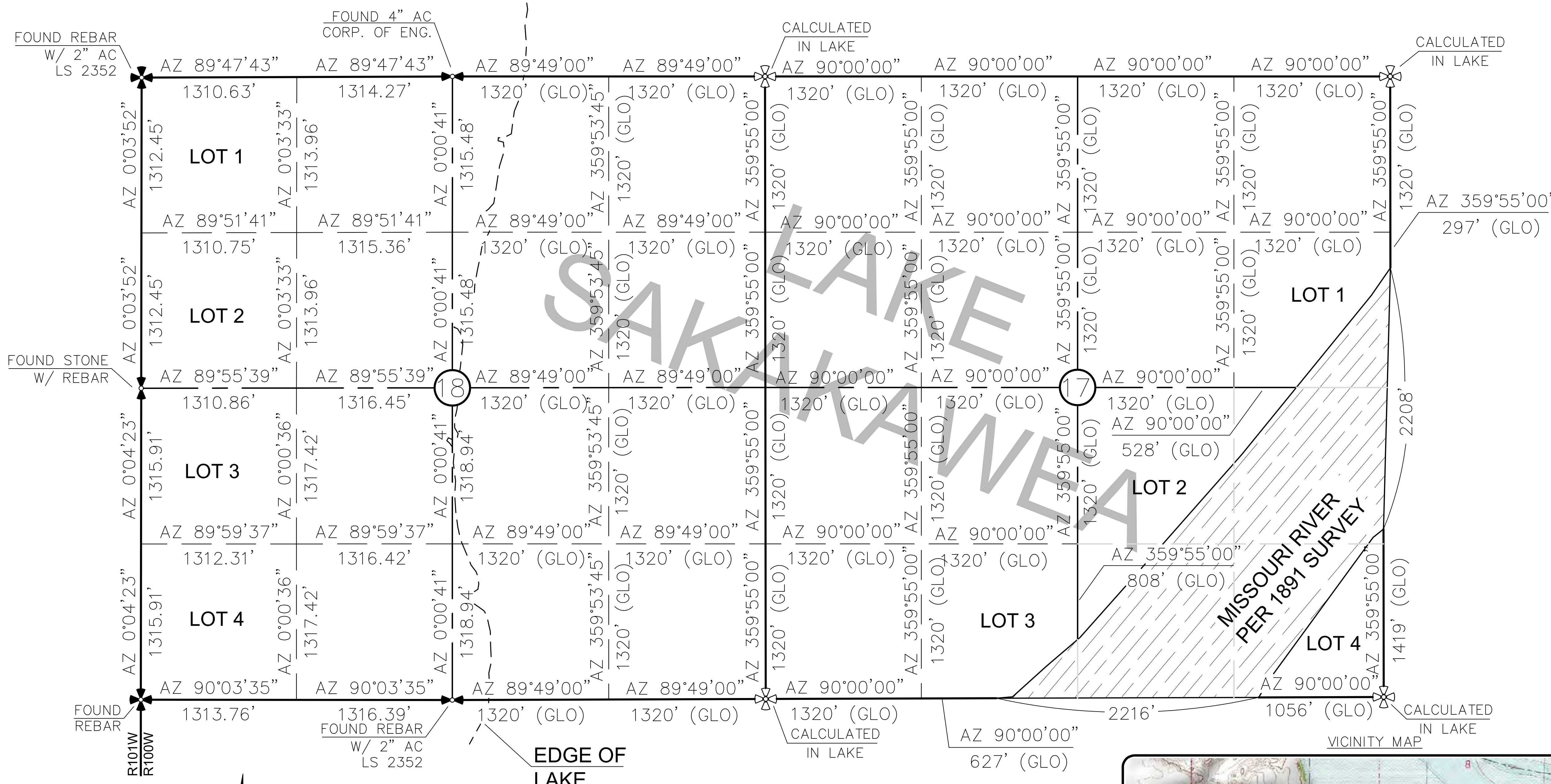
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,968.0	1,968.0	Pierre			
4,616.3	4,615.0	Greenhorn			
5,022.3	5,021.0	Mowry			
5,449.3	5,448.0	Dakota			
6,448.3	6,447.0	Rierdon			
6,786.3	6,785.0	Dunham Salt			
6,897.3	6,896.0	Dunham Salt Base			
6,994.3	6,993.0	Spearfish			
7,249.3	7,248.0	Pine Salt			
7,297.3	7,296.0	Pine Salt Base			
7,342.3	7,341.0	Opeche Salt			
7,372.3	7,371.0	Opeche Salt Base			
7,574.3	7,573.0	Broom Creek (Top of Minnelusa Gp.)			
7,654.3	7,653.0	Amsden			
7,822.3	7,821.0	Tyler			
8,013.3	8,012.0	Otter (Base of Minnelusa Gp.)			
8,368.3	8,367.0	Kibbey			
8,518.3	8,517.0	Charles Salt			
9,142.3	9,141.0	UB			
9,217.3	9,216.0	Base Last Salt			
9,265.3	9,264.0	Ratcliffe			
9,441.3	9,440.0	Mission Canyon			
10,003.3	10,002.0	Lodgepole			
10,209.3	10,208.0	Lodgepole Fracture Zone (LP_B)			
10,748.5	10,698.0	False Bakken			
10,764.2	10,708.0	Upper Bakken			
10,787.3	10,722.0	Middle Bakken			
10,875.8	10,767.0	Lower Bakken			
10,911.5	10,781.0	Pronghorn			
10,948.7	10,793.0	Three Forks			
11,000.4	10,805.0	Three Forks Target Top			
11,497.5	10,815.0	Three Forks Target Base			
11,675.0	10,816.0	Base Target/Claystone			

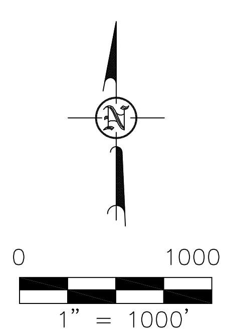
Plan Annotations

Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates +N/-S (usft)	+E/-W (usft)	Comment
2,500.0	2,500.0	0.0	0.0	Start Build 2.00
2,650.0	2,649.9	0.0	-3.9	Start 851.2 hold at 2650.0 MD
3,501.2	3,500.0	0.0	-48.5	Start DLS 2.00 TFO 180.00
3,651.2	3,649.9	0.0	-52.4	Start 6685.4 hold at 3651.2 MD
10,336.6	10,335.3	0.0	-52.4	Start Build 12.00
11,084.1	10,812.8	-347.4	271.5	Start 15.0 hold at 11084.1 MD
11,099.1	10,812.8	-358.3	281.8	Start DLS 3.00 TFO -90.13
12,655.6	10,821.5	-965.6	1,668.4	Start 8382.8 hold at 12655.6 MD
21,038.5	10,865.4	-1,010.0	10,051.0	TD at 21038.5

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA





— MONUMENT — RECOVERED

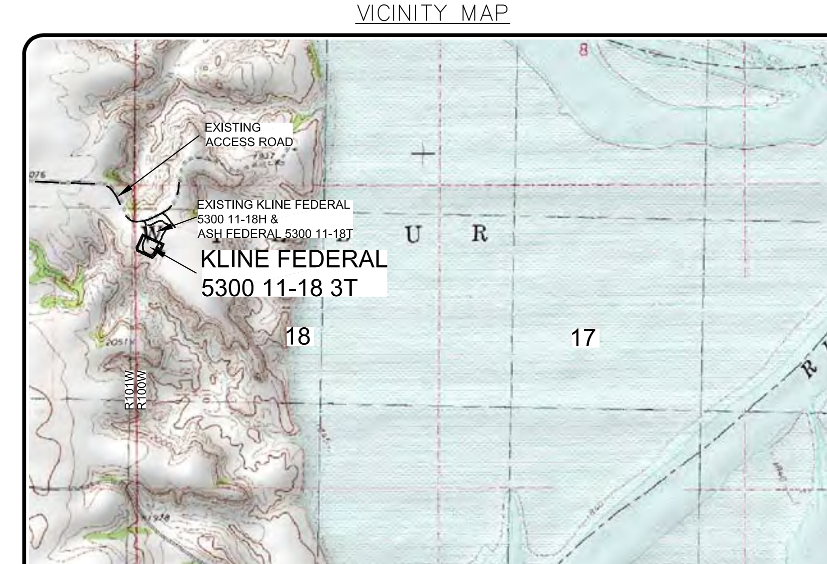
— MONUMENT — NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 6/19/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 6/19/14
NORTH DAKOTA

VICINITY MAP



Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 17 & 18, T153N, R100W
MKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
Checked By: D.D.K. Date: APRIL 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
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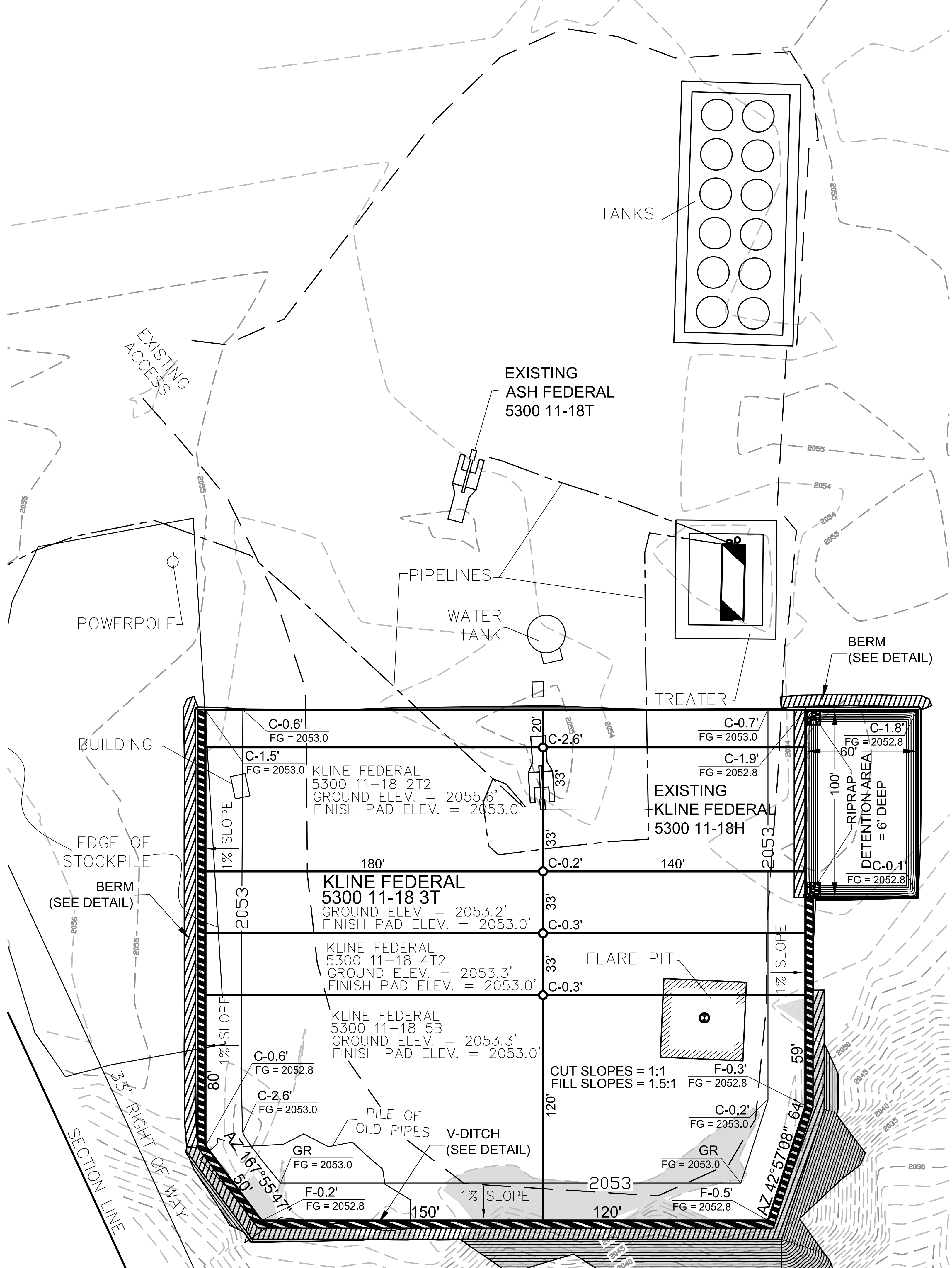
Other offices in Minnesota, North Dakota and South Dakota

INTERSTATE ENGINEERING
Professionals you need, people you trust

SHEET NO. 2/8

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

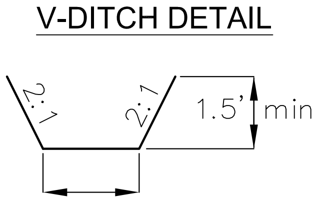


NOTE 1: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE 2: All existing facilities to be removed on construction expansion.

NOTE 3: Cuttings will be hauled to approved disposal site.

V-DITCH DETAIL



— BERM

— DITCH

Proposed Contours

Original Contours

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REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 6/19/14
NORTH DAKOTA

0 60
1" = 60'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

3/8
SHEET NO.

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Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

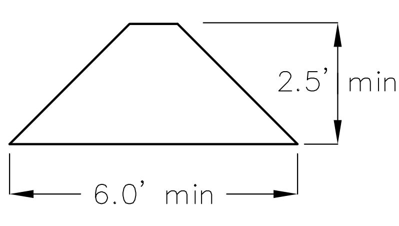
WELL SITE ELEVATION	2053.2
WELL PAD ELEVATION	2053.0

EXCAVATION	1,906
PLUS PIT	0
	1,906
EMBANKMENT	869
PLUS SHRINKAGE (30%)	261
	1,130
STOCKPILE PIT	0
STOCKPILE TOP SOIL (6")	1,934
BERMS	883 LF = 286 CY
DITCHES	727 LF = 111 CY
DETENTION AREA	1,112 CY
ADDITIONAL MATERIAL NEEDED	221
DISTURBED AREA FROM PAD	2.40 ACRES

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
1020' FNL
290' FWL

BERM DETAIL



DITCH DETAIL

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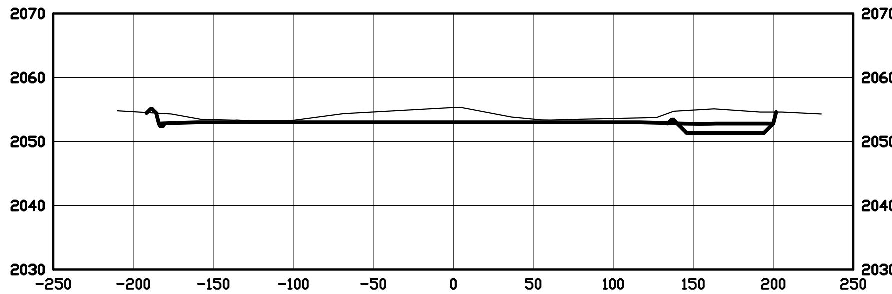
OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS

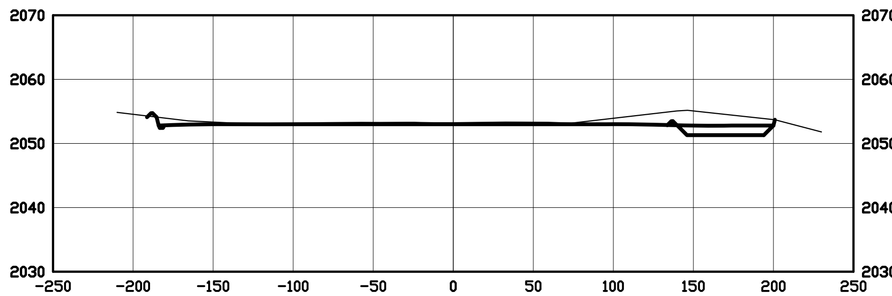
CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



STA. 2+82

REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS−3880
DATE: 6/19/14
NORTH DAKOTA



STA. 2+00

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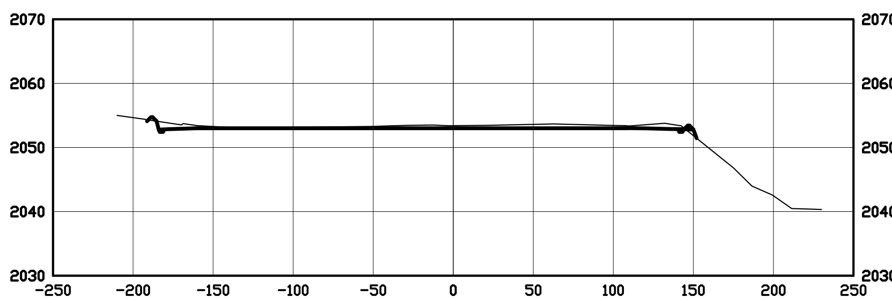
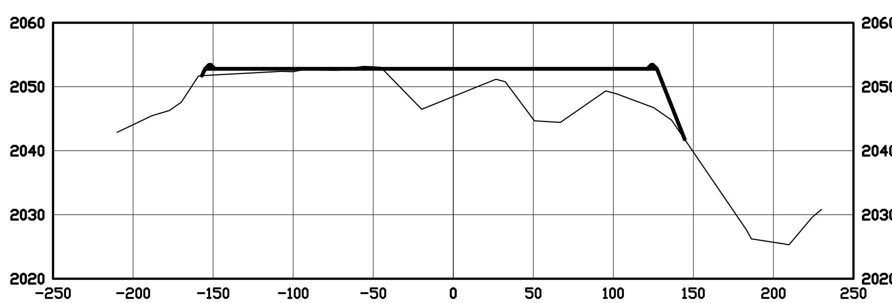


STA. 1+00



STA. 0+00

SCALE
HORIZ 1"=120'
VERT 1"=30'

7/8
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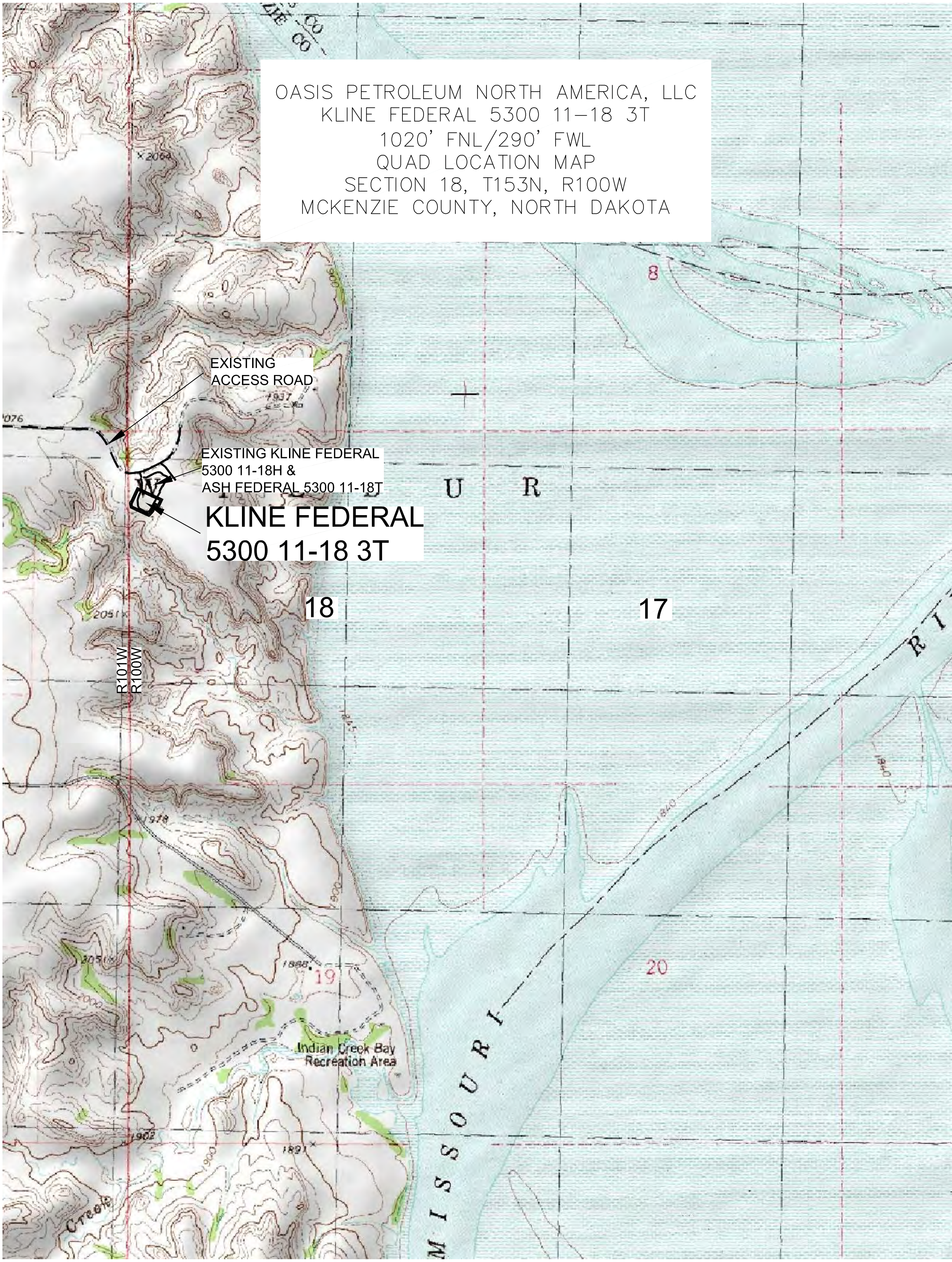
OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS



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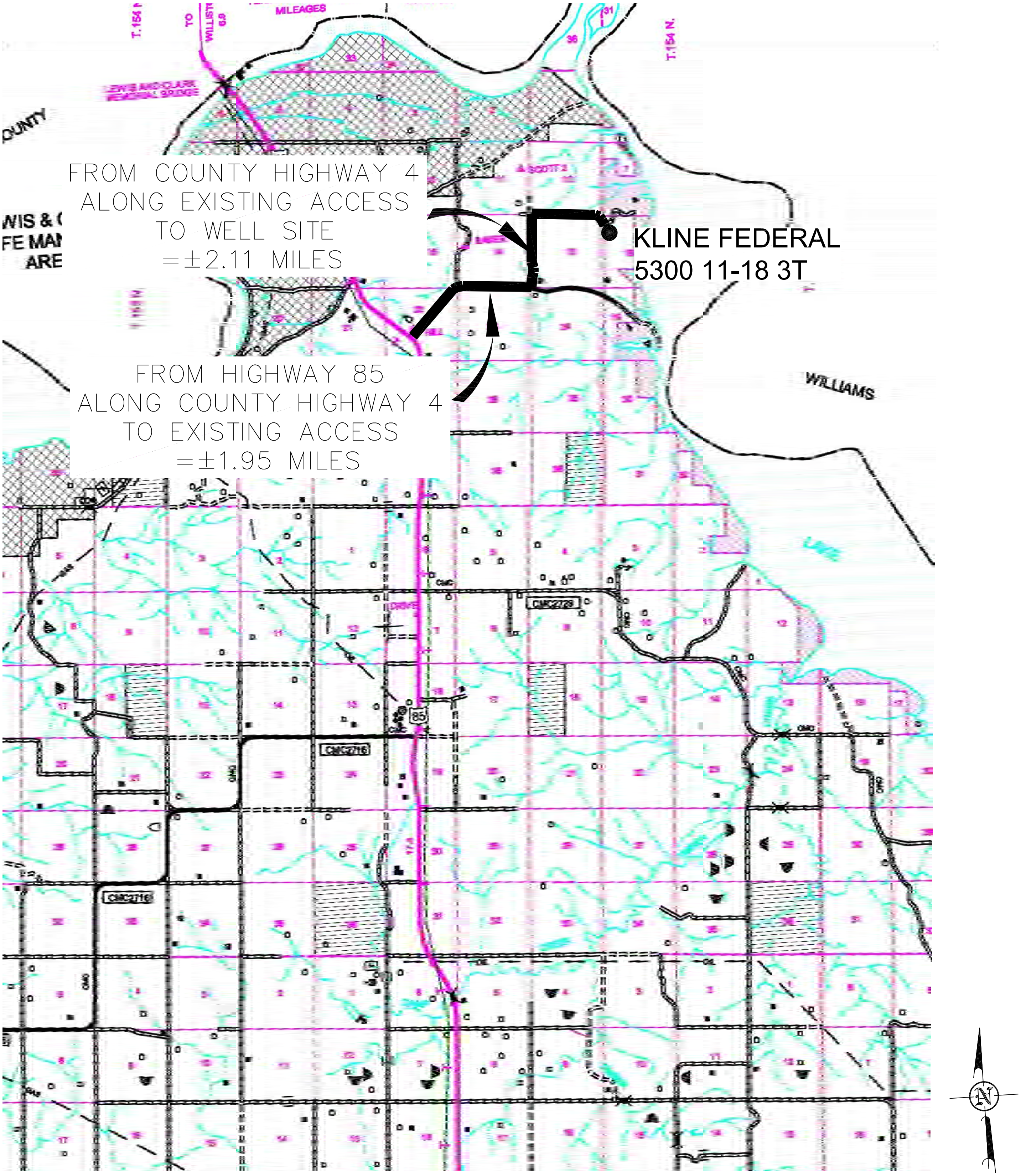
OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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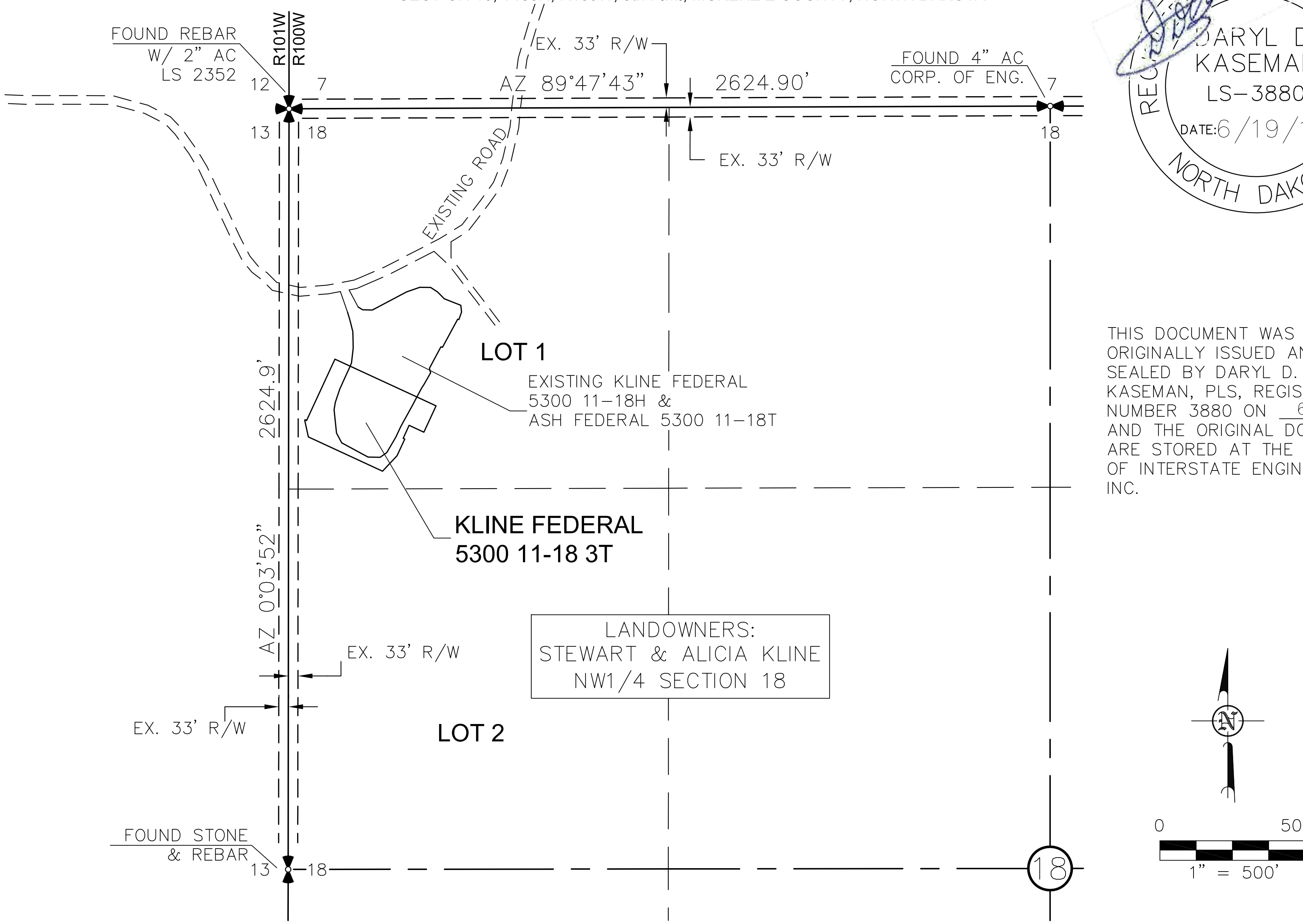
OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By:	B.H.H.	Project No.:	S14-09-127.01
Checked By:	D.D.K.	Date:	APRIL 2014

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS

ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 3T"
1020 FEET FROM NORTH LINE AND 290 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 6/19/14
NORTH DAKOTA

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NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

Revision No.	Date	By	Description
REV 1	6/16/14	BHH	MOVED WELLS

OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-127.01
Checked By: D.D.K. Date: APRIL 2014

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SHEET NO. 4/8

GAS CAPTURE PLAN AFFIDAVIT

STATE OF TEXAS §
§
COUNTY OF HARRIS §

Robert Eason, being duly sworn, states as follows:

1. He is employed by Oasis Petroleum North America LLC ("Oasis") as Marketing Manager, is over the age of 21 and has personal knowledge of the matters set forth in this affidavit.

2. This affidavit is submitted in conjunction with the Application for Permit to Drill for the Kline Federal 5300 11-18 3T well, with a surface location in the NW NW of Section 18, Township 153 North, Range 100 West, McKenzie County, North Dakota (the "Well").

3. Oasis currently anticipates that gas to be produced from the Well will be gathered by Hiland Partners (the "Gathering Company"). Oasis has advised the Gathering Company of its intent to drill the Well and has advised the Gathering Company that it currently anticipates that the Well will be completed in May 2015, with an initial gas production rate of approximately 640 mcf/day.

Robert Eason

Robert Eason
Marketing Manager

Subscribed and sworn to before me this 24 day of June, 2014.

DEBORAH ANN MANGUM
Notary Public, State of Texas
My Commission Expires
March 28, 2015

Deborah Ann Mangum

Notary Public in and for the State of Texas
My Commission expires: 3.28.2015

GAS CAPTURE PLAN – OASIS PETROLEUM

Kline Federal 5300 11-18 3T

Section 18-T153N-R100W
Baker Field
McKenzie County, North Dakota

Anticipated first flow date	May-15
Gas Gatherer:	Hiland Partners
Gas to be processed at*:	Hiland Operated Watford City Plant
Maximum Daily Capacity of Existing Gas Line*:	55,000 MCFD
Current Throughput of Existing Gas Line*:	33,000 MCFD
Anticipated Daily Capacity of Existing Gas Line at Date of First Gas Sales*:	66,000 MCFD
Anticipated Throughput of Existing Gas Line at Date of First Gas Sales*:	65,000 MCFD
Gas Gatherer's Issues or Expansion Plans for the Area*:	Line looping and compression
Map:	Attached
Affidavit:	Attached

*Provided by Gatherer

Flowback Strategy

Total Number of Wells at Location: 10

Multi-Well Start-up Plan: Initial production from the 1st new well at the CTB is anticipated in May 2015 with each following well making 1st production aproximately every 5th day thereafter

Estimated Flow Rate:	Kline Federal 5300 11-18 3T (well only)		Kline DSU (10 wells)	
	MCFD	BOPD	MCFD	BOPD
30 Days:	640	711	5,100	5,646
60 Days:	549	610	6,302	6,982
180 Days:	329	365	3,543	3,918

Oasis Flaring Percentage

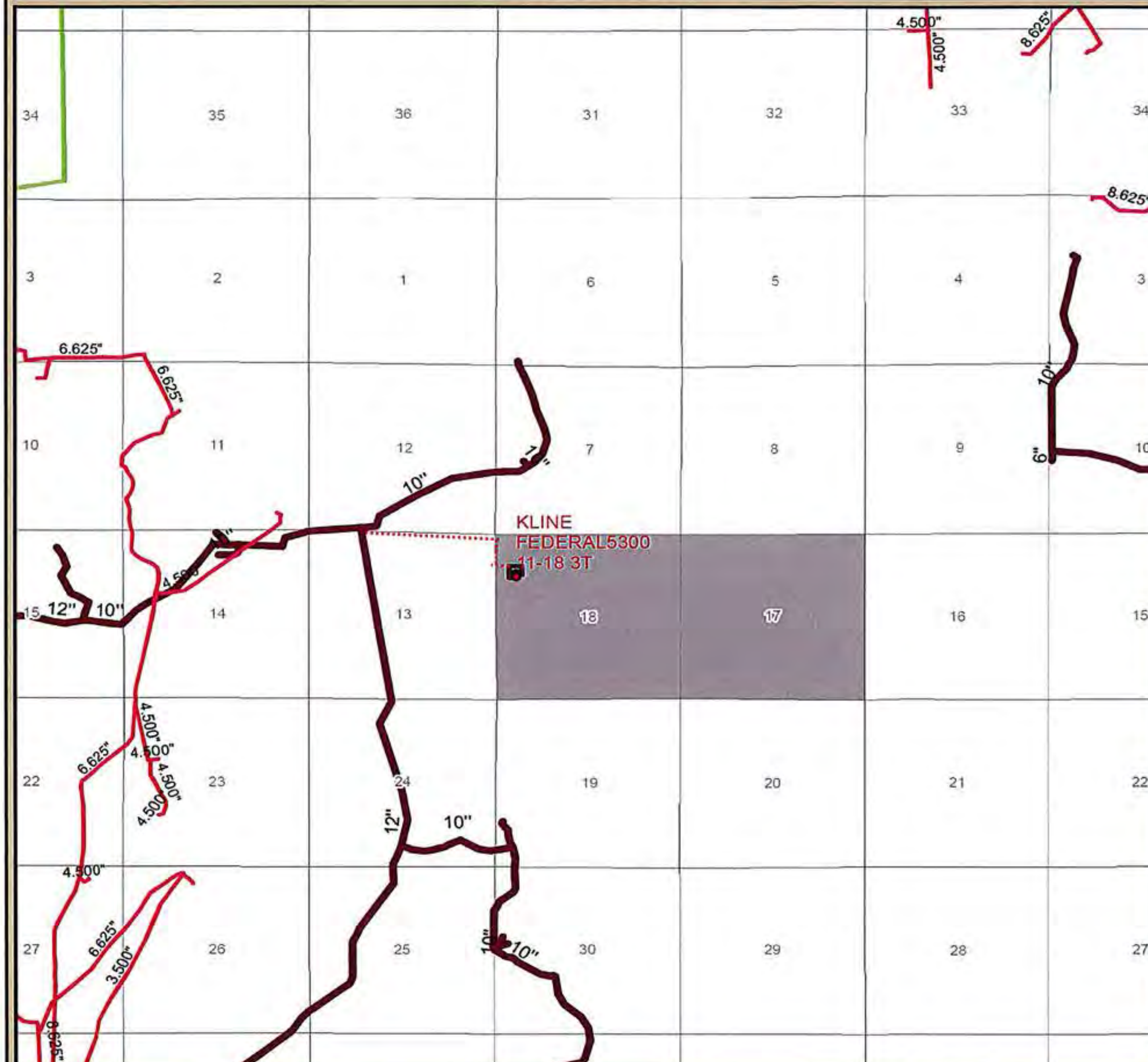
	Statewide	Baker Field
Oasis % of Gas Flared:	**12%**	**6%**

Average over the last 6 months

Alternatives to Flaring

Source: Oasis Marketing (281) 404-9435

Gas Capture Plan - Detail View
KLINE FEDERAL5300 11-18 3T
Section 18 T153N R100W
McKenzie County, North Dakota

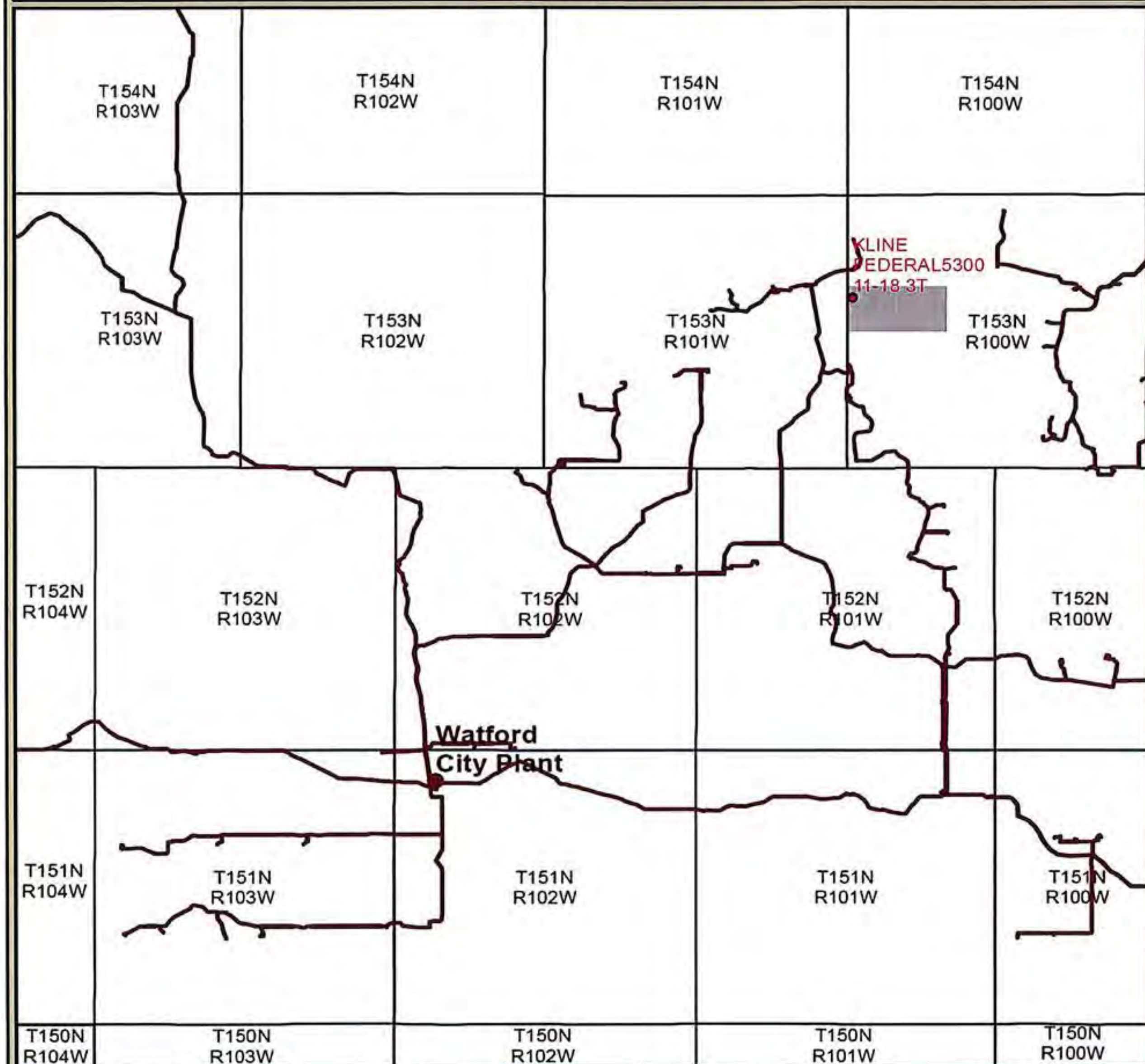


Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant

- Proposed Well
- Proposed CTB
- Hiland Gas Line
- Oneok Gas Line
- Williston Basin Interstate

OASIS PETROLEUM

Gas Capture Plan - Overview
KLINE FEDERAL5300 11-18 3T
Section 18 T153N R100W
McKenzie County, North Dakota



Proposed Well
Hiland Gas Line
Processing Plant

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant

OASIS PETROLEUM

OASIS PETROLEUM

8/20/2014

Mineral Resources Permit Manager
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: Kline Federal 5300 11-18 2T2
Kline Federal 5300 11-18 3T
Kline Federal 5300 11-18 4T2
Kline Federal 5300 11-18 5B
Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Kline Federal 5300 11-18 2T2, Kline Federal 5300 11-18 3T, Kline Federal 5300 11-18 4T2 and Kline Federal 5300 11-18 5B. The request was made to Aaron Chisolm (address@co.mckenzie.nd.us) in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

Heather McCowan

Heather McCowan
Regulatory Assistant
Oasis Petroleum North America, LLC

Hello Taylor,

They will be hauled to the JMAC Resources Disposal
5009 139th Ave NW, Williston, ND 58801
(701) 774-8511

Thanks,

Heather McCowan

Regulatory Assistant | 1001 Fannin, Suite 1500, Houston, Texas 77002 | 281-404-9563 Direct | hmccowan@oasispetroleum.com

From: Roth, Taylor J. [mailto:tjroth@nd.gov]
Sent: Wednesday, August 20, 2014 9:59 AM
To: Heather McCowan
Subject: RE: Kline Federal pad

Heather,

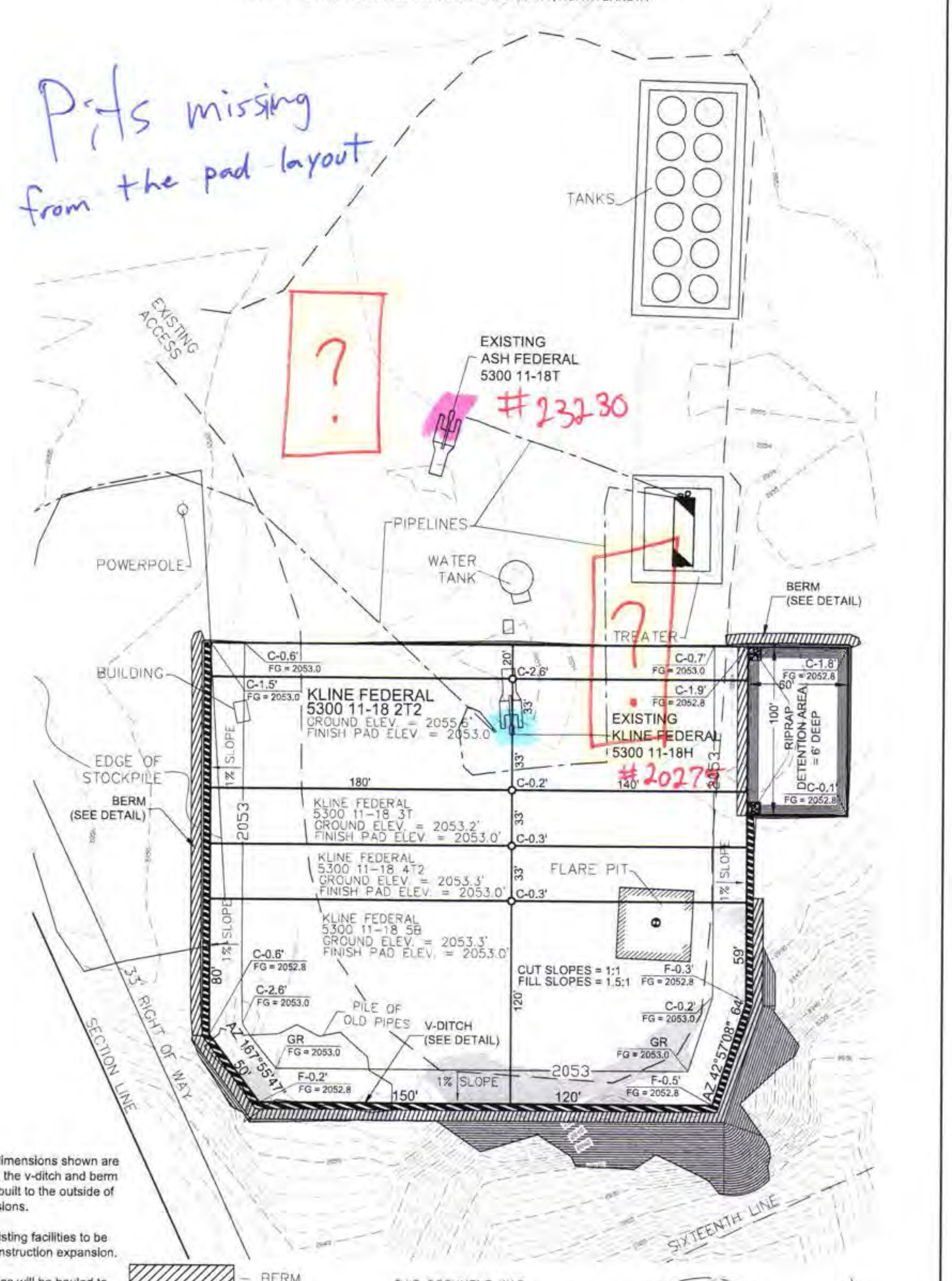
What will Oasis be doing with the cuttings on this pad?

Thank you very much,

Taylor J. Roth
Survey & Permitting Technician
NDIC, Dept. Mineral Resources
Oil and Gas Division
701-328-1720 (direct)
tjroth@nd.gov

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 11-18 2T2"
960 FEET FROM NORTH LINE AND 318 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE 1: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE 2: All existing facilities to be removed on construction expansion.

NOTE 3: Cuttings will be hauled to approved disposal site.

V-DITCH DETAIL

2:1 2:1 1.5' min
2.0' min

— BERM
— DITCH
——— Proposed Contours
– – – Original Contours

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REGISTERED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE 6/19/14
NORTH DAKOTA

0 60
1" = 60'

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"
800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

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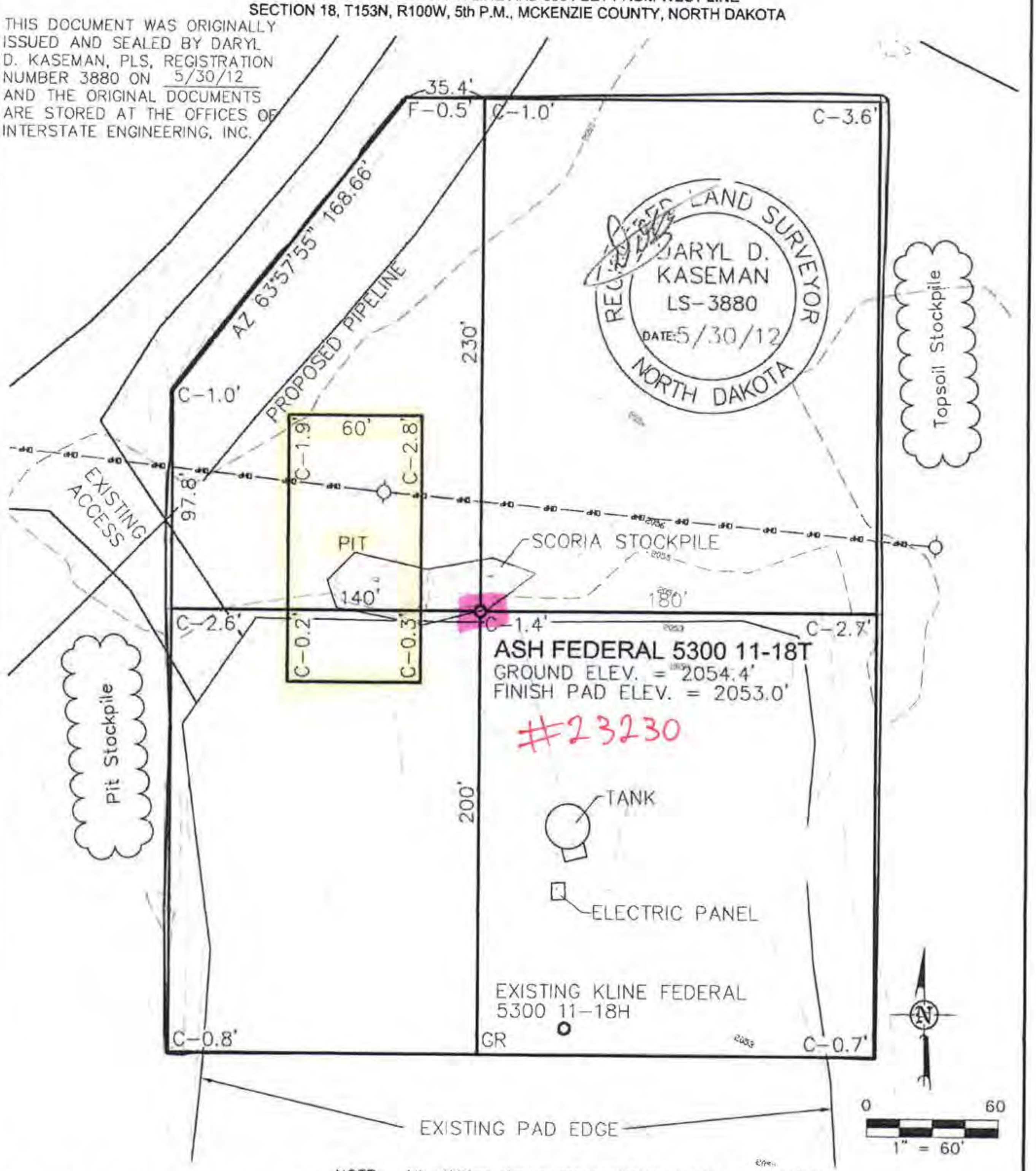

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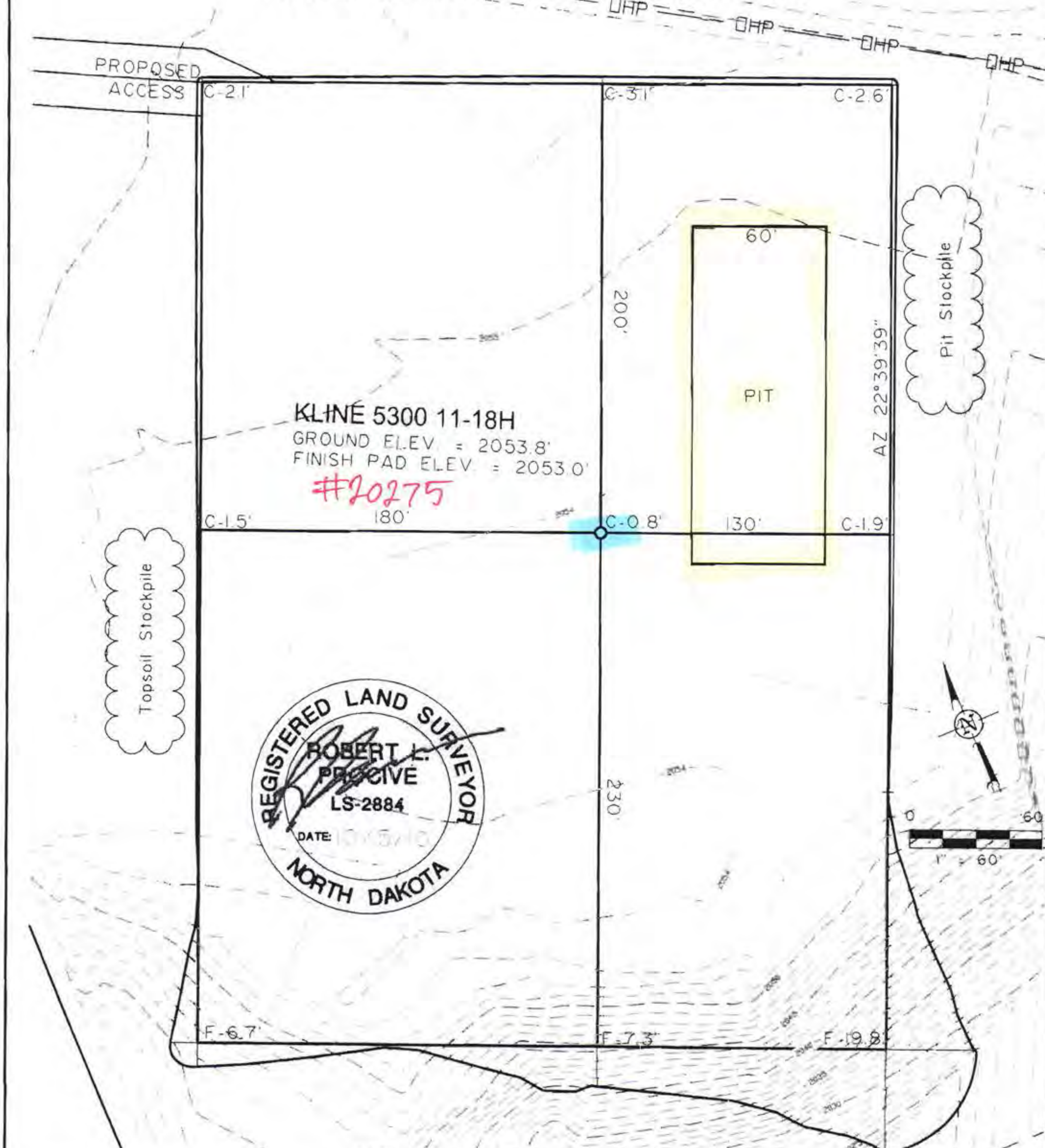
PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 202 HOUSTON, TX 77002
"KLINE 5300 11-18H"
990 FEET FROM NORTH LINE AND 305 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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			Drawn By: J.J.S. Project No.: S10-9-190				
			Checked By: A.J.H./R.L.P. Date: OCT. 2010				

Oasis Petroleum

Indian Hills
153N-100W-17/18
Kline Federal 5300 11-18 3T

Wellbore #1
Design #6

Anticollision Report

30 June, 2014

Precision Wellbore Placement

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Reference Design #6

Filter type:	NO GLOBAL FILTER: Using user defined selection & filtering criteria		
Interpolation Method:	Stations	**Error Model:**	ISCWSA
Depth Range:	Unlimited	**Scan Method:**	Closest Approach 3D
Results Limited by:	Maximum center-center distance of 2,000.0 usft	**Error Surface:**	Elliptical Conic
Warning Levels Evaluated at:	2.00 **Sigma**	**Casing Method:**	Not applied

Survey Tool Program **Date** 06/30/14

From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description
0.0	21,038.5	Design #6 (Wellbore #1)	MWD	MWD - Standard

Summary

Site Name / Offset Well - Wellbore - Design	Reference Measured Depth (usft)	Offset Measured Depth (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Separation Factor	Warning
153N-100W-17/18						
Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1	2,555.1	2,529.2	222.4	216.2	35.569	CC
Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1	2,600.0	2,573.9	222.6	216.1	34.527	ES
Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1	21,038.5	20,482.0	1,261.2	719.9	2.330	SF
Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6	2,500.0	2,500.0	65.7	54.7	5.993	CC
Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6	21,038.5	20,717.6	500.4	-84.1	0.856	Level 1, ES, SF
Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1	2,564.1	2,540.4	27.4	20.9	4.220	CC
Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1	20,900.0	20,650.0	496.5	-76.8	0.866	Level 1, ES, SF
Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6	2,500.0	2,500.0	33.6	22.6	3.064	CC
Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6	21,038.5	21,373.9	501.2	-83.6	0.857	Level 1, ES, SF
Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6	2,500.0	2,500.0	65.9	55.0	6.018	CC
Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6	21,038.5	21,197.2	503.5	-77.1	0.867	Level 1, ES, SF

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1

Offset Site Error: 0.0 usft
Offset Well Error: 0.0 usft

Survey Program: 2261-MWD, 13302-MWD

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
300.0	300.0	2,180.0	2,183.0	0.5	0.0	15.10	218.9	59.1	1,920.4	1,919.9	0.53	3,596.174	
400.0	400.0	2,180.0	2,183.0	0.8	0.0	15.10	218.9	59.1	1,821.2	1,820.4	0.76	2,400.104	
500.0	500.0	2,180.0	2,183.0	1.0	0.0	15.10	218.9	59.1	1,722.0	1,721.0	0.98	1,750.784	
600.0	600.0	2,180.0	2,183.0	1.2	0.0	15.10	218.9	59.1	1,622.9	1,621.7	1.21	1,343.116	
700.0	700.0	2,180.0	2,183.0	1.4	0.0	15.10	218.9	59.1	1,524.0	1,522.5	1.43	1,063.410	
800.0	800.0	2,180.0	2,183.0	1.7	0.0	15.10	218.9	59.1	1,425.1	1,423.5	1.66	859.635	
900.0	900.0	2,180.0	2,183.0	1.9	0.0	15.10	218.9	59.1	1,326.5	1,324.6	1.88	704.612	
1,000.0	1,000.0	2,180.0	2,183.0	2.1	0.0	15.10	218.9	59.1	1,228.1	1,226.0	2.11	582.763	
1,100.0	1,100.0	2,180.0	2,183.0	2.3	0.0	15.10	218.9	59.1	1,130.0	1,127.6	2.33	484.520	
1,200.0	1,200.0	2,180.0	2,183.0	2.6	0.0	15.10	218.9	59.1	1,032.2	1,029.6	2.56	403.690	
1,300.0	1,300.0	2,180.0	2,183.0	2.8	0.0	15.10	218.9	59.1	934.9	932.1	2.78	336.092	
1,400.0	1,400.0	2,180.0	2,183.0	3.0	0.0	15.10	218.9	59.1	838.2	835.2	3.01	278.813	
1,500.0	1,500.0	2,180.0	2,183.0	3.2	0.0	15.10	218.9	59.1	742.5	739.2	3.23	229.776	
1,600.0	1,600.0	2,180.0	2,183.0	3.5	0.0	15.10	218.9	59.1	647.9	644.5	3.46	187.487	
1,700.0	1,700.0	2,180.0	2,183.0	3.7	0.0	15.10	218.9	59.1	555.4	551.7	3.68	150.886	
1,800.0	1,800.0	2,180.0	2,183.0	3.9	0.0	15.10	218.9	59.1	465.9	462.0	3.91	119.286	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1 — **Offset Site Error:** 0.0 usft

Survey Program: 2261-MWD, 13302-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
1,900.0	1,900.0	2,180.0	2,183.0	4.1	0.0	15.10	218.9	59.1	381.6	377.5	4.13	92.397	
2,000.0	2,000.0	2,180.0	2,183.0	4.4	0.0	15.10	218.9	59.1	307.0	302.6	4.36	70.489	
2,100.0	2,100.0	2,180.0	2,183.0	4.6	0.0	15.10	218.9	59.1	250.7	246.1	4.58	54.735	
2,200.0	2,200.0	2,180.0	2,183.0	4.8	0.0	15.10	218.9	59.1	226.8	222.0	4.80	47.205	
2,300.0	2,300.0	2,277.2	2,280.2	5.0	0.1	15.22	217.7	59.2	225.6	220.5	5.16	43.757	
2,400.0	2,400.0	2,376.3	2,379.3	5.3	0.3	15.33	216.0	59.2	224.0	218.4	5.59	40.080	
2,500.0	2,500.0	2,475.2	2,478.2	5.5	0.5	15.32	214.8	58.8	222.7	216.7	6.02	36.970	
2,555.1	2,555.1	2,529.2	2,532.2	5.6	0.7	105.40	214.5	58.5	222.4	216.2	6.25	35.569 CC	
2,600.0	2,600.0	2,573.9	2,576.8	5.7	0.8	105.62	214.4	58.2	222.6	216.1	6.45	34.527 ES	
2,650.0	2,649.9	2,623.8	2,626.7	5.8	0.9	106.01	214.3	57.6	223.0	216.3	6.66	33.500	
2,700.0	2,699.9	2,673.7	2,676.7	5.9	1.0	106.48	214.3	56.9	223.5	216.7	6.87	32.546	
2,800.0	2,799.7	2,773.2	2,776.1	6.1	1.2	107.33	214.5	55.2	224.7	217.4	7.29	30.809	
2,900.0	2,899.6	2,872.4	2,875.3	6.3	1.4	108.10	215.1	53.4	226.3	218.6	7.72	29.308	
3,000.0	2,999.5	2,971.7	2,974.6	6.5	1.6	108.80	216.0	51.5	228.2	220.1	8.16	27.984	
3,100.0	3,099.3	3,071.1	3,073.9	6.7	1.9	109.41	217.3	49.3	230.4	221.9	8.58	26.843	
3,200.0	3,199.2	3,170.9	3,173.8	7.0	2.1	110.06	218.7	47.3	232.8	223.8	9.00	25.870	
3,300.0	3,299.0	3,271.0	3,273.9	7.2	2.3	110.74	219.9	45.5	235.2	225.8	9.41	24.992	
3,400.0	3,398.9	3,370.3	3,373.1	7.4	2.5	111.43	221.2	43.9	237.7	227.9	9.83	24.173	
3,501.2	3,500.0	3,471.8	3,474.6	7.6	2.7	112.13	222.6	42.3	240.4	230.1	10.27	23.408	
3,600.0	3,598.7	3,570.5	3,573.3	7.8	2.9	112.48	223.8	40.7	242.3	231.6	10.68	22.691	
3,651.2	3,649.9	3,622.1	3,624.9	7.9	3.0	22.37	224.4	40.0	242.7	231.9	10.80	22.471	
3,700.0	3,698.7	3,671.1	3,673.8	8.0	3.1	22.17	224.9	39.2	242.9	231.9	11.00	22.075	
3,800.0	3,798.7	3,770.6	3,773.4	8.2	3.3	21.77	226.0	37.9	243.4	231.9	11.44	21.281	
3,900.0	3,898.7	3,870.6	3,873.3	8.5	3.5	21.39	227.1	36.6	243.9	232.1	11.87	20.546	
4,000.0	3,998.7	3,970.8	3,973.5	8.7	3.7	21.06	228.1	35.4	244.4	232.1	12.31	19.862	
4,100.0	4,098.7	4,070.6	4,073.3	8.9	3.9	20.76	229.1	34.4	245.0	232.2	12.74	19.225	
4,200.0	4,198.7	4,170.4	4,173.1	9.1	4.1	20.47	230.1	33.5	245.6	232.4	13.18	18.630	
4,300.0	4,298.7	4,270.6	4,273.3	9.4	4.4	20.20	231.1	32.6	246.2	232.6	13.62	18.079	
4,400.0	4,398.7	4,371.0	4,373.7	9.6	4.6	20.02	231.8	32.1	246.7	232.7	14.05	17.556	
4,500.0	4,498.7	4,471.1	4,473.7	9.8	4.8	19.88	232.4	31.6	247.1	232.6	14.49	17.058	
4,600.0	4,598.7	4,570.7	4,573.4	10.0	5.0	19.74	233.0	31.2	247.5	232.6	14.92	16.588	
4,700.0	4,698.7	4,670.3	4,672.9	10.2	5.2	19.56	233.8	30.7	248.2	232.8	15.36	16.159	
4,800.0	4,798.7	4,769.4	4,772.1	10.5	5.4	19.31	235.0	30.0	249.0	233.2	15.79	15.768	
4,900.0	4,898.7	4,869.6	4,872.2	10.7	5.6	19.17	236.2	29.7	250.1	233.8	16.23	15.408	
5,000.0	4,998.7	4,969.5	4,972.1	10.9	5.8	19.12	237.2	29.8	251.1	234.4	16.67	15.066	
5,100.0	5,098.7	5,069.7	5,072.3	11.1	6.0	19.15	238.1	30.3	252.1	235.0	17.10	14.741	
5,200.0	5,198.7	5,171.2	5,173.9	11.4	6.3	19.24	238.7	30.9	252.8	235.2	17.53	14.417	
5,300.0	5,298.7	5,273.5	5,276.1	11.6	6.5	19.31	238.4	31.1	252.6	234.6	17.96	14.060	
5,400.0	5,398.7	5,373.3	5,375.9	11.8	6.7	19.48	237.6	31.6	252.0	233.6	18.39	13.701	
5,500.0	5,498.7	5,472.3	5,474.9	12.0	6.9	19.57	237.2	31.9	251.7	232.9	18.82	13.373	
5,519.2	5,517.9	5,491.3	5,493.9	12.1	6.9	19.58	237.1	31.9	251.7	232.8	18.90	13.315	
5,600.0	5,598.7	5,571.4	5,574.0	12.2	7.1	19.59	237.2	32.0	251.8	232.6	19.25	13.080	
5,700.0	5,698.7	5,672.0	5,674.6	12.5	7.3	19.52	237.6	31.8	252.1	232.4	19.68	12.807	
5,798.1	5,796.8	5,770.2	5,772.8	12.7	7.5	19.39	237.7	31.3	252.0	231.9	20.11	12.533	
5,800.0	5,798.7	5,772.1	5,774.7	12.7	7.5	19.39	237.7	31.3	252.0	231.9	20.11	12.528	
5,900.0	5,898.7	5,871.4	5,874.0	12.9	7.7	19.17	238.2	30.4	252.2	231.6	20.55	12.272	
6,000.0	5,998.7	5,970.6	5,973.2	13.1	7.9	19.01	238.8	29.9	252.6	231.6	20.98	12.036	
6,100.0	6,098.7	6,069.4	6,072.0	13.4	8.1	18.87	239.8	29.5	253.4	232.0	21.42	11.830	
6,200.0	6,198.7	6,167.3	6,169.9	13.6	8.3	18.71	241.3	29.3	254.9	233.0	21.86	11.660	
6,300.0	6,298.7	6,265.3	6,267.9	13.8	8.5	18.56	243.7	29.4	257.1	234.8	22.29	11.534	
6,400.0	6,398.7	6,365.4	6,368.0	14.0	8.8	18.44	246.4	29.7	259.8	237.0	22.73	11.427	
6,500.0	6,498.7	6,463.4	6,465.9	14.3	9.0	18.41	249.2	30.5	262.7	239.6	23.17	11.341	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1

Offset Site Error: 0.0 usft

Survey Program: 2261-MWD, 13302-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
6,600.0	6,598.7	6,561.8	6,564.2	14.5	9.2	18.45	252.5	31.8	266.4	242.8	23.61	11.283	
6,700.0	6,698.7	6,660.8	6,663.2	14.7	9.4	18.51	256.5	33.4	270.7	246.7	24.05	11.257	
6,800.0	6,798.7	6,766.9	6,769.2	14.9	9.6	18.41	259.5	34.0	273.6	249.1	24.49	11.170	
6,900.0	6,898.7	6,866.7	6,869.0	15.2	9.8	18.20	261.7	33.6	275.5	250.6	24.92	11.054	
7,000.0	6,998.7	6,965.3	6,967.5	15.4	10.0	18.00	264.3	33.5	277.9	252.6	25.36	10.961	
7,100.0	7,098.7	7,065.4	7,067.6	15.6	10.3	17.86	266.9	33.6	280.5	254.7	25.79	10.874	
7,200.0	7,198.7	7,165.6	7,167.8	15.8	10.5	17.72	269.5	33.7	283.0	256.8	26.23	10.790	
7,300.0	7,298.7	7,265.4	7,267.5	16.0	10.7	17.63	271.9	34.0	285.4	258.8	26.66	10.705	
7,400.0	7,398.7	7,364.8	7,366.8	16.3	10.9	17.57	274.5	34.5	288.1	261.0	27.10	10.628	
7,500.0	7,498.7	7,464.8	7,466.8	16.5	11.1	17.64	277.0	35.7	290.8	263.3	27.54	10.558	
7,600.0	7,598.7	7,564.8	7,566.8	16.7	11.3	17.78	279.4	37.2	293.5	265.5	27.98	10.491	
7,700.0	7,698.7	7,665.5	7,667.4	16.9	11.5	17.93	281.6	38.7	296.1	267.7	28.41	10.422	
7,800.0	7,798.7	7,764.5	7,766.4	17.2	11.7	18.12	283.8	40.5	298.7	269.9	28.85	10.356	
7,900.0	7,898.7	7,868.0	7,869.9	17.4	12.0	18.28	285.6	41.9	300.9	271.6	29.28	10.273	
8,000.0	7,998.7	7,969.2	7,971.1	17.6	12.2	18.40	286.7	43.0	302.2	272.5	29.71	10.171	
8,100.0	8,098.7	8,069.5	8,071.4	17.8	12.4	18.56	287.5	44.1	303.3	273.2	30.15	10.063	
8,200.0	8,198.7	8,169.6	8,171.4	18.1	12.6	18.71	288.3	45.3	304.5	273.9	30.58	9.957	
8,300.0	8,298.7	8,270.2	8,272.1	18.3	12.8	18.82	289.1	46.1	305.4	274.4	31.01	9.850	
8,400.0	8,398.7	8,370.5	8,372.4	18.5	13.0	18.90	289.7	46.8	306.2	274.7	31.43	9.740	
8,500.0	8,498.7	8,470.3	8,472.1	18.7	13.2	18.98	290.3	47.4	307.0	275.1	31.87	9.633	
8,600.0	8,598.7	8,570.1	8,571.9	19.0	13.4	19.10	290.9	48.3	307.8	275.5	32.30	9.530	
8,700.0	8,698.7	8,669.4	8,671.2	19.2	13.6	19.29	291.5	49.6	308.8	276.1	32.74	9.433	
8,800.0	8,798.7	8,769.3	8,771.1	19.4	13.8	19.49	292.2	51.0	310.0	276.8	33.17	9.345	
8,900.0	8,898.7	8,869.8	8,871.6	19.6	14.0	19.72	292.8	52.6	311.1	277.5	33.61	9.257	
9,000.0	8,998.7	8,970.1	8,971.9	19.9	14.2	19.91	293.3	53.9	312.0	278.0	34.03	9.167	
9,100.0	9,098.7	9,070.4	9,072.2	20.1	14.5	20.13	293.7	55.3	312.8	278.4	34.47	9.076	
9,200.0	9,198.7	9,170.6	9,172.4	20.3	14.7	20.37	294.0	56.7	313.6	278.7	34.90	8.986	
9,300.0	9,298.7	9,270.4	9,272.2	20.5	14.9	20.58	294.3	58.1	314.4	279.0	35.33	8.899	
9,400.0	9,398.7	9,370.9	9,372.7	20.8	15.1	20.79	294.6	59.5	315.1	279.4	35.76	8.812	
9,500.0	9,498.7	9,471.5	9,473.2	21.0	15.3	21.01	294.6	60.7	315.6	279.4	36.19	8.721	
9,600.0	9,598.7	9,571.3	9,573.0	21.2	15.5	21.20	294.8	61.9	316.2	279.6	36.62	8.634	
9,700.0	9,698.7	9,671.6	9,673.3	21.4	15.7	21.38	294.8	63.0	316.6	279.5	37.05	8.544	
9,800.0	9,798.7	9,771.0	9,772.7	21.6	15.9	21.57	294.9	64.2	317.1	279.7	37.49	8.460	
9,900.0	9,898.7	9,871.2	9,872.9	21.9	16.1	21.75	295.2	65.3	317.8	279.9	37.92	8.380	
10,000.0	9,998.7	9,970.9	9,972.6	22.1	16.3	21.87	295.4	66.2	318.3	280.0	38.35	8.301	
10,100.0	10,098.7	10,070.1	10,071.8	22.3	16.5	21.95	296.0	66.9	319.1	280.3	38.78	8.228	
10,200.0	10,198.7	10,169.6	10,171.3	22.5	16.7	22.03	296.7	67.7	320.1	280.9	39.22	8.162	
10,300.0	10,298.7	10,269.2	10,270.9	22.8	17.0	22.11	297.6	68.5	321.2	281.6	39.66	8.100	
10,336.6	10,335.3	10,305.5	10,307.2	22.9	17.0	22.13	298.0	68.8	321.7	281.9	39.81	8.079	
10,350.0	10,348.7	10,318.4	10,320.1	22.9	17.1	-114.86	298.1	68.9	321.9	282.1	39.88	8.073	
10,375.0	10,373.7	10,341.1	10,342.7	22.9	17.1	-114.84	298.2	69.6	322.9	283.0	39.96	8.082	
10,400.0	10,398.5	10,360.4	10,362.0	23.0	17.1	-114.75	298.3	71.0	324.8	284.8	40.02	8.116	
10,425.0	10,423.2	10,377.0	10,378.5	23.0	17.2	-114.53	298.3	72.9	327.7	287.6	40.06	8.179	
10,450.0	10,447.6	10,394.7	10,396.0	23.1	17.2	-114.19	298.5	75.9	331.7	291.6	40.10	8.271	
10,475.0	10,471.8	10,408.0	10,408.9	23.1	17.3	-113.70	298.7	78.8	336.8	296.7	40.12	8.396	
10,500.0	10,495.5	10,429.5	10,429.7	23.1	17.3	-113.17	299.1	84.5	343.0	302.8	40.16	8.541	
10,525.0	10,518.8	10,449.2	10,448.5	23.2	17.4	-112.58	299.5	90.4	350.0	309.8	40.19	8.709	
10,550.0	10,541.7	10,471.0	10,469.1	23.2	17.4	-112.05	300.0	97.2	357.7	317.5	40.23	8.893	
10,575.0	10,563.9	10,487.3	10,484.5	23.3	17.4	-111.42	300.6	102.6	366.2	326.0	40.26	9.098	
10,600.0	10,585.5	10,502.0	10,498.3	23.3	17.5	-110.68	301.2	107.8	375.6	335.4	40.29	9.323	
10,625.0	10,606.5	10,523.3	10,517.9	23.4	17.5	-110.06	302.1	116.0	385.8	345.4	40.35	9.562	
10,650.0	10,626.7	10,542.3	10,535.1	23.4	17.6	-109.29	302.7	124.1	396.6	356.1	40.41	9.813	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1 — **Offset Site Error:** 0.0 usft

Survey Program: 2261-MWD, 13302-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
10,675.0	10,646.1	10,560.5	10,551.3	23.5	17.6	-108.43	303.3	132.4	407.9	367.5	40.49	10.074	
10,700.0	10,664.6	10,577.3	10,565.9	23.6	17.7	-107.44	303.8	140.6	420.0	379.4	40.60	10.345	
10,725.0	10,682.3	10,595.0	10,580.8	23.6	17.7	-106.37	304.3	150.1	432.6	391.9	40.72	10.623	
10,750.0	10,698.9	10,609.2	10,592.5	23.7	17.8	-105.07	304.7	158.2	445.9	405.0	40.88	10.908	
10,775.0	10,714.6	10,626.0	10,605.9	23.8	17.8	-103.79	305.1	168.3	459.7	418.7	41.06	11.197	
10,800.0	10,729.3	10,641.1	10,617.7	23.9	17.9	-102.38	305.5	177.7	474.0	432.8	41.26	11.490	
10,825.0	10,742.9	10,657.0	10,630.0	24.0	18.0	-100.97	305.9	187.8	488.8	447.3	41.48	11.785	
10,850.0	10,755.3	10,672.5	10,641.6	24.2	18.0	-99.48	306.3	198.0	504.0	462.3	41.72	12.081	
10,875.0	10,766.6	10,688.0	10,652.7	24.3	18.1	-97.89	306.7	208.8	519.5	477.6	41.97	12.378	
10,900.0	10,776.8	10,704.5	10,664.0	24.5	18.2	-96.29	306.9	220.8	535.4	493.1	42.24	12.675	
10,925.0	10,785.7	10,721.6	10,675.2	24.7	18.3	-94.66	307.1	233.7	551.5	509.0	42.52	12.971	
10,950.0	10,793.4	10,737.7	10,685.5	24.9	18.4	-92.97	307.2	246.1	567.8	525.0	42.80	13.267	
10,975.0	10,799.8	10,751.0	10,693.9	25.1	18.5	-91.13	307.2	256.5	584.3	541.2	43.06	13.568	
11,000.0	10,804.9	10,763.4	10,701.5	25.3	18.6	-89.20	307.3	266.3	601.1	557.7	43.32	13.874	
11,025.0	10,808.8	10,774.0	10,707.7	25.5	18.6	-87.10	307.5	274.9	618.1	574.5	43.56	14.190	
11,050.0	10,811.4	10,786.6	10,714.6	25.8	18.7	-85.10	307.6	285.3	635.4	591.6	43.78	14.512	
11,075.0	10,812.6	10,807.1	10,725.3	26.0	18.9	-83.63	307.7	302.9	652.6	608.6	44.05	14.815	
11,084.1	10,812.8	10,814.6	10,728.9	26.1	19.0	-83.09	307.7	309.4	658.9	614.8	44.15	14.924	
11,099.1	10,812.8	10,825.8	10,734.2	26.3	19.1	-83.81	307.6	319.3	669.2	624.8	44.40	15.073	
11,200.0	10,813.4	10,881.1	10,757.2	27.5	19.6	-86.88	306.8	369.6	738.0	691.9	46.10	16.010	
11,300.0	10,813.9	10,927.7	10,772.2	28.9	20.2	-88.59	307.2	413.7	805.3	757.4	47.95	16.796	
11,400.0	10,814.5	10,994.7	10,787.3	30.6	21.1	-89.99	309.1	478.9	870.9	820.7	50.20	17.347	
11,500.0	10,815.0	11,066.8	10,796.1	32.4	22.3	-90.63	311.3	550.3	932.7	879.9	52.81	17.663	
11,600.0	10,815.6	11,150.4	10,795.7	34.3	23.7	-90.50	314.5	633.9	991.0	935.2	55.85	17.743	
11,700.0	10,816.1	11,246.7	10,792.8	36.3	25.6	-90.23	317.0	730.1	1,043.9	984.5	59.43	17.566	
11,800.0	10,816.7	11,352.5	10,792.8	38.5	27.8	-90.16	318.7	835.9	1,091.5	1,028.0	63.50	17.188	
11,900.0	10,817.3	11,439.2	10,795.5	40.7	29.8	-90.26	319.1	922.5	1,133.3	1,065.8	67.49	16.792	
12,000.0	10,817.9	11,521.8	10,798.3	43.0	31.7	-90.36	320.3	1,005.1	1,171.2	1,099.7	71.55	16.370	
12,100.0	10,818.4	11,657.4	10,798.3	45.3	35.0	-90.28	321.3	1,140.6	1,203.9	1,127.0	76.92	15.652	
12,200.0	10,819.0	11,818.0	10,795.1	47.6	39.1	-90.06	314.2	1,301.0	1,226.2	1,143.1	83.14	14.749	
12,300.0	10,819.5	11,927.2	10,796.2	50.0	41.9	-90.07	306.1	1,409.9	1,240.5	1,152.3	88.15	14.073	
12,400.0	10,820.1	12,019.3	10,798.7	52.3	44.4	-90.14	299.3	1,501.8	1,249.6	1,156.9	92.63	13.490	
12,500.0	10,820.6	12,122.4	10,802.3	54.7	47.1	-90.26	291.9	1,604.5	1,253.7	1,156.5	97.28	12.888	
12,600.0	10,821.2	12,210.4	10,805.4	57.0	49.5	-90.37	285.9	1,692.2	1,253.0	1,151.6	101.35	12.363	
12,655.6	10,821.5	12,261.7	10,807.2	58.3	50.9	-90.43	282.7	1,743.4	1,250.6	1,147.0	103.59	12.073	
12,700.0	10,821.7	12,303.0	10,808.9	59.4	52.0	-90.50	280.3	1,784.6	1,248.3	1,142.4	105.86	11.791	
12,800.0	10,822.2	12,378.6	10,811.1	61.8	54.1	-90.58	276.6	1,860.1	1,243.9	1,133.4	110.56	11.251	
12,900.0	10,822.7	12,467.2	10,811.8	64.2	56.6	-90.59	273.8	1,948.6	1,241.3	1,125.7	115.65	10.733	
13,000.0	10,823.3	12,552.1	10,812.4	66.6	58.9	-90.61	271.7	2,033.5	1,239.4	1,118.7	120.68	10.270	
13,085.8	10,823.7	12,623.6	10,812.9	68.8	60.9	-90.61	271.0	2,105.0	1,239.0	1,114.0	124.99	9.913	
13,100.0	10,823.8	12,635.8	10,812.9	69.2	61.3	-90.61	270.9	2,117.2	1,239.0	1,113.3	125.71	9.856	
13,200.0	10,824.3	12,723.3	10,812.6	71.7	63.7	-90.57	271.1	2,204.7	1,239.7	1,108.8	130.87	9.473	
13,300.0	10,824.8	12,822.0	10,812.3	74.3	66.5	-90.53	272.0	2,303.4	1,241.1	1,104.7	136.37	9.101	
13,400.0	10,825.4	12,927.9	10,812.0	76.8	69.5	-90.49	272.5	2,409.3	1,242.1	1,100.0	142.11	8.740	
13,500.0	10,825.9	13,019.0	10,811.5	79.5	72.0	-90.45	272.9	2,500.4	1,243.1	1,095.7	147.45	8.431	
13,600.0	10,826.4	13,130.7	10,810.8	82.1	75.2	-90.39	273.9	2,612.1	1,244.5	1,091.1	153.39	8.114	
13,700.0	10,826.9	13,241.4	10,811.7	84.7	77.4	-90.40	273.4	2,722.8	1,244.6	1,086.2	158.40	7.857	
13,781.3	10,827.4	13,319.2	10,813.4	86.9	78.7	-90.46	272.6	2,800.5	1,244.2	1,082.3	161.99	7.681	
13,800.0	10,827.5	13,333.0	10,813.7	87.4	78.9	-90.47	272.6	2,814.4	1,244.3	1,081.6	162.65	7.650	
13,900.0	10,828.0	13,384.2	10,815.9	90.1	79.4	-90.56	273.5	2,865.5	1,246.7	1,080.7	165.99	7.511	
14,000.0	10,828.5	13,489.1	10,818.9	92.8	80.6	-90.67	278.2	2,970.3	1,251.8	1,081.8	169.97	7.365	
14,100.0	10,829.0	13,606.1	10,818.5	95.5	82.0	-90.62	280.8	3,087.2	1,254.4	1,080.2	174.22	7.200	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1

Offset Site Error: 0.0 usft

Survey Program: 2261-MWD, 13302-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
14,200.0	10,829.6	13,707.4	10,817.9	98.3	83.3	-90.57	282.9	3,188.4	1,257.0	1,078.6	178.37	7.047	
14,300.0	10,830.1	13,841.2	10,815.8	101.0	85.2	-90.44	284.4	3,322.2	1,258.8	1,075.8	183.09	6.876	
14,400.0	10,830.6	13,962.2	10,813.8	103.8	87.0	-90.32	282.6	3,443.2	1,257.9	1,070.1	187.75	6.700	
14,500.0	10,831.1	14,060.1	10,814.2	106.5	88.6	-90.32	280.6	3,541.0	1,256.4	1,064.2	192.14	6.539	
14,600.0	10,831.6	14,169.7	10,813.0	109.3	90.4	-90.24	277.4	3,650.6	1,253.9	1,057.1	196.80	6.371	
14,700.0	10,832.2	14,253.4	10,811.8	112.1	91.8	-90.16	275.7	3,734.2	1,252.4	1,051.3	201.09	6.228	
14,800.0	10,832.7	14,365.0	10,811.4	114.9	93.8	-90.12	273.1	3,845.8	1,250.5	1,044.6	205.93	6.073	
14,900.0	10,833.2	14,461.9	10,812.0	117.6	95.6	-90.12	270.6	3,942.7	1,248.4	1,037.9	210.56	5.929	
15,000.0	10,833.7	14,549.0	10,813.5	120.4	97.2	-90.17	269.1	4,029.8	1,247.2	1,032.1	215.06	5.799	
15,100.0	10,834.3	14,650.3	10,814.7	123.2	99.2	-90.20	267.5	4,131.1	1,246.2	1,026.3	219.88	5.668	
15,200.0	10,834.8	14,742.7	10,814.1	126.1	101.0	-90.15	266.7	4,223.4	1,245.8	1,021.2	224.58	5.547	
15,248.3	10,835.0	14,786.8	10,813.3	127.4	101.9	-90.10	266.3	4,267.6	1,245.7	1,018.9	226.86	5.491	
15,300.0	10,835.3	14,830.5	10,812.5	128.9	102.8	-90.06	266.2	4,311.2	1,245.9	1,016.6	229.23	5.435	
15,400.0	10,835.8	14,919.7	10,813.2	131.7	104.6	-90.07	266.3	4,400.5	1,246.5	1,012.6	233.93	5.328	
15,500.0	10,836.4	15,003.6	10,815.7	134.5	106.4	-90.16	267.8	4,484.2	1,248.8	1,010.2	238.56	5.235	
15,600.0	10,836.9	15,128.6	10,818.3	137.4	109.0	-90.25	269.3	4,609.2	1,250.5	1,006.4	244.10	5.123	
15,700.0	10,837.4	15,230.9	10,817.6	140.2	111.2	-90.19	269.1	4,711.5	1,250.8	1,001.6	249.20	5.019	
15,800.0	10,837.9	15,310.8	10,817.0	143.0	113.0	-90.15	269.7	4,791.4	1,252.2	998.3	253.85	4.933	
15,900.0	10,838.5	15,383.9	10,816.7	145.9	114.6	-90.12	271.9	4,864.5	1,255.7	997.3	258.36	4.860	
16,000.0	10,839.0	15,551.7	10,816.6	148.7	118.4	-90.07	274.0	5,032.3	1,257.4	992.4	265.04	4.744	
16,100.0	10,839.5	15,659.4	10,817.8	151.6	120.9	-90.10	271.9	5,139.9	1,256.0	985.6	270.41	4.645	
16,200.0	10,840.0	15,755.7	10,817.8	154.4	123.1	-90.08	270.0	5,236.2	1,254.6	979.1	275.55	4.553	
16,300.0	10,840.5	15,851.9	10,818.9	157.3	125.4	-90.11	268.1	5,332.4	1,253.2	972.5	280.71	4.464	
16,400.0	10,841.1	15,948.1	10,820.9	160.1	127.7	-90.18	266.7	5,428.5	1,252.3	966.4	285.89	4.380	
16,500.0	10,841.6	16,097.3	10,825.4	163.0	131.3	-90.34	263.6	5,577.6	1,251.3	958.9	292.37	4.280	
16,600.0	10,842.1	16,218.3	10,823.0	165.9	134.2	-90.21	256.2	5,698.3	1,245.6	947.4	298.22	4.177	
16,700.0	10,842.6	16,309.2	10,819.5	168.7	136.5	-90.02	250.0	5,788.9	1,239.4	936.1	303.35	4.086	
16,800.0	10,843.2	16,400.1	10,819.8	171.6	138.7	-90.01	244.7	5,879.6	1,234.1	925.6	308.50	4.000	
16,900.0	10,843.7	16,498.6	10,823.7	174.5	141.2	-90.17	239.6	5,978.0	1,229.5	915.6	313.85	3.917	
17,000.0	10,844.2	16,582.4	10,826.2	177.3	143.3	-90.27	235.8	6,061.6	1,225.4	906.6	318.84	3.843	
17,100.0	10,844.7	16,714.5	10,828.6	180.2	146.6	-90.35	229.0	6,193.5	1,220.9	895.8	325.08	3.756	
17,200.0	10,845.3	16,814.5	10,826.5	183.1	149.1	-90.23	222.5	6,293.3	1,214.9	884.4	330.52	3.676	
17,300.0	10,845.8	16,923.5	10,825.7	186.0	151.9	-90.17	214.5	6,402.0	1,208.1	871.9	336.21	3.593	
17,400.0	10,846.3	17,012.3	10,826.8	188.8	154.2	-90.19	208.1	6,490.6	1,201.4	860.0	341.40	3.519	
17,500.0	10,846.8	17,101.3	10,827.0	191.7	156.5	-90.19	202.9	6,579.4	1,196.0	849.4	346.60	3.451	
17,600.0	10,847.4	17,187.7	10,827.1	194.6	158.7	-90.17	198.4	6,665.7	1,191.4	839.7	351.75	3.387	
17,700.0	10,847.9	17,279.4	10,829.0	197.5	161.1	-90.24	194.8	6,757.3	1,188.0	831.0	357.04	3.327	
17,800.0	10,848.4	17,367.8	10,829.7	200.4	163.4	-90.25	191.7	6,845.6	1,185.1	822.8	362.25	3.271	
17,900.0	10,848.9	17,475.1	10,831.0	203.3	166.2	-90.28	188.5	6,952.9	1,182.6	814.6	367.98	3.214	
18,000.0	10,849.5	17,560.0	10,831.4	206.2	168.5	-90.28	186.0	7,037.7	1,180.2	807.1	373.12	3.163	
18,100.0	10,850.0	17,656.2	10,832.1	209.0	171.0	-90.30	184.3	7,133.9	1,178.9	800.4	378.57	3.114	
18,200.0	10,850.5	17,754.0	10,831.1	211.9	173.6	-90.22	182.4	7,231.7	1,177.5	793.5	384.07	3.066	
18,300.0	10,851.0	17,850.2	10,831.3	214.8	176.2	-90.21	181.3	7,327.9	1,176.9	787.3	389.54	3.021	
18,400.0	10,851.5	17,945.9	10,830.3	217.7	178.7	-90.13	180.1	7,423.5	1,176.2	781.2	395.00	2.978	
18,431.7	10,851.7	17,972.8	10,829.9	218.6	179.4	-90.11	179.9	7,450.5	1,176.2	779.5	396.65	2.965	
18,500.0	10,852.1	18,032.9	10,829.2	220.6	181.0	-90.06	179.9	7,510.5	1,176.4	776.2	400.24	2.939	
18,600.0	10,852.6	18,132.7	10,828.7	223.5	183.7	-90.01	179.9	7,610.4	1,177.0	771.2	405.83	2.900	
18,700.0	10,853.1	18,213.4	10,827.4	226.4	185.9	-89.92	180.7	7,691.1	1,178.5	767.6	410.91	2.868	
18,800.0	10,853.6	18,305.6	10,827.8	229.3	188.3	-89.92	182.9	7,783.2	1,181.4	765.1	416.29	2.838	
18,900.0	10,854.2	18,387.4	10,827.7	232.2	190.5	-89.90	185.4	7,865.0	1,185.1	763.7	421.41	2.812	
19,000.0	10,854.7	18,471.3	10,827.4	235.1	192.8	-89.86	189.6	7,948.8	1,190.5	764.0	426.58	2.791	
19,100.0	10,855.2	18,616.5	10,829.6	238.0	196.7	-89.93	195.4	8,093.8	1,195.3	761.8	433.41	2.758	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Ash Federal 5300 11-18T - Wellbore #1 - Wellbore #1

Survey Program: 2261-MWD, 13302-MWD

Offset Site Error: 0.0 usft
Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
19,200.0	10,855.7	18,724.8	10,831.7	240.9	199.7	-90.00	196.8	8,202.1	1,197.2	757.9	439.26	2.725	
19,300.0	10,856.3	18,822.6	10,834.2	243.8	202.3	-90.10	197.7	8,299.9	1,198.6	753.7	444.84	2.694	
19,400.0	10,856.8	18,915.0	10,836.8	246.7	204.8	-90.20	199.0	8,392.2	1,200.5	750.2	450.27	2.666	
19,500.0	10,857.3	19,022.6	10,840.5	249.6	207.8	-90.35	200.8	8,499.7	1,202.8	746.6	456.11	2.637	
19,600.0	10,857.8	19,138.6	10,844.2	252.5	210.9	-90.49	201.4	8,615.6	1,203.8	741.6	462.19	2.605	
19,700.0	10,858.4	19,239.6	10,846.0	255.4	213.7	-90.56	201.0	8,716.6	1,203.9	736.1	467.88	2.573	
19,800.0	10,858.9	19,338.8	10,846.2	258.3	216.4	-90.54	200.6	8,815.9	1,204.1	730.6	473.52	2.543	
19,900.0	10,859.4	19,434.9	10,845.6	261.2	219.1	-90.49	200.6	8,912.0	1,204.6	725.5	479.09	2.514	
20,000.0	10,859.9	19,537.4	10,844.1	264.1	221.9	-90.39	200.5	9,014.4	1,205.0	720.2	484.85	2.485	
20,100.0	10,860.4	19,626.7	10,842.1	267.0	224.4	-90.27	200.7	9,103.8	1,205.8	715.5	490.24	2.460	
20,200.0	10,861.0	19,706.0	10,841.1	269.9	226.6	-90.21	202.0	9,183.0	1,207.9	712.6	495.35	2.439	
20,300.0	10,861.5	19,780.4	10,840.4	272.8	228.6	-90.15	204.8	9,257.3	1,212.1	711.8	500.33	2.423	
20,400.0	10,862.0	19,855.7	10,840.4	275.7	230.7	-90.13	209.0	9,332.5	1,218.3	713.0	505.32	2.411	
20,500.0	10,862.5	19,951.5	10,840.7	278.6	233.3	-90.12	215.6	9,428.1	1,225.7	714.8	510.88	2.399	
20,600.0	10,863.1	20,057.4	10,840.9	281.6	236.3	-90.11	222.7	9,533.7	1,232.9	716.2	516.71	2.386	
20,700.0	10,863.6	20,159.3	10,841.1	284.5	239.1	-90.09	228.9	9,635.5	1,239.5	717.1	522.45	2.373	
20,800.0	10,864.1	20,255.4	10,840.9	287.4	241.7	-90.06	234.8	9,731.4	1,246.2	718.2	528.03	2.360	
20,900.0	10,864.6	20,363.6	10,839.9	290.3	244.7	-89.98	241.6	9,839.4	1,253.0	719.1	533.94	2.347	
21,000.0	10,865.2	20,475.4	10,837.7	293.2	247.8	-89.86	247.4	9,951.0	1,258.7	718.7	539.97	2.331	
21,038.5	10,865.4	20,482.0	10,837.6	294.3	248.0	-89.85	247.7	9,957.6	1,261.2	719.9	541.27	2.330 SF	

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6 **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
0.0	0.0	0.0	0.0	0.0	0.0	24.43	59.8	27.2	65.7				
100.0	100.0	100.0	100.0	0.1	0.1	24.43	59.8	27.2	65.7	65.5	0.17	389.542	
200.0	200.0	200.0	200.0	0.3	0.3	24.43	59.8	27.2	65.7	65.0	0.62	106.239	
300.0	300.0	300.0	300.0	0.5	0.5	24.43	59.8	27.2	65.7	64.6	1.07	61.507	
400.0	400.0	400.0	400.0	0.8	0.8	24.43	59.8	27.2	65.7	64.1	1.52	43.283	
500.0	500.0	500.0	500.0	1.0	1.0	24.43	59.8	27.2	65.7	63.7	1.97	33.389	
600.0	600.0	600.0	600.0	1.2	1.2	24.43	59.8	27.2	65.7	63.3	2.42	27.177	
700.0	700.0	700.0	700.0	1.4	1.4	24.43	59.8	27.2	65.7	62.8	2.87	22.914	
800.0	800.0	800.0	800.0	1.7	1.7	24.43	59.8	27.2	65.7	62.4	3.32	19.807	
900.0	900.0	900.0	900.0	1.9	1.9	24.43	59.8	27.2	65.7	61.9	3.76	17.442	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	24.43	59.8	27.2	65.7	61.5	4.21	15.582	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	24.43	59.8	27.2	65.7	61.0	4.66	14.080	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	24.43	59.8	27.2	65.7	60.6	5.11	12.842	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	24.43	59.8	27.2	65.7	60.1	5.56	11.804	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	24.43	59.8	27.2	65.7	59.7	6.01	10.922	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	24.43	59.8	27.2	65.7	59.2	6.46	10.162	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	24.43	59.8	27.2	65.7	58.8	6.91	9.501	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	24.43	59.8	27.2	65.7	58.3	7.36	8.921	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	24.43	59.8	27.2	65.7	57.9	7.81	8.407	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	24.43	59.8	27.2	65.7	57.4	8.26	7.950	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	24.43	59.8	27.2	65.7	57.0	8.71	7.540	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	24.43	59.8	27.2	65.7	56.5	9.16	7.170	
2,200.0	2,200.0	2,200.0	2,200.0	4.8	4.8	24.43	59.8	27.2	65.7	56.1	9.61	6.834	
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.0	24.43	59.8	27.2	65.7	55.6	10.06	6.529	
2,400.0	2,400.0	2,400.0	2,400.0	5.3	5.3	24.43	59.8	27.2	65.7	55.2	10.51	6.249	
2,500.0	2,500.0	2,500.0	2,500.0	5.5	5.5	24.43	59.8	27.2	65.7	54.7	10.96	5.993	CC
2,600.0	2,600.0	2,598.9	2,598.9	5.7	5.7	117.07	59.8	28.9	67.2	55.8	11.38	5.905	
2,650.0	2,649.9	2,648.2	2,648.1	5.8	5.8	120.18	59.8	31.0	69.3	57.7	11.57	5.984	
2,700.0	2,699.9	2,697.9	2,697.8	5.9	5.9	123.76	59.8	33.6	72.0	60.3	11.78	6.118	
2,800.0	2,799.7	2,797.3	2,797.1	6.1	6.1	130.11	59.8	38.8	78.4	66.2	12.18	6.434	
2,900.0	2,899.6	2,896.8	2,896.4	6.3	6.3	135.46	59.8	44.0	85.5	72.9	12.58	6.793	
3,000.0	2,999.5	2,996.3	2,995.7	6.5	6.5	139.96	59.8	49.2	93.2	80.2	12.99	7.177	
3,100.0	3,099.3	3,095.7	3,095.0	6.7	6.7	143.75	59.8	54.4	101.5	88.1	13.40	7.572	
3,200.0	3,199.2	3,195.2	3,194.3	7.0	6.9	146.97	59.8	59.6	110.1	96.3	13.81	7.970	
3,300.0	3,299.0	3,294.6	3,293.7	7.2	7.2	149.71	59.8	64.8	119.0	104.8	14.23	8.365	
3,400.0	3,398.9	3,394.1	3,393.0	7.4	7.4	152.06	59.8	70.0	128.2	113.5	14.64	8.752	
3,501.2	3,500.0	3,494.7	3,493.5	7.6	7.6	154.13	59.8	75.3	137.6	122.5	15.07	9.134	
3,600.0	3,598.7	3,597.5	3,596.2	7.8	7.8	155.46	59.8	79.1	144.0	128.5	15.48	9.302	
3,651.2	3,649.9	3,651.2	3,649.9	7.9	7.9	65.63	59.8	79.6	144.9	129.2	15.67	9.247	
3,700.0	3,698.7	3,700.0	3,698.7	8.0	8.0	65.63	59.8	79.6	144.9	129.0	15.87	9.129	
3,800.0	3,798.7	3,800.0	3,798.7	8.2	8.2	65.63	59.8	79.6	144.9	128.6	16.31	8.880	
3,900.0	3,898.7	3,900.0	3,898.7	8.5	8.5	65.63	59.8	79.6	144.9	128.1	16.76	8.645	
4,000.0	3,998.7	4,000.0	3,998.7	8.7	8.7	65.63	59.8	79.6	144.9	127.7	17.20	8.421	
4,100.0	4,098.7	4,100.0	4,098.7	8.9	8.9	65.63	59.8	79.6	144.9	127.2	17.65	8.209	
4,200.0	4,198.7	4,200.0	4,198.7	9.1	9.1	65.63	59.8	79.6	144.9	126.8	18.09	8.007	
4,300.0	4,298.7	4,300.0	4,298.7	9.4	9.4	65.63	59.8	79.6	144.9	126.3	18.54	7.814	
4,400.0	4,398.7	4,400.0	4,398.7	9.6	9.6	65.63	59.8	79.6	144.9	125.9	18.99	7.631	
4,500.0	4,498.7	4,500.0	4,498.7	9.8	9.8	65.63	59.8	79.6	144.9	125.4	19.43	7.456	
4,600.0	4,598.7	4,600.0	4,598.7	10.0	10.0	65.63	59.8	79.6	144.9	125.0	19.88	7.289	
4,700.0	4,698.7	4,700.0	4,698.7	10.2	10.2	65.63	59.8	79.6	144.9	124.6	20.32	7.129	
4,800.0	4,798.7	4,800.0	4,798.7	10.5	10.5	65.63	59.8	79.6	144.9	124.1	20.77	6.975	
4,900.0	4,898.7	4,900.0	4,898.7	10.7	10.7	65.63	59.8	79.6	144.9	123.7	21.22	6.829	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
5,000.0	4,998.7	5,000.0	4,998.7	10.9	10.9	65.63	59.8	79.6	144.9	123.2	21.66	6.688	
5,100.0	5,098.7	5,100.0	5,098.7	11.1	11.1	65.63	59.8	79.6	144.9	122.8	22.11	6.553	
5,200.0	5,198.7	5,200.0	5,198.7	11.4	11.4	65.63	59.8	79.6	144.9	122.3	22.56	6.423	
5,300.0	5,298.7	5,300.0	5,298.7	11.6	11.6	65.63	59.8	79.6	144.9	121.9	23.00	6.298	
5,400.0	5,398.7	5,400.0	5,398.7	11.8	11.8	65.63	59.8	79.6	144.9	121.4	23.45	6.178	
5,500.0	5,498.7	5,500.0	5,498.7	12.0	12.0	65.63	59.8	79.6	144.9	121.0	23.90	6.062	
5,600.0	5,598.7	5,600.0	5,598.7	12.2	12.2	65.63	59.8	79.6	144.9	120.5	24.34	5.951	
5,700.0	5,698.7	5,700.0	5,698.7	12.5	12.5	65.63	59.8	79.6	144.9	120.1	24.79	5.844	
5,800.0	5,798.7	5,800.0	5,798.7	12.7	12.7	65.63	59.8	79.6	144.9	119.6	25.24	5.740	
5,900.0	5,898.7	5,900.0	5,898.7	12.9	12.9	65.63	59.8	79.6	144.9	119.2	25.69	5.640	
6,000.0	5,998.7	6,000.0	5,998.7	13.1	13.1	65.63	59.8	79.6	144.9	118.7	26.13	5.543	
6,100.0	6,098.7	6,100.0	6,098.7	13.4	13.4	65.63	59.8	79.6	144.9	118.3	26.58	5.450	
6,200.0	6,198.7	6,200.0	6,198.7	13.6	13.6	65.63	59.8	79.6	144.9	117.8	27.03	5.360	
6,300.0	6,298.7	6,300.0	6,298.7	13.8	13.8	65.63	59.8	79.6	144.9	117.4	27.48	5.273	
6,400.0	6,398.7	6,400.0	6,398.7	14.0	14.0	65.63	59.8	79.6	144.9	117.0	27.93	5.188	
6,500.0	6,498.7	6,500.0	6,498.7	14.3	14.3	65.63	59.8	79.6	144.9	116.5	28.37	5.106	
6,600.0	6,598.7	6,600.0	6,598.7	14.5	14.5	65.63	59.8	79.6	144.9	116.1	28.82	5.027	
6,700.0	6,698.7	6,700.0	6,698.7	14.7	14.7	65.63	59.8	79.6	144.9	115.6	29.27	4.950	
6,800.0	6,798.7	6,800.0	6,798.7	14.9	14.9	65.63	59.8	79.6	144.9	115.2	29.72	4.875	
6,900.0	6,898.7	6,900.0	6,898.7	15.2	15.2	65.63	59.8	79.6	144.9	114.7	30.16	4.803	
7,000.0	6,998.7	7,000.0	6,998.7	15.4	15.4	65.63	59.8	79.6	144.9	114.3	30.61	4.733	
7,100.0	7,098.7	7,100.0	7,098.7	15.6	15.6	65.63	59.8	79.6	144.9	113.8	31.06	4.664	
7,200.0	7,198.7	7,200.0	7,198.7	15.8	15.8	65.63	59.8	79.6	144.9	113.4	31.51	4.598	
7,300.0	7,298.7	7,300.0	7,298.7	16.0	16.0	65.63	59.8	79.6	144.9	112.9	31.96	4.533	
7,400.0	7,398.7	7,400.0	7,398.7	16.3	16.3	65.63	59.8	79.6	144.9	112.5	32.41	4.471	
7,500.0	7,498.7	7,500.0	7,498.7	16.5	16.5	65.63	59.8	79.6	144.9	112.0	32.85	4.410	
7,600.0	7,598.7	7,600.0	7,598.7	16.7	16.7	65.63	59.8	79.6	144.9	111.6	33.30	4.350	
7,700.0	7,698.7	7,700.0	7,698.7	16.9	16.9	65.63	59.8	79.6	144.9	111.1	33.75	4.293	
7,800.0	7,798.7	7,800.0	7,798.7	17.2	17.2	65.63	59.8	79.6	144.9	110.7	34.20	4.236	
7,900.0	7,898.7	7,900.0	7,898.7	17.4	17.4	65.63	59.8	79.6	144.9	110.2	34.65	4.181	
8,000.0	7,998.7	8,000.0	7,998.7	17.6	17.6	65.63	59.8	79.6	144.9	109.8	35.10	4.128	
8,100.0	8,098.7	8,100.0	8,098.7	17.8	17.8	65.63	59.8	79.6	144.9	109.3	35.54	4.076	
8,200.0	8,198.7	8,200.0	8,198.7	18.1	18.1	65.63	59.8	79.6	144.9	108.9	35.99	4.025	
8,300.0	8,298.7	8,300.0	8,298.7	18.3	18.3	65.63	59.8	79.6	144.9	108.4	36.44	3.976	
8,400.0	8,398.7	8,400.0	8,398.7	18.5	18.5	65.63	59.8	79.6	144.9	108.0	36.89	3.927	
8,500.0	8,498.7	8,500.0	8,498.7	18.7	18.7	65.63	59.8	79.6	144.9	107.5	37.34	3.880	
8,600.0	8,598.7	8,600.0	8,598.7	19.0	19.0	65.63	59.8	79.6	144.9	107.1	37.79	3.834	
8,700.0	8,698.7	8,700.0	8,698.7	19.2	19.2	65.63	59.8	79.6	144.9	106.6	38.24	3.789	
8,800.0	8,798.7	8,800.0	8,798.7	19.4	19.4	65.63	59.8	79.6	144.9	106.2	38.68	3.745	
8,900.0	8,898.7	8,900.0	8,898.7	19.6	19.6	65.63	59.8	79.6	144.9	105.7	39.13	3.702	
9,000.0	8,998.7	9,000.0	8,998.7	19.9	19.9	65.63	59.8	79.6	144.9	105.3	39.58	3.660	
9,100.0	9,098.7	9,100.0	9,098.7	20.1	20.1	65.63	59.8	79.6	144.9	104.8	40.03	3.619	
9,200.0	9,198.7	9,200.0	9,198.7	20.3	20.3	65.63	59.8	79.6	144.9	104.4	40.48	3.579	
9,300.0	9,298.7	9,300.0	9,298.7	20.5	20.5	65.63	59.8	79.6	144.9	103.9	40.93	3.540	
9,400.0	9,398.7	9,400.0	9,398.7	20.8	20.8	65.63	59.8	79.6	144.9	103.5	41.38	3.501	
9,500.0	9,498.7	9,500.0	9,498.7	21.0	21.0	65.63	59.8	79.6	144.9	103.1	41.83	3.464	
9,600.0	9,598.7	9,600.0	9,598.7	21.2	21.2	65.63	59.8	79.6	144.9	102.6	42.27	3.427	
9,700.0	9,698.7	9,700.0	9,698.7	21.4	21.4	65.63	59.8	79.6	144.9	102.2	42.72	3.391	
9,800.0	9,798.7	9,800.0	9,798.7	21.6	21.6	65.63	59.8	79.6	144.9	101.7	43.17	3.356	
9,900.0	9,898.7	9,900.0	9,898.7	21.9	21.9	65.63	59.8	79.6	144.9	101.3	43.62	3.321	
10,000.0	9,998.7	10,000.0	9,998.7	22.1	22.1	65.63	59.8	79.6	144.9	100.8	44.07	3.287	
10,100.0	10,098.7	10,100.0	10,098.7	22.3	22.3	65.63	59.8	79.6	144.9	100.4	44.52	3.254	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6 **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
10,200.0	10,198.7	10,200.0	10,198.7	22.5	22.5	65.63	59.8	79.6	144.9	99.9	44.97	3.222	
10,300.0	10,298.7	10,300.0	10,298.7	22.8	22.8	65.63	59.8	79.6	144.9	99.5	45.42	3.190	
10,336.6	10,335.3	10,336.6	10,335.3	22.9	22.9	65.63	59.8	79.6	144.9	99.3	45.58	3.179	
10,350.0	10,348.7	10,350.0	10,348.7	22.9	22.9	-71.45	59.8	79.6	144.8	99.2	45.65	3.172	
10,375.0	10,373.7	10,373.2	10,371.9	22.9	22.9	-71.92	59.7	79.7	144.5	98.7	45.74	3.158	
10,400.0	10,398.5	10,394.5	10,393.2	23.0	23.0	-72.50	59.3	80.4	144.3	98.5	45.81	3.149	
10,406.4	10,404.8	10,400.0	10,398.7	23.0	23.0	-72.67	59.1	80.8	144.3	98.4	45.82	3.148	
10,425.0	10,423.2	10,415.8	10,414.4	23.0	23.0	-73.17	58.3	82.1	144.3	98.4	45.87	3.146	
10,450.0	10,447.6	10,437.2	10,435.6	23.1	23.1	-73.91	56.9	84.5	144.6	98.7	45.93	3.149	
10,475.0	10,471.8	10,458.6	10,456.6	23.1	23.1	-74.72	55.1	87.8	145.3	99.3	46.00	3.158	
10,500.0	10,495.5	10,480.0	10,477.5	23.1	23.2	-75.59	52.7	91.9	146.2	100.1	46.06	3.173	
10,525.0	10,518.8	10,500.0	10,496.8	23.2	23.2	-76.45	50.1	96.4	147.4	101.2	46.13	3.195	
10,550.0	10,541.7	10,523.0	10,518.7	23.2	23.3	-77.49	46.6	102.5	148.9	102.7	46.21	3.222	
10,575.0	10,563.9	10,544.5	10,538.9	23.3	23.3	-78.50	42.8	109.0	150.7	104.4	46.31	3.255	
10,600.0	10,585.5	10,566.2	10,558.8	23.3	23.4	-79.53	38.6	116.3	152.9	106.5	46.41	3.295	
10,625.0	10,606.5	10,587.8	10,578.3	23.4	23.4	-80.57	33.9	124.4	155.4	108.9	46.53	3.341	
10,650.0	10,626.7	10,609.5	10,597.4	23.4	23.5	-81.61	28.7	133.3	158.3	111.6	46.66	3.393	
10,675.0	10,646.1	10,631.3	10,616.1	23.5	23.6	-82.64	23.2	143.0	161.5	114.7	46.80	3.451	
10,700.0	10,664.6	10,653.1	10,634.3	23.6	23.6	-83.65	17.2	153.4	165.1	118.1	46.96	3.515	
10,725.0	10,682.3	10,675.0	10,652.0	23.6	23.7	-84.64	10.7	164.6	169.0	121.8	47.13	3.585	
10,750.0	10,698.9	10,696.9	10,669.1	23.7	23.8	-85.58	3.9	176.4	173.2	125.9	47.32	3.661	
10,775.0	10,714.6	10,718.9	10,685.6	23.8	23.9	-86.49	-3.4	189.0	177.8	130.3	47.52	3.742	
10,800.0	10,729.3	10,741.0	10,701.5	23.9	24.0	-87.35	-11.1	202.3	182.7	135.0	47.74	3.827	
10,825.0	10,742.9	10,763.2	10,716.8	24.0	24.1	-88.16	-19.2	216.3	188.0	140.0	47.98	3.918	
10,850.0	10,755.3	10,785.5	10,731.3	24.2	24.2	-88.91	-27.6	230.9	193.5	145.3	48.23	4.012	
10,875.0	10,766.6	10,807.9	10,745.1	24.3	24.3	-89.61	-36.4	246.2	199.3	150.8	48.50	4.109	
10,900.0	10,776.8	10,830.4	10,758.1	24.5	24.5	-90.26	-45.6	262.2	205.4	156.6	48.80	4.209	
10,925.0	10,785.7	10,853.1	10,770.3	24.7	24.6	-90.84	-55.2	278.7	211.8	162.7	49.11	4.312	
10,950.0	10,793.4	10,875.9	10,781.6	24.9	24.8	-91.37	-65.1	295.8	218.4	168.9	49.45	4.416	
10,975.0	10,799.8	10,898.9	10,792.1	25.1	25.0	-91.84	-75.3	313.5	225.2	175.4	49.81	4.521	
11,000.0	10,804.9	10,922.0	10,801.6	25.3	25.2	-92.26	-85.8	331.8	232.2	182.0	50.20	4.626	
11,025.0	10,808.8	10,945.3	10,810.2	25.5	25.4	-92.62	-96.7	350.6	239.4	188.8	50.61	4.730	
11,050.0	10,811.4	10,968.9	10,817.8	25.8	25.6	-92.94	-107.8	369.9	246.8	195.7	51.05	4.834	
11,075.0	10,812.6	10,992.7	10,824.3	26.0	25.8	-93.21	-119.3	389.7	254.3	202.7	51.51	4.936	
11,084.1	10,812.8	11,001.4	10,826.4	26.1	25.9	-93.29	-123.5	397.0	257.0	205.3	51.69	4.973	
11,099.1	10,812.8	11,015.9	10,829.6	26.3	26.1	-93.94	-130.6	409.3	261.6	209.6	51.96	5.034	
11,200.0	10,813.4	11,117.1	10,839.5	27.5	27.3	-95.42	-180.8	496.3	289.1	234.8	54.32	5.323	
11,300.0	10,813.9	11,200.0	10,839.9	28.9	28.5	-94.99	-221.0	568.8	312.4	255.4	56.98	5.483	
11,400.0	10,814.5	11,285.6	10,840.4	30.6	29.9	-94.62	-259.2	645.5	334.9	274.9	60.01	5.580	
11,500.0	10,815.0	11,368.3	10,840.8	32.4	31.3	-94.32	-292.8	721.0	356.5	293.2	63.25	5.636	
11,600.0	10,815.6	11,450.4	10,841.3	34.3	32.9	-94.06	-322.9	797.4	377.2	310.5	66.67	5.658	
11,700.0	10,816.1	11,531.9	10,841.7	36.3	34.5	-93.83	-349.5	874.5	397.0	326.7	70.21	5.654	
11,800.0	10,816.7	11,612.9	10,842.1	38.5	36.2	-93.63	-372.6	952.1	415.8	341.9	73.84	5.631	
11,900.0	10,817.3	11,700.0	10,842.6	40.7	38.1	-93.44	-393.8	1,036.5	433.7	356.0	77.64	5.585	
12,000.0	10,817.9	11,773.5	10,843.0	43.0	39.8	-93.30	-408.7	1,108.5	450.4	369.3	81.16	5.550	
12,100.0	10,818.4	11,853.2	10,843.4	45.3	41.6	-93.16	-421.6	1,187.1	466.2	381.4	84.78	5.499	
12,200.0	10,819.0	11,932.4	10,843.9	47.6	43.4	-93.04	-431.3	1,265.8	481.0	392.6	88.33	5.445	
12,300.0	10,819.5	12,011.4	10,844.3	50.0	45.3	-92.93	-437.6	1,344.4	494.6	402.8	91.77	5.390	
12,400.0	10,820.1	12,100.0	10,844.8	52.3	47.4	-92.83	-440.9	1,433.0	507.3	412.0	95.32	5.322	
12,500.0	10,820.6	12,184.0	10,845.2	54.7	49.4	-92.75	-441.6	1,517.0	517.5	418.9	98.62	5.247	
12,600.0	10,821.2	12,283.8	10,845.7	57.0	51.8	-92.70	-442.4	1,616.8	522.7	420.6	102.17	5.116	
12,655.6	10,821.5	12,339.5	10,846.0	58.3	53.2	-92.69	-442.8	1,672.5	523.4	419.3	104.07	5.029	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6

Offset Site Error: 0.0 usft

Survey Program: 0-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
12,700.0	10,821.7	12,383.8	10,846.2	59.4	54.3	-92.69	-443.2	1,716.8	523.3	416.9	106.36	4.920	
12,800.0	10,822.2	12,483.8	10,846.8	61.8	56.9	-92.69	-444.0	1,816.8	523.0	411.4	111.58	4.687	
12,900.0	10,822.7	12,583.8	10,847.3	64.2	59.4	-92.69	-444.8	1,916.8	522.7	405.9	116.86	4.473	
13,000.0	10,823.3	12,683.8	10,847.8	66.6	62.0	-92.69	-445.6	2,016.8	522.4	400.2	122.20	4.275	
13,100.0	10,823.8	12,783.8	10,848.3	69.2	64.7	-92.69	-446.4	2,116.8	522.2	394.6	127.58	4.093	
13,200.0	10,824.3	12,883.8	10,848.9	71.7	67.3	-92.69	-447.2	2,216.8	521.9	388.9	133.00	3.924	
13,300.0	10,824.8	12,983.8	10,849.4	74.3	70.0	-92.70	-448.0	2,316.8	521.6	383.2	138.45	3.768	
13,400.0	10,825.4	13,083.8	10,849.9	76.8	72.7	-92.70	-448.8	2,416.8	521.3	377.4	143.94	3.622	
13,500.0	10,825.9	13,183.8	10,850.4	79.5	75.4	-92.70	-449.6	2,516.8	521.1	371.6	149.45	3.486	
13,600.0	10,826.4	13,283.8	10,851.0	82.1	78.1	-92.70	-450.4	2,616.8	520.8	365.8	154.99	3.360	
13,700.0	10,826.9	13,383.8	10,851.5	84.7	80.9	-92.70	-451.2	2,716.8	520.5	360.0	160.56	3.242	
13,800.0	10,827.5	13,483.8	10,852.0	87.4	83.6	-92.70	-452.0	2,816.8	520.2	354.1	166.14	3.131	
13,900.0	10,828.0	13,583.8	10,852.5	90.1	86.4	-92.70	-452.8	2,916.8	520.0	348.2	171.74	3.028	
14,000.0	10,828.5	13,683.8	10,853.0	92.8	89.2	-92.71	-453.6	3,016.8	519.7	342.3	177.36	2.930	
14,100.0	10,829.0	13,783.8	10,853.6	95.5	92.0	-92.71	-454.5	3,116.7	519.4	336.4	182.99	2.839	
14,200.0	10,829.6	13,883.8	10,854.1	98.3	94.8	-92.71	-455.3	3,216.7	519.1	330.5	188.64	2.752	
14,300.0	10,830.1	13,983.8	10,854.6	101.0	97.6	-92.71	-456.1	3,316.7	518.9	324.6	194.30	2.671	
14,400.0	10,830.6	14,083.8	10,855.1	103.8	100.4	-92.71	-456.9	3,416.7	518.6	318.6	199.97	2.593	
14,500.0	10,831.1	14,183.8	10,855.7	106.5	103.2	-92.71	-457.7	3,516.7	518.3	312.7	205.65	2.520	
14,600.0	10,831.6	14,283.8	10,856.2	109.3	106.0	-92.71	-458.5	3,616.7	518.0	306.7	211.34	2.451	
14,700.0	10,832.2	14,383.8	10,856.7	112.1	108.9	-92.72	-459.3	3,716.7	517.8	300.7	217.03	2.386	
14,800.0	10,832.7	14,483.8	10,857.2	114.9	111.7	-92.72	-460.1	3,816.7	517.5	294.8	222.74	2.323	
14,900.0	10,833.2	14,583.8	10,857.8	117.6	114.5	-92.72	-460.9	3,916.7	517.2	288.8	228.46	2.264	
15,000.0	10,833.7	14,683.8	10,858.3	120.4	117.4	-92.72	-461.7	4,016.7	517.0	282.8	234.18	2.208	
15,100.0	10,834.3	14,783.8	10,858.8	123.2	120.2	-92.72	-462.5	4,116.7	516.7	276.8	239.91	2.154	
15,200.0	10,834.8	14,883.8	10,859.3	126.1	123.1	-92.72	-463.3	4,216.7	516.4	270.8	245.64	2.102	
15,300.0	10,835.3	14,983.8	10,859.9	128.9	126.0	-92.72	-464.1	4,316.7	516.1	264.7	251.38	2.053	
15,400.0	10,835.8	15,083.8	10,860.4	131.7	128.8	-92.73	-464.9	4,416.7	515.9	258.7	257.12	2.006	
15,500.0	10,836.4	15,183.8	10,860.9	134.5	131.7	-92.73	-465.7	4,516.7	515.6	252.7	262.87	1.961	
15,600.0	10,836.9	15,283.8	10,861.4	137.4	134.5	-92.73	-466.5	4,616.7	515.3	246.7	268.63	1.918	
15,700.0	10,837.4	15,383.8	10,861.9	140.2	137.4	-92.73	-467.3	4,716.7	515.0	240.6	274.39	1.877	
15,800.0	10,837.9	15,483.8	10,862.5	143.0	140.3	-92.73	-468.1	4,816.7	514.8	234.6	280.15	1.837	
15,900.0	10,838.5	15,583.8	10,863.0	145.9	143.2	-92.73	-468.9	4,916.7	514.5	228.6	285.92	1.799	
16,000.0	10,839.0	15,683.8	10,863.5	148.7	146.0	-92.73	-469.7	5,016.7	514.2	222.5	291.69	1.763	
16,100.0	10,839.5	15,783.8	10,864.0	151.6	148.9	-92.74	-470.5	5,116.6	513.9	216.5	297.46	1.728	
16,200.0	10,840.0	15,883.8	10,864.6	154.4	151.8	-92.74	-471.3	5,216.6	513.7	210.4	303.24	1.694	
16,300.0	10,840.5	15,983.8	10,865.1	157.3	154.7	-92.74	-472.1	5,316.6	513.4	204.4	309.02	1.661	
16,400.0	10,841.1	16,083.8	10,865.6	160.1	157.6	-92.74	-472.9	5,416.6	513.1	198.3	314.80	1.630	
16,500.0	10,841.6	16,183.8	10,866.1	163.0	160.4	-92.74	-473.7	5,516.6	512.8	192.3	320.58	1.600	
16,600.0	10,842.1	16,283.8	10,866.7	165.9	163.3	-92.74	-474.6	5,616.6	512.6	186.2	326.37	1.570	
16,700.0	10,842.6	16,383.8	10,867.2	168.7	166.2	-92.74	-475.4	5,716.6	512.3	180.1	332.16	1.542	
16,800.0	10,843.2	16,483.8	10,867.7	171.6	169.1	-92.75	-476.2	5,816.6	512.0	174.1	337.95	1.515	
16,900.0	10,843.7	16,583.8	10,868.2	174.5	172.0	-92.75	-477.0	5,916.6	511.7	168.0	343.75	1.489	Level 3
17,000.0	10,844.2	16,683.8	10,868.8	177.3	174.9	-92.75	-477.8	6,016.6	511.5	161.9	349.54	1.463	Level 3
17,100.0	10,844.7	16,783.8	10,869.3	180.2	177.8	-92.75	-478.6	6,116.6	511.2	155.8	355.34	1.439	Level 3
17,200.0	10,845.3	16,883.8	10,869.8	183.1	180.7	-92.75	-479.4	6,216.6	510.9	149.8	361.14	1.415	Level 3
17,300.0	10,845.8	16,983.8	10,870.3	186.0	183.6	-92.75	-480.2	6,316.6	510.6	143.7	366.94	1.392	Level 3
17,400.0	10,846.3	17,083.8	10,870.8	188.8	186.5	-92.76	-481.0	6,416.6	510.4	137.6	372.75	1.369	Level 3
17,500.0	10,846.8	17,183.8	10,871.4	191.7	189.4	-92.76	-481.8	6,516.6	510.1	131.5	378.55	1.347	Level 3
17,600.0	10,847.4	17,283.8	10,871.9	194.6	192.3	-92.76	-482.6	6,616.6	509.8	125.5	384.36	1.326	Level 3
17,700.0	10,847.9	17,383.8	10,872.4	197.5	195.2	-92.76	-483.4	6,716.6	509.5	119.4	390.17	1.306	Level 3
17,800.0	10,848.4	17,483.8	10,872.9	200.4	198.1	-92.76	-484.2	6,816.6	509.3	113.3	395.98	1.286	Level 3

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18 2T2 - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
17,900.0	10,848.9	17,583.8	10,873.5	203.3	201.0	-92.76	-485.0	6,916.6	509.0	107.2	401.79	1.267	Level 3
18,000.0	10,849.5	17,683.8	10,874.0	206.2	203.9	-92.76	-485.8	7,016.6	508.7	101.1	407.60	1.248	Level 2
18,100.0	10,850.0	17,783.8	10,874.5	209.0	206.8	-92.77	-486.6	7,116.5	508.4	95.0	413.41	1.230	Level 2
18,200.0	10,850.5	17,883.8	10,875.0	211.9	209.7	-92.77	-487.4	7,216.5	508.2	88.9	419.23	1.212	Level 2
18,300.0	10,851.0	17,983.8	10,875.6	214.8	212.6	-92.77	-488.2	7,316.5	507.9	82.9	425.04	1.195	Level 2
18,400.0	10,851.5	18,083.8	10,876.1	217.7	215.5	-92.77	-489.0	7,416.5	507.6	76.8	430.86	1.178	Level 2
18,500.0	10,852.1	18,183.8	10,876.6	220.6	218.4	-92.77	-489.8	7,516.5	507.3	70.7	436.68	1.162	Level 2
18,600.0	10,852.6	18,283.8	10,877.1	223.5	221.3	-92.77	-490.6	7,616.5	507.1	64.6	442.50	1.146	Level 2
18,700.0	10,853.1	18,383.8	10,877.7	226.4	224.2	-92.77	-491.4	7,716.5	506.8	58.5	448.31	1.130	Level 2
18,800.0	10,853.6	18,483.8	10,878.2	229.3	227.2	-92.78	-492.2	7,816.5	506.5	52.4	454.14	1.115	Level 2
18,900.0	10,854.2	18,583.8	10,878.7	232.2	230.1	-92.78	-493.0	7,916.5	506.2	46.3	459.96	1.101	Level 2
19,000.0	10,854.7	18,683.8	10,879.2	235.1	233.0	-92.78	-493.8	8,016.5	506.0	40.2	465.78	1.086	Level 2
19,100.0	10,855.2	18,783.8	10,879.7	238.0	235.9	-92.78	-494.7	8,116.5	505.7	34.1	471.60	1.072	Level 2
19,200.0	10,855.7	18,883.8	10,880.3	240.9	238.8	-92.78	-495.5	8,216.5	505.4	28.0	477.43	1.059	Level 2
19,300.0	10,856.3	18,983.8	10,880.8	243.8	241.7	-92.78	-496.3	8,316.5	505.2	21.9	483.25	1.045	Level 2
19,400.0	10,856.8	19,083.8	10,881.3	246.7	244.6	-92.79	-497.1	8,416.5	504.9	15.8	489.08	1.032	Level 2
19,500.0	10,857.3	19,183.8	10,881.8	249.6	247.5	-92.79	-497.9	8,516.5	504.6	9.7	494.90	1.020	Level 2
19,600.0	10,857.8	19,283.8	10,882.4	252.5	250.4	-92.79	-498.7	8,616.5	504.3	3.6	500.73	1.007	Level 2
19,700.0	10,858.4	19,383.8	10,882.9	255.4	253.4	-92.79	-499.5	8,716.5	504.1	-2.5	506.56	0.995	Level 1
19,800.0	10,858.9	19,483.8	10,883.4	258.3	256.3	-92.79	-500.3	8,816.5	503.8	-8.6	512.38	0.983	Level 1
19,900.0	10,859.4	19,583.8	10,883.9	261.2	259.2	-92.79	-501.1	8,916.5	503.5	-14.7	518.21	0.972	Level 1
20,000.0	10,859.9	19,683.8	10,884.5	264.1	262.1	-92.79	-501.9	9,016.5	503.2	-20.8	524.04	0.960	Level 1
20,100.0	10,860.4	19,783.8	10,885.0	267.0	265.0	-92.80	-502.7	9,116.4	503.0	-26.9	529.87	0.949	Level 1
20,200.0	10,861.0	19,883.8	10,885.5	269.9	267.9	-92.80	-503.5	9,216.4	502.7	-33.0	535.70	0.938	Level 1
20,300.0	10,861.5	19,983.8	10,886.0	272.8	270.9	-92.80	-504.3	9,316.4	502.4	-39.1	541.53	0.928	Level 1
20,400.0	10,862.0	20,083.8	10,886.6	275.7	273.8	-92.80	-505.1	9,416.4	502.1	-45.2	547.36	0.917	Level 1
20,500.0	10,862.5	20,183.8	10,887.1	278.6	276.7	-92.80	-505.9	9,516.4	501.9	-51.3	553.20	0.907	Level 1
20,600.0	10,863.1	20,283.8	10,887.6	281.6	279.6	-92.80	-506.7	9,616.4	501.6	-57.4	559.03	0.897	Level 1
20,700.0	10,863.6	20,383.8	10,888.1	284.5	282.5	-92.81	-507.5	9,716.4	501.3	-63.6	564.86	0.887	Level 1
20,800.0	10,864.1	20,483.8	10,888.7	287.4	285.4	-92.81	-508.3	9,816.4	501.0	-69.7	570.69	0.878	Level 1
20,900.0	10,864.6	20,583.8	10,889.2	290.3	288.4	-92.81	-509.1	9,916.4	500.8	-75.8	576.53	0.869	Level 1
21,000.0	10,865.2	20,683.8	10,889.7	293.2	291.3	-92.81	-509.9	10,016.4	500.5	-81.9	582.36	0.859	Level 1
21,034.5	10,865.3	20,717.6	10,889.9	294.2	292.3	-92.81	-510.2	10,050.2	500.4	-84.0	584.35	0.856	Level 1
21,038.5	10,865.4	20,717.6	10,889.9	294.3	292.3	-92.81	-510.2	10,050.2	500.4	-84.1	584.47	0.856	Level 1, ES, SF

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1 — **Offset Site Error:** 0.0 usft

Survey Program: 2175-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
200.0	200.0	2,090.0	2,090.0	0.3	0.0	25.89	29.4	14.3	1,914.3	1,914.0	0.31	6,190.000	
300.0	300.0	2,090.0	2,090.0	0.5	0.0	25.89	29.4	14.3	1,814.3	1,813.8	0.53	3,397.431	
400.0	400.0	2,090.0	2,090.0	0.8	0.0	25.89	29.4	14.3	1,714.3	1,713.6	0.76	2,259.283	
500.0	500.0	2,090.0	2,090.0	1.0	0.0	25.89	29.4	14.3	1,614.3	1,613.3	0.98	1,641.328	
600.0	600.0	2,090.0	2,090.0	1.2	0.0	25.89	29.4	14.3	1,514.4	1,513.1	1.21	1,253.273	
700.0	700.0	2,090.0	2,090.0	1.4	0.0	25.89	29.4	14.3	1,414.4	1,412.9	1.43	986.947	
800.0	800.0	2,090.0	2,090.0	1.7	0.0	25.89	29.4	14.3	1,314.4	1,312.7	1.66	792.838	
900.0	900.0	2,090.0	2,090.0	1.9	0.0	25.89	29.4	14.3	1,214.4	1,212.6	1.88	645.081	
1,000.0	1,000.0	2,090.0	2,090.0	2.1	0.0	25.89	29.4	14.3	1,114.5	1,112.4	2.11	528.845	
1,100.0	1,100.0	2,090.0	2,090.0	2.3	0.0	25.89	29.4	14.3	1,014.5	1,012.2	2.33	435.018	
1,200.0	1,200.0	2,090.0	2,090.0	2.6	0.0	25.89	29.4	14.3	914.6	912.0	2.56	357.690	
1,300.0	1,300.0	2,090.0	2,090.0	2.8	0.0	25.89	29.4	14.3	814.7	811.9	2.78	292.864	
1,400.0	1,400.0	2,090.0	2,090.0	3.0	0.0	25.89	29.4	14.3	714.7	711.7	3.01	237.738	
1,500.0	1,500.0	2,090.0	2,090.0	3.2	0.0	25.89	29.4	14.3	614.9	611.6	3.23	190.290	
1,600.0	1,600.0	2,090.0	2,090.0	3.5	0.0	25.89	29.4	14.3	515.0	511.6	3.46	149.028	
1,700.0	1,700.0	2,090.0	2,090.0	3.7	0.0	25.89	29.4	14.3	415.3	411.6	3.68	112.827	
1,800.0	1,800.0	2,090.0	2,090.0	3.9	0.0	25.89	29.4	14.3	315.7	311.8	3.91	80.833	
1,900.0	1,900.0	2,090.0	2,090.0	4.1	0.0	25.89	29.4	14.3	216.5	212.3	4.13	52.412	
2,000.0	2,000.0	2,090.0	2,090.0	4.4	0.0	25.89	29.4	14.3	118.6	114.2	4.36	27.230	
2,100.0	2,100.0	2,090.0	2,090.0	4.6	0.0	25.89	29.4	14.3	35.5	31.0	4.58	7.759	
2,139.6	2,139.6	2,115.7	2,115.7	4.7	0.0	25.95	29.3	14.3	32.6	27.9	4.70	6.939	
2,200.0	2,200.0	2,176.5	2,176.5	4.8	0.1	26.58	28.6	14.3	32.0	27.1	4.91	6.524	
2,300.0	2,300.0	2,276.5	2,276.4	5.0	0.3	27.89	27.0	14.3	30.5	25.2	5.34	5.715	
2,400.0	2,400.0	2,376.4	2,376.4	5.3	0.5	28.23	25.6	13.8	29.1	23.3	5.78	5.034	
2,500.0	2,500.0	2,476.4	2,476.3	5.5	0.7	26.63	24.8	12.5	27.8	21.6	6.23	4.463	
2,564.1	2,564.1	2,540.4	2,540.3	5.6	0.9	116.05	24.6	11.3	27.4	20.9	6.50	4.220 CC	
2,600.0	2,600.0	2,576.3	2,576.2	5.7	1.0	116.67	24.6	10.6	27.5	20.9	6.65	4.139	
2,650.0	2,649.9	2,626.3	2,626.2	5.8	1.1	118.37	24.7	9.4	28.1	21.2	6.86	4.092	
2,700.0	2,699.9	2,676.3	2,676.2	5.9	1.2	120.16	25.0	8.0	28.9	21.8	7.07	4.080	
2,800.0	2,799.7	2,776.2	2,776.1	6.1	1.4	122.37	25.5	4.4	30.2	22.7	7.50	4.022	
2,900.0	2,899.6	2,875.8	2,875.6	6.3	1.6	126.27	26.4	2.4	32.7	24.8	7.91	4.138	
3,000.0	2,999.5	2,975.9	2,975.7	6.5	1.8	130.02	27.3	0.7	35.7	27.3	8.32	4.286	
3,100.0	3,099.3	3,076.1	3,075.8	6.7	2.1	132.96	27.9	-1.5	38.1	29.4	8.75	4.359	
3,200.0	3,199.2	3,176.2	3,175.9	7.0	2.3	135.03	28.4	-4.3	40.2	31.0	9.17	4.380	
3,300.0	3,299.0	3,276.1	3,275.7	7.2	2.5	137.14	28.7	-7.0	42.2	32.6	9.59	4.400	
3,400.0	3,398.9	3,375.9	3,375.6	7.4	2.7	139.20	29.1	-9.5	44.5	34.5	10.00	4.454	
3,501.2	3,500.0	3,477.0	3,476.6	7.6	2.9	141.09	29.6	-11.8	47.1	36.7	10.42	4.523	
3,600.0	3,598.7	3,575.8	3,575.4	7.8	3.1	141.59	30.2	-13.9	48.6	37.7	10.83	4.484	
3,651.2	3,649.9	3,627.0	3,626.6	7.9	3.2	50.74	30.5	-15.1	48.2	37.2	10.99	4.388	
3,700.0	3,698.7	3,675.7	3,675.3	8.0	3.3	49.64	30.8	-16.1	47.6	36.4	11.19	4.254	
3,800.0	3,798.7	3,775.4	3,775.0	8.2	3.5	47.52	31.5	-18.0	46.7	35.0	11.61	4.017	
3,900.0	3,898.7	3,875.5	3,875.0	8.5	3.7	45.41	32.3	-19.6	46.0	34.0	12.04	3.823	
4,000.0	3,998.7	3,975.3	3,974.8	8.7	3.9	43.23	33.1	-21.3	45.5	33.0	12.48	3.645	
4,100.0	4,098.7	4,075.2	4,074.7	8.9	4.1	41.23	34.1	-22.6	45.3	32.4	12.91	3.509	
4,144.5	4,143.2	4,119.7	4,119.2	9.0	4.2	40.46	34.4	-23.0	45.3	32.2	13.10	3.456	
4,200.0	4,198.7	4,175.1	4,174.6	9.1	4.3	39.57	34.9	-23.5	45.3	32.0	13.34	3.397	
4,300.0	4,298.7	4,275.2	4,274.7	9.4	4.6	37.92	35.8	-24.5	45.3	31.6	13.77	3.293	
4,349.2	4,347.9	4,324.4	4,323.9	9.5	4.7	37.05	36.2	-25.1	45.3	31.3	13.98	3.241	
4,400.0	4,398.7	4,375.2	4,374.7	9.6	4.8	36.16	36.6	-25.6	45.3	31.1	14.21	3.192	
4,500.0	4,498.7	4,475.2	4,474.7	9.8	5.0	34.66	37.3	-26.6	45.4	30.8	14.64	3.101	
4,600.0	4,598.7	4,575.2	4,574.6	10.0	5.2	33.49	38.0	-27.3	45.5	30.4	15.07	3.020	
4,700.0	4,698.7	4,675.2	4,674.7	10.2	5.4	32.40	38.5	-28.0	45.5	30.0	15.50	2.937	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1 — **Offset Site Error:** 0.0 usft

Survey Program: 2175-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
4,783.6	4,782.3	4,758.8	4,758.3	10.4	5.6	31.49	38.8	-28.6	45.5	29.6	15.87	2.868	
4,800.0	4,798.7	4,775.2	4,774.7	10.5	5.6	31.30	38.9	-28.8	45.5	29.6	15.94	2.856	
4,900.0	4,898.7	4,875.2	4,874.6	10.7	5.8	30.10	39.5	-29.5	45.7	29.3	16.37	2.791	
5,000.0	4,998.7	4,975.2	4,974.7	10.9	6.0	29.27	39.9	-30.0	45.7	28.9	16.81	2.721	
5,100.0	5,098.7	5,075.2	5,074.7	11.1	6.2	28.11	40.4	-30.8	45.8	28.6	17.24	2.659	
5,200.0	5,198.7	5,175.5	5,174.9	11.4	6.4	26.93	40.7	-31.7	45.7	28.0	17.68	2.583	
5,300.0	5,298.7	5,275.6	5,275.1	11.6	6.6	26.60	40.2	-32.3	44.9	26.8	18.11	2.479	
5,400.0	5,398.7	5,375.6	5,375.1	11.8	6.9	26.36	39.5	-32.8	44.1	25.5	18.54	2.376	
5,500.0	5,498.7	5,475.9	5,475.3	12.0	7.1	26.21	38.6	-33.4	43.0	24.0	18.98	2.266	
5,600.0	5,598.7	5,576.0	5,575.4	12.2	7.3	25.95	37.1	-34.4	41.2	21.8	19.41	2.124	
5,700.0	5,698.7	5,675.9	5,675.3	12.5	7.5	26.31	35.5	-34.9	39.6	19.7	19.84	1.995	
5,800.0	5,798.7	5,775.7	5,775.1	12.7	7.7	26.36	34.3	-35.4	38.3	18.1	20.27	1.891	
5,900.0	5,898.7	5,875.6	5,875.0	12.9	7.9	25.78	33.8	-36.1	37.5	16.8	20.70	1.812	
6,000.0	5,998.7	5,975.3	5,974.7	13.1	8.1	24.32	33.8	-37.1	37.1	15.9	21.14	1.753	
6,001.5	6,000.2	5,976.8	5,976.2	13.1	8.1	24.29	33.8	-37.2	37.1	15.9	21.15	1.752	
6,100.0	6,098.7	6,075.2	6,074.6	13.4	8.3	22.12	34.5	-38.4	37.3	15.7	21.58	1.728	
6,200.0	6,198.7	6,175.0	6,174.4	13.6	8.5	20.07	35.6	-39.4	37.9	15.9	22.02	1.720	
6,300.0	6,298.7	6,274.8	6,274.1	13.8	8.7	18.56	37.0	-40.0	39.0	16.6	22.46	1.738	
6,400.0	6,398.7	6,374.6	6,374.0	14.0	9.0	17.37	38.8	-40.3	40.7	17.8	22.90	1.777	
6,500.0	6,498.7	6,474.5	6,473.8	14.3	9.2	16.41	40.8	-40.4	42.6	19.2	23.33	1.824	
6,600.0	6,598.7	6,574.2	6,573.5	14.5	9.4	15.40	43.3	-40.5	45.0	21.2	23.77	1.891	
6,700.0	6,698.7	6,674.1	6,673.4	14.7	9.6	14.52	46.1	-40.5	47.7	23.5	24.21	1.969	
6,800.0	6,798.7	6,773.9	6,773.1	14.9	9.8	13.90	49.2	-40.2	50.7	26.1	24.65	2.058	
6,900.0	6,898.7	6,874.3	6,873.5	15.2	10.0	14.37	51.6	-39.2	53.3	28.2	25.09	2.125	
7,000.0	6,998.7	6,974.7	6,973.9	15.4	10.2	15.06	53.5	-38.0	55.4	29.9	25.52	2.172	
7,100.0	7,098.7	7,075.4	7,074.5	15.6	10.4	15.87	53.9	-37.1	56.0	30.1	25.95	2.160	
7,200.0	7,198.7	7,175.8	7,174.9	15.8	10.6	16.54	54.0	-36.4	56.3	30.0	26.38	2.136	
7,300.0	7,298.7	7,276.4	7,275.6	16.0	10.8	16.85	52.7	-36.4	55.1	28.3	26.81	2.055	
7,400.0	7,398.7	7,376.5	7,375.6	16.3	11.0	16.98	51.2	-36.8	53.5	26.3	27.24	1.965	
7,500.0	7,498.7	7,476.4	7,475.5	16.5	11.2	16.83	49.6	-37.4	51.8	24.2	27.67	1.873	
7,600.0	7,598.7	7,576.3	7,575.4	16.7	11.5	16.50	48.2	-38.1	50.3	22.2	28.11	1.790	
7,700.0	7,698.7	7,676.1	7,675.1	16.9	11.7	16.10	47.2	-38.8	49.1	20.6	28.54	1.720	
7,800.0	7,798.7	7,775.9	7,775.0	17.2	11.9	16.08	46.5	-39.0	48.4	19.5	28.98	1.671	
7,882.5	7,881.2	7,858.1	7,857.2	17.4	12.1	16.22	46.2	-39.0	48.2	18.8	29.34	1.642	
7,900.0	7,898.7	7,875.6	7,874.6	17.4	12.1	16.24	46.2	-38.9	48.2	18.8	29.41	1.638	
8,000.0	7,998.7	7,975.4	7,974.4	17.6	12.3	16.42	46.6	-38.7	48.6	18.7	29.85	1.627	
8,100.0	8,098.7	8,075.3	8,074.4	17.8	12.5	16.74	47.1	-38.2	49.2	18.9	30.28	1.624	
8,200.0	8,198.7	8,175.2	8,174.3	18.1	12.7	16.91	47.8	-37.9	49.9	19.2	30.72	1.626	
8,300.0	8,298.7	8,275.0	8,274.0	18.3	12.9	17.34	48.6	-37.2	51.0	19.8	31.15	1.636	
8,400.0	8,398.7	8,375.0	8,374.0	18.5	13.1	17.93	49.7	-36.3	52.3	20.7	31.58	1.654	
8,500.0	8,498.7	8,475.0	8,474.0	18.7	13.4	18.46	50.8	-35.4	53.6	21.5	32.02	1.673	
8,600.0	8,598.7	8,575.0	8,574.0	19.0	13.6	18.99	51.9	-34.5	54.9	22.4	32.45	1.691	
8,700.0	8,698.7	8,675.0	8,674.0	19.2	13.8	19.32	52.9	-33.8	56.1	23.2	32.88	1.706	
8,800.0	8,798.7	8,775.0	8,774.0	19.4	14.0	19.55	54.0	-33.2	57.3	24.0	33.31	1.720	
8,900.0	8,898.7	8,874.8	8,873.8	19.6	14.2	19.85	55.1	-32.5	58.6	24.9	33.75	1.737	
9,000.0	8,998.7	8,974.7	8,973.7	19.9	14.4	20.02	56.6	-31.8	60.2	26.1	34.18	1.762	
9,100.0	9,098.7	9,074.7	9,073.7	20.1	14.6	20.41	58.0	-30.8	61.9	27.3	34.62	1.789	
9,200.0	9,198.7	9,174.7	9,173.7	20.3	14.8	20.66	59.5	-30.0	63.6	28.6	35.05	1.815	
9,300.0	9,298.7	9,274.8	9,273.8	20.5	15.0	21.08	60.8	-29.0	65.1	29.7	35.48	1.836	
9,400.0	9,398.7	9,374.7	9,373.7	20.8	15.2	21.32	62.0	-28.2	66.6	30.6	35.92	1.853	
9,500.0	9,498.7	9,474.7	9,473.6	21.0	15.4	21.70	63.3	-27.2	68.2	31.8	36.35	1.875	
9,600.0	9,598.7	9,574.8	9,573.7	21.2	15.6	21.91	64.7	-26.4	69.7	32.9	36.78	1.895	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1

Offset Site Error: 0.0 usft

Survey Program: 2175-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
9,700.0	9,698.7	9,674.9	9,673.8	21.4	15.8	22.08	65.9	-25.7	71.1	33.9	37.21	1.910	
9,800.0	9,798.7	9,774.9	9,773.8	21.6	16.1	22.28	66.9	-25.0	72.3	34.6	37.64	1.920	
9,900.0	9,898.7	9,874.9	9,873.8	21.9	16.3	22.50	67.9	-24.3	73.5	35.4	38.08	1.929	
10,000.0	9,998.7	9,974.8	9,973.7	22.1	16.5	22.60	69.0	-23.7	74.7	36.2	38.51	1.940	
10,100.0	10,098.7	10,074.7	10,073.5	22.3	16.7	22.74	70.2	-23.0	76.2	37.2	38.95	1.955	
10,200.0	10,198.7	10,174.6	10,173.5	22.5	16.9	22.87	71.6	-22.2	77.7	38.3	39.38	1.973	
10,300.0	10,298.7	10,275.1	10,274.0	22.8	17.1	23.49	72.7	-20.8	79.3	39.5	39.82	1.992	
10,336.6	10,335.3	10,314.5	10,313.2	22.9	17.2	25.68	71.3	-18.1	79.1	39.2	39.98	1.980	
10,350.0	10,348.7	10,329.1	10,327.6	22.9	17.2	-109.93	69.9	-16.4	78.8	38.7	40.08	1.966	
10,375.0	10,373.7	10,356.1	10,354.1	22.9	17.3	-107.03	66.2	-12.2	78.0	37.9	40.16	1.943	
10,400.0	10,398.5	10,382.7	10,379.7	23.0	17.3	-103.90	61.1	-7.1	77.2	36.9	40.25	1.917	
10,425.0	10,423.2	10,408.6	10,404.1	23.0	17.4	-100.74	55.0	-1.3	76.3	36.0	40.33	1.892	
10,450.0	10,447.6	10,434.5	10,428.2	23.1	17.4	-97.68	48.1	5.2	75.6	35.2	40.42	1.870	
10,475.0	10,471.8	10,460.7	10,452.1	23.1	17.5	-94.77	40.0	12.1	74.8	34.3	40.52	1.845	
10,500.0	10,495.5	10,486.7	10,475.5	23.1	17.5	-92.07	31.0	19.4	73.8	33.2	40.61	1.816	
10,525.0	10,518.8	10,512.2	10,497.9	23.2	17.6	-89.75	21.5	26.8	72.6	31.9	40.71	1.784	
10,550.0	10,541.7	10,536.9	10,519.3	23.2	17.6	-87.92	11.7	34.3	71.5	30.7	40.80	1.752	
10,575.0	10,563.9	10,561.0	10,539.9	23.3	17.7	-86.54	1.8	42.1	70.5	29.6	40.90	1.724	
10,600.0	10,585.5	10,584.6	10,559.4	23.3	17.8	-85.27	-8.3	50.6	69.9	28.9	41.00	1.706	
10,625.0	10,606.5	10,608.5	10,578.4	23.4	17.8	-83.83	-19.1	60.2	69.7	28.6	41.10	1.696	
10,632.3	10,612.5	10,615.5	10,583.8	23.4	17.9	-83.37	-22.5	63.2	69.7	28.5	41.13	1.694	
10,650.0	10,626.7	10,632.3	10,596.4	23.4	17.9	-82.14	-30.7	70.7	69.7	28.6	41.20	1.693	
10,675.0	10,646.1	10,656.1	10,613.3	23.5	18.0	-80.20	-43.0	82.1	70.1	28.8	41.28	1.698	
10,700.0	10,664.6	10,679.8	10,629.1	23.6	18.1	-78.10	-55.9	94.3	70.7	29.4	41.34	1.710	
10,725.0	10,682.3	10,703.0	10,643.5	23.6	18.2	-76.04	-68.9	106.8	71.6	30.2	41.39	1.730	
10,750.0	10,698.9	10,726.1	10,657.0	23.7	18.4	-74.13	-82.2	120.0	72.9	31.5	41.43	1.760	
10,775.0	10,714.6	10,748.9	10,669.6	23.8	18.5	-72.52	-95.5	133.6	74.6	33.1	41.47	1.798	
10,800.0	10,729.3	10,771.4	10,680.9	23.9	18.7	-70.98	-108.9	147.7	76.6	35.1	41.51	1.846	
10,825.0	10,742.9	10,793.9	10,691.0	24.0	18.8	-69.39	-122.5	162.5	79.0	37.5	41.54	1.903	
10,850.0	10,755.3	10,815.8	10,699.5	24.2	19.0	-67.74	-136.2	177.4	81.8	40.3	41.54	1.970	
10,875.0	10,766.6	10,837.0	10,706.6	24.3	19.2	-66.15	-149.4	192.3	85.1	43.6	41.53	2.050	
10,900.0	10,776.8	10,860.9	10,713.6	24.5	19.4	-64.77	-164.4	209.5	88.6	47.1	41.57	2.132	
10,925.0	10,785.7	10,883.8	10,719.9	24.7	19.7	-63.95	-178.8	226.2	91.9	50.2	41.70	2.204	
10,950.0	10,793.4	10,905.8	10,725.1	24.9	19.9	-63.37	-192.6	242.5	95.3	53.4	41.90	2.273	
10,975.0	10,799.8	10,926.2	10,728.7	25.1	20.2	-62.85	-205.4	258.0	99.0	56.9	42.11	2.350	
11,000.0	10,804.9	10,946.8	10,731.3	25.3	20.4	-62.34	-218.1	274.0	103.2	60.8	42.36	2.435	
11,025.0	10,808.8	10,967.9	10,732.9	25.5	20.7	-61.93	-230.9	290.7	107.7	65.0	42.67	2.524	
11,050.0	10,811.4	10,991.1	10,734.0	25.8	21.0	-61.90	-244.9	309.2	112.1	69.0	43.12	2.600	
11,075.0	10,812.6	11,014.2	10,734.9	26.0	21.3	-62.32	-258.6	327.7	116.3	72.5	43.73	2.659	
11,084.1	10,812.8	11,022.6	10,735.1	26.1	21.5	-62.58	-263.6	334.5	117.7	73.7	43.98	2.677	
11,099.1	10,812.8	11,036.5	10,735.4	26.3	21.7	-63.27	-271.7	345.7	120.2	75.7	44.47	2.702	
11,200.0	10,813.4	11,129.0	10,736.8	27.5	23.1	-67.15	-324.2	421.9	137.2	89.4	47.82	2.869	
11,300.0	10,813.9	11,216.0	10,736.9	28.9	24.7	-69.78	-368.9	496.5	156.1	104.9	51.17	3.051	
11,400.0	10,814.5	11,298.0	10,736.9	30.6	26.2	-72.04	-404.8	570.2	178.4	123.8	54.65	3.265	
11,500.0	10,815.0	11,378.8	10,737.7	32.4	27.9	-74.36	-433.3	645.7	204.5	146.2	58.35	3.505	
11,600.0	10,815.6	11,455.0	10,738.7	34.3	29.5	-76.35	-453.7	719.1	234.2	172.2	62.03	3.776	
11,700.0	10,816.1	11,528.6	10,739.1	36.3	31.0	-77.91	-467.3	791.5	267.4	201.7	65.66	4.072	
11,800.0	10,816.7	11,608.8	10,740.1	38.5	32.8	-79.46	-476.0	871.2	302.8	233.3	69.49	4.357	
11,900.0	10,817.3	11,700.0	10,739.8	40.7	34.8	-80.45	-484.8	961.9	335.0	261.5	73.53	4.556	
12,000.0	10,817.9	11,789.4	10,739.2	43.0	36.9	-81.12	-491.6	1,051.0	364.2	286.7	77.51	4.699	
12,100.0	10,818.4	11,884.9	10,739.3	45.3	39.2	-81.72	-498.0	1,146.4	389.2	307.6	81.63	4.768	
12,200.0	10,819.0	11,977.4	10,739.4	47.6	41.4	-82.12	-503.8	1,238.7	409.7	324.1	85.62	4.785	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1 **Offset Site Error:** 0.0 usft

Survey Program: 2175-MWD **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
12,300.0	10,819.5	12,070.5	10,739.6	50.0	43.7	-82.44	-507.6	1,331.7	427.2	337.6	89.54	4.771	
12,400.0	10,820.1	12,165.3	10,739.7	52.3	46.1	-82.63	-511.4	1,426.5	439.6	346.3	93.37	4.709	
12,500.0	10,820.6	12,256.4	10,741.4	54.7	48.5	-82.94	-512.6	1,517.5	449.3	352.3	96.99	4.632	
12,600.0	10,821.2	12,356.0	10,743.9	57.0	51.0	-83.28	-513.2	1,617.1	454.4	353.8	100.66	4.515	
12,655.6	10,821.5	12,411.6	10,744.8	58.3	52.5	-83.36	-513.7	1,672.7	455.0	352.4	102.61	4.435	
12,700.0	10,821.7	12,454.8	10,745.7	59.4	53.6	-83.43	-513.9	1,715.8	454.9	350.0	104.93	4.335	
12,800.0	10,822.2	12,565.5	10,748.3	61.8	56.6	-83.68	-515.2	1,826.5	454.2	343.6	110.61	4.106	
12,900.0	10,822.7	12,672.0	10,750.2	64.2	59.5	-83.80	-519.3	1,932.9	450.7	334.5	116.20	3.878	
13,000.0	10,823.3	12,763.0	10,751.9	66.6	61.9	-83.92	-521.9	2,023.9	448.2	326.8	121.40	3.691	
13,100.0	10,823.8	12,859.1	10,753.4	69.2	64.6	-84.04	-523.2	2,120.0	447.2	320.5	126.78	3.528	
13,200.0	10,824.3	12,955.8	10,754.3	71.7	67.2	-84.09	-524.3	2,216.6	446.6	314.4	132.19	3.378	
13,227.2	10,824.5	12,981.7	10,754.7	72.4	67.9	-84.12	-524.5	2,242.5	446.5	312.9	133.66	3.341	
13,300.0	10,824.8	13,051.4	10,756.4	74.3	69.8	-84.30	-524.4	2,312.2	446.8	309.2	137.64	3.246	
13,400.0	10,825.4	13,149.6	10,759.0	76.8	72.5	-84.57	-523.9	2,410.4	447.7	304.5	143.18	3.127	
13,500.0	10,825.9	13,247.5	10,760.4	79.5	75.2	-84.70	-523.1	2,508.2	449.0	300.3	148.71	3.019	
13,600.0	10,826.4	13,342.9	10,762.2	82.1	77.9	-84.89	-521.4	2,603.6	451.1	296.9	154.21	2.925	
13,700.0	10,826.9	13,440.3	10,762.6	84.7	80.6	-84.91	-519.1	2,701.0	454.1	294.3	159.74	2.842	
13,800.0	10,827.5	13,542.0	10,761.8	87.4	83.4	-84.78	-516.7	2,802.6	457.0	291.7	165.37	2.764	
13,900.0	10,828.0	13,643.5	10,760.5	90.1	86.3	-84.58	-514.8	2,904.1	459.6	288.6	171.00	2.688	
14,000.0	10,828.5	13,740.9	10,759.4	92.8	89.0	-84.41	-512.9	3,001.5	462.1	285.6	176.53	2.618	
14,100.0	10,829.0	13,835.4	10,761.1	95.5	91.6	-84.60	-509.8	3,096.0	465.8	283.8	182.07	2.558	
14,200.0	10,829.6	13,933.4	10,760.3	98.3	94.4	-84.48	-506.1	3,193.8	470.3	282.6	187.64	2.506	
14,300.0	10,830.1	14,035.6	10,759.7	101.0	97.2	-84.40	-502.1	3,296.0	474.8	281.4	193.35	2.456	
14,400.0	10,830.6	14,139.6	10,760.2	103.8	100.2	-84.43	-498.9	3,399.9	478.4	279.2	199.15	2.402	
14,500.0	10,831.1	14,243.0	10,760.8	106.5	103.1	-84.49	-496.4	3,503.3	481.3	276.3	204.96	2.348	
14,600.0	10,831.6	14,345.4	10,758.2	109.3	106.0	-84.13	-495.0	3,605.7	483.5	272.9	210.61	2.296	
14,700.0	10,832.2	14,451.8	10,755.7	112.1	109.1	-83.79	-494.3	3,712.0	485.0	268.7	216.38	2.242	
14,800.0	10,832.7	14,545.9	10,755.5	114.9	111.8	-83.72	-493.8	3,806.1	486.1	264.2	221.89	2.191	
14,900.0	10,833.2	14,637.9	10,756.2	117.6	114.4	-83.78	-491.7	3,898.1	488.9	261.5	227.39	2.150	
15,000.0	10,833.7	14,741.5	10,758.6	120.4	117.3	-84.04	-488.5	4,001.6	492.4	259.1	233.32	2.110	
15,100.0	10,834.3	14,846.6	10,759.9	123.2	120.3	-84.17	-486.5	4,106.6	494.7	255.4	239.24	2.068	
15,200.0	10,834.8	14,955.7	10,759.1	126.1	123.5	-84.02	-486.4	4,215.7	495.4	250.3	245.18	2.021	
15,205.1	10,834.8	14,960.6	10,759.1	126.2	123.6	-84.00	-486.4	4,220.7	495.4	250.0	245.46	2.018	
15,300.0	10,835.3	15,053.1	10,758.5	128.9	126.3	-83.88	-486.7	4,313.1	495.7	245.0	250.79	1.977	
15,400.0	10,835.8	15,156.6	10,757.9	131.7	129.3	-83.75	-487.3	4,416.6	495.8	239.2	256.58	1.932	
15,500.0	10,836.4	15,256.5	10,756.8	134.5	132.2	-83.56	-488.3	4,516.5	495.6	233.3	262.24	1.890	
15,536.8	10,836.6	15,292.4	10,756.3	135.6	133.2	-83.49	-488.6	4,552.4	495.5	231.2	264.29	1.875	
15,600.0	10,836.9	15,354.9	10,755.7	137.4	135.0	-83.38	-488.9	4,614.9	495.6	227.7	267.85	1.850	
15,700.0	10,837.4	15,455.2	10,755.9	140.2	137.9	-83.33	-489.4	4,715.2	495.7	222.1	273.59	1.812	
15,800.0	10,837.9	15,553.2	10,756.1	143.0	140.7	-83.31	-489.8	4,813.3	495.9	216.6	279.27	1.776	
15,900.0	10,838.5	15,651.3	10,756.5	145.9	143.6	-83.30	-489.6	4,911.4	496.6	211.7	284.96	1.743	
16,000.0	10,839.0	15,753.2	10,757.8	148.7	146.5	-83.40	-489.3	5,013.2	497.4	206.5	290.83	1.710	
16,100.0	10,839.5	15,858.8	10,759.8	151.6	149.6	-83.57	-489.6	5,118.8	497.4	200.5	296.84	1.676	
16,200.0	10,840.0	15,965.2	10,759.9	154.4	152.7	-83.50	-491.7	5,225.1	496.0	193.3	302.76	1.638	
16,300.0	10,840.5	16,069.1	10,760.3	157.3	155.7	-83.44	-494.8	5,329.0	493.7	185.0	308.61	1.600	
16,400.0	10,841.1	16,169.3	10,760.4	160.1	158.6	-83.36	-498.2	5,429.1	490.8	176.5	314.34	1.561	
16,500.0	10,841.6	16,273.2	10,761.6	163.0	161.6	-83.39	-501.9	5,533.0	487.7	167.5	320.25	1.523	
16,600.0	10,842.1	16,379.8	10,764.4	165.9	164.7	-83.60	-506.8	5,639.4	483.5	157.1	326.35	1.481	Level 3
16,700.0	10,842.6	16,482.8	10,768.0	168.7	167.7	-83.89	-512.6	5,742.2	478.1	145.7	332.38	1.438	Level 3
16,800.0	10,843.2	16,577.6	10,770.5	171.6	170.5	-84.08	-517.7	5,836.9	473.0	134.9	338.12	1.399	Level 3
16,900.0	10,843.7	16,673.1	10,772.1	174.5	173.3	-84.16	-521.9	5,932.3	469.0	125.2	343.83	1.364	Level 3
17,000.0	10,844.2	16,774.4	10,774.0	177.3	176.2	-84.28	-526.3	6,033.4	465.1	115.4	349.72	1.330	Level 3

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 3T	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 11-18H - Wellbore #1 - Wellbore #1 — Offset Site Error: 0.0 usft

Survey Program: 2175-MWD — Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
17,100.0	10,844.7	16,866.4	10,774.8	180.2	178.9	-84.28	-529.6	6,125.4	462.1	106.8	355.29	1.301	Level 3
17,200.0	10,845.3	16,959.5	10,774.8	183.1	181.6	-84.21	-531.4	6,218.5	460.6	99.8	360.83	1.277	Level 3
17,261.5	10,845.6	17,017.9	10,775.0	184.9	183.3	-84.19	-532.0	6,276.8	460.4	96.1	364.29	1.264	Level 3
17,300.0	10,845.8	17,055.4	10,775.2	186.0	184.4	-84.19	-532.1	6,314.4	460.5	94.0	366.50	1.256	Level 3
17,400.0	10,846.3	17,160.8	10,775.7	188.8	187.5	-84.18	-533.2	6,419.7	460.0	87.5	372.44	1.235	Level 2
17,500.0	10,846.8	17,259.6	10,776.6	191.7	190.4	-84.22	-534.5	6,518.6	459.1	80.9	378.23	1.214	Level 2
17,600.0	10,847.4	17,360.7	10,777.9	194.6	193.3	-84.30	-535.7	6,619.6	458.5	74.3	384.11	1.194	Level 2
17,700.0	10,847.9	17,464.1	10,779.0	197.5	196.3	-84.35	-537.7	6,723.0	457.0	66.9	390.04	1.172	Level 2
17,800.0	10,848.4	17,564.8	10,779.4	200.4	199.3	-84.32	-539.9	6,823.6	455.3	59.5	395.84	1.150	Level 2
17,900.0	10,848.9	17,666.0	10,779.6	203.3	202.2	-84.25	-542.6	6,924.8	453.2	51.6	401.64	1.128	Level 2
18,000.0	10,849.5	17,764.3	10,779.5	206.2	205.1	-84.15	-545.1	7,023.1	451.3	43.9	407.32	1.108	Level 2
18,100.0	10,850.0	17,862.5	10,779.3	209.0	207.9	-84.04	-547.2	7,121.3	449.7	36.7	413.00	1.089	Level 2
18,200.0	10,850.5	17,957.9	10,778.7	211.9	210.7	-83.89	-548.8	7,216.7	448.7	30.2	418.56	1.072	Level 2
18,300.0	10,851.0	18,060.9	10,779.0	214.8	213.7	-83.85	-549.6	7,319.6	448.6	24.2	424.42	1.057	Level 2
18,400.0	10,851.5	18,157.9	10,779.6	217.7	216.5	-83.85	-551.2	7,416.6	447.4	17.3	430.14	1.040	Level 2
18,414.9	10,851.6	18,171.4	10,779.6	218.1	216.9	-83.84	-551.3	7,430.1	447.4	16.4	430.95	1.038	Level 2
18,500.0	10,852.1	18,254.3	10,779.5	220.6	219.4	-83.77	-551.5	7,513.0	447.7	12.0	435.78	1.027	Level 2
18,600.0	10,852.6	18,356.8	10,779.7	223.5	222.3	-83.73	-551.7	7,615.6	448.0	6.4	441.62	1.015	Level 2
18,646.6	10,852.8	18,403.0	10,779.0	224.8	223.7	-83.62	-552.2	7,661.8	448.0	3.7	444.22	1.008	Level 2
18,700.0	10,853.1	18,453.4	10,778.1	226.4	225.2	-83.47	-552.4	7,712.1	448.1	1.0	447.11	1.002	Level 2
18,800.0	10,853.6	18,551.9	10,776.4	229.3	228.0	-83.20	-552.2	7,810.6	449.1	-3.5	452.63	0.992	Level 1
18,900.0	10,854.2	18,649.9	10,776.0	232.2	230.9	-83.09	-551.6	7,908.6	450.3	-8.0	458.28	0.983	Level 1
19,000.0	10,854.7	18,749.5	10,776.5	235.1	233.8	-83.11	-550.6	8,008.2	451.9	-12.2	464.08	0.974	Level 1
19,100.0	10,855.2	18,852.3	10,776.9	238.0	236.8	-83.12	-549.8	8,111.0	453.2	-16.8	469.96	0.964	Level 1
19,200.0	10,855.7	18,945.1	10,777.0	240.9	239.5	-83.10	-548.8	8,203.8	454.8	-20.7	475.53	0.956	Level 1
19,300.0	10,856.3	19,043.8	10,777.1	243.8	242.3	-83.08	-546.7	8,302.5	457.6	-23.7	481.26	0.951	Level 1
19,400.0	10,856.8	19,146.8	10,777.4	246.7	245.3	-83.09	-544.7	8,405.5	460.0	-27.1	487.15	0.944	Level 1
19,500.0	10,857.3	19,245.5	10,777.7	249.6	248.2	-83.10	-543.0	8,504.1	462.3	-30.6	492.92	0.938	Level 1
19,600.0	10,857.8	19,346.8	10,777.7	252.5	251.1	-83.06	-541.3	8,605.4	464.5	-34.2	498.71	0.931	Level 1
19,700.0	10,858.4	19,443.3	10,777.5	255.4	253.9	-83.01	-539.5	8,701.9	467.1	-37.3	504.35	0.926	Level 1
19,800.0	10,858.9	19,542.6	10,777.6	258.3	256.8	-83.00	-537.0	8,801.2	470.1	-40.0	510.12	0.922	Level 1
19,900.0	10,859.4	19,646.3	10,777.4	261.2	259.8	-82.95	-535.0	8,904.9	472.6	-43.4	515.97	0.916	Level 1
20,000.0	10,859.9	19,744.8	10,777.5	264.1	262.7	-82.94	-533.3	9,003.4	474.9	-46.8	521.72	0.910	Level 1
20,100.0	10,860.4	19,843.0	10,777.9	267.0	265.6	-82.96	-531.0	9,101.5	477.8	-49.7	527.48	0.906	Level 1
20,200.0	10,861.0	19,944.3	10,777.9	269.9	268.5	-82.94	-529.0	9,202.7	480.3	-52.9	533.28	0.901	Level 1
20,300.0	10,861.5	20,043.1	10,778.2	272.8	271.4	-82.96	-526.6	9,301.6	483.3	-55.8	539.07	0.896	Level 1
20,400.0	10,862.0	20,144.8	10,778.4	275.7	274.3	-82.96	-524.5	9,403.2	485.9	-59.1	544.91	0.892	Level 1
20,500.0	10,862.5	20,245.4	10,778.1	278.6	277.3	-82.89	-522.6	9,503.8	488.3	-62.3	550.65	0.887	Level 1
20,600.0	10,863.1	20,345.8	10,777.5	281.6	280.2	-82.79	-521.0	9,604.1	490.6	-65.7	556.35	0.882	Level 1
20,700.0	10,863.6	20,446.4	10,777.2	284.5	283.1	-82.73	-519.3	9,704.8	492.9	-69.2	562.09	0.877	Level 1
20,800.0	10,864.1	20,547.5	10,776.2	287.4	286.1	-82.59	-518.0	9,805.9	494.9	-72.8	567.75	0.872	Level 1
20,900.0	10,864.6	20,650.0	10,774.2	290.3	289.0	-82.31	-517.2	9,908.3	496.5	-76.8	573.29	0.866	Level 1, ES, SF
21,000.0	10,865.2	20,650.0	10,774.2	293.2	289.0	-82.31	-517.2	9,908.3	507.9	-68.3	576.18	0.881	Level 1
21,038.5	10,865.4	20,650.0	10,774.2	294.3	289.0	-82.31	-517.2	9,908.3	517.4	-59.9	577.30	0.896	Level 1

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Gyrodata, Inc.
Anticollision Report

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6

Offset Site Error: 0.0 usft

Survey Program: 0-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
0.0	0.0	0.0	0.0	0.0	0.0	-154.87	-30.4	-14.3	33.6				
100.0	100.0	100.0	100.0	0.1	0.1	-154.87	-30.4	-14.3	33.6	33.4	0.17	199.178	
200.0	200.0	200.0	200.0	0.3	0.3	-154.87	-30.4	-14.3	33.6	33.0	0.62	54.321	
300.0	300.0	300.0	300.0	0.5	0.5	-154.87	-30.4	-14.3	33.6	32.5	1.07	31.449	
400.0	400.0	400.0	400.0	0.8	0.8	-154.87	-30.4	-14.3	33.6	32.1	1.52	22.131	
500.0	500.0	500.0	500.0	1.0	1.0	-154.87	-30.4	-14.3	33.6	31.6	1.97	17.072	
600.0	600.0	600.0	600.0	1.2	1.2	-154.87	-30.4	-14.3	33.6	31.2	2.42	13.896	
700.0	700.0	700.0	700.0	1.4	1.4	-154.87	-30.4	-14.3	33.6	30.7	2.87	11.716	
800.0	800.0	800.0	800.0	1.7	1.7	-154.87	-30.4	-14.3	33.6	30.3	3.32	10.128	
900.0	900.0	900.0	900.0	1.9	1.9	-154.87	-30.4	-14.3	33.6	29.8	3.76	8.918	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-154.87	-30.4	-14.3	33.6	29.4	4.21	7.967	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-154.87	-30.4	-14.3	33.6	28.9	4.66	7.199	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-154.87	-30.4	-14.3	33.6	28.5	5.11	6.566	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-154.87	-30.4	-14.3	33.6	28.0	5.56	6.036	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-154.87	-30.4	-14.3	33.6	27.6	6.01	5.584	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-154.87	-30.4	-14.3	33.6	27.1	6.46	5.196	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-154.87	-30.4	-14.3	33.6	26.7	6.91	4.858	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-154.87	-30.4	-14.3	33.6	26.2	7.36	4.561	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-154.87	-30.4	-14.3	33.6	25.8	7.81	4.299	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-154.87	-30.4	-14.3	33.6	25.3	8.26	4.065	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-154.87	-30.4	-14.3	33.6	24.9	8.71	3.855	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-154.87	-30.4	-14.3	33.6	24.4	9.16	3.666	
2,200.0	2,200.0	2,200.0	2,200.0	4.8	4.8	-154.87	-30.4	-14.3	33.6	24.0	9.61	3.494	
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.0	-154.87	-30.4	-14.3	33.6	23.5	10.06	3.338	
2,400.0	2,400.0	2,400.0	2,400.0	5.3	5.3	-154.87	-30.4	-14.3	33.6	23.1	10.51	3.195	
2,500.0	2,500.0	2,500.0	2,500.0	5.5	5.5	-154.87	-30.4	-14.3	33.6	22.6	10.96	3.064	CC
2,600.0	2,600.0	2,599.0	2,599.0	5.7	5.7	-65.84	-31.3	-15.7	34.3	22.9	11.37	3.013	
2,650.0	2,649.9	2,648.5	2,648.5	5.8	5.8	-66.99	-32.3	-17.6	35.1	23.6	11.56	3.038	
2,700.0	2,699.9	2,698.5	2,698.4	5.9	5.9	-68.31	-33.6	-19.9	36.2	24.4	11.76	3.079	
2,800.0	2,799.7	2,798.5	2,798.2	6.1	6.1	-70.73	-36.2	-24.4	38.4	26.3	12.15	3.160	
2,900.0	2,899.6	2,898.4	2,898.0	6.3	6.3	-72.87	-38.9	-28.9	40.7	28.1	12.56	3.239	
3,000.0	2,999.5	2,998.4	2,997.9	6.5	6.5	-74.79	-41.5	-33.5	43.0	30.0	12.97	3.316	
3,100.0	3,099.3	3,098.4	3,097.7	6.7	6.7	-76.51	-44.1	-38.0	45.4	32.0	13.38	3.389	
3,200.0	3,199.2	3,198.3	3,197.5	7.0	6.9	-78.06	-46.7	-42.5	47.8	34.0	13.80	3.460	
3,300.0	3,299.0	3,298.3	3,297.3	7.2	7.1	-79.46	-49.3	-47.0	50.2	36.0	14.23	3.527	
3,400.0	3,398.9	3,398.3	3,397.2	7.4	7.3	-80.73	-51.9	-51.6	52.6	38.0	14.66	3.591	
3,501.2	3,500.0	3,499.5	3,498.2	7.6	7.5	-81.90	-54.6	-56.2	55.2	40.1	15.10	3.653	
3,600.0	3,598.7	3,598.2	3,596.8	7.8	7.7	-81.33	-57.2	-60.6	57.9	42.3	15.51	3.730	
3,651.2	3,649.9	3,649.4	3,647.9	7.9	7.8	-169.78	-58.5	-63.0	59.5	43.9	15.59	3.817	
3,700.0	3,698.7	3,698.1	3,696.6	8.0	7.9	-167.95	-59.8	-65.2	61.2	45.4	15.78	3.876	
3,800.0	3,798.7	3,797.9	3,796.3	8.2	8.2	-164.52	-62.4	-69.7	64.8	48.6	16.20	3.999	
3,900.0	3,898.7	3,899.5	3,897.8	8.5	8.4	-162.34	-64.2	-72.8	67.4	50.8	16.63	4.054	
4,000.0	3,998.7	4,000.4	3,998.7	8.7	8.6	-162.07	-64.5	-73.3	67.8	50.7	17.06	3.971	
4,100.0	4,098.7	4,100.4	4,098.7	8.9	8.8	-162.07	-64.5	-73.3	67.8	50.2	17.50	3.871	
4,200.0	4,198.7	4,200.4	4,198.7	9.1	9.0	-162.07	-64.5	-73.3	67.8	49.8	17.94	3.776	
4,300.0	4,298.7	4,300.4	4,298.7	9.4	9.2	-162.07	-64.5	-73.3	67.8	49.4	18.39	3.685	
4,400.0	4,398.7	4,400.4	4,398.7	9.6	9.4	-162.07	-64.5	-73.3	67.8	48.9	18.83	3.598	
4,500.0	4,498.7	4,500.4	4,498.7	9.8	9.7	-162.07	-64.5	-73.3	67.8	48.5	19.27	3.515	
4,600.0	4,598.7	4,600.4	4,598.7	10.0	9.9	-162.07	-64.5	-73.3	67.8	48.0	19.72	3.436	
4,700.0	4,698.7	4,700.4	4,698.7	10.2	10.1	-162.07	-64.5	-73.3	67.8	47.6	20.16	3.360	
4,800.0	4,798.7	4,800.4	4,798.7	10.5	10.3	-162.07	-64.5	-73.3	67.8	47.1	20.61	3.288	
4,900.0	4,898.7	4,900.4	4,898.7	10.7	10.5	-162.07	-64.5	-73.3	67.8	46.7	21.05	3.218	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
5,000.0	4,998.7	5,000.4	4,998.7	10.9	10.8	-162.07	-64.5	-73.3	67.8	46.3	21.49	3.152	
5,100.0	5,098.7	5,100.4	5,098.7	11.1	11.0	-162.07	-64.5	-73.3	67.8	45.8	21.94	3.088	
5,200.0	5,198.7	5,200.4	5,198.7	11.4	11.2	-162.07	-64.5	-73.3	67.8	45.4	22.38	3.027	
5,300.0	5,298.7	5,300.4	5,298.7	11.6	11.4	-162.07	-64.5	-73.3	67.8	44.9	22.83	2.968	
5,400.0	5,398.7	5,400.4	5,398.7	11.8	11.6	-162.07	-64.5	-73.3	67.8	44.5	23.28	2.911	
5,500.0	5,498.7	5,500.4	5,498.7	12.0	11.9	-162.07	-64.5	-73.3	67.8	44.0	23.72	2.856	
5,600.0	5,598.7	5,600.4	5,598.7	12.2	12.1	-162.07	-64.5	-73.3	67.8	43.6	24.17	2.803	
5,700.0	5,698.7	5,700.4	5,698.7	12.5	12.3	-162.07	-64.5	-73.3	67.8	43.1	24.61	2.753	
5,800.0	5,798.7	5,800.4	5,798.7	12.7	12.5	-162.07	-64.5	-73.3	67.8	42.7	25.06	2.704	
5,900.0	5,898.7	5,900.4	5,898.7	12.9	12.8	-162.07	-64.5	-73.3	67.8	42.2	25.50	2.656	
6,000.0	5,998.7	6,000.4	5,998.7	13.1	13.0	-162.07	-64.5	-73.3	67.8	41.8	25.95	2.611	
6,100.0	6,098.7	6,100.4	6,098.7	13.4	13.2	-162.07	-64.5	-73.3	67.8	41.4	26.40	2.567	
6,200.0	6,198.7	6,200.4	6,198.7	13.6	13.4	-162.07	-64.5	-73.3	67.8	40.9	26.84	2.524	
6,300.0	6,298.7	6,300.4	6,298.7	13.8	13.6	-162.07	-64.5	-73.3	67.8	40.5	27.29	2.483	
6,400.0	6,398.7	6,400.4	6,398.7	14.0	13.9	-162.07	-64.5	-73.3	67.8	40.0	27.74	2.443	
6,500.0	6,498.7	6,500.4	6,498.7	14.3	14.1	-162.07	-64.5	-73.3	67.8	39.6	28.18	2.404	
6,600.0	6,598.7	6,600.4	6,598.7	14.5	14.3	-162.07	-64.5	-73.3	67.8	39.1	28.63	2.366	
6,700.0	6,698.7	6,700.4	6,698.7	14.7	14.5	-162.07	-64.5	-73.3	67.8	38.7	29.08	2.330	
6,800.0	6,798.7	6,800.4	6,798.7	14.9	14.7	-162.07	-64.5	-73.3	67.8	38.2	29.52	2.295	
6,900.0	6,898.7	6,900.4	6,898.7	15.2	15.0	-162.07	-64.5	-73.3	67.8	37.8	29.97	2.260	
7,000.0	6,998.7	7,000.4	6,998.7	15.4	15.2	-162.07	-64.5	-73.3	67.8	37.3	30.42	2.227	
7,100.0	7,098.7	7,100.4	7,098.7	15.6	15.4	-162.07	-64.5	-73.3	67.8	36.9	30.87	2.195	
7,200.0	7,198.7	7,200.4	7,198.7	15.8	15.6	-162.07	-64.5	-73.3	67.8	36.4	31.31	2.164	
7,300.0	7,298.7	7,300.4	7,298.7	16.0	15.9	-162.07	-64.5	-73.3	67.8	36.0	31.76	2.133	
7,400.0	7,398.7	7,400.4	7,398.7	16.3	16.1	-162.07	-64.5	-73.3	67.8	35.5	32.21	2.104	
7,500.0	7,498.7	7,500.4	7,498.7	16.5	16.3	-162.07	-64.5	-73.3	67.8	35.1	32.66	2.075	
7,600.0	7,598.7	7,600.4	7,598.7	16.7	16.5	-162.07	-64.5	-73.3	67.8	34.6	33.10	2.047	
7,700.0	7,698.7	7,700.4	7,698.7	16.9	16.7	-162.07	-64.5	-73.3	67.8	34.2	33.55	2.019	
7,800.0	7,798.7	7,800.4	7,798.7	17.2	17.0	-162.07	-64.5	-73.3	67.8	33.8	34.00	1.993	
7,900.0	7,898.7	7,900.4	7,898.7	17.4	17.2	-162.07	-64.5	-73.3	67.8	33.3	34.45	1.967	
8,000.0	7,998.7	8,000.4	7,998.7	17.6	17.4	-162.07	-64.5	-73.3	67.8	32.9	34.89	1.942	
8,100.0	8,098.7	8,100.4	8,098.7	17.8	17.6	-162.07	-64.5	-73.3	67.8	32.4	35.34	1.917	
8,200.0	8,198.7	8,200.4	8,198.7	18.1	17.9	-162.07	-64.5	-73.3	67.8	32.0	35.79	1.893	
8,300.0	8,298.7	8,300.4	8,298.7	18.3	18.1	-162.07	-64.5	-73.3	67.8	31.5	36.24	1.870	
8,400.0	8,398.7	8,400.4	8,398.7	18.5	18.3	-162.07	-64.5	-73.3	67.8	31.1	36.69	1.847	
8,500.0	8,498.7	8,500.4	8,498.7	18.7	18.5	-162.07	-64.5	-73.3	67.8	30.6	37.13	1.825	
8,600.0	8,598.7	8,600.4	8,598.7	19.0	18.8	-162.07	-64.5	-73.3	67.8	30.2	37.58	1.803	
8,700.0	8,698.7	8,700.4	8,698.7	19.2	19.0	-162.07	-64.5	-73.3	67.8	29.7	38.03	1.782	
8,800.0	8,798.7	8,800.4	8,798.7	19.4	19.2	-162.07	-64.5	-73.3	67.8	29.3	38.48	1.761	
8,900.0	8,898.7	8,900.4	8,898.7	19.6	19.4	-162.07	-64.5	-73.3	67.8	28.8	38.93	1.741	
9,000.0	8,998.7	9,000.4	8,998.7	19.9	19.6	-162.07	-64.5	-73.3	67.8	28.4	39.37	1.721	
9,100.0	9,098.7	9,100.4	9,098.7	20.1	19.9	-162.07	-64.5	-73.3	67.8	27.9	39.82	1.701	
9,200.0	9,198.7	9,200.4	9,198.7	20.3	20.1	-162.07	-64.5	-73.3	67.8	27.5	40.27	1.682	
9,300.0	9,298.7	9,300.4	9,298.7	20.5	20.3	-162.07	-64.5	-73.3	67.8	27.0	40.72	1.664	
9,400.0	9,398.7	9,400.4	9,398.7	20.8	20.5	-162.07	-64.5	-73.3	67.8	26.6	41.17	1.646	
9,500.0	9,498.7	9,500.4	9,498.7	21.0	20.8	-162.07	-64.5	-73.3	67.8	26.1	41.61	1.628	
9,600.0	9,598.7	9,600.4	9,598.7	21.2	21.0	-162.07	-64.5	-73.3	67.8	25.7	42.06	1.611	
9,700.0	9,698.7	9,700.4	9,698.7	21.4	21.2	-162.07	-64.5	-73.3	67.8	25.2	42.51	1.594	
9,800.0	9,798.7	9,800.4	9,798.7	21.6	21.4	-162.07	-64.5	-73.3	67.8	24.8	42.96	1.577	
9,900.0	9,898.7	9,900.4	9,898.7	21.9	21.7	-162.07	-64.5	-73.3	67.8	24.3	43.41	1.561	
10,000.0	9,998.7	10,000.4	9,998.7	22.1	21.9	-162.07	-64.5	-73.3	67.8	23.9	43.86	1.545	
10,100.0	10,098.7	10,100.4	10,098.7	22.3	22.1	-162.07	-64.5	-73.3	67.8	23.4	44.30	1.529	

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
10,200.0	10,198.7	10,200.4	10,198.7	22.5	22.3	-162.07	-64.5	-73.3	67.8	23.0	44.75	1.514	
10,300.0	10,298.7	10,300.4	10,298.7	22.8	22.6	-162.07	-64.5	-73.3	67.8	22.5	45.20	1.499	Level 3
10,336.6	10,335.3	10,337.0	10,335.3	22.9	22.6	-162.07	-64.5	-73.3	67.8	22.4	45.37	1.493	Level 3
10,350.0	10,348.7	10,350.4	10,348.7	22.9	22.7	61.08	-64.5	-73.3	67.7	22.1	45.52	1.486	Level 3
10,375.0	10,373.7	10,374.4	10,372.7	22.9	22.7	62.02	-64.6	-73.2	67.1	21.5	45.59	1.472	Level 3
10,400.0	10,398.5	10,397.2	10,395.5	23.0	22.8	63.24	-65.5	-72.7	66.6	21.0	45.64	1.460	Level 3
10,425.0	10,423.2	10,420.1	10,418.2	23.0	22.8	64.66	-67.4	-71.6	66.4	20.7	45.68	1.454	Level 3
10,435.8	10,433.8	10,430.0	10,428.1	23.0	22.8	65.33	-68.5	-71.0	66.4	20.7	45.69	1.453	Level 3
10,450.0	10,447.6	10,443.0	10,441.0	23.1	22.8	66.26	-70.2	-70.1	66.4	20.7	45.71	1.453	Level 3
10,475.0	10,471.8	10,466.0	10,463.5	23.1	22.9	68.02	-74.0	-68.0	66.7	21.0	45.75	1.458	Level 3
10,500.0	10,495.5	10,489.1	10,485.9	23.1	22.9	69.91	-78.8	-65.3	67.3	21.5	45.80	1.469	Level 3
10,525.0	10,518.8	10,512.2	10,508.1	23.2	23.0	71.90	-84.5	-62.2	68.2	22.3	45.86	1.487	Level 3
10,550.0	10,541.7	10,535.3	10,529.9	23.2	23.0	73.95	-91.1	-58.5	69.4	23.5	45.94	1.511	
10,575.0	10,563.9	10,558.5	10,551.4	23.3	23.0	76.03	-98.7	-54.3	71.0	24.9	46.04	1.541	
10,600.0	10,585.5	10,581.7	10,572.5	23.3	23.1	78.11	-107.3	-49.5	72.9	26.7	46.15	1.579	
10,625.0	10,606.5	10,605.1	10,593.2	23.4	23.1	80.15	-116.7	-44.3	75.1	28.8	46.28	1.623	
10,650.0	10,626.7	10,628.4	10,613.4	23.4	23.2	82.12	-127.0	-38.6	77.7	31.3	46.41	1.675	
10,675.0	10,646.1	10,651.8	10,633.0	23.5	23.2	84.01	-138.2	-32.4	80.7	34.1	46.56	1.733	
10,700.0	10,664.6	10,675.0	10,651.7	23.6	23.2	85.75	-150.1	-25.8	84.0	37.3	46.71	1.798	
10,725.0	10,682.3	10,698.9	10,670.3	23.6	23.3	87.44	-163.2	-18.5	87.6	40.8	46.87	1.870	
10,750.0	10,698.9	10,722.5	10,688.0	23.7	23.3	88.96	-176.9	-10.9	91.6	44.5	47.03	1.947	
10,775.0	10,714.6	10,746.2	10,704.9	23.8	23.4	90.34	-191.4	-2.9	95.8	48.6	47.20	2.030	
10,800.0	10,729.3	10,769.9	10,721.0	23.9	23.5	91.60	-206.7	5.6	100.4	53.0	47.37	2.119	
10,825.0	10,742.9	10,793.8	10,736.2	24.0	23.6	92.72	-222.7	14.5	105.2	57.6	47.56	2.212	
10,850.0	10,755.3	10,817.7	10,750.6	24.2	23.7	93.71	-239.5	23.7	110.2	62.5	47.76	2.308	
10,875.0	10,766.6	10,841.8	10,764.1	24.3	23.8	94.58	-256.9	33.4	115.5	67.5	47.97	2.408	
10,900.0	10,776.8	10,865.9	10,776.6	24.5	23.9	95.33	-274.9	43.4	121.0	72.8	48.21	2.510	
10,925.0	10,785.7	10,890.1	10,788.1	24.7	24.0	95.97	-293.6	53.8	126.7	78.2	48.47	2.614	
10,950.0	10,793.4	10,914.5	10,798.5	24.9	24.2	96.51	-312.8	64.4	132.5	83.8	48.75	2.718	
10,975.0	10,799.8	10,939.0	10,807.9	25.1	24.3	96.96	-332.6	75.4	138.5	89.5	49.06	2.823	
11,000.0	10,804.9	10,963.6	10,816.1	25.3	24.5	97.33	-352.9	86.6	144.6	95.2	49.40	2.928	
11,025.0	10,808.8	10,988.4	10,823.1	25.5	24.7	97.61	-373.7	98.2	150.9	101.1	49.77	3.031	
11,050.0	10,811.4	11,013.3	10,828.9	25.8	24.9	97.83	-394.9	109.9	157.2	107.0	50.17	3.133	
11,075.0	10,812.6	11,038.4	10,833.5	26.0	25.1	97.97	-416.4	121.8	163.6	113.0	50.59	3.233	
11,084.1	10,812.8	11,047.5	10,834.9	26.1	25.1	98.01	-424.4	126.2	165.9	115.2	50.75	3.269	
11,099.1	10,812.8	11,062.7	10,836.7	26.3	25.3	98.45	-437.6	133.6	169.7	118.7	50.99	3.329	
11,200.0	10,813.4	11,166.6	10,839.7	27.5	26.3	98.11	-528.1	184.2	196.4	143.3	53.10	3.698	
11,300.0	10,813.9	11,274.6	10,840.3	28.9	27.6	97.14	-619.9	241.2	223.2	167.6	55.65	4.011	
11,400.0	10,814.5	11,384.5	10,840.9	30.6	29.1	96.40	-709.8	304.4	249.7	191.2	58.50	4.268	
11,500.0	10,815.0	11,496.4	10,841.6	32.4	30.8	95.81	-797.4	373.9	275.6	214.0	61.60	4.474	
11,600.0	10,815.6	11,610.2	10,842.2	34.3	32.8	95.33	-882.2	449.8	300.9	236.0	64.92	4.635	
11,700.0	10,816.1	11,726.1	10,842.9	36.3	34.9	94.94	-963.7	532.0	325.5	257.0	68.42	4.757	
11,800.0	10,816.7	11,843.9	10,843.6	38.5	37.2	94.62	-1,041.4	620.7	349.2	277.1	72.05	4.846	
11,900.0	10,817.3	11,963.9	10,844.3	40.7	39.7	94.35	-1,114.7	715.6	372.0	296.2	75.77	4.909	
12,000.0	10,817.9	12,085.9	10,845.1	43.0	42.3	94.13	-1,183.0	816.7	393.7	314.1	79.53	4.950	
12,100.0	10,818.4	12,209.9	10,845.8	45.3	45.1	93.94	-1,245.7	923.7	414.2	330.9	83.32	4.971	
12,200.0	10,819.0	12,336.0	10,846.6	47.6	47.9	93.78	-1,302.1	1,036.4	433.5	346.4	87.10	4.977	
12,300.0	10,819.5	12,464.0	10,847.4	50.0	50.8	93.64	-1,351.6	1,154.4	451.4	360.5	90.83	4.969	
12,400.0	10,820.1	12,593.8	10,848.1	52.3	53.8	93.53	-1,393.7	1,277.2	467.8	373.3	94.47	4.952	
12,500.0	10,820.6	12,725.4	10,848.9	54.7	56.9	93.43	-1,427.8	1,404.2	482.6	384.6	98.00	4.924	
12,600.0	10,821.2	12,858.6	10,849.7	57.0	59.9	93.36	-1,453.2	1,534.9	495.7	394.3	101.40	4.889	
12,655.6	10,821.5	12,933.3	10,850.1	58.3	61.6	93.32	-1,463.5	1,609.0	502.2	399.0	103.21	4.866	

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
12,700.0	10,821.7	12,993.3	10,850.4	59.4	63.0	93.28	-1,469.7	1,668.6	506.6	400.7	105.88	4.784	
12,800.0	10,822.2	13,129.4	10,851.1	61.8	66.0	93.25	-1,476.7	1,804.6	511.2	399.1	112.02	4.563	
12,900.0	10,822.7	13,235.4	10,851.7	64.2	68.4	93.25	-1,477.2	1,910.5	511.1	393.7	117.44	4.352	
13,000.0	10,823.3	13,335.4	10,852.2	66.6	70.7	93.25	-1,477.6	2,010.5	511.0	388.2	122.76	4.163	
13,100.0	10,823.8	13,435.4	10,852.7	69.2	73.0	93.25	-1,478.0	2,110.5	510.9	382.8	128.13	3.987	
13,200.0	10,824.3	13,535.4	10,853.3	71.7	75.4	93.25	-1,478.5	2,210.5	510.8	377.2	133.54	3.825	
13,300.0	10,824.8	13,635.4	10,853.8	74.3	77.8	93.25	-1,478.9	2,310.5	510.6	371.7	138.99	3.674	
13,400.0	10,825.4	13,735.4	10,854.3	76.8	80.2	93.25	-1,479.3	2,410.5	510.5	366.1	144.47	3.534	
13,500.0	10,825.9	13,835.4	10,854.8	79.5	82.7	93.25	-1,479.7	2,510.5	510.4	360.4	149.98	3.403	
13,600.0	10,826.4	13,935.4	10,855.4	82.1	85.2	93.25	-1,480.1	2,610.5	510.3	354.8	155.51	3.281	
13,700.0	10,826.9	14,035.4	10,855.9	84.7	87.7	93.25	-1,480.5	2,710.5	510.2	349.1	161.06	3.167	
13,800.0	10,827.5	14,135.4	10,856.4	87.4	90.2	93.25	-1,480.9	2,810.5	510.0	343.4	166.64	3.061	
13,900.0	10,828.0	14,235.4	10,856.9	90.1	92.8	93.25	-1,481.3	2,910.5	509.9	337.7	172.23	2.961	
14,000.0	10,828.5	14,335.4	10,857.5	92.8	95.4	93.26	-1,481.7	3,010.5	509.8	331.9	177.85	2.867	
14,100.0	10,829.0	14,435.4	10,858.0	95.5	98.0	93.26	-1,482.1	3,110.5	509.7	326.2	183.47	2.778	
14,200.0	10,829.6	14,535.4	10,858.5	98.3	100.7	93.26	-1,482.5	3,210.5	509.6	320.4	189.11	2.694	
14,300.0	10,830.1	14,635.4	10,859.0	101.0	103.3	93.26	-1,482.9	3,310.5	509.4	314.7	194.76	2.616	
14,400.0	10,830.6	14,735.4	10,859.6	103.8	106.0	93.26	-1,483.3	3,410.5	509.3	308.9	200.43	2.541	
14,500.0	10,831.1	14,835.4	10,860.1	106.5	108.7	93.26	-1,483.8	3,510.5	509.2	303.1	206.10	2.471	
14,600.0	10,831.6	14,935.4	10,860.6	109.3	111.3	93.26	-1,484.2	3,610.5	509.1	297.3	211.79	2.404	
14,700.0	10,832.2	15,035.4	10,861.1	112.1	114.0	93.26	-1,484.6	3,710.5	508.9	291.5	217.48	2.340	
14,800.0	10,832.7	15,135.4	10,861.6	114.9	116.8	93.26	-1,485.0	3,810.5	508.8	285.6	223.18	2.280	
14,900.0	10,833.2	15,235.4	10,862.2	117.6	119.5	93.26	-1,485.4	3,910.5	508.7	279.8	228.89	2.222	
15,000.0	10,833.7	15,335.4	10,862.7	120.4	122.2	93.26	-1,485.8	4,010.5	508.6	274.0	234.61	2.168	
15,100.0	10,834.3	15,435.4	10,863.2	123.2	125.0	93.26	-1,486.2	4,110.5	508.5	268.1	240.33	2.116	
15,200.0	10,834.8	15,535.4	10,863.7	126.1	127.7	93.26	-1,486.6	4,210.5	508.3	262.3	246.06	2.066	
15,300.0	10,835.3	15,635.4	10,864.3	128.9	130.5	93.27	-1,487.0	4,310.5	508.2	256.4	251.80	2.018	
15,400.0	10,835.8	15,735.4	10,864.8	131.7	133.2	93.27	-1,487.4	4,410.5	508.1	250.6	257.54	1.973	
15,500.0	10,836.4	15,835.4	10,865.3	134.5	136.0	93.27	-1,487.8	4,510.4	508.0	244.7	263.29	1.929	
15,600.0	10,836.9	15,935.4	10,865.8	137.4	138.8	93.27	-1,488.2	4,610.4	507.9	238.8	269.04	1.888	
15,700.0	10,837.4	16,035.4	10,866.4	140.2	141.6	93.27	-1,488.6	4,710.4	507.7	232.9	274.79	1.848	
15,800.0	10,837.9	16,135.4	10,866.9	143.0	144.4	93.27	-1,489.1	4,810.4	507.6	227.1	280.55	1.809	
15,900.0	10,838.5	16,235.4	10,867.4	145.9	147.2	93.27	-1,489.5	4,910.4	507.5	221.2	286.31	1.772	
16,000.0	10,839.0	16,335.4	10,867.9	148.7	150.0	93.27	-1,489.9	5,010.4	507.4	215.3	292.08	1.737	
16,100.0	10,839.5	16,435.4	10,868.5	151.6	152.8	93.27	-1,490.3	5,110.4	507.2	209.4	297.85	1.703	
16,200.0	10,840.0	16,535.4	10,869.0	154.4	155.6	93.27	-1,490.7	5,210.4	507.1	203.5	303.62	1.670	
16,300.0	10,840.5	16,635.4	10,869.5	157.3	158.4	93.27	-1,491.1	5,310.4	507.0	197.6	309.40	1.639	
16,400.0	10,841.1	16,735.4	10,870.0	160.1	161.2	93.27	-1,491.5	5,410.4	506.9	191.7	315.18	1.608	
16,500.0	10,841.6	16,835.4	10,870.5	163.0	164.1	93.27	-1,491.9	5,510.4	506.8	185.8	320.96	1.579	
16,600.0	10,842.1	16,935.4	10,871.1	165.9	166.9	93.28	-1,492.3	5,610.4	506.6	179.9	326.74	1.551	
16,700.0	10,842.6	17,035.4	10,871.6	168.7	169.7	93.28	-1,492.7	5,710.4	506.5	174.0	332.53	1.523	
16,800.0	10,843.2	17,135.4	10,872.1	171.6	172.6	93.28	-1,493.1	5,810.4	506.4	168.1	338.32	1.497	Level 3
16,900.0	10,843.7	17,235.4	10,872.6	174.5	175.4	93.28	-1,493.5	5,910.4	506.3	162.2	344.11	1.471	Level 3
17,000.0	10,844.2	17,335.4	10,873.2	177.3	178.2	93.28	-1,493.9	6,010.4	506.1	156.2	349.90	1.447	Level 3
17,100.0	10,844.7	17,435.4	10,873.7	180.2	181.1	93.28	-1,494.3	6,110.4	506.0	150.3	355.70	1.423	Level 3
17,200.0	10,845.3	17,535.4	10,874.2	183.1	183.9	93.28	-1,494.8	6,210.4	505.9	144.4	361.49	1.399	Level 3
17,300.0	10,845.8	17,635.4	10,874.7	186.0	186.8	93.28	-1,495.2	6,310.4	505.8	138.5	367.29	1.377	Level 3
17,400.0	10,846.3	17,735.4	10,875.3	188.8	189.6	93.28	-1,495.6	6,410.4	505.7	132.6	373.09	1.355	Level 3
17,500.0	10,846.8	17,835.4	10,875.8	191.7	192.5	93.28	-1,496.0	6,510.4	505.5	126.6	378.90	1.334	Level 3
17,600.0	10,847.4	17,935.4	10,876.3	194.6	195.3	93.28	-1,496.4	6,610.4	505.4	120.7	384.70	1.314	Level 3
17,700.0	10,847.9	18,035.4	10,876.8	197.5	198.2	93.28	-1,496.8	6,710.4	505.3	114.8	390.50	1.294	Level 3
17,800.0	10,848.4	18,135.4	10,877.4	200.4	201.1	93.28	-1,497.2	6,810.4	505.2	108.9	396.31	1.275	Level 3

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	TVD Reference:	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 11-18 3T	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Wellbore #1	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #6	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 4T2 - Wellbore #1 - Design #6 — Offset Site Error: 0.0 usft

Survey Program: 0-MWD — Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
17,900.0	10,848.9	18,235.4	10,877.9	203.3	203.9	93.29	-1,497.6	6,910.4	505.1	102.9	402.12	1.256	Level 3
18,000.0	10,849.5	18,335.4	10,878.4	206.2	206.8	93.29	-1,498.0	7,010.4	504.9	97.0	407.93	1.238	Level 2
18,100.0	10,850.0	18,435.4	10,878.9	209.0	209.7	93.29	-1,498.4	7,110.4	504.8	91.1	413.74	1.220	Level 2
18,200.0	10,850.5	18,535.4	10,879.4	211.9	212.5	93.29	-1,498.8	7,210.4	504.7	85.1	419.55	1.203	Level 2
18,300.0	10,851.0	18,635.4	10,880.0	214.8	215.4	93.29	-1,499.2	7,310.4	504.6	79.2	425.36	1.186	Level 2
18,400.0	10,851.5	18,735.4	10,880.5	217.7	218.3	93.29	-1,499.6	7,410.4	504.4	73.3	431.18	1.170	Level 2
18,500.0	10,852.1	18,835.4	10,881.0	220.6	221.1	93.29	-1,500.1	7,510.4	504.3	67.3	436.99	1.154	Level 2
18,600.0	10,852.6	18,935.4	10,881.5	223.5	224.0	93.29	-1,500.5	7,610.4	504.2	61.4	442.81	1.139	Level 2
18,700.0	10,853.1	19,035.4	10,882.1	226.4	226.9	93.29	-1,500.9	7,710.4	504.1	55.5	448.63	1.124	Level 2
18,800.0	10,853.6	19,135.4	10,882.6	229.3	229.8	93.29	-1,501.3	7,810.4	504.0	49.5	454.45	1.109	Level 2
18,900.0	10,854.2	19,235.4	10,883.1	232.2	232.7	93.29	-1,501.7	7,910.4	503.8	43.6	460.26	1.095	Level 2
19,000.0	10,854.7	19,335.4	10,883.6	235.1	235.5	93.29	-1,502.1	8,010.4	503.7	37.6	466.08	1.081	Level 2
19,100.0	10,855.2	19,435.4	10,884.2	238.0	238.4	93.30	-1,502.5	8,110.4	503.6	31.7	471.91	1.067	Level 2
19,200.0	10,855.7	19,535.4	10,884.7	240.9	241.3	93.30	-1,502.9	8,210.4	503.5	25.8	477.73	1.054	Level 2
19,300.0	10,856.3	19,635.4	10,885.2	243.8	244.2	93.30	-1,503.3	8,310.4	503.4	19.8	483.55	1.041	Level 2
19,400.0	10,856.8	19,735.4	10,885.7	246.7	247.1	93.30	-1,503.7	8,410.4	503.2	13.9	489.37	1.028	Level 2
19,500.0	10,857.3	19,835.4	10,886.3	249.6	250.0	93.30	-1,504.1	8,510.4	503.1	7.9	495.20	1.016	Level 2
19,600.0	10,857.8	19,935.4	10,886.8	252.5	252.8	93.30	-1,504.5	8,610.4	503.0	2.0	501.02	1.004	Level 2
19,700.0	10,858.4	20,035.4	10,887.3	255.4	255.7	93.30	-1,504.9	8,710.4	502.9	-4.0	506.85	0.992	Level 1
19,800.0	10,858.9	20,135.4	10,887.8	258.3	258.6	93.30	-1,505.4	8,810.4	502.7	-9.9	512.67	0.981	Level 1
19,900.0	10,859.4	20,235.4	10,888.3	261.2	261.5	93.30	-1,505.8	8,910.3	502.6	-15.9	518.50	0.969	Level 1
20,000.0	10,859.9	20,335.4	10,888.9	264.1	264.4	93.30	-1,506.2	9,010.3	502.5	-21.8	524.33	0.958	Level 1
20,100.0	10,860.4	20,435.4	10,889.4	267.0	267.3	93.30	-1,506.6	9,110.3	502.4	-27.8	530.15	0.948	Level 1
20,200.0	10,861.0	20,535.4	10,889.9	269.9	270.2	93.30	-1,507.0	9,210.3	502.3	-33.7	535.98	0.937	Level 1
20,300.0	10,861.5	20,635.4	10,890.4	272.8	273.1	93.30	-1,507.4	9,310.3	502.1	-39.7	541.81	0.927	Level 1
20,400.0	10,862.0	20,735.4	10,891.0	275.7	276.0	93.31	-1,507.8	9,410.3	502.0	-45.6	547.64	0.917	Level 1
20,500.0	10,862.5	20,835.4	10,891.5	278.6	278.9	93.31	-1,508.2	9,510.3	501.9	-51.6	553.47	0.907	Level 1
20,600.0	10,863.1	20,935.4	10,892.0	281.6	281.8	93.31	-1,508.6	9,610.3	501.8	-57.5	559.30	0.897	Level 1
20,700.0	10,863.6	21,035.4	10,892.5	284.5	284.7	93.31	-1,509.0	9,710.3	501.7	-63.5	565.13	0.888	Level 1
20,800.0	10,864.1	21,135.4	10,893.1	287.4	287.6	93.31	-1,509.4	9,810.3	501.5	-69.4	570.96	0.878	Level 1
20,900.0	10,864.6	21,235.4	10,893.6	290.3	290.5	93.31	-1,509.8	9,910.3	501.4	-75.4	576.79	0.869	Level 1
21,000.0	10,865.2	21,335.4	10,894.1	293.2	293.4	93.31	-1,510.2	10,010.3	501.3	-81.3	582.62	0.860	Level 1
21,038.5	10,865.4	21,373.9	10,894.3	294.3	294.5	93.31	-1,510.4	10,048.8	501.2	-83.6	584.87	0.857	Level 1, ES, SF

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
0.0	0.0	0.0	0.0	0.0	0.0	-155.03	-59.8	-27.8	65.9				
100.0	100.0	100.0	100.0	0.1	0.1	-155.03	-59.8	-27.8	65.9	65.8	0.17	391.195	
200.0	200.0	200.0	200.0	0.3	0.3	-155.03	-59.8	-27.8	65.9	65.3	0.62	106.690	
300.0	300.0	300.0	300.0	0.5	0.5	-155.03	-59.8	-27.8	65.9	64.9	1.07	61.768	
400.0	400.0	400.0	400.0	0.8	0.8	-155.03	-59.8	-27.8	65.9	64.4	1.52	43.466	
500.0	500.0	500.0	500.0	1.0	1.0	-155.03	-59.8	-27.8	65.9	64.0	1.97	33.531	
600.0	600.0	600.0	600.0	1.2	1.2	-155.03	-59.8	-27.8	65.9	63.5	2.42	27.293	
700.0	700.0	700.0	700.0	1.4	1.4	-155.03	-59.8	-27.8	65.9	63.1	2.87	23.012	
800.0	800.0	800.0	800.0	1.7	1.7	-155.03	-59.8	-27.8	65.9	62.6	3.32	19.891	
900.0	900.0	900.0	900.0	1.9	1.9	-155.03	-59.8	-27.8	65.9	62.2	3.76	17.516	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-155.03	-59.8	-27.8	65.9	61.7	4.21	15.648	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-155.03	-59.8	-27.8	65.9	61.3	4.66	14.140	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-155.03	-59.8	-27.8	65.9	60.8	5.11	12.897	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-155.03	-59.8	-27.8	65.9	60.4	5.56	11.854	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-155.03	-59.8	-27.8	65.9	59.9	6.01	10.968	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-155.03	-59.8	-27.8	65.9	59.5	6.46	10.205	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-155.03	-59.8	-27.8	65.9	59.0	6.91	9.541	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-155.03	-59.8	-27.8	65.9	58.6	7.36	8.959	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-155.03	-59.8	-27.8	65.9	58.1	7.81	8.443	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-155.03	-59.8	-27.8	65.9	57.7	8.26	7.984	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-155.03	-59.8	-27.8	65.9	57.2	8.71	7.572	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-155.03	-59.8	-27.8	65.9	56.8	9.16	7.200	
2,200.0	2,200.0	2,200.0	2,200.0	4.8	4.8	-155.03	-59.8	-27.8	65.9	56.3	9.61	6.863	
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.0	-155.03	-59.8	-27.8	65.9	55.9	10.06	6.556	
2,400.0	2,400.0	2,400.0	2,400.0	5.3	5.3	-155.03	-59.8	-27.8	65.9	55.4	10.51	6.276	
2,500.0	2,500.0	2,500.0	2,500.0	5.5	5.5	-155.03	-59.8	-27.8	65.9	55.0	10.96	6.018	CC
2,600.0	2,600.0	2,598.3	2,598.3	5.7	5.7	-67.39	-61.4	-27.3	66.5	55.1	11.36	5.851	
2,650.0	2,649.9	2,647.3	2,647.3	5.8	5.8	-70.26	-63.3	-26.5	67.3	55.8	11.55	5.830	
2,700.0	2,699.9	2,697.1	2,697.0	5.9	5.8	-73.70	-65.8	-25.7	68.6	56.8	11.73	5.843	
2,800.0	2,799.7	2,796.7	2,796.5	6.1	6.0	-80.17	-70.7	-23.9	71.8	59.7	12.12	5.924	
2,900.0	2,899.6	2,896.4	2,896.0	6.3	6.2	-86.02	-75.6	-22.1	75.8	63.3	12.51	6.064	
3,000.0	2,999.5	2,996.0	2,995.5	6.5	6.4	-91.23	-80.5	-20.3	80.6	67.7	12.90	6.248	
3,100.0	3,099.3	3,095.6	3,094.9	6.7	6.6	-95.83	-85.4	-18.5	86.0	72.7	13.30	6.464	
3,200.0	3,199.2	3,195.3	3,194.4	7.0	6.8	-99.87	-90.3	-16.7	91.8	78.1	13.70	6.701	
3,300.0	3,299.0	3,294.9	3,293.9	7.2	7.0	-103.41	-95.2	-15.0	98.1	84.0	14.11	6.951	
3,400.0	3,398.9	3,394.5	3,393.4	7.4	7.2	-106.52	-100.1	-13.2	104.6	90.1	14.51	7.209	
3,501.2	3,500.0	3,495.4	3,494.2	7.6	7.4	-109.28	-105.0	-11.4	111.6	96.6	14.93	7.472	
3,600.0	3,598.7	3,597.8	3,596.5	7.8	7.6	-111.06	-108.6	-10.1	116.4	101.0	15.35	7.583	
3,651.2	3,649.9	3,651.2	3,649.9	7.9	7.7	158.71	-109.0	-9.9	117.0	101.4	15.59	7.506	
3,700.0	3,698.7	3,700.0	3,698.7	8.0	7.8	158.71	-109.0	-9.9	117.0	101.2	15.80	7.407	
3,800.0	3,798.7	3,800.0	3,798.7	8.2	8.0	158.71	-109.0	-9.9	117.0	100.8	16.24	7.206	
3,900.0	3,898.7	3,900.0	3,898.7	8.5	8.2	158.71	-109.0	-9.9	117.0	100.3	16.68	7.016	
4,000.0	3,998.7	4,000.0	3,998.7	8.7	8.5	158.71	-109.0	-9.9	117.0	99.9	17.12	6.835	
4,100.0	4,098.7	4,100.0	4,098.7	8.9	8.7	158.71	-109.0	-9.9	117.0	99.4	17.56	6.663	
4,200.0	4,198.7	4,200.0	4,198.7	9.1	8.9	158.71	-109.0	-9.9	117.0	99.0	18.00	6.499	
4,300.0	4,298.7	4,300.0	4,298.7	9.4	9.1	158.71	-109.0	-9.9	117.0	98.6	18.45	6.343	
4,400.0	4,398.7	4,400.0	4,398.7	9.6	9.3	158.71	-109.0	-9.9	117.0	98.1	18.89	6.194	
4,500.0	4,498.7	4,500.0	4,498.7	9.8	9.6	158.71	-109.0	-9.9	117.0	97.7	19.33	6.052	
4,600.0	4,598.7	4,600.0	4,598.7	10.0	9.8	158.71	-109.0	-9.9	117.0	97.2	19.78	5.916	
4,700.0	4,698.7	4,700.0	4,698.7	10.2	10.0	158.71	-109.0	-9.9	117.0	96.8	20.22	5.786	
4,800.0	4,798.7	4,800.0	4,798.7	10.5	10.2	158.71	-109.0	-9.9	117.0	96.3	20.67	5.662	
4,900.0	4,898.7	4,900.0	4,898.7	10.7	10.4	158.71	-109.0	-9.9	117.0	95.9	21.11	5.543	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6

Offset Site Error: 0.0 usft

Survey Program: 0-MWD

Offset Well Error: 0.0 usft

Reference		Offset		Semi Major Axis					Distance				
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
5,000.0	4,998.7	5,000.0	4,998.7	10.9	10.7	158.71	-109.0	-9.9	117.0	95.5	21.56	5.428	
5,100.0	5,098.7	5,100.0	5,098.7	11.1	10.9	158.71	-109.0	-9.9	117.0	95.0	22.00	5.319	
5,200.0	5,198.7	5,200.0	5,198.7	11.4	11.1	158.71	-109.0	-9.9	117.0	94.6	22.45	5.213	
5,300.0	5,298.7	5,300.0	5,298.7	11.6	11.3	158.71	-109.0	-9.9	117.0	94.1	22.89	5.112	
5,400.0	5,398.7	5,400.0	5,398.7	11.8	11.6	158.71	-109.0	-9.9	117.0	93.7	23.34	5.014	
5,500.0	5,498.7	5,500.0	5,498.7	12.0	11.8	158.71	-109.0	-9.9	117.0	93.2	23.78	4.920	
5,600.0	5,598.7	5,600.0	5,598.7	12.2	12.0	158.71	-109.0	-9.9	117.0	92.8	24.23	4.830	
5,700.0	5,698.7	5,700.0	5,698.7	12.5	12.2	158.71	-109.0	-9.9	117.0	92.3	24.67	4.742	
5,800.0	5,798.7	5,800.0	5,798.7	12.7	12.4	158.71	-109.0	-9.9	117.0	91.9	25.12	4.658	
5,900.0	5,898.7	5,900.0	5,898.7	12.9	12.7	158.71	-109.0	-9.9	117.0	91.4	25.56	4.577	
6,000.0	5,998.7	6,000.0	5,998.7	13.1	12.9	158.71	-109.0	-9.9	117.0	91.0	26.01	4.498	
6,100.0	6,098.7	6,100.0	6,098.7	13.4	13.1	158.71	-109.0	-9.9	117.0	90.6	26.46	4.423	
6,200.0	6,198.7	6,200.0	6,198.7	13.6	13.3	158.71	-109.0	-9.9	117.0	90.1	26.90	4.349	
6,300.0	6,298.7	6,300.0	6,298.7	13.8	13.6	158.71	-109.0	-9.9	117.0	89.7	27.35	4.278	
6,400.0	6,398.7	6,400.0	6,398.7	14.0	13.8	158.71	-109.0	-9.9	117.0	89.2	27.80	4.209	
6,500.0	6,498.7	6,500.0	6,498.7	14.3	14.0	158.71	-109.0	-9.9	117.0	88.8	28.24	4.143	
6,600.0	6,598.7	6,600.0	6,598.7	14.5	14.2	158.71	-109.0	-9.9	117.0	88.3	28.69	4.078	
6,700.0	6,698.7	6,700.0	6,698.7	14.7	14.4	158.71	-109.0	-9.9	117.0	87.9	29.14	4.016	
6,800.0	6,798.7	6,800.0	6,798.7	14.9	14.7	158.71	-109.0	-9.9	117.0	87.4	29.58	3.955	
6,900.0	6,898.7	6,900.0	6,898.7	15.2	14.9	158.71	-109.0	-9.9	117.0	87.0	30.03	3.896	
7,000.0	6,998.7	7,000.0	6,998.7	15.4	15.1	158.71	-109.0	-9.9	117.0	86.5	30.48	3.839	
7,100.0	7,098.7	7,100.0	7,098.7	15.6	15.3	158.71	-109.0	-9.9	117.0	86.1	30.93	3.784	
7,200.0	7,198.7	7,200.0	7,198.7	15.8	15.6	158.71	-109.0	-9.9	117.0	85.6	31.37	3.730	
7,300.0	7,298.7	7,300.0	7,298.7	16.0	15.8	158.71	-109.0	-9.9	117.0	85.2	31.82	3.677	
7,400.0	7,398.7	7,400.0	7,398.7	16.3	16.0	158.71	-109.0	-9.9	117.0	84.7	32.27	3.626	
7,500.0	7,498.7	7,500.0	7,498.7	16.5	16.2	158.71	-109.0	-9.9	117.0	84.3	32.71	3.577	
7,600.0	7,598.7	7,600.0	7,598.7	16.7	16.5	158.71	-109.0	-9.9	117.0	83.8	33.16	3.528	
7,700.0	7,698.7	7,700.0	7,698.7	16.9	16.7	158.71	-109.0	-9.9	117.0	83.4	33.61	3.481	
7,800.0	7,798.7	7,800.0	7,798.7	17.2	16.9	158.71	-109.0	-9.9	117.0	83.0	34.06	3.436	
7,900.0	7,898.7	7,900.0	7,898.7	17.4	17.1	158.71	-109.0	-9.9	117.0	82.5	34.51	3.391	
8,000.0	7,998.7	8,000.0	7,998.7	17.6	17.3	158.71	-109.0	-9.9	117.0	82.1	34.95	3.348	
8,100.0	8,098.7	8,100.0	8,098.7	17.8	17.6	158.71	-109.0	-9.9	117.0	81.6	35.40	3.305	
8,200.0	8,198.7	8,200.0	8,198.7	18.1	17.8	158.71	-109.0	-9.9	117.0	81.2	35.85	3.264	
8,300.0	8,298.7	8,300.0	8,298.7	18.3	18.0	158.71	-109.0	-9.9	117.0	80.7	36.30	3.224	
8,400.0	8,398.7	8,400.0	8,398.7	18.5	18.2	158.71	-109.0	-9.9	117.0	80.3	36.74	3.184	
8,500.0	8,498.7	8,500.0	8,498.7	18.7	18.5	158.71	-109.0	-9.9	117.0	79.8	37.19	3.146	
8,600.0	8,598.7	8,600.0	8,598.7	19.0	18.7	158.71	-109.0	-9.9	117.0	79.4	37.64	3.109	
8,700.0	8,698.7	8,700.0	8,698.7	19.2	18.9	158.71	-109.0	-9.9	117.0	78.9	38.09	3.072	
8,800.0	8,798.7	8,800.0	8,798.7	19.4	19.1	158.71	-109.0	-9.9	117.0	78.5	38.54	3.036	
8,900.0	8,898.7	8,900.0	8,898.7	19.6	19.4	158.71	-109.0	-9.9	117.0	78.0	38.98	3.001	
9,000.0	8,998.7	9,000.0	8,998.7	19.9	19.6	158.71	-109.0	-9.9	117.0	77.6	39.43	2.967	
9,100.0	9,098.7	9,100.0	9,098.7	20.1	19.8	158.71	-109.0	-9.9	117.0	77.1	39.88	2.934	
9,200.0	9,198.7	9,200.0	9,198.7	20.3	20.0	158.71	-109.0	-9.9	117.0	76.7	40.33	2.901	
9,300.0	9,298.7	9,300.0	9,298.7	20.5	20.3	158.71	-109.0	-9.9	117.0	76.2	40.78	2.870	
9,400.0	9,398.7	9,400.0	9,398.7	20.8	20.5	158.71	-109.0	-9.9	117.0	75.8	41.22	2.838	
9,500.0	9,498.7	9,500.0	9,498.7	21.0	20.7	158.71	-109.0	-9.9	117.0	75.3	41.67	2.808	
9,600.0	9,598.7	9,600.0	9,598.7	21.2	20.9	158.71	-109.0	-9.9	117.0	74.9	42.12	2.778	
9,700.0	9,698.7	9,700.0	9,698.7	21.4	21.2	158.71	-109.0	-9.9	117.0	74.4	42.57	2.749	
9,800.0	9,798.7	9,800.0	9,798.7	21.6	21.4	158.71	-109.0	-9.9	117.0	74.0	43.02	2.720	
9,900.0	9,898.7	9,900.0	9,898.7	21.9	21.6	158.71	-109.0	-9.9	117.0	73.5	43.47	2.692	
10,000.0	9,998.7	10,000.0	9,998.7	22.1	21.8	158.71	-109.0	-9.9	117.0	73.1	43.91	2.664	
10,100.0	10,098.7	10,100.0	10,098.7	22.3	22.0	158.71	-109.0	-9.9	117.0	72.6	44.36	2.638	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6

Offset Site Error: 0.0 usft

Survey Program: 0-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
10,200.0	10,198.7	10,200.0	10,198.7	22.5	22.3	158.71	-109.0	-9.9	117.0	72.2	44.81	2.611	
10,249.4	10,248.1	10,249.4	10,248.1	22.7	22.4	158.71	-109.0	-9.9	117.0	72.0	45.03	2.598	
10,300.0	10,298.7	10,294.8	10,293.5	22.8	22.5	158.67	-109.4	-9.7	117.6	72.3	45.24	2.599	
10,336.6	10,335.3	10,325.0	10,323.6	22.9	22.5	158.51	-111.4	-8.5	120.3	74.9	45.38	2.651	
10,350.0	10,348.7	10,334.8	10,333.3	22.9	22.6	21.39	-112.4	-8.0	121.7	76.3	45.39	2.681	
10,375.0	10,373.7	10,354.6	10,352.9	22.9	22.6	21.27	-115.0	-6.5	124.1	78.7	45.39	2.734	
10,400.0	10,398.5	10,375.0	10,372.9	23.0	22.6	21.28	-118.3	-4.6	126.3	81.0	45.32	2.787	
10,425.0	10,423.2	10,394.1	10,391.5	23.0	22.7	21.40	-122.1	-2.4	128.3	83.2	45.17	2.842	
10,450.0	10,447.6	10,413.8	10,410.4	23.1	22.7	21.63	-126.7	0.3	130.2	85.2	44.95	2.896	
10,475.0	10,471.8	10,433.4	10,429.1	23.1	22.7	21.98	-132.0	3.3	131.8	87.2	44.66	2.952	
10,500.0	10,495.5	10,452.9	10,447.4	23.1	22.8	22.43	-137.8	6.7	133.3	89.0	44.30	3.008	
10,525.0	10,518.8	10,475.0	10,467.8	23.2	22.8	23.06	-145.3	11.0	134.6	90.7	43.90	3.065	
10,550.0	10,541.7	10,491.9	10,483.1	23.2	22.9	23.63	-151.5	14.6	135.6	92.2	43.44	3.122	
10,575.0	10,563.9	10,511.4	10,500.4	23.3	22.9	24.38	-159.2	19.1	136.5	93.6	42.94	3.180	
10,600.0	10,585.5	10,530.8	10,517.2	23.3	23.0	25.22	-167.6	23.9	137.3	94.9	42.42	3.237	
10,625.0	10,606.5	10,550.0	10,533.5	23.4	23.0	26.16	-176.4	29.0	137.9	96.0	41.88	3.293	
10,650.0	10,626.7	10,569.6	10,549.7	23.4	23.1	27.21	-186.0	34.5	138.4	97.1	41.35	3.348	
10,675.0	10,646.1	10,589.0	10,565.2	23.5	23.1	28.34	-196.0	40.3	138.8	98.0	40.82	3.400	
10,700.0	10,664.6	10,608.3	10,580.3	23.6	23.2	29.57	-206.5	46.4	139.1	98.7	40.33	3.448	
10,725.0	10,682.3	10,627.7	10,594.8	23.6	23.3	30.89	-217.6	52.8	139.2	99.4	39.89	3.491	
10,750.0	10,698.9	10,647.1	10,608.8	23.7	23.3	32.31	-229.2	59.5	139.4	99.9	39.51	3.527	
10,775.0	10,714.6	10,666.5	10,622.3	23.8	23.4	33.82	-241.3	66.4	139.5	100.2	39.22	3.556	
10,800.0	10,729.3	10,685.9	10,635.2	23.9	23.5	35.43	-253.8	73.7	139.5	100.5	39.04	3.574	
10,825.0	10,742.9	10,705.3	10,647.5	24.0	23.6	37.12	-266.8	81.2	139.6	100.7	38.97	3.583	
10,850.0	10,755.3	10,725.0	10,659.4	24.2	23.7	38.93	-280.5	89.1	139.8	100.7	39.05	3.579	
10,875.0	10,766.6	10,744.3	10,670.4	24.3	23.8	40.77	-294.2	97.0	139.9	100.7	39.25	3.566	
10,900.0	10,776.8	10,763.9	10,680.8	24.5	23.9	42.72	-308.5	105.3	140.2	100.6	39.59	3.541	
10,925.0	10,785.7	10,783.5	10,690.7	24.7	24.0	44.73	-323.2	113.8	140.6	100.5	40.09	3.507	
10,950.0	10,793.4	10,803.2	10,699.8	24.9	24.1	46.81	-338.4	122.5	141.1	100.4	40.71	3.466	
10,975.0	10,799.8	10,825.0	10,709.1	25.1	24.3	49.18	-355.4	132.4	141.8	100.2	41.55	3.413	
11,000.0	10,804.9	10,842.9	10,716.0	25.3	24.4	51.14	-369.7	140.6	142.7	100.3	42.33	3.370	
11,025.0	10,808.8	10,862.9	10,723.0	25.5	24.6	53.37	-385.9	150.0	143.7	100.5	43.28	3.321	
11,050.0	10,811.4	10,883.0	10,729.2	25.8	24.7	55.62	-402.5	159.5	145.0	100.7	44.30	3.274	
11,075.0	10,812.6	10,903.2	10,734.7	26.0	24.9	57.89	-419.4	169.3	146.6	101.2	45.37	3.231	
11,084.1	10,812.8	10,910.6	10,736.5	26.1	25.0	58.72	-425.6	172.8	147.2	101.5	45.77	3.217	
11,099.1	10,812.8	10,922.9	10,739.2	26.3	25.1	60.15	-436.0	178.8	148.5	102.0	46.44	3.197	
11,200.0	10,813.4	11,007.7	10,749.5	27.5	25.9	66.71	-508.7	220.9	166.0	115.9	50.07	3.315	
11,300.0	10,813.9	11,108.1	10,750.2	28.9	27.0	70.02	-595.0	272.2	192.9	139.8	53.12	3.632	
11,400.0	10,814.5	11,214.9	10,750.8	30.6	28.4	72.61	-683.7	331.6	220.4	164.1	56.32	3.913	
11,500.0	10,815.0	11,323.6	10,751.4	32.4	30.0	74.62	-770.6	396.9	247.8	188.1	59.64	4.155	
11,600.0	10,815.6	11,434.3	10,752.0	34.3	31.9	76.20	-855.0	468.5	274.9	211.8	63.09	4.358	
11,700.0	10,816.1	11,547.1	10,752.7	36.3	33.9	77.49	-936.6	546.3	301.6	235.0	66.65	4.525	
11,800.0	10,816.7	11,662.0	10,753.4	38.5	36.1	78.55	-1,014.9	630.5	327.8	257.5	70.31	4.662	
11,900.0	10,817.3	11,779.1	10,754.1	40.7	38.5	79.43	-1,089.3	720.9	353.2	279.2	74.03	4.771	
12,000.0	10,817.9	11,898.5	10,754.8	43.0	41.1	80.18	-1,159.3	817.5	377.8	300.0	77.78	4.857	
12,100.0	10,818.4	12,020.0	10,755.5	45.3	43.7	80.81	-1,224.3	920.2	401.4	319.8	81.55	4.922	
12,200.0	10,819.0	12,143.9	10,756.3	47.6	46.5	81.35	-1,283.6	1,028.9	423.9	338.6	85.31	4.969	
12,300.0	10,819.5	12,269.9	10,757.0	50.0	49.4	81.81	-1,336.6	1,143.2	445.1	356.1	89.00	5.001	
12,400.0	10,820.1	12,398.1	10,757.8	52.3	52.4	82.21	-1,382.7	1,262.9	465.0	372.4	92.62	5.021	
12,500.0	10,820.6	12,528.5	10,758.5	54.7	55.4	82.56	-1,421.2	1,387.4	483.5	387.3	96.14	5.029	
12,600.0	10,821.2	12,660.9	10,759.3	57.0	58.5	82.85	-1,451.5	1,516.2	500.3	400.8	99.53	5.027	
12,655.6	10,821.5	12,735.3	10,759.7	58.3	60.2	83.00	-1,464.6	1,589.5	508.9	407.6	101.34	5.022	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6

Offset Site Error: 0.0 usft

Survey Program: 0-MWD

Offset Well Error: 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
12,700.0	10,821.7	12,795.2	10,760.1	59.4	61.6	83.11	-1,473.1	1,648.8	514.9	410.9	104.00	4.951	
12,800.0	10,822.2	12,931.7	10,760.8	61.8	64.7	83.27	-1,485.4	1,784.7	523.4	413.2	110.13	4.752	
12,900.0	10,822.7	13,058.8	10,761.5	64.2	67.5	83.30	-1,488.3	1,911.7	525.0	408.9	116.06	4.523	
13,000.0	10,823.3	13,158.8	10,762.0	66.6	69.8	83.29	-1,488.6	2,011.7	524.7	403.4	121.35	4.324	
13,100.0	10,823.8	13,258.8	10,762.5	69.2	72.1	83.29	-1,488.8	2,111.7	524.4	397.8	126.68	4.140	
13,200.0	10,824.3	13,358.8	10,763.1	71.7	74.5	83.29	-1,489.1	2,211.7	524.2	392.1	132.06	3.969	
13,300.0	10,824.8	13,458.8	10,763.6	74.3	76.9	83.28	-1,489.4	2,311.7	523.9	386.4	137.47	3.811	
13,400.0	10,825.4	13,558.8	10,764.1	76.8	79.3	83.28	-1,489.6	2,411.7	523.7	380.7	142.92	3.664	
13,500.0	10,825.9	13,658.8	10,764.6	79.5	81.8	83.28	-1,489.9	2,511.7	523.4	375.0	148.39	3.527	
13,600.0	10,826.4	13,758.8	10,765.2	82.1	84.3	83.27	-1,490.2	2,611.7	523.1	369.2	153.89	3.399	
13,700.0	10,826.9	13,858.8	10,765.7	84.7	86.8	83.27	-1,490.4	2,711.7	522.9	363.5	159.42	3.280	
13,800.0	10,827.5	13,958.8	10,766.2	87.4	89.3	83.27	-1,490.7	2,811.7	522.6	357.6	164.96	3.168	
13,900.0	10,828.0	14,058.8	10,766.7	90.1	91.9	83.26	-1,491.0	2,911.7	522.3	351.8	170.52	3.063	
14,000.0	10,828.5	14,158.8	10,767.3	92.8	94.5	83.26	-1,491.2	3,011.7	522.1	346.0	176.10	2.965	
14,100.0	10,829.0	14,258.8	10,767.8	95.5	97.1	83.26	-1,491.5	3,111.7	521.8	340.1	181.69	2.872	
14,200.0	10,829.6	14,358.8	10,768.3	98.3	99.7	83.25	-1,491.7	3,211.7	521.6	334.3	187.30	2.785	
14,300.0	10,830.1	14,458.8	10,768.8	101.0	102.4	83.25	-1,492.0	3,311.7	521.3	328.4	192.92	2.702	
14,400.0	10,830.6	14,558.8	10,769.3	103.8	105.0	83.25	-1,492.3	3,411.7	521.0	322.5	198.55	2.624	
14,500.0	10,831.1	14,658.8	10,769.9	106.5	107.7	83.24	-1,492.5	3,511.7	520.8	316.6	204.19	2.550	
14,600.0	10,831.6	14,758.8	10,770.4	109.3	110.4	83.24	-1,492.8	3,611.6	520.5	310.7	209.84	2.480	
14,700.0	10,832.2	14,858.8	10,770.9	112.1	113.1	83.24	-1,493.1	3,711.6	520.2	304.7	215.50	2.414	
14,800.0	10,832.7	14,958.8	10,771.4	114.9	115.8	83.23	-1,493.3	3,811.6	520.0	298.8	221.17	2.351	
14,900.0	10,833.2	15,058.8	10,772.0	117.6	118.5	83.23	-1,493.6	3,911.6	519.7	292.9	226.85	2.291	
15,000.0	10,833.7	15,158.8	10,772.5	120.4	121.3	83.23	-1,493.9	4,011.6	519.4	286.9	232.53	2.234	
15,100.0	10,834.3	15,258.8	10,773.0	123.2	124.0	83.22	-1,494.1	4,111.6	519.2	281.0	238.22	2.179	
15,200.0	10,834.8	15,358.8	10,773.5	126.1	126.7	83.22	-1,494.4	4,211.6	518.9	275.0	243.92	2.127	
15,300.0	10,835.3	15,458.8	10,774.1	128.9	129.5	83.22	-1,494.6	4,311.6	518.7	269.0	249.62	2.078	
15,400.0	10,835.8	15,558.8	10,774.6	131.7	132.3	83.21	-1,494.9	4,411.6	518.4	263.1	255.33	2.030	
15,500.0	10,836.4	15,658.8	10,775.1	134.5	135.0	83.21	-1,495.2	4,511.6	518.1	257.1	261.04	1.985	
15,600.0	10,836.9	15,758.8	10,775.6	137.4	137.8	83.21	-1,495.4	4,611.6	517.9	251.1	266.76	1.941	
15,700.0	10,837.4	15,858.8	10,776.2	140.2	140.6	83.20	-1,495.7	4,711.6	517.6	245.1	272.48	1.900	
15,800.0	10,837.9	15,958.8	10,776.7	143.0	143.4	83.20	-1,496.0	4,811.6	517.3	239.1	278.20	1.860	
15,900.0	10,838.5	16,058.7	10,777.2	145.9	146.2	83.20	-1,496.2	4,911.6	517.1	233.1	283.93	1.821	
16,000.0	10,839.0	16,158.7	10,777.7	148.7	149.0	83.19	-1,496.5	5,011.6	516.8	227.1	289.66	1.784	
16,100.0	10,839.5	16,258.7	10,778.2	151.6	151.8	83.19	-1,496.8	5,111.6	516.5	221.1	295.40	1.749	
16,200.0	10,840.0	16,358.7	10,778.8	154.4	154.6	83.18	-1,497.0	5,211.6	516.3	215.1	301.14	1.714	
16,300.0	10,840.5	16,458.7	10,779.3	157.3	157.4	83.18	-1,497.3	5,311.6	516.0	209.1	306.88	1.682	
16,400.0	10,841.1	16,558.7	10,779.8	160.1	160.3	83.18	-1,497.6	5,411.6	515.8	203.1	312.62	1.650	
16,500.0	10,841.6	16,658.7	10,780.3	163.0	163.1	83.17	-1,497.8	5,511.6	515.5	197.1	318.37	1.619	
16,600.0	10,842.1	16,758.7	10,780.9	165.9	165.9	83.17	-1,498.1	5,611.6	515.2	191.1	324.12	1.590	
16,700.0	10,842.6	16,858.7	10,781.4	168.7	168.7	83.17	-1,498.3	5,711.6	515.0	185.1	329.87	1.561	
16,800.0	10,843.2	16,958.7	10,781.9	171.6	171.6	83.16	-1,498.6	5,811.6	514.7	179.1	335.63	1.534	
16,900.0	10,843.7	17,058.7	10,782.4	174.5	174.4	83.16	-1,498.9	5,911.6	514.4	173.1	341.38	1.507	
17,000.0	10,844.2	17,158.7	10,783.0	177.3	177.3	83.16	-1,499.1	6,011.6	514.2	167.0	347.14	1.481	Level 3
17,100.0	10,844.7	17,258.7	10,783.5	180.2	180.1	83.15	-1,499.4	6,111.6	513.9	161.0	352.90	1.456	Level 3
17,200.0	10,845.3	17,358.7	10,784.0	183.1	182.9	83.15	-1,499.7	6,211.6	513.6	155.0	358.66	1.432	Level 3
17,300.0	10,845.8	17,458.7	10,784.5	186.0	185.8	83.15	-1,499.9	6,311.6	513.4	149.0	364.43	1.409	Level 3
17,400.0	10,846.3	17,558.7	10,785.1	188.8	188.6	83.14	-1,500.2	6,411.6	513.1	142.9	370.19	1.386	Level 3
17,500.0	10,846.8	17,658.7	10,785.6	191.7	191.5	83.14	-1,500.5	6,511.6	512.9	136.9	375.96	1.364	Level 3
17,600.0	10,847.4	17,758.7	10,786.1	194.6	194.4	83.14	-1,500.7	6,611.6	512.6	130.9	381.73	1.343	Level 3
17,700.0	10,847.9	17,858.7	10,786.6	197.5	197.2	83.13	-1,501.0	6,711.6	512.3	124.8	387.50	1.322	Level 3
17,800.0	10,848.4	17,958.7	10,787.1	200.4	200.1	83.13	-1,501.2	6,811.6	512.1	118.8	393.27	1.302	Level 3

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 21-18 5B - Wellbore #1 - Design #6 — **Offset Site Error:** 0.0 usft

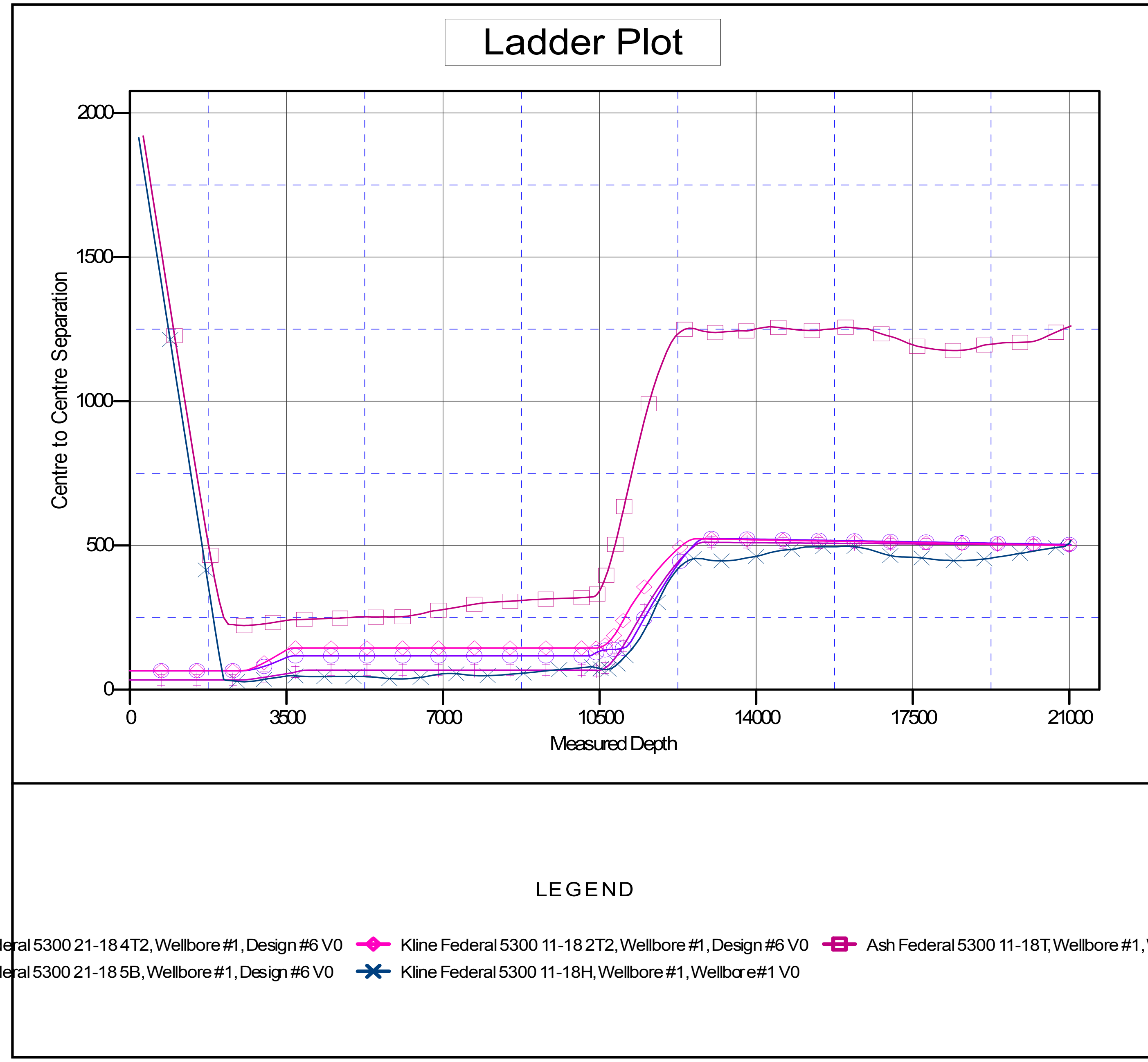
Survey Program: 0-MWD — **Offset Well Error:** 0.0 usft

Reference Measured Depth (usft)	Reference Vertical Depth (usft)	Offset Measured Depth (usft)	Offset Vertical Depth (usft)	Semi Major Axis Reference (usft)	Semi Major Axis Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
17,900.0	10,848.9	18,058.7	10,787.7	203.3	202.9	83.13	-1,501.5	6,911.6	511.8	112.8	399.04	1.283	Level 3
18,000.0	10,849.5	18,158.7	10,788.2	206.2	205.8	83.12	-1,501.8	7,011.6	511.5	106.7	404.82	1.264	Level 3
18,100.0	10,850.0	18,258.7	10,788.7	209.0	208.7	83.12	-1,502.0	7,111.6	511.3	100.7	410.59	1.245	Level 2
18,200.0	10,850.5	18,358.7	10,789.2	211.9	211.5	83.11	-1,502.3	7,211.6	511.0	94.6	416.37	1.227	Level 2
18,300.0	10,851.0	18,458.7	10,789.8	214.8	214.4	83.11	-1,502.6	7,311.6	510.7	88.6	422.14	1.210	Level 2
18,400.0	10,851.5	18,558.7	10,790.3	217.7	217.3	83.11	-1,502.8	7,411.6	510.5	82.6	427.92	1.193	Level 2
18,500.0	10,852.1	18,658.7	10,790.8	220.6	220.2	83.10	-1,503.1	7,511.6	510.2	76.5	433.70	1.176	Level 2
18,600.0	10,852.6	18,758.7	10,791.3	223.5	223.0	83.10	-1,503.4	7,611.6	510.0	70.5	439.48	1.160	Level 2
18,700.0	10,853.1	18,858.7	10,791.9	226.4	225.9	83.10	-1,503.6	7,711.6	509.7	64.4	445.26	1.145	Level 2
18,800.0	10,853.6	18,958.7	10,792.4	229.3	228.8	83.09	-1,503.9	7,811.6	509.4	58.4	451.04	1.129	Level 2
18,900.0	10,854.2	19,058.7	10,792.9	232.2	231.7	83.09	-1,504.2	7,911.6	509.2	52.3	456.83	1.115	Level 2
19,000.0	10,854.7	19,158.7	10,793.4	235.1	234.5	83.09	-1,504.4	8,011.6	508.9	46.3	462.61	1.100	Level 2
19,100.0	10,855.2	19,258.7	10,794.0	238.0	237.4	83.08	-1,504.7	8,111.6	508.6	40.2	468.39	1.086	Level 2
19,200.0	10,855.7	19,358.7	10,794.5	240.9	240.3	83.08	-1,504.9	8,211.6	508.4	34.2	474.18	1.072	Level 2
19,300.0	10,856.3	19,458.7	10,795.0	243.8	243.2	83.07	-1,505.2	8,311.6	508.1	28.2	479.96	1.059	Level 2
19,400.0	10,856.8	19,558.7	10,795.5	246.7	246.1	83.07	-1,505.5	8,411.5	507.9	22.1	485.75	1.045	Level 2
19,500.0	10,857.3	19,658.7	10,796.0	249.6	249.0	83.07	-1,505.7	8,511.5	507.6	16.1	491.54	1.033	Level 2
19,600.0	10,857.8	19,758.7	10,796.6	252.5	251.9	83.06	-1,506.0	8,611.5	507.3	10.0	497.32	1.020	Level 2
19,700.0	10,858.4	19,858.7	10,797.1	255.4	254.7	83.06	-1,506.3	8,711.5	507.1	3.9	503.11	1.008	Level 2
19,800.0	10,858.9	19,958.7	10,797.6	258.3	257.6	83.06	-1,506.5	8,811.5	506.8	-2.1	508.90	0.996	Level 1
19,900.0	10,859.4	20,058.7	10,798.1	261.2	260.5	83.05	-1,506.8	8,911.5	506.5	-8.2	514.69	0.984	Level 1
20,000.0	10,859.9	20,158.7	10,798.7	264.1	263.4	83.05	-1,507.1	9,011.5	506.3	-14.2	520.48	0.973	Level 1
20,100.0	10,860.4	20,258.7	10,799.2	267.0	266.3	83.05	-1,507.3	9,111.5	506.0	-20.3	526.27	0.961	Level 1
20,200.0	10,861.0	20,358.7	10,799.7	269.9	269.2	83.04	-1,507.6	9,211.5	505.7	-26.3	532.06	0.951	Level 1
20,300.0	10,861.5	20,458.7	10,800.2	272.8	272.1	83.04	-1,507.8	9,311.5	505.5	-32.4	537.85	0.940	Level 1
20,400.0	10,862.0	20,558.7	10,800.8	275.7	275.0	83.03	-1,508.1	9,411.5	505.2	-38.4	543.64	0.929	Level 1
20,500.0	10,862.5	20,658.7	10,801.3	278.6	277.9	83.03	-1,508.4	9,511.5	505.0	-44.5	549.44	0.919	Level 1
20,600.0	10,863.1	20,758.7	10,801.8	281.6	280.8	83.03	-1,508.6	9,611.5	504.7	-50.5	555.23	0.909	Level 1
20,700.0	10,863.6	20,858.7	10,802.3	284.5	283.7	83.02	-1,508.9	9,711.5	504.4	-56.6	561.02	0.899	Level 1
20,800.0	10,864.1	20,958.7	10,802.9	287.4	286.6	83.02	-1,509.2	9,811.5	504.2	-62.6	566.81	0.889	Level 1
20,900.0	10,864.6	21,058.7	10,803.4	290.3	289.5	83.02	-1,509.4	9,911.5	503.9	-68.7	572.61	0.880	Level 1
21,000.0	10,865.2	21,158.7	10,803.9	293.2	292.4	83.01	-1,509.7	10,011.5	503.6	-74.8	578.40	0.871	Level 1
21,038.5	10,865.4	21,197.2	10,804.1	294.3	293.5	83.01	-1,509.8	10,050.0	503.5	-77.1	580.63	0.867	Level 1, ES, SF

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Reference Depths are relative to WELL @ 2078.0usft (Original Well Ele	Coordinates are relative to: Kline Federal 5300 11-18 3T
Offset Depths are relative to Offset Datum	Coordinate System is US State Plane 1983, North Dakota Northern Zone
Central Meridian is 100° 30' 0.000 W	Grid Convergence at Surface is: -2.31°



Company:	Oasis Petroleum	**Local Co-ordinate Reference:**	Well Kline Federal 5300 11-18 3T
Project:	Indian Hills	**TVD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	**MD Reference:**	WELL @ 2078.0usft (Original Well Elev)
Site Error:	0.0 usft	**North Reference:**	True
Reference Well:	Kline Federal 5300 11-18 3T	**Survey Calculation Method:**	Minimum Curvature
Well Error:	0.0 usft	**Output errors are at**	2.00 sigma
Reference Wellbore	Wellbore #1	**Database:**	EDM 5000.1 Single User Db
Reference Design:	Design #6	**Offset TVD Reference:**	Offset Datum

Reference Depths are relative to WELL @ 2078.0usft (Original Well Ele	Coordinates are relative to: Kline Federal 5300 11-18 3T
Offset Depths are relative to Offset Datum	Coordinate System is US State Plane 1983, North Dakota Northern Zone
Central Meridian is 100° 30' 0.000 W	Grid Convergence at Surface is: -2.31°